DSA using Java

Course Notes

Contains 23 PDF lessons

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Module

Java YouTube Course

Topic

Java

Subtopic

Java

Lesson

Java_Notes

Lesson 1 of 23

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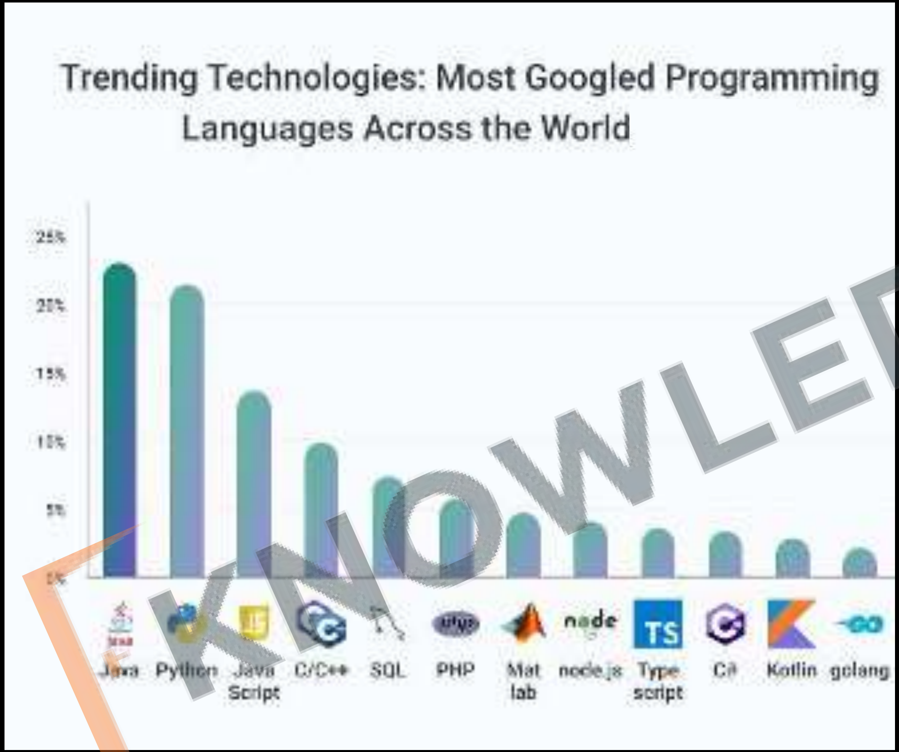
1. Introduction to Java

1. Why you must learn Java
2. What is a Programming Language
3. What is an Algorithm
4. What is Syntax
5. History of Java
6. Magic of Byte Code
7. How Java Changed the Internet
8. Java Buzzwords
9. Object Oriented Programming

1.1 Why you must learn Java



1. One of the most popular language. Java currently runs on 60,00,00,00,000 devices.
2. Wide Usage (Web-apps, backend, Mobile apps, enterprise software).
3. High paying and a lot of Jobs
4. Object Oriented
5. Rich APIs and Community Support

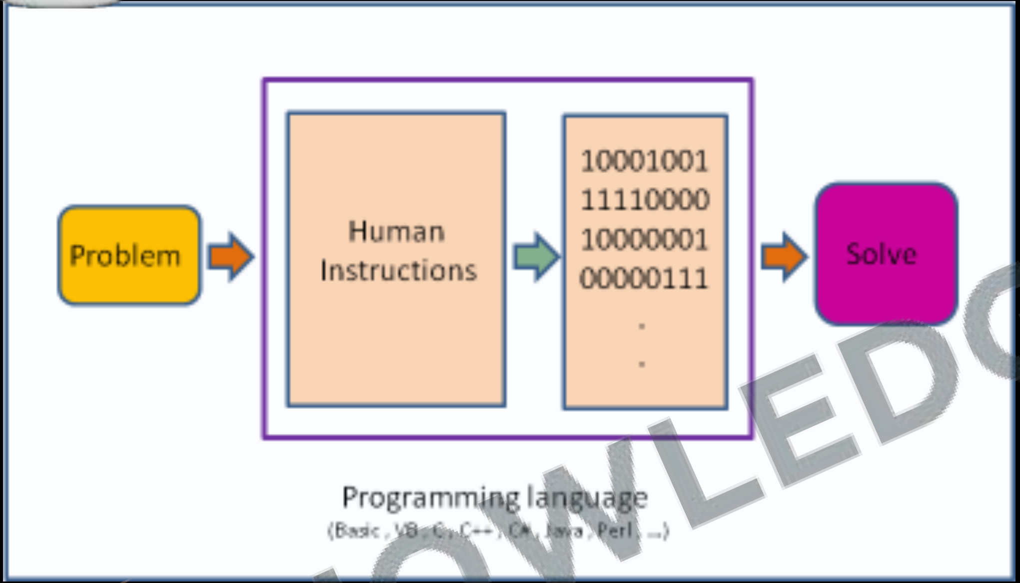
1.2 What is a Programming Language

Humans use natural language (like Hindi/English) to communicate

1.2 What is a Programming Language

Computers only understand 0/1 or on/off

1.2 What is a Programming Language



- Giving instructions to a computer
- **Instructions**: Tells computer what to do. These instructions are called code.
- Human instructions are given in High level languages.

Compiler converts high level languages to low level languages or machine code.

1.3 What is an Algorithm

1.3 What is an Algorithm

An algorithm is a step-by-step procedure for solving a problem or performing a task.

1.4 What is Syntax

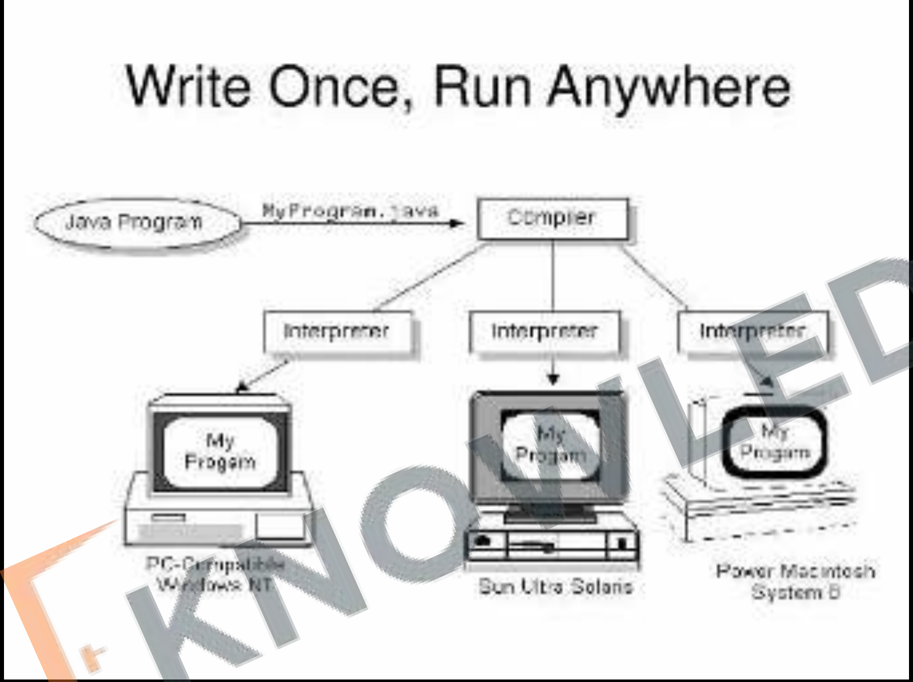
- Structure of words in a sentence.
- Rules of the language.
- For programming exact syntax must be followed.

1.5 History of Java

Developed by James Gosling at Sun Microsystems (Early 1990s):

Originally named 'Oak', later renamed Java in 1995.

1.5 History of Java

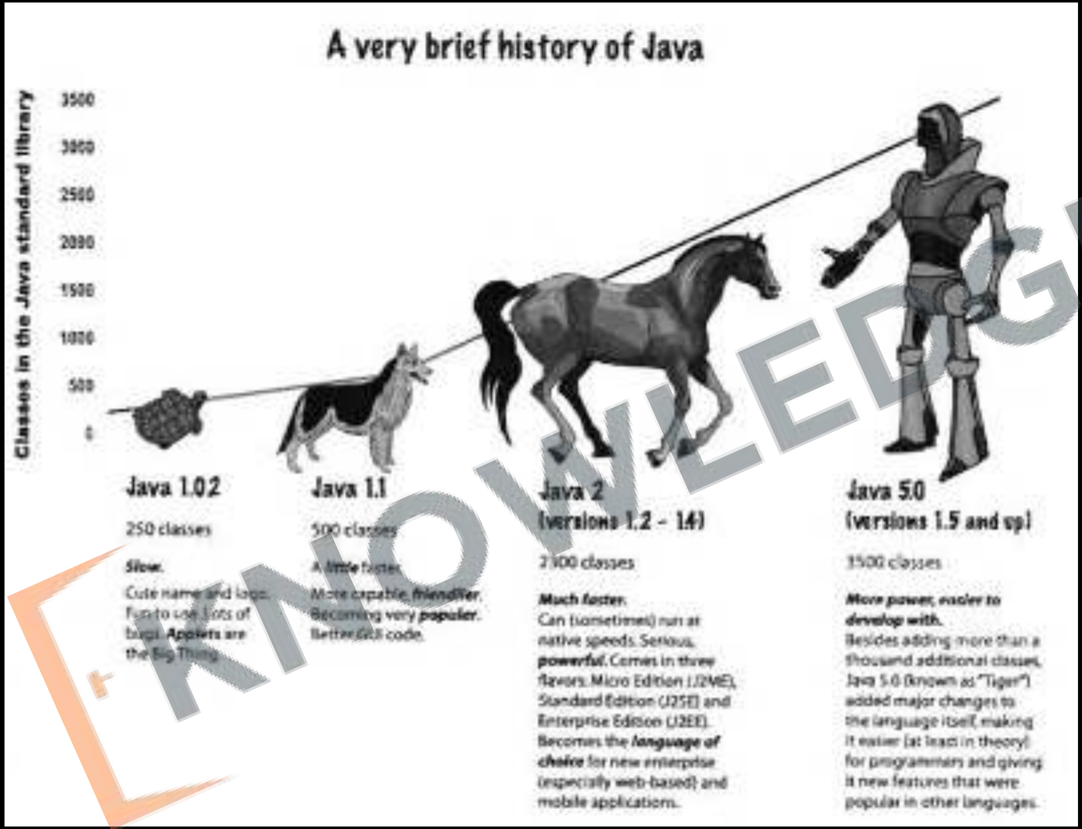


First Release (1995): Introduced "Write Once, Run Anywhere" concept with cross-platform compatibility.

1.5 History of Java

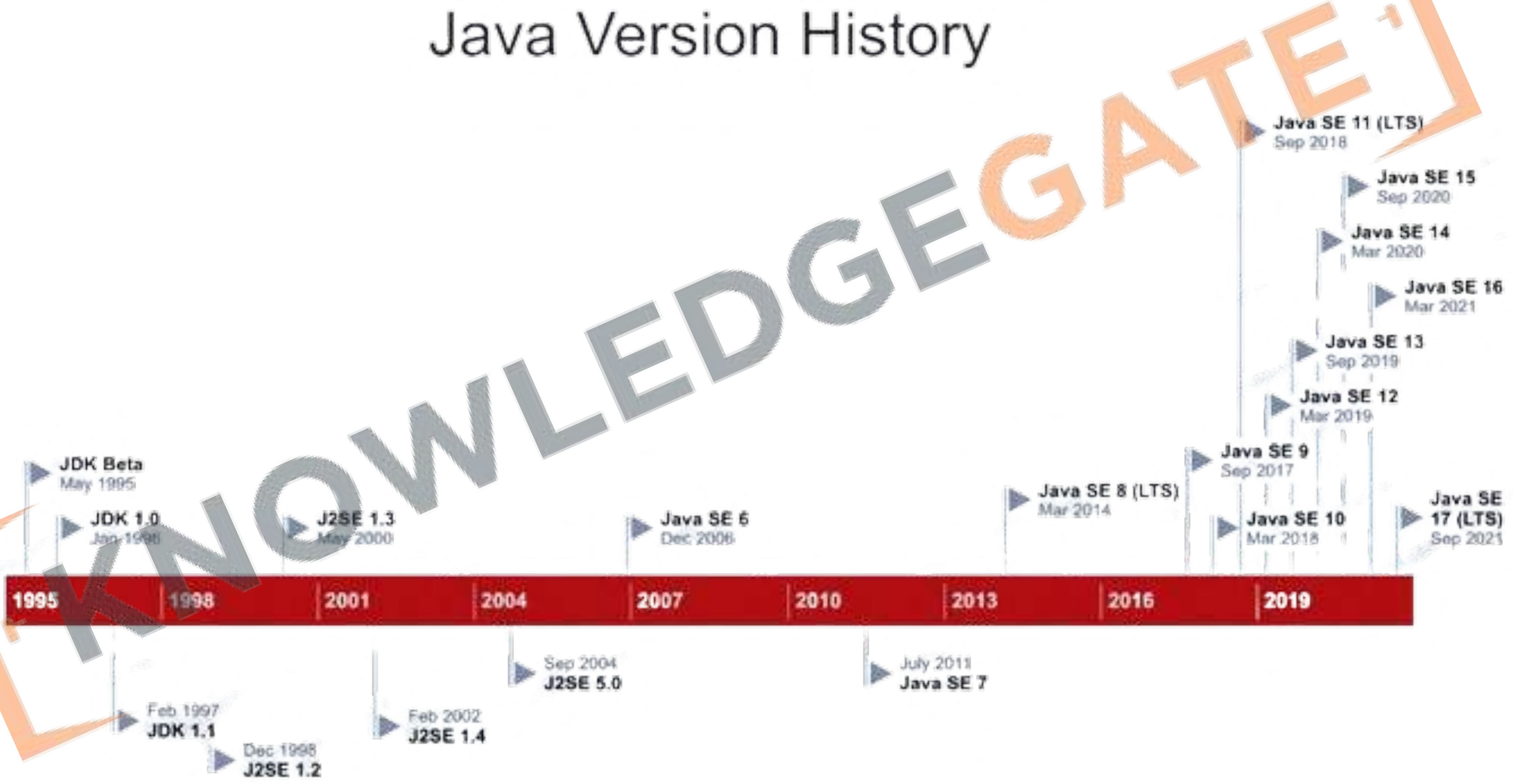
Developed with vision of backward compatibility. Should not break with new version release.

1.5 History of Java

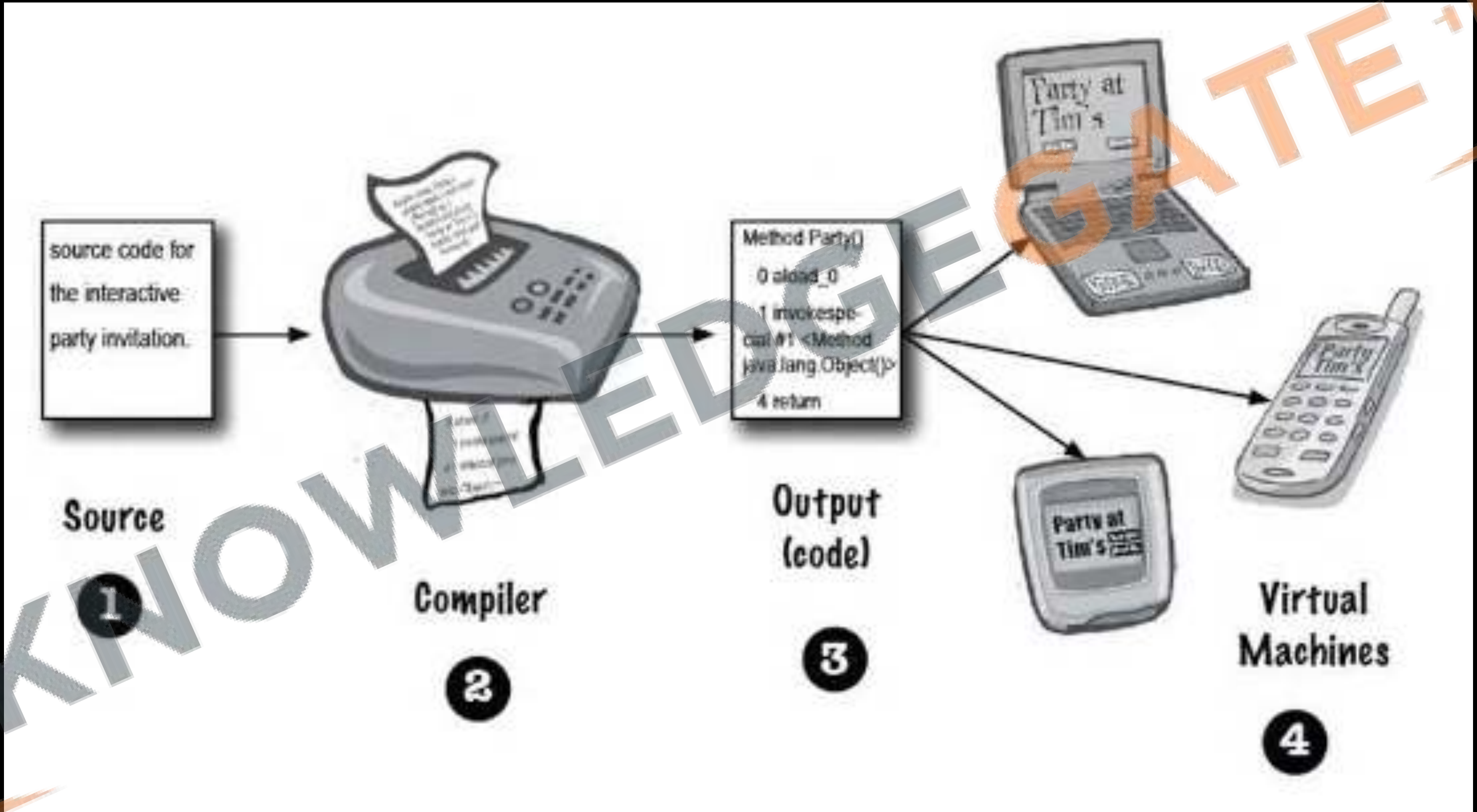


Rapid Growth and Diversification (Late 1990s - 2010s): Expanded from web applets to server-side applications; standardized into different editions for various computing platforms.

1.5 History of Java



1.6 Magic of Byte Code



1.7 How Java Changed the Internet

Portability with Write Once Run Anywhere

Security because of Code running on Virtual Machine

1.8 Java Buzzwords

Robust

Java is robust due to its strong memory management, exception handling, and type-checking mechanisms, which help in preventing system crashes and ensuring reliable performance.

1.8 Java Buzzwords

Multithreaded

Multithreading in programming is the ability of a CPU to execute multiple threads concurrently, allowing for more efficient processing and task management.

1.8 Java Buzzwords

Architecture Neutral

Java is architecturally neutral because its compiled code (bytecode) can run on any device with a Java Virtual Machine (JVM), regardless of the underlying hardware architecture.

1.8 Java Buzzwords

Interpreted and High Performance

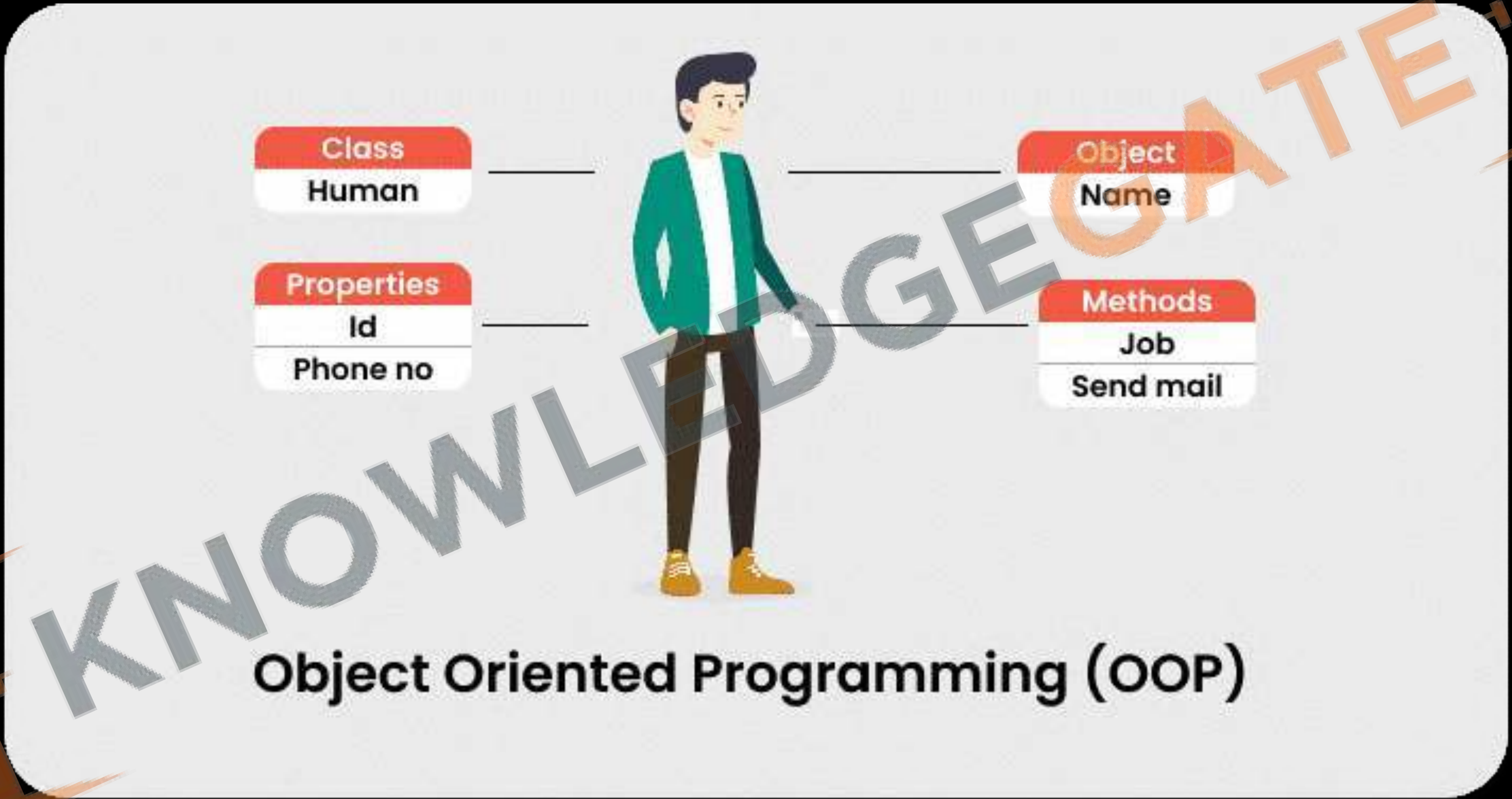
Java combines high performance with interpretability, as its bytecode is interpreted by the Java Virtual Machine (JVM), which employs Just-In-Time (JIT) compilation for efficient and fast execution.

1.8 Java Buzzwords

Distributed

Java is inherently distributed, designed to facilitate network-based application development and interaction, seamlessly integrating with Internet protocols and remote method invocation.

1.9 Object Oriented Programming



1.9 Object Oriented Programming

Revision

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2. Java Basics

1. Installing JDK
2. First Class using Text Editor
3. Compiling and Running
4. Anatomy of a Class
5. File Extensions
6. JDK vs JVM vs JRE
7. Showing Output
8. Importance of the main method
9. Installing IDE(Intellij Idea)
10. Creating first Project

2.1 Installing JDK

- Search JDK Download
- Make sure to download from the oracle website.
- Download the latest version

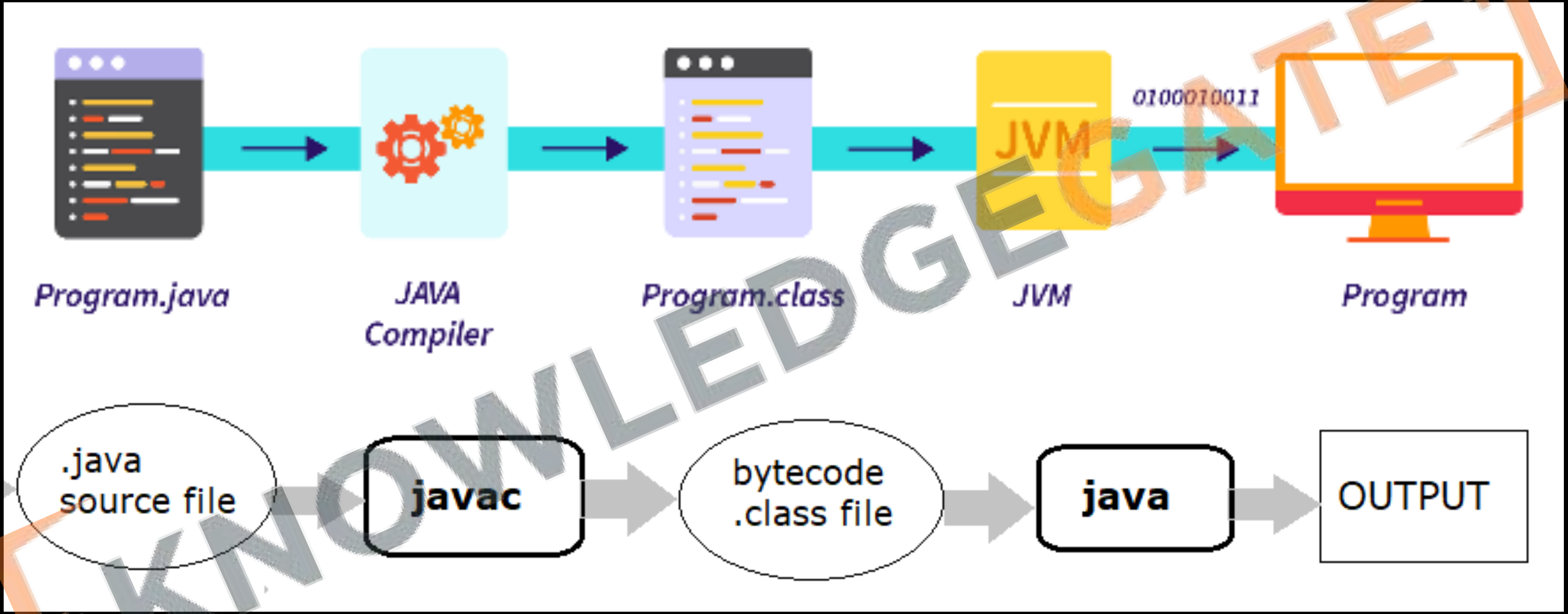
2.2 First Class using Text Editor

First.java

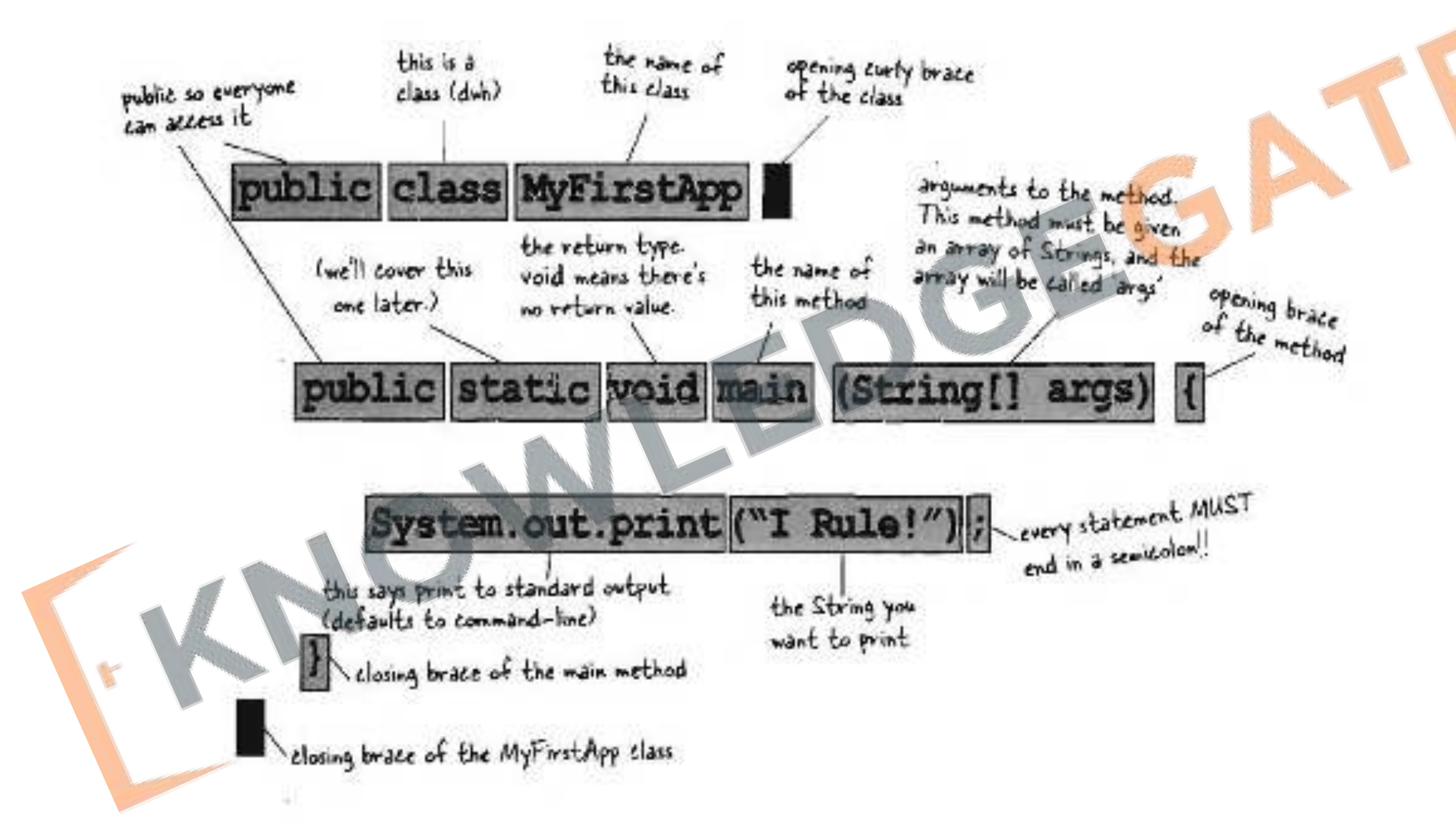
```
import java.lang.*;

public class First {
  public static void main(String[] args) {
    System.out.println("Welcome to KGCoding");
  }
}
```

2.3 Compiling and Running



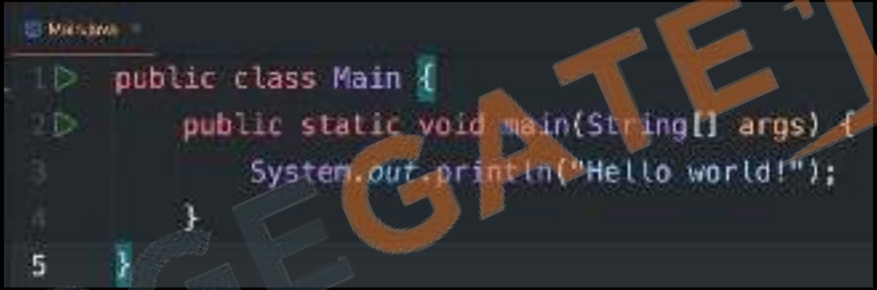
2.4 Anatomy of a Class



2.5 File Extensions

.Java

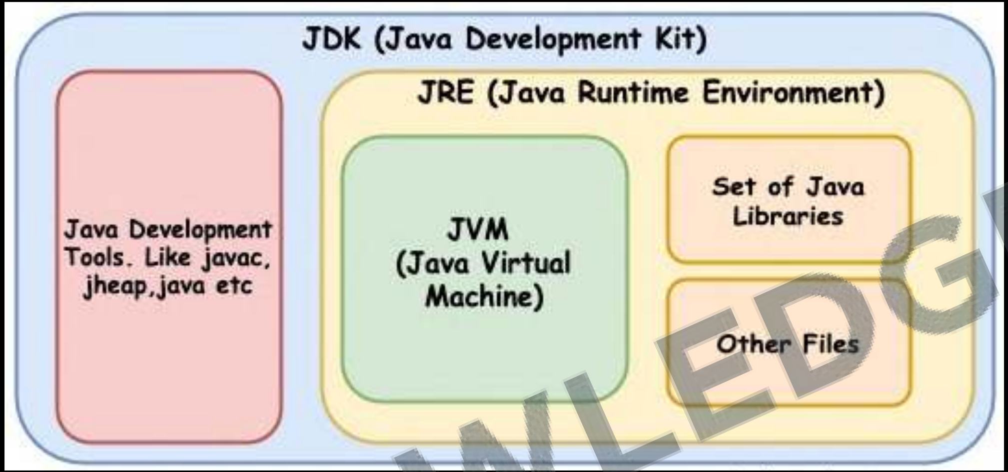
- Contains Java Source Code
- High Level Human Readable
- Used for Development
- File is editable



.Class

- Contains Java Bytecode
- For consumption of JVM
- Used for Execution
- Not meant to be edited

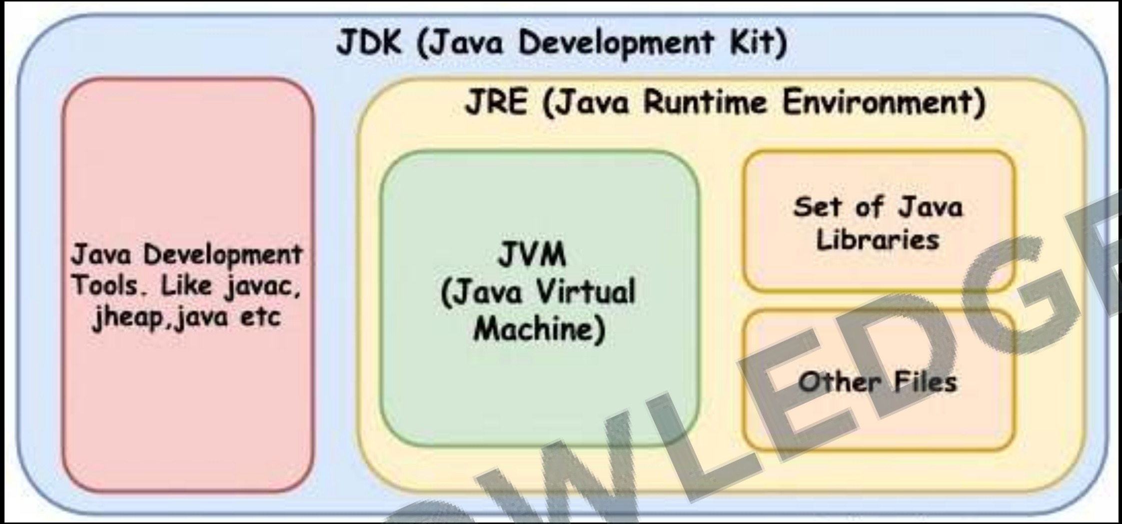
2.6 JDK vs JVM vs JRE



.JDK

- It's a software development kit required to develop Java applications.
- Includes the JRE, an interpreter/loader (Java), a compiler (javac), a doc generator (Javadoc), and other tools needed for Java development.
- Essentially, JDK is a superset of JRE.

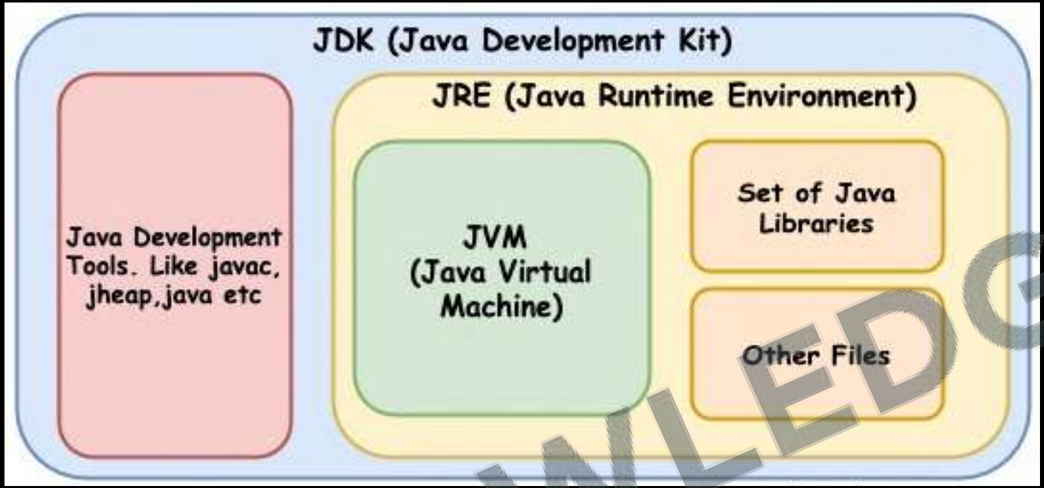
2.6 JDK vs JVM vs JRE



.JRE

- It's a part of the JDK but can be downloaded separately.
- Provides the libraries, the JVM, and other components to run applications
- Does not have tools and utilities for developers like compilers or debuggers.

2.6 JDK vs JVM vs JRE



.JVM

- It's a part of JRE and responsible for executing the bytecode.
- Ensures Java's write-once-run-anywhere capability.
- Not platform-independent: a different JVM is needed for each type of OS.

2.7 Showing Output

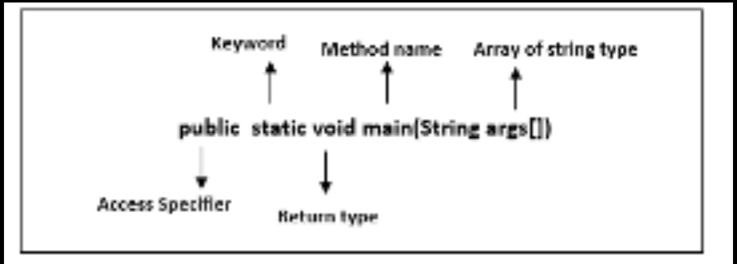
Example	Result

`System.out.print("one");`
`System.out.print("two");`
`System.out.println("three");`
	onetwothree

`System.out.println("one");`
`System.out.println("two");`
`System.out.println("three");`
	one
	two
	three

`System.out.println();`	[new line]

2.8 Importance of the main method



- Entry Point: It's the entry point of a Java program, where the execution starts. Without the main method, the Java Virtual Machine (JVM) does not know where to begin running the code.
- Public and Static: The main method must be public and static, ensuring it's accessible to the JVM without needing to instantiate the class.
- Fixed Signature: The main method has a fixed signature: public static void main(String[] args). Deviating from this signature means the JVM won't recognize it as the starting point.

2.9 What is IDE

1. IDE stands for Integrated Development Environment.
2. Software suite that consolidates basic tools required for software development.
3. Central hub for coding, finding problems, and testing.
4. Designed to improve developer efficiency.

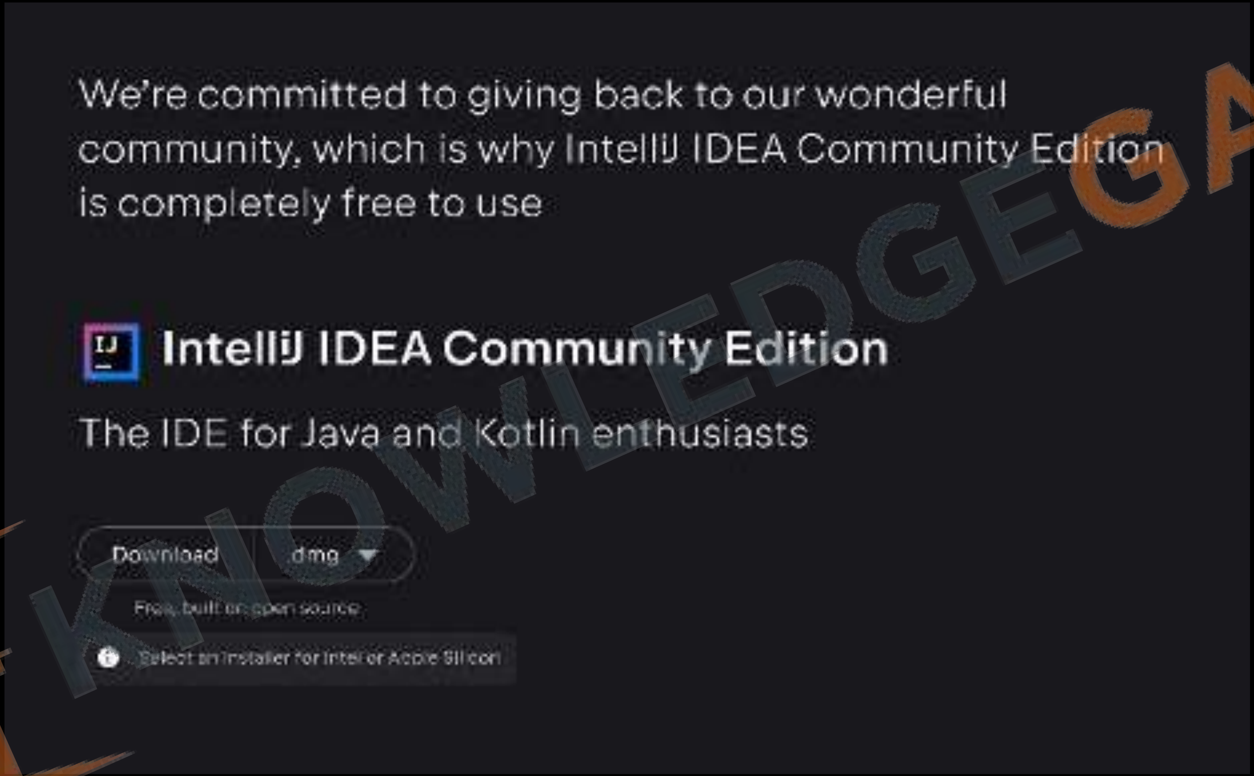
2.9 Need of IDE

1. Streamlines development.
2. Increases productivity.
3. Simplifies complex tasks.
4. Offers a unified workspace.
5. IDE Features
    1. Code Autocomplete
    2. Syntax Highlighting
    3. Version Control
    4. Error Checking

2.9 Installing IDE(Intellij Idea)

- Search Intellij IDEA
- Make sure to download the community Version.
- Keep Your Software up to date.

2.9 Installing IDE(Intellij Idea)



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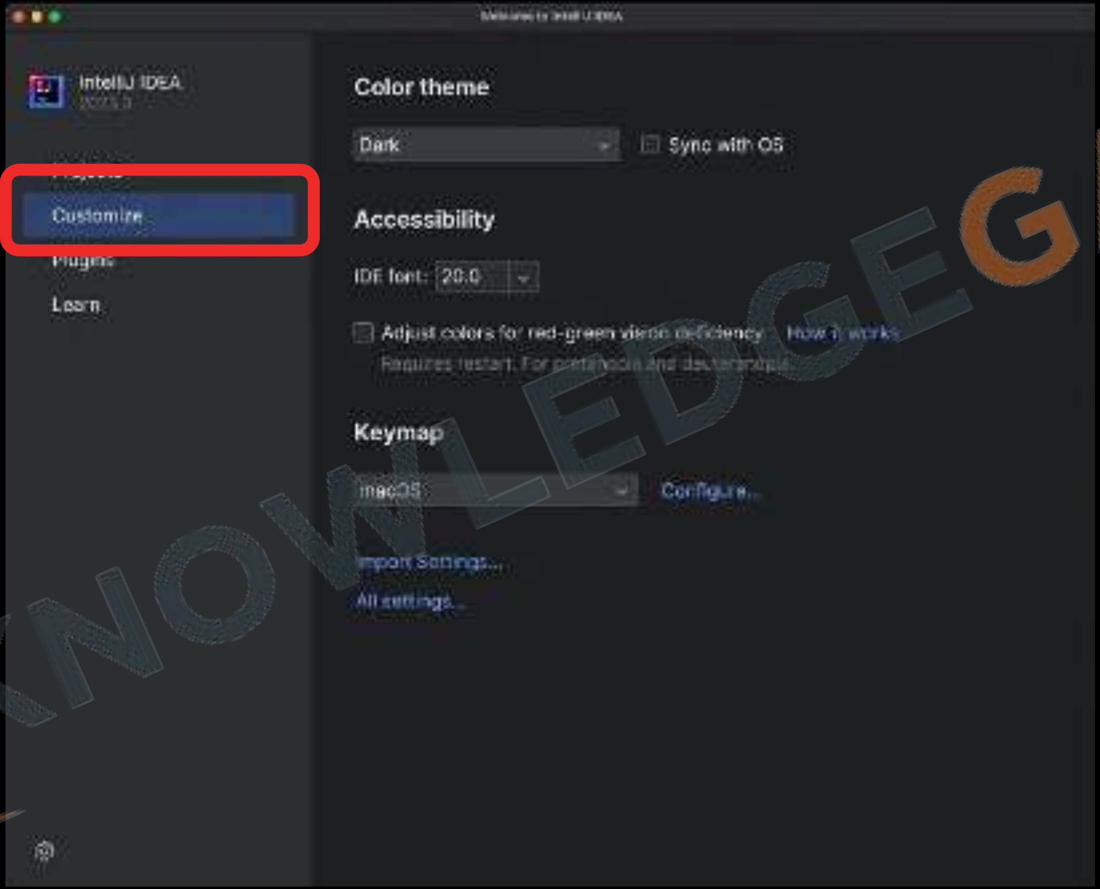

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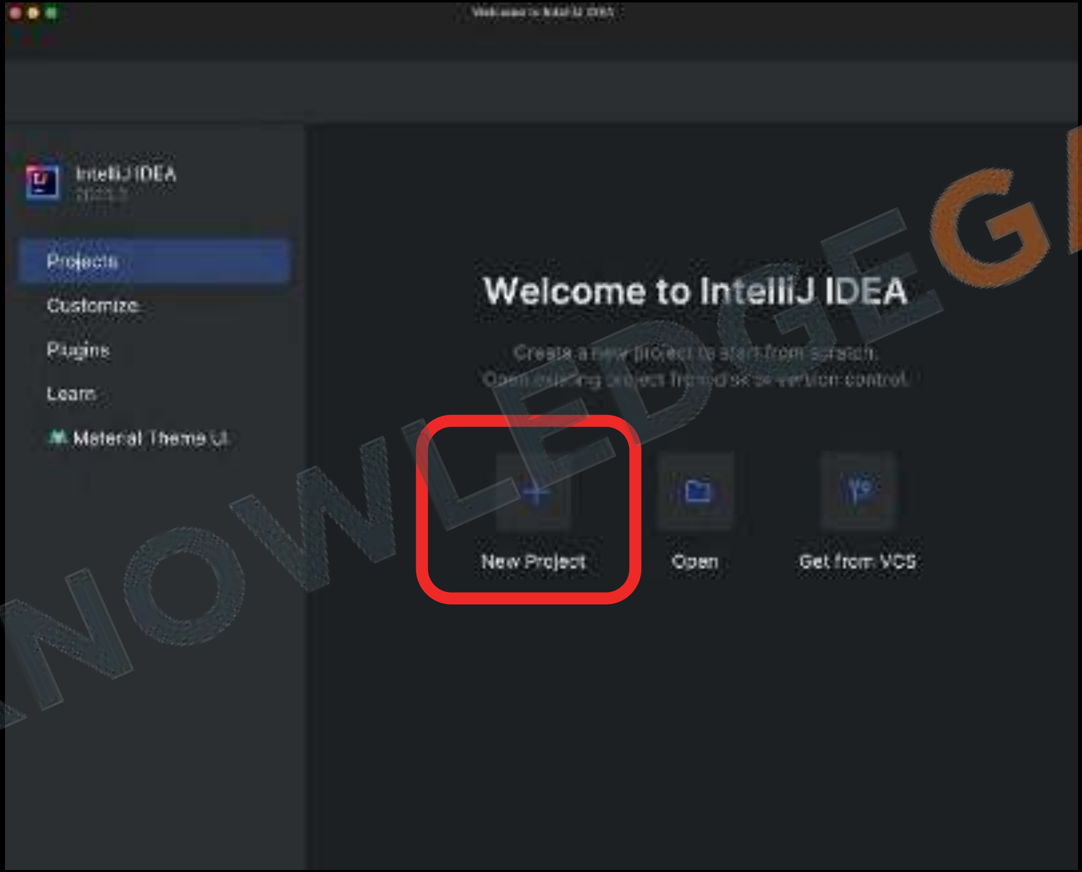

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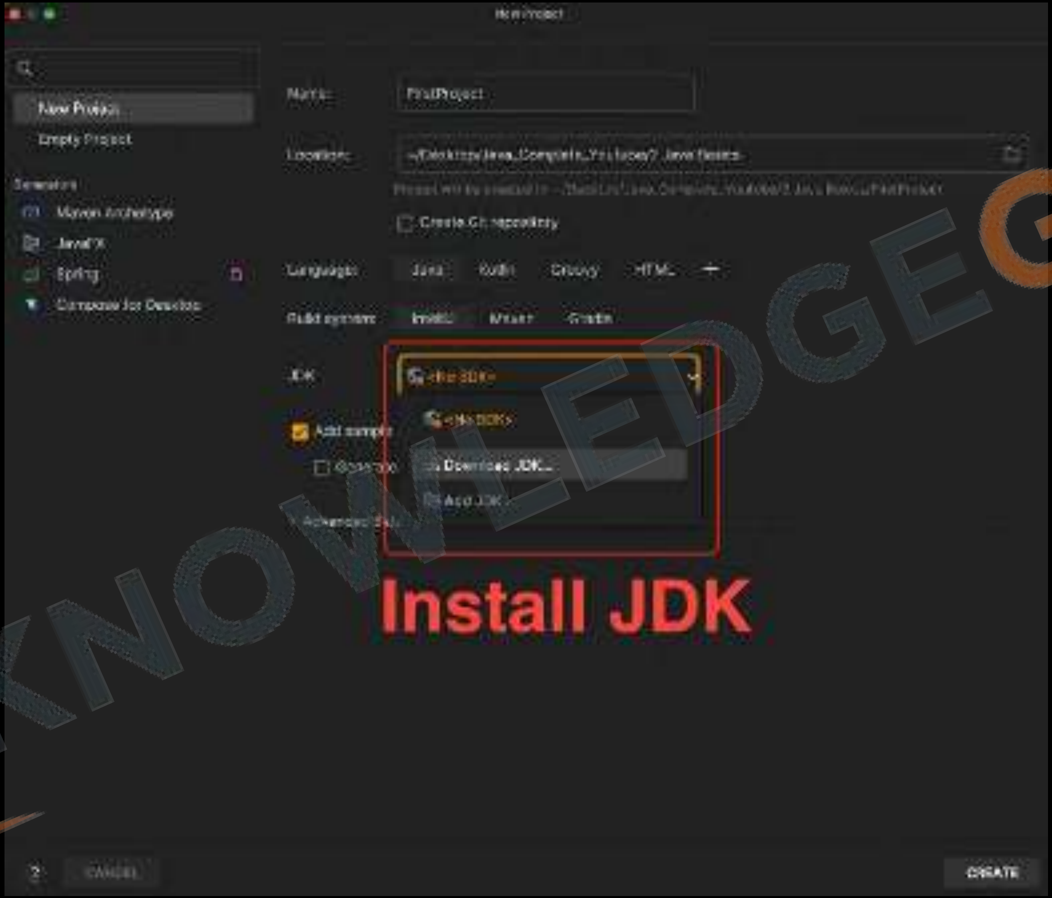
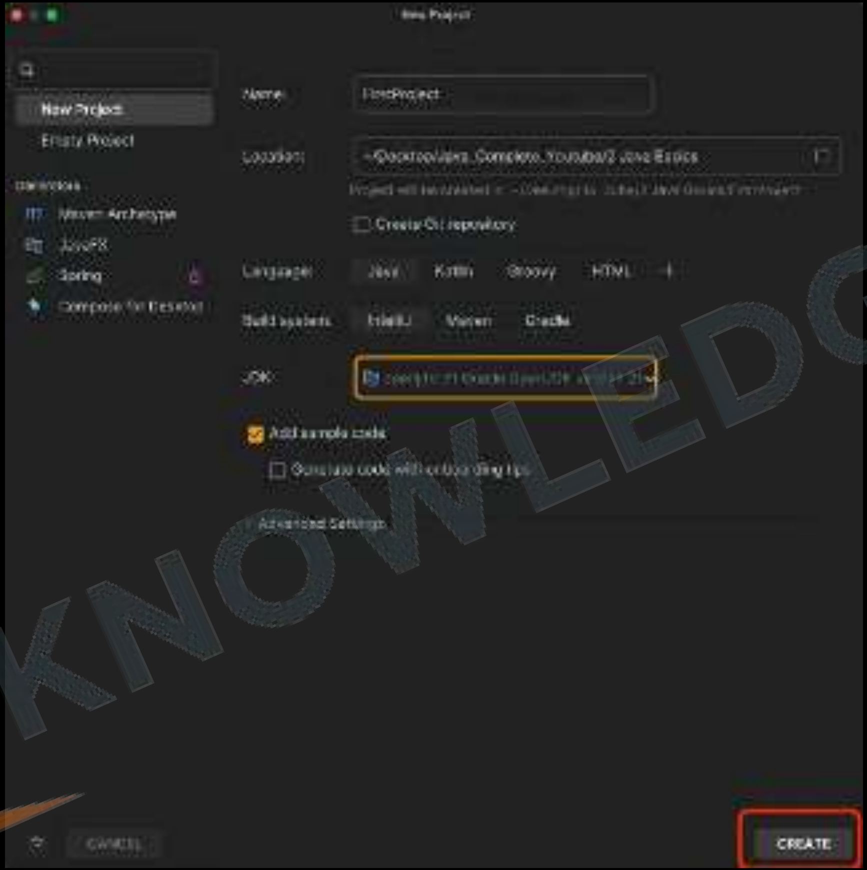


2.10 Creating first Project

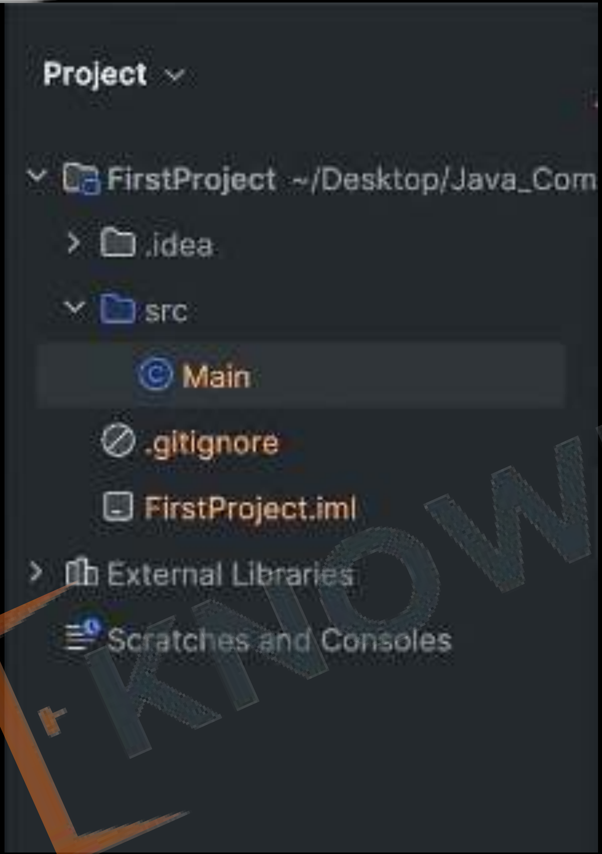


2.10 First Java Class

Main.java

```java
public class Main {
    public static void main(String[] args) {
        System.out.println("Hello world!");
    }
}
```

2.10 Project Structure



Revision

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CHALLENGE

Tasks:

1. Create a class to output "good morning" using a text editor and check output.
2. Create a new Project in Intelij Idea and output "subscribe" on the console.
3. Show the following patterns:

Practice Exercise

Java Basics

Answer in True/False:

1. Computers understand high level languages like Java, C.
2. An Algorithm is a set of instructions to accomplish a task.
3. Computer is smart enough to ignore incorrect syntax.
4. Java was first released in 1992.
5. Java was named over a person who made good coffee.
6. ByteCode is platform independent.
7. JDK is a part of JRE.
8. It’s optional to declare main method as public.
9. .class file contains machine code.
10. println adds a new line at the end of the line.

Practice Exercise

Java Basics

Answer in True/False:

1. Computers understand high level languages like Java, C.	False
2. An Algorithm is a set of instructions to accomplish a task.	True
3. Computer is smart enough to ignore incorrect syntax.	False
4. Java was first released in 1992.	False
5. Java was named over a person who made good coffee.	False
6. ByteCode is platform independent.	True
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8. It’s optional to declare main method as public.	False
9. .class file contains machine code.	False
10. println adds a new line at the end of the line.	True

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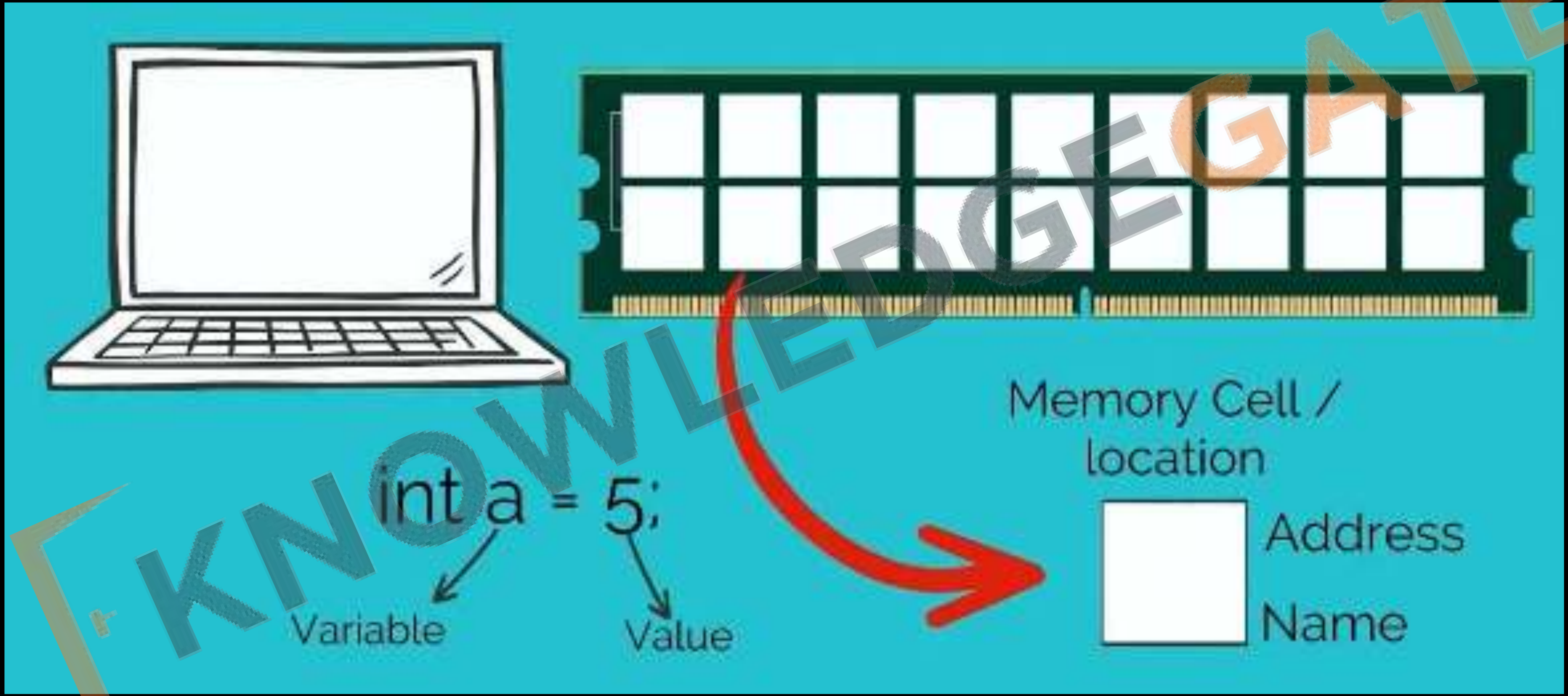
3 Data Types, Variables & Input

1. Variables
2. Data Types
3. Naming Conventions
4. Literals
5. Keywords
6. Escape Sequences
7. User Input
8. Type Conversion and Casting

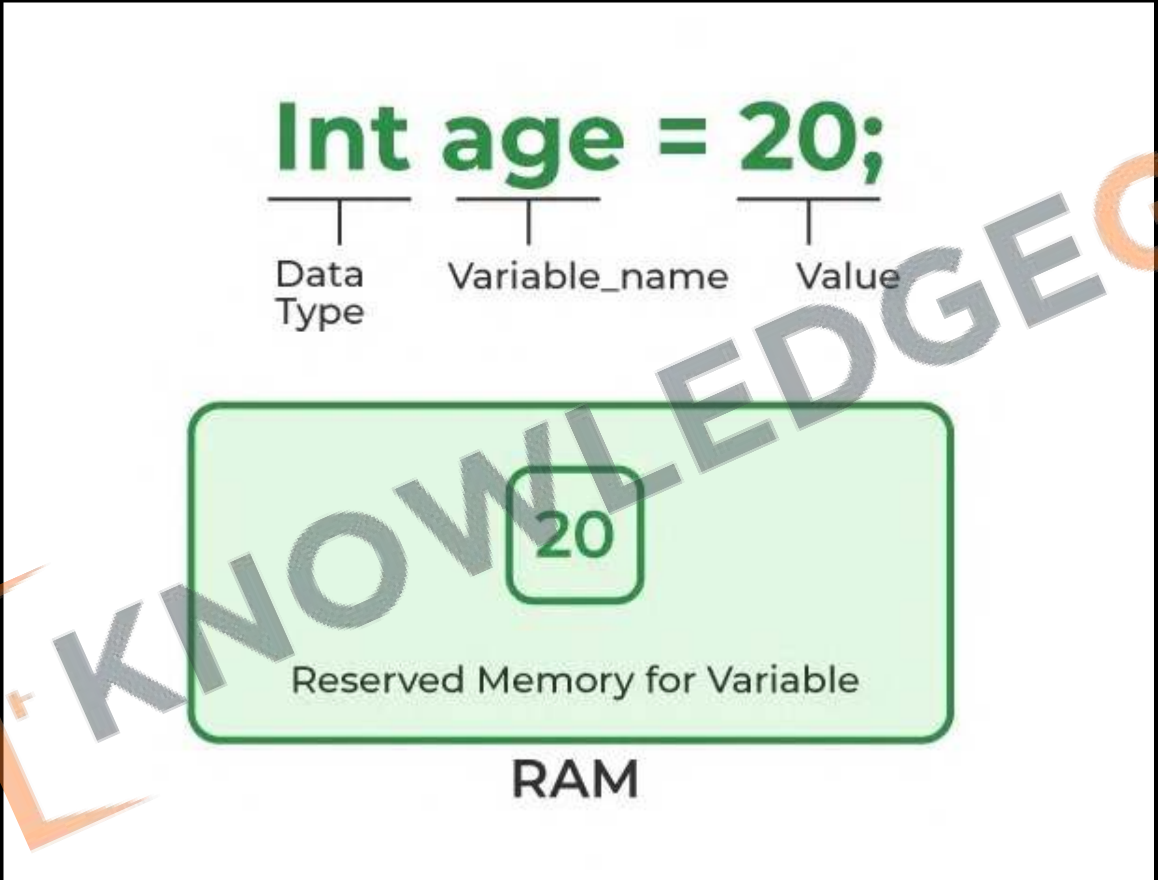
3.1. What are Variables?

Variables are like containers used for storing data values.

3.1. Memory Allocation



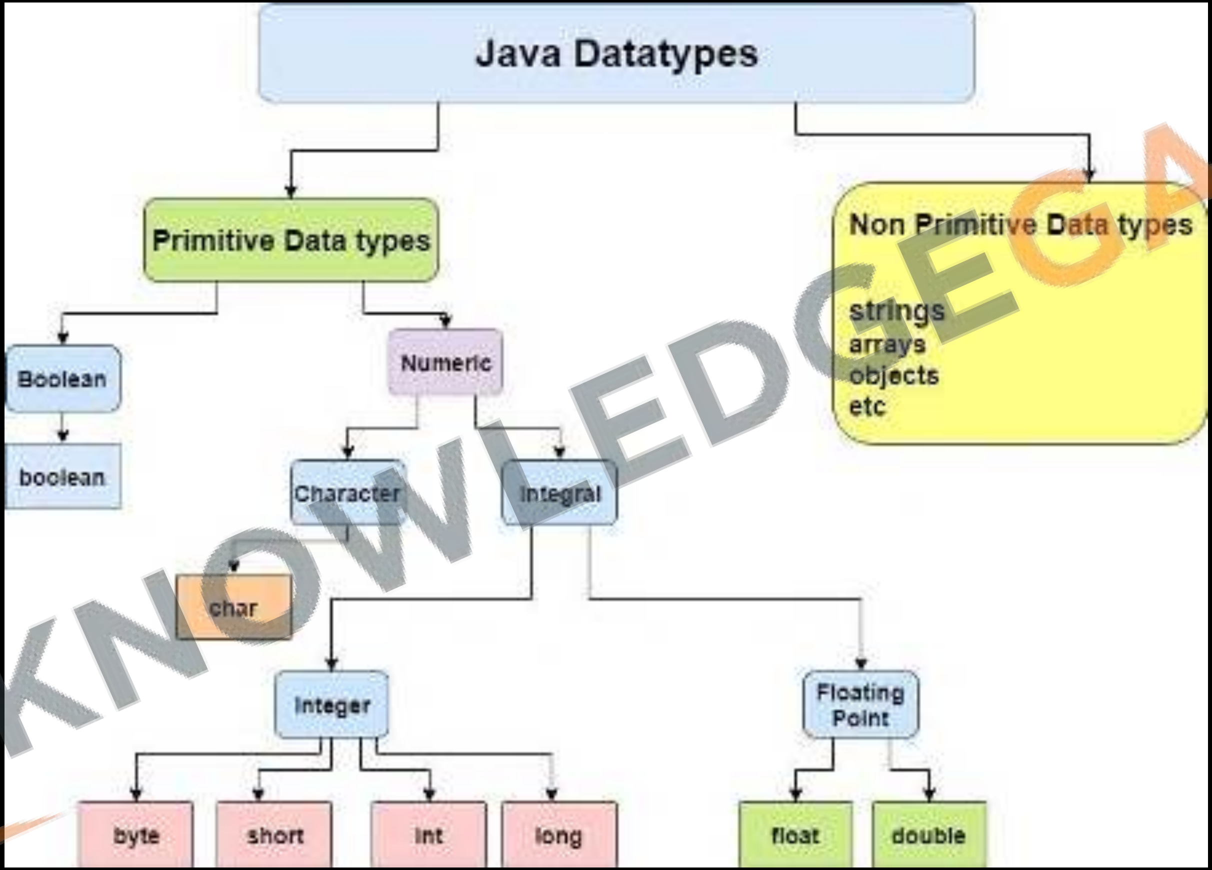
3.1. Variable Declaration



3.2 Data Types

	Size	Default Value	Type of Value Stored
byte	1 byte	0	Integral
short	2 byte	0	Integral
int	4 byte	0	Integral
long	8 byte	0L	Integral
char	2 byte	'\u0000'	Character
float	4 byte	0.0f	Decimal
double	8 byte	0.0d	Decimal
boolean	1 bit (till JDK 1.3 it uses 1 byte)	false	True or False

3.2 Data Types



3.3. Naming Conventions

camelCase

- Start with a lowercase letter. Capitalize the first letter of each subsequent word.
- Example: myVariableName

snake_case

- Start with an lowercase letter. Separate words with underscore
- Example: my_variable_name

Kebab-case

- All lowercase letters. Separate words with hyphens. Example: my-variable-name

Keep a Good and Short Name

- Choose names that are descriptive but not too long. It should make it easy to understand the variable's purpose.
- Example: age, firstName, isMarried

3.3. Java Identifier Rules

1. The only allowed characters for identifiers are all alphanumeric characters([A-Z],[a-z],[0-9]), '$' (dollar sign) and '_' (underscore).
2. Can't use keywords or reserved words
3. Identifiers should not start with digits([0-9]).
4. Java identifiers are case-sensitive.
5. There is no limit on the length of the identifier but it is advisable to use an optimum length of 4 – 15 letters only.

1)_$_(valid)
2)Ca$h(valid)
3)Java2share(valid)
4)all@hands(invalid)
5)123abc (invalid)
6)Total# (invalid)
7)Int (valid)
8)Integer(valid)
9)int (invalid)
10) tot123

3.4 Literals

Integer literals -> 10, 5, -8 etc...

Floating-point literals -> 1.2, 0.25, -1.999, etc...

Boolean literals -> true, false

Character literals -> 'a', 'A', 'N', 'q', etc...

String literals -> "hi", "hello", "What's up?"

3.5 Keywords


abstract
default super switch strictfp
void static break protected
volatile
case implements assert
throw native throws
synchronized
final
class impact catch
transtient double
try
new enum goto long
const byte char float
short interface return package
interface continue public
finally import boolean
private instanceof int extends

3.6 Escape Sequences

Escape Sequence	Description
\t	Insert a tab in the text at this point.
\b	Insert a backspace in the text at this point.
\n	Insert a newline in the text at this point.
\'	Insert a single quote character in the text at this point.
\"	Insert a double quote character in the text at this point.
\\	Insert a backslash character in the text at this point.

CHALLENGE

Task:

4. Show the following patterns using single print statement:

3.7 User Input

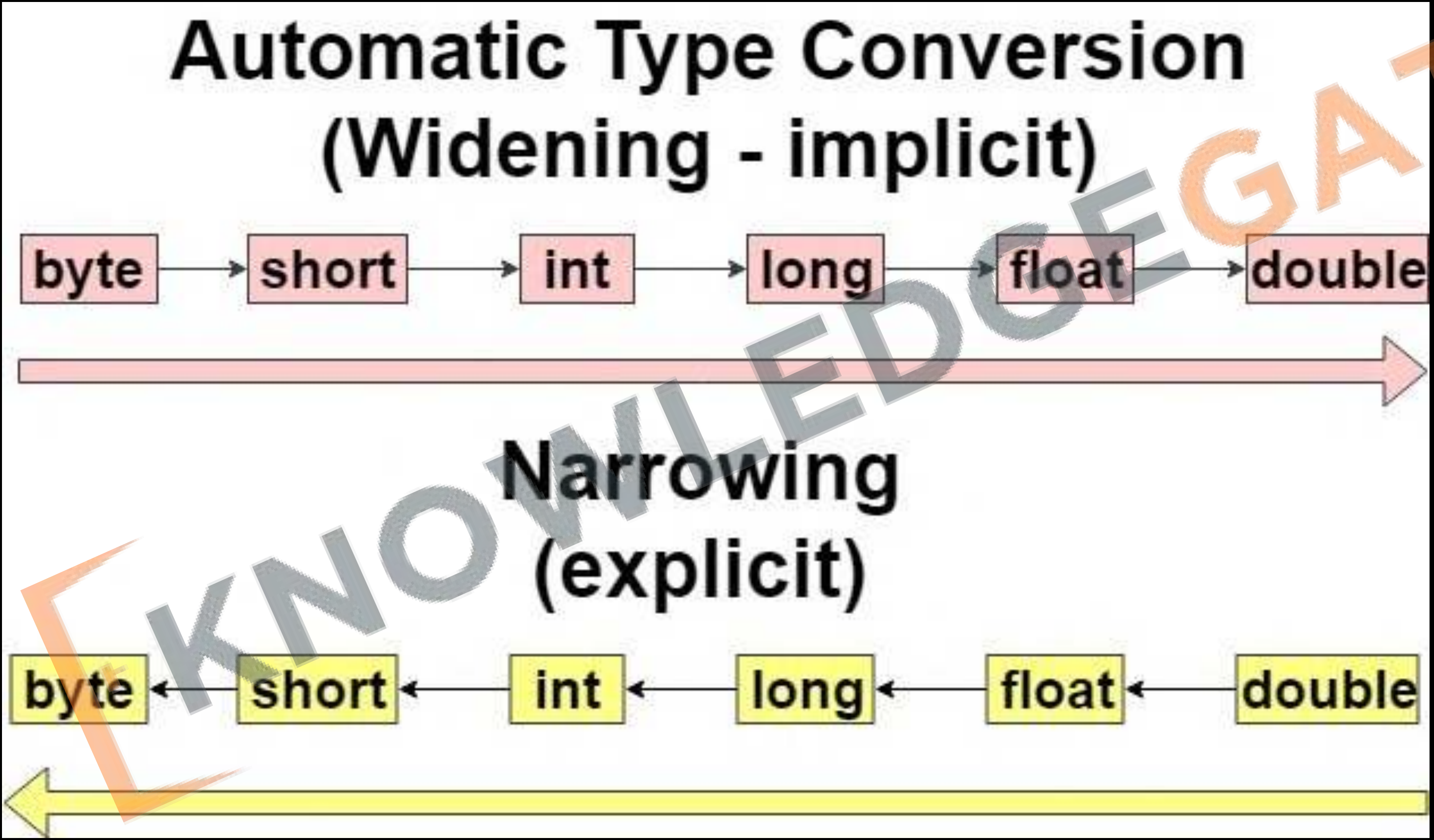
```
import java.util.Scanner;

public class Main {
    public static void main(String[] args) {
        Scanner scanner = new Scanner(System.in);
        System.out.print("Enter your name:");
        String name = scanner.nextLine();
        System.out.println("Welcome " + name);
    }
}
```

NAME
nextInt();
nextDouble();
nextFloat();
nextLong();
nextShort();
next();
nextLine();

CHALLENGE

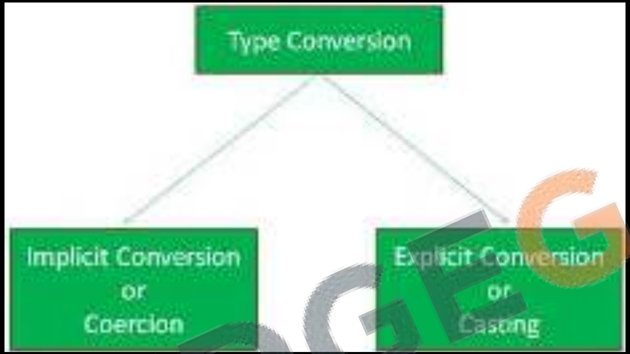
5. Create a program to input name of the person and respond with "Welcome NAME to KG Coding"
6. Create a program to add two numbers.

3.8 Type Conversion and Casting



3.8 Type Conversion and Casting

```
// implicit
long big = 45;
float dec = 3;
double d = 3.4f;
```




```
// explicit
float eDec = (float) 3.4;
long eBig = (long) 3.4;
int eInt = (int) 3.4;
```

Revision

1. Variables
2. Data Types
3. Naming Conventions
4. Literals
5. Keywords
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Practice Exercise

Data Types, Variables & Input

Answer in True/False:

1. In Java, a variable's name can start with a number.
2. char in Java can store a single character.
3. Class names in Java typically start with a lowercase letter.
4. 100L is a valid long literal in Java.
5. \d is an escape sequence in Java for a digit character.
6. Scanner class is used for reading console input.
7. In Java, an int can be automatically converted to a byte.
8. Java variable names are case-sensitive.
9. Scanner class can be used to read both primitive data types and strings.
10. Explicit casting is required to convert a double to an int.

Practice Exercise

Data Types, Variables & Input

Answer in True/False:

1. In Java, a variable's name can start with a number. False
2. char in Java can store a single character. True
3. Class names in Java typically start with a lowercase letter. False
4. 100L is a valid long literal in Java. True
5. \d is an escape sequence in Java for a digit character. False
6. Scanner class is used for reading console input. True
7. In Java, an int can be automatically converted to a byte. False
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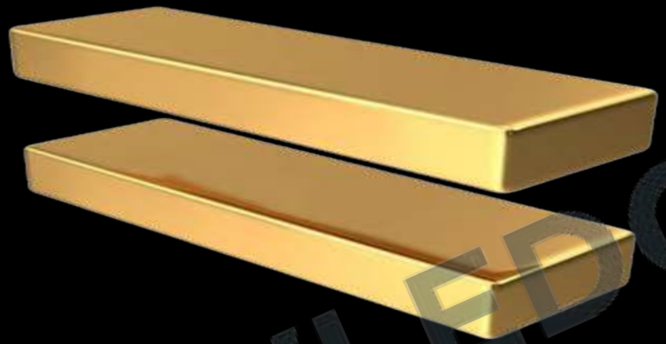
4. Operators, If-else & Number System

1. Assignment Operator
2. Arithmetic Operators
3. Order of Operation
4. Shorthand Operators
5. Unary Operators
6. If-else
7. Relational Operators
8. Logical Operators
9. Operator Precedence
10. Intro to Number System
11. Intro to Bitwise Operators

4.1 Assignment Operator



Assigns the right-hand operand's value to the left-hand operand.
Example: int a = 5;.

CHALLENGE

7. Create a program to swap two numbers.

4.2 Arithmetic Operators

Operators	Meaning	Example	Result
+	Addition	4+2	6
-	Subtraction	4-2	2
*	Multiplication	4*2	8
/	Division	4/2	2
%	Modulus operator to get remainder in integer division	5%2	1

4.3 Order of Operation

B	O	D	M	A	S
Bracket	Order	Divide	Multiply	Add	Subtract

( )	$\sqrt{x}$ x^2	÷ or x		+ or –	
Parentheses	Exponents	Multiply	Divide	Add	Subtract
P	E	M	D	A	S

4.4 Shorthand Operators

Operator symbol	Name of the operator	Example	Equivalent construct
+=	Addition assignment	x += 4;	x = x + 4;
-=	Subtraction assignment	x -= 4;	x = x - 4;
*=	Multiplication assignment	x *= 4;	x = x * 4;
/=	Division assignment	x /= 4;	x = x / 4;
%=	Remainder assignment	x %= 4;	x = x % 4;

4.5 Unary Operators

Operator	Description	Example
-	Converts a positive value to a negative	x = -y
Pre Increment	Increment the value by 1 and then use it in our statement	x = ++y
Pre Decrement	Decrement the value by 1 and then use it in our statement	x = --y
Post Increment	Use current value in the statement and then increment it by 1	x = y++
Post Decrement	Use current value in the statement and then decrement it by 1	x = y--

CHALLENGE

8. Create a program that takes two numbers and shows result of all arithmetic operators (+,-,*,/,%).
9. Create a program to calculate product of two floating points numbers.
10. Create a program to calculate Perimeter of a rectangle.
    Perimeter of rectangle ABCD = A+B+C+D
11. Create a program to calculate the Area of a Triangle.
    Area of triangle = ½*B*H
12. Create a program to calculate simple interest.
    Simple Interest = (P x T x R)/100
13. Create a program to calculate Compound interest.
    Compound Interest = P(1 + R/100)t
14. Create a program to convert Fahrenheit to Celsius
    °C = (°F - 32) × 5/9

4.6 if-else

1. Syntax: Uses if () {} to check a condition.
2. What is if: Executes block if condition is true, skips if false.
3. What is else: Executes a block when the if condition is false.
4. Curly Braces can be omitted for single statements, but not recommended.
5. If-else Ladder: Multiple if and else if blocks; only one executes.
6. Use Variables: Can store conditions in variables for use in if statements.

4.7 Relational Operators

<, >, <=, >=, ==, !=

Equality

- == Checks value equality.

Inequality

- != Checks value inequality.

Relational

- > Greater than.
- < Less than.
- >= Greater than or equal to.
- <= Less than or equal to.

Order of Relational operators is less than arithmetic operators

4.8 Logical Operators





AND

Or

NOT

1. Types: && (AND), || (OR), ! (NOT)
2. AND (&&): All conditions must be true for the result to be true.
3. OR (||): Only one condition must be true for the result to be true.
4. NOT (!): Inverts the Boolean value of a condition.
5. Lower Priority than Math and Comparison operators

CHALLENGE

15. Create a program that determines if a number is positive, negative, or zero.
16. Create a program that determines if a number is odd or even.
17. Create a program that determines the greatest of the three numbers.
18. Create a program that determines if a given year is a leap year (considering conditions like divisible by 4 but not 100, unless also divisible by 400).
19. Create a program that calculates grades based on marks

    A -> above 90% B -> above 75%

    C -> above 60% D -> above 30%

    F -> below 30%
20. Create a program that categorize a person into different age groups

    Child -> below 13 Teen -> below 20

    Adult -> below 60 Senior-> above 60

4.9 Operator Precedence

Operator Type	Category	Precedence	Associativity
Unary	postfix	a++, a--	Right to left
	prefix	++a, --a, +a, -a, ~, !	Right to left
Arithmetic	Multiplication	*, /, %	Left to Right
	Addition	+, -	Left to Right
Shift	Shift	<<, >>, >>>	Left to Right
Relational	Comparison	<, >, <=, >=, instanceOf	Left to Right
	equality	==, !=	Left to Right
Bitwise	Bitwise AND	&	Left to Right
	Bitwise exclusive OR	^	Left to Right
	Bitwise inclusive OR	|	Left to Right
Logical	Logical AND	&&	Left to Right
	Logical OR	||	Left to Right
Ternary	Ternary	? :	Right to Left
Assignment	assignment	=, +=, -=, *=, /=, %=, &=, ^=, |=, <<=, >>=, >>>=	Right to Left

Operator Precedence: Determines the evaluation order of operators in an expression based on their priority levels.

Associativity: Defines the order of operation for operators with the same precedence, usually left-to-right or right-to-left.

4.10 Intro to Number System

Number System	Base or Radix	Digits or symbols
Binary	2	0,1
Octal	8	0,1,2,3,4,5,6,7
Hexadecimal	16	0,1,2,3,4,5,6,7,8,9,A,B,C,D,E,F

Number System	Base or Radix	Digits or symbols
Unary	1	0/1
Decimal	10	0,1,2,3,4,5,6,7,8,9

4.10 Intro to Number System

Decimal Number System

- Decimal Number System is a base-10 system that has ten digits: 0, 1, 2, 3, 4, 5, 6, 7, 8, 9.
- The decimal number system is said to be of base, or radix, 10 because it uses 10 digits and the coefficients are multiplied by powers of 10.
- This is the base that we often use in our day to day life.
- Example: $(7,392)_{10}$, where 7,392 is a shorthand notation for what should be written as

$$7 * 10^3 + 3 * 10^2 + 9 * 10^1 + 2 * 10^0$$

4.10 Intro to Number System

Binary Number System

- The coefficients of the binary number system have only two possible values: 0 and 1.
- Each coefficient a_j is multiplied by a power of the radix, e.g., 2^j, and the results are added to obtain the decimal equivalent of the number.

Example: $(11010.11)_2$ value in Decimal?

$$1 * 2^4 + 1 * 2^3 + 0 * 2^2 + 1 * 2^1 + 0 * 2^0 + 1 * 2^{-1} + 1 * 2^{-2} = 26.75$$

4.10 Intro to Number System

Decimal	Binary
0	0000
1	0001
2	0010
3	0011
4	0100
5	0101
6	0110
7	0111
8	1000
9	1001
10	1010
11	1011
12	1100
13	1101
14	1110
15	1111

4.11 Intro to Bitwise Operators

1.AND Operator (&): Performs on two integers. Each bit of the output is 1 if the corresponding bits of both operands are 1, otherwise 0.
2.OR Operator (|): Performs on two integers. Each bit of the output is 0 if the corresponding bits of both operands are 0, otherwise 1.
3.XOR Operator (^): Performs on two integers. Each bit of the output is 1 if the corresponding bits of the operands are different.
4.NOT Operator (~): Performs a bitwise complement. It inverts the bits of its operand (0 becomes 1, and 1 becomes 0).
5.Left Shift Operator (<<): Shifts the left operand's bits to the left by the number of positions specified by the right operand, filling the new rightmost bits with zeros.
6.Right Shift Operator (>>): Shifts the left operand's bits to the right. If the left operand is positive, zeros are filled into the new leftmost bits; if negative, ones are filled in.

CHALLENGE

21. Create a program that shows bitwise AND of two numbers.
22. Create a program that shows bitwise OR of two numbers.
23. Create a program that shows bitwise XOR of two numbers.
24. Create a program that shows bitwise compliment of a number.
25. Create a program that shows use of left shift operator.
26. Create a program that shows use of right shift operator.
27. Write a program to check if a given number is even or odd using bitwise operators.

Revision

1. Assignment Operator
2. Arithmetic Operators
3. Order of Operation
4. Shorthand Operators
5. Unary Operators
6. If-else
7. Relational Operators
8. Logical Operators
9. Operator Precedence
10. Intro to Number System
11. Intro to Bitwise Operators

Practice Exercise

Operators, If-else & Number System

Answer in True/False:

1. In Java, && and || operators perform short-circuit evaluation.
2. In an if-else statement, the else block executes only when the if condition is false.
3. Java allows an if statement without the else part.
4. The ^ operator in Java is used for exponentiation.
5. Unary minus operator can be used to negate the value of a variable in Java.
6. a += b is equivalent to a = a + b in Java.
7. In Java, the binary number system uses base 10.
8. The number 1010 in binary is equivalent to 10 in decimal.
9. & and | are logical operators in Java.
10. In Java, a >> 2 shifts the binary bits of a to the left by 2 positions.

Practice Exercise

Operators, If-else & Number System

Answer in True/False:

1. In Java, && and || operators perform short-circuit evaluation. True
2. In an if-else statement, the else block executes only when the if condition is false. True
3. Java allows an if statement without the else part. True
4. The ^ operator in Java is used for exponentiation. False
5. Unary minus operator can be used to negate the value of a variable in Java. True
6. a += b is equivalent to a = a + b in Java. True
7. In Java, the binary number system uses base 10. False
8. The number 1010 in binary is equivalent to 10 in decimal. True
9. & and | are logical operators in Java. False
10. In Java, a >> 2 shifts the binary bits of a to the left by 2 positions. False

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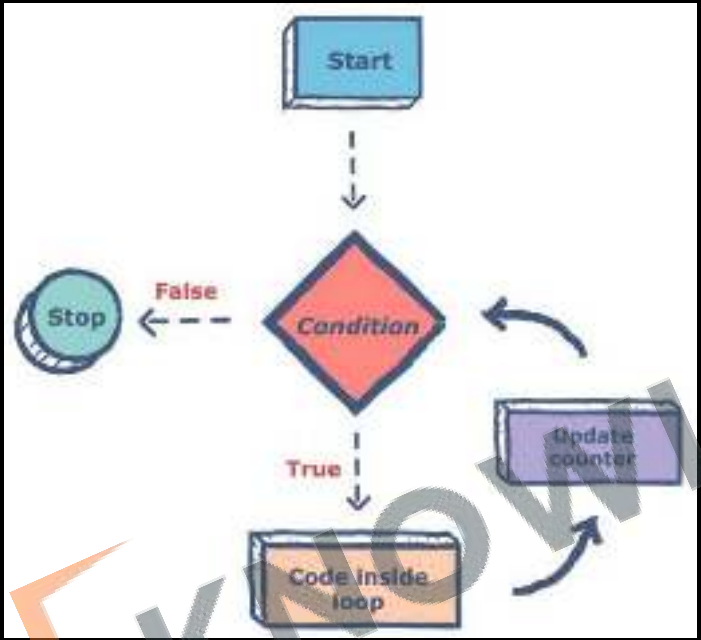
5 While Loop, Methods & Arrays

1. Comments
2. While Loop
3. Methods
4. Return statement
5. Arguments
6. Arrays
7. 2D Arrays

5.1 Java Comments

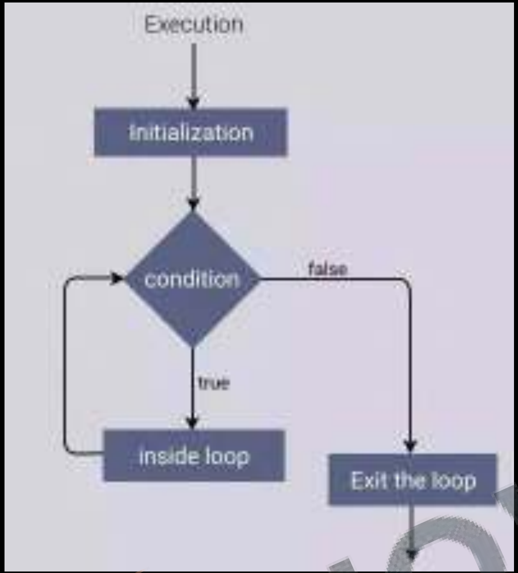
1. Used to add notes in Java code
2. Not displayed on the web page
3. Helpful for code organization
4. Syntax:
    1. Single Line: //
    2. Multi Line: /* */
    3. Java Docs: /** */

5.2 What is a Loop?



1. Code that runs multiple times based on a condition.
2. Repeated execution of code.
3. Loops automate repetitive tasks.
4. Types of Loops: while, for, while, do-while.
5. Iterations: Number of times the loop runs.

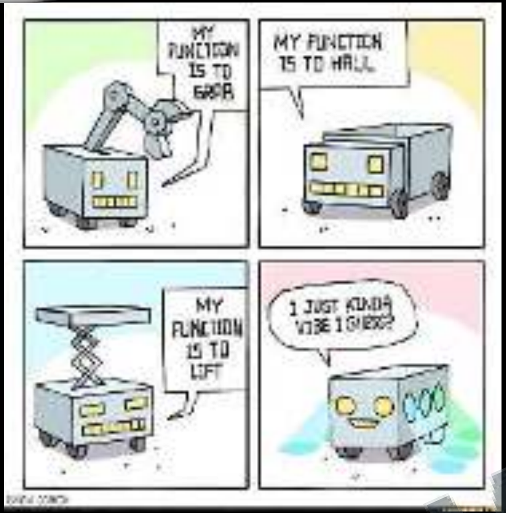
5.2 While Loop




```
while (condition) {
  // Body of the loop
}
```

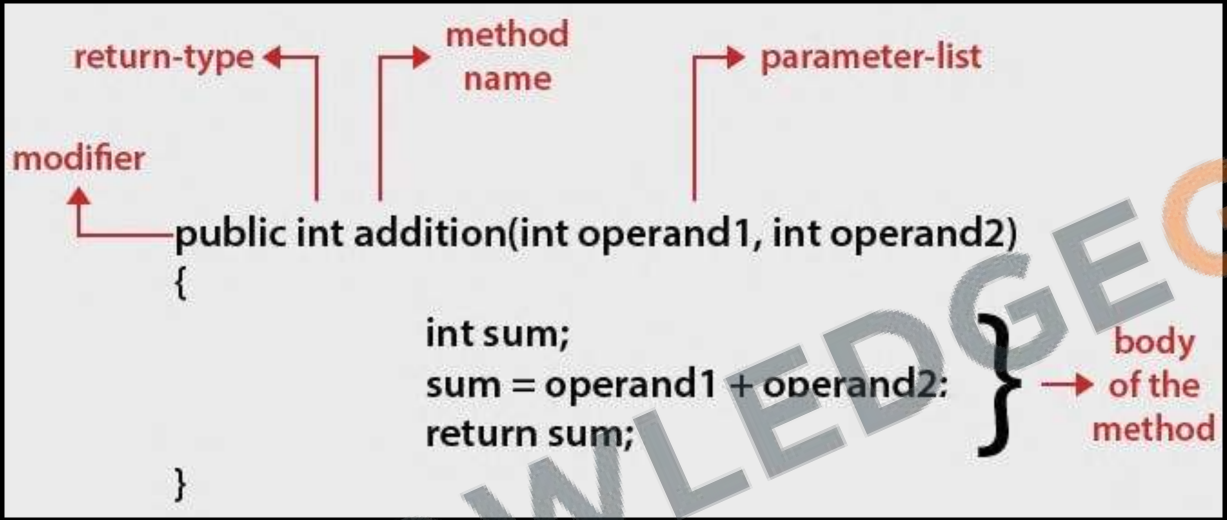

1. Iterations: Number of times the loop runs.
2. Used for non-standard conditions.
3. Repeating a block of code while a condition is true.
4. Remember: Always include an update to avoid infinite loops.

5.3 What are Functions / Methods?



1. Definition: Blocks of reusable code.
2. DRY Principle: "Don't Repeat Yourself" it Encourages code reusability.
3. Usage: Organizes code and performs specific tasks.
4. Naming Rules: Same as variable names: camelCase
5. Example: ”Beta Gas band kar de"

5.3 Method Syntax



1. Follows same rules as variable names.
2. Use () to contain parameters.
3. Invoke by using the function name followed by ().
4. Fundamental for code organization and reusability.

5.4 Return statement

1. Sends a value back from a function.
2. Example: "Ek glass paani laao"
3. What Can Be Returned: Value, variable, calculation, etc.
4. Return ends the function immediately.
5. Function calls make code jump around.
6. Prefer returning values over using global variables.

5.5 Arguments vs Parameters

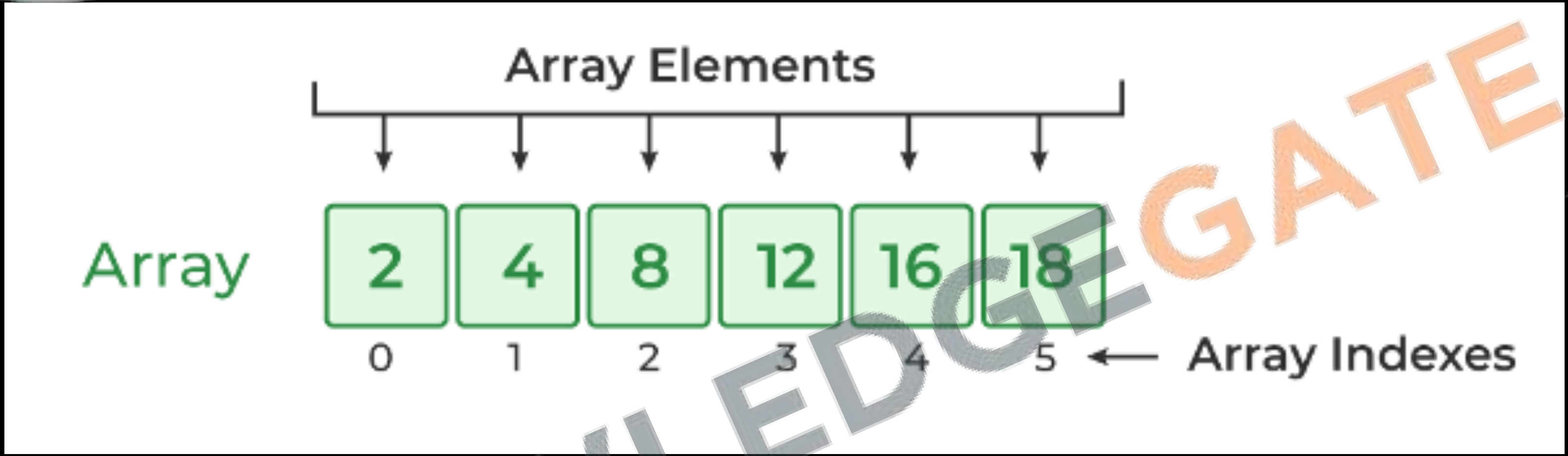
Parameter Function Return

1. Input values that a function takes.
2. Parameters put value into function, while return gets value out.
3. Example: "Ek packet dahi laao"
4. Naming Convention: Same as variable names.
5. Parameter vs Argument
6. Examples: System.out.print, Scanner.nextInt(), are functions we have already used
7. Multiple Parameters: Functions can take more than one.
8. Default Value: Can set a default value for a parameter.

CHALLENGE

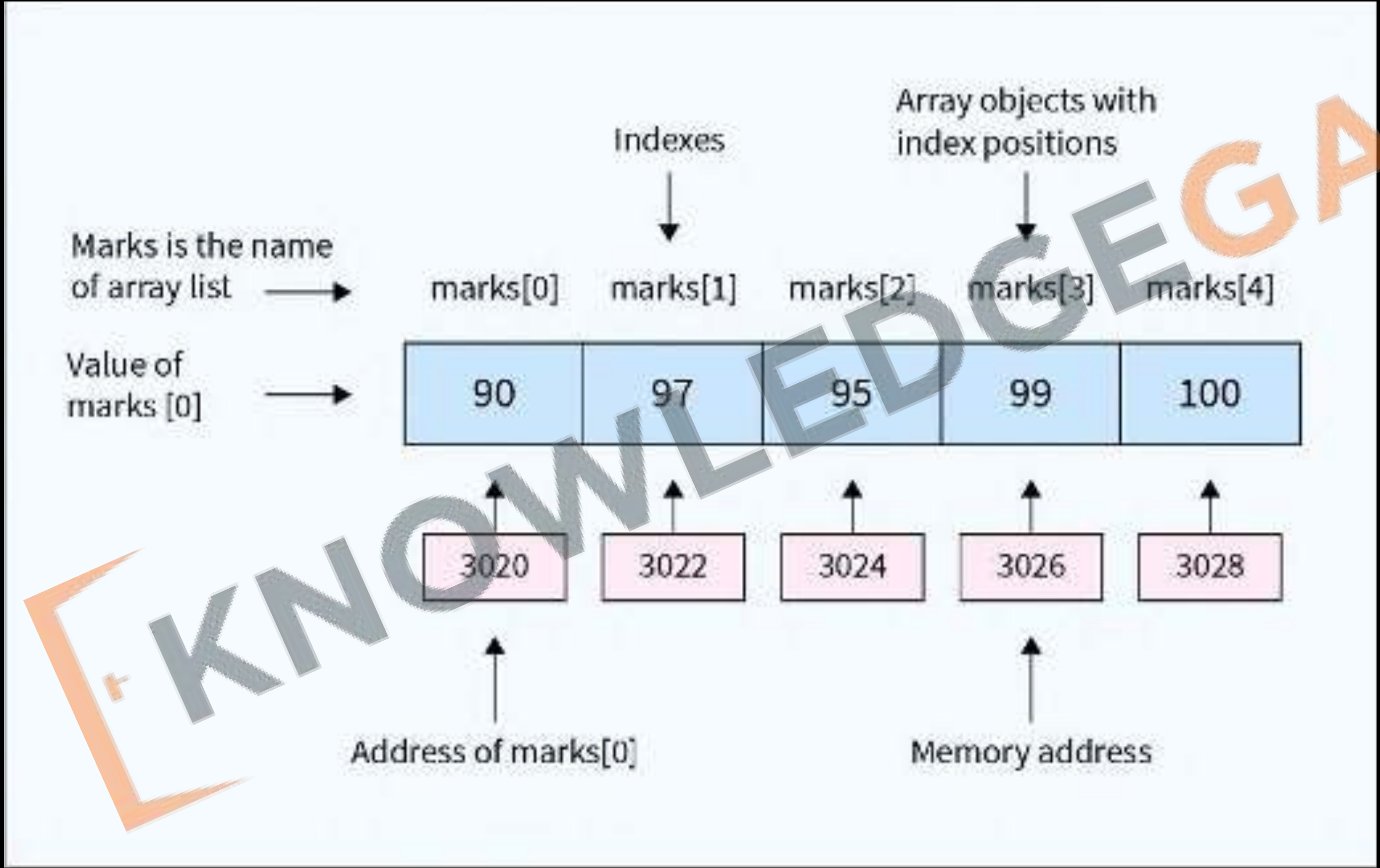
28. Develop a program that prints the multiplication table for a given number.
29. Create a program to sum all odd numbers from 1 to a specified number N.
30. Write a function that calculates the factorial of a given number.
31. Create a program that computes the sum of the digits of an integer.
32. Create a program to find the Least Common Multiple (LCM) of two numbers.
33. Create a program to find the Greatest Common Divisor (GCD) of two integers.
34. Create a program to check whether a given number is prime.
35. Create a program to reverse the digits of a number.
36. Create a program to print the Fibonacci series up to a certain number.
37. Create a program to check if a number is an Armstrong number.
38. Create a program to verify if a number is a palindrome.
39. Create a program that print patterns:

5.6 What is an Array?

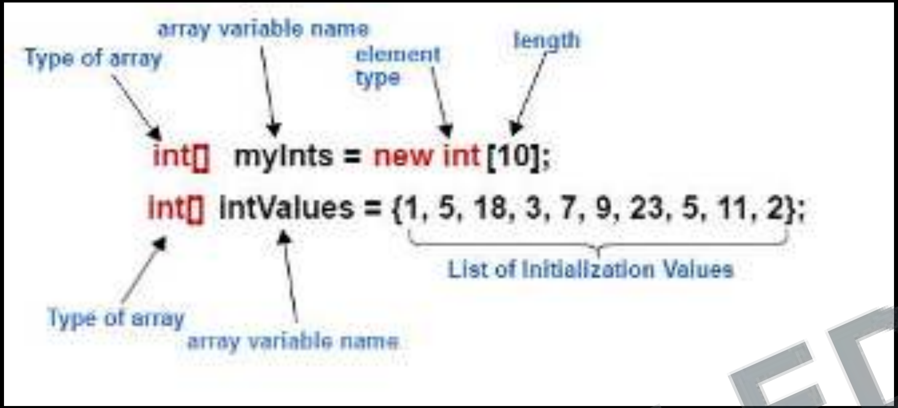


1. An Array is just a list of values.
2. Index: Starts with 0.
3. Arrays are used for storing multiple values in a single variable.

5.6 Array Memory

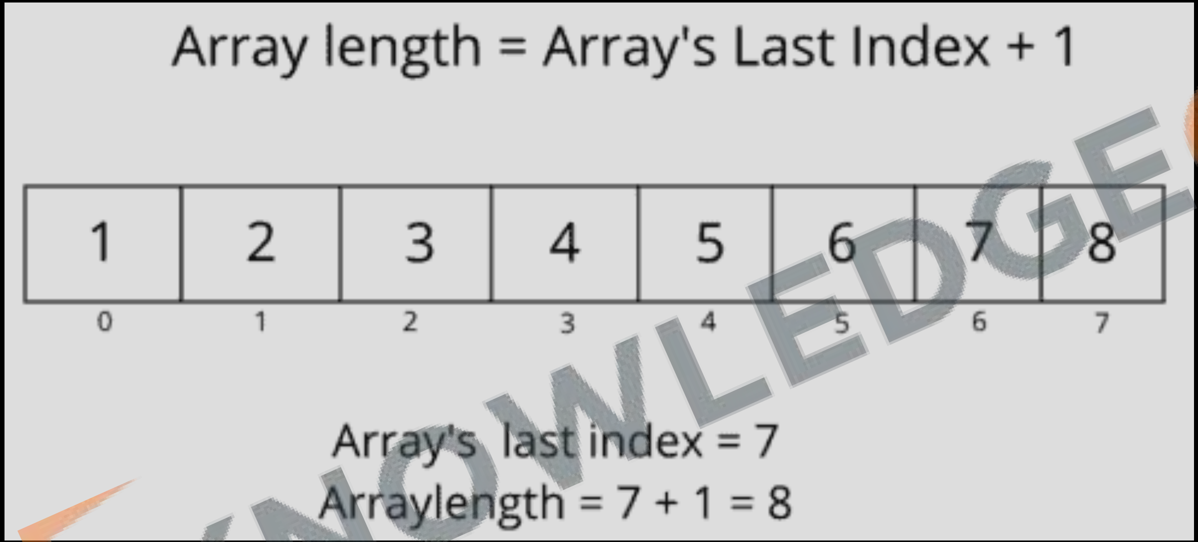


5.6 Array Syntax



1. Use {} to create a new array, {} brackets enclose list of values
2. Arrays can be saved to a variable.
3. Accessing Values: Use [] with index.
4. Syntax Rules:
   - Brackets start and end the array.
   - Values separated by commas.
   - Can span multiple lines.

5.6 Array Length

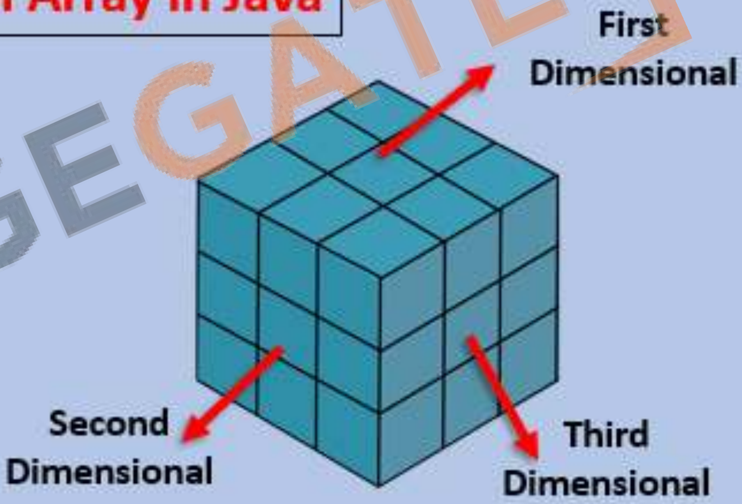


5.7 2D Arrays

Types of Multidimensional Array in Java

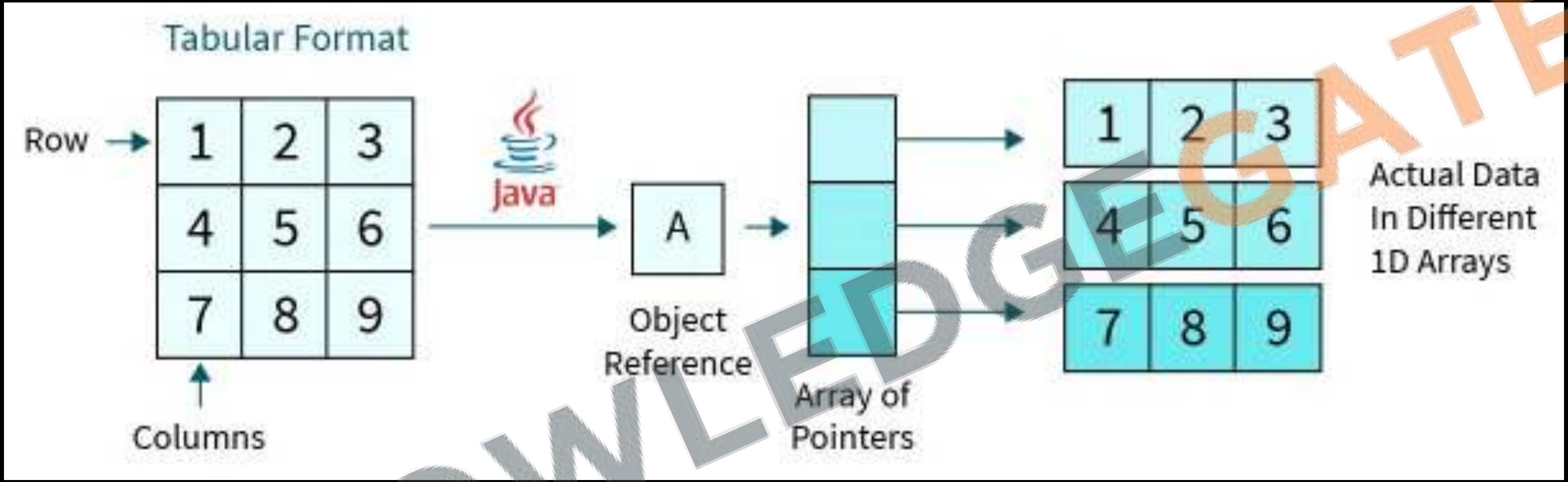
	Column 0	Column 1	Column 2
Row 0	X[0][0]	X[0][1]	X[0][2]
Row 1	X[1][0]	X[1][1]	X[1][2]
Row 2	X[2][0]	X[2][1]	X[2][2]

2D-Array



3D-Array

5.7 2D Arrays



Tabular Format
Row
1 2 3
4 5 6
7 8 9
Columns
Java
A
Object Reference
Array of Pointers
1 2 3
4 5 6
7 8 9
Actual Data In Different 1D Arrays

5.7 2D Arrays

```java
int[][] numArr = new int[2][3];
int[][] inArray = {{1, 2, 5}, {8, 9, 4}};

numArr[0][0] = 5;
```

CHALLENGE

40. Create a program to find the sum and average of all elements in an array.
41. Create a program to find number of occurrences of an element in an array.
42. Create a program to find the maximum and minimum element in an array.
43. Create a program to check if the given array is sorted.
44. Create a program to return a new array deleting a specific element.
45. Create a program to reverse an array.
46. Create a program to check is the array is palindrome or not.
47. Create a program to merge two sorted arrays.
48. Create a program to search an element in a 2-D array.
49. Create a program to do sum and average of all elements in a 2-D array.
50. Create a program to find the sum of two diagonal elements.

Revision

1. Comments
2. While Loop
3. Methods
4. Return statement
5. Arguments
6. Arrays
7. 2D Arrays

Practice Exercise

While Loop, Methods & Arrays

Answer in True/False:

1. A 'while' loop in Java continues to execute as long as its condition is true.
2. The body of a 'while' loop will execute at least once, regardless of the condition.
3. A 'while' loop cannot be used for iterating over an array.
4. Infinite loops are not possible with 'while' loops in Java.
5. A method in Java can return more than one value at a time.
6. It's mandatory for every Java method to have a return type.
7. The size of an array in Java can be modified after it is created.
8. Arrays in Java can contain elements of different data types.
9. An array is a reference type in Java.
10. Java arrays are zero-indexed, meaning the first element has an index of 0.

Practice Exercise

While Loop, Methods & Arrays

Answer in True/False:

1. A 'while' loop in Java continues to execute as long as its condition is true. True
2. The body of a 'while' loop will execute at least once, regardless of the condition. False
3. A 'while' loop cannot be used for iterating over an array. False
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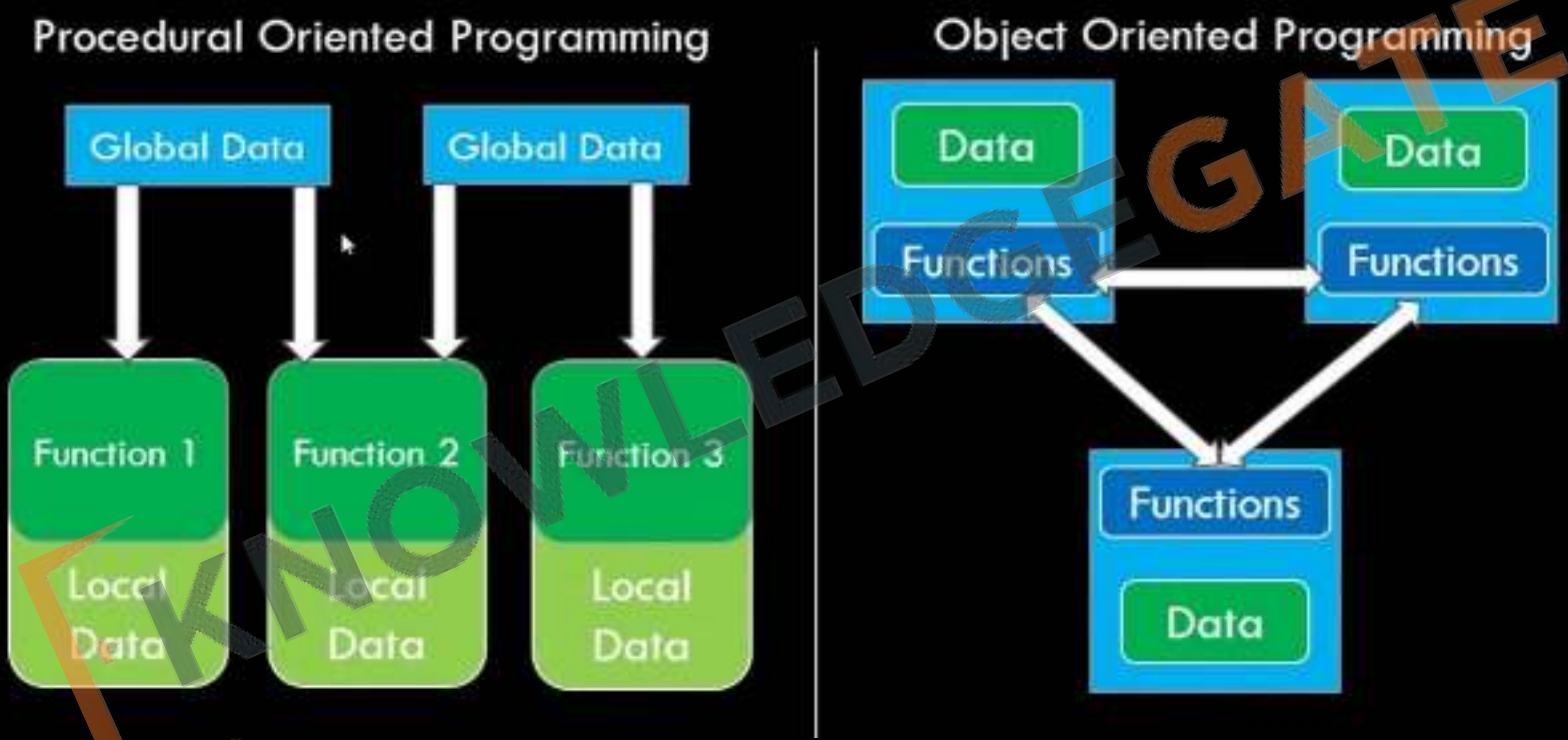
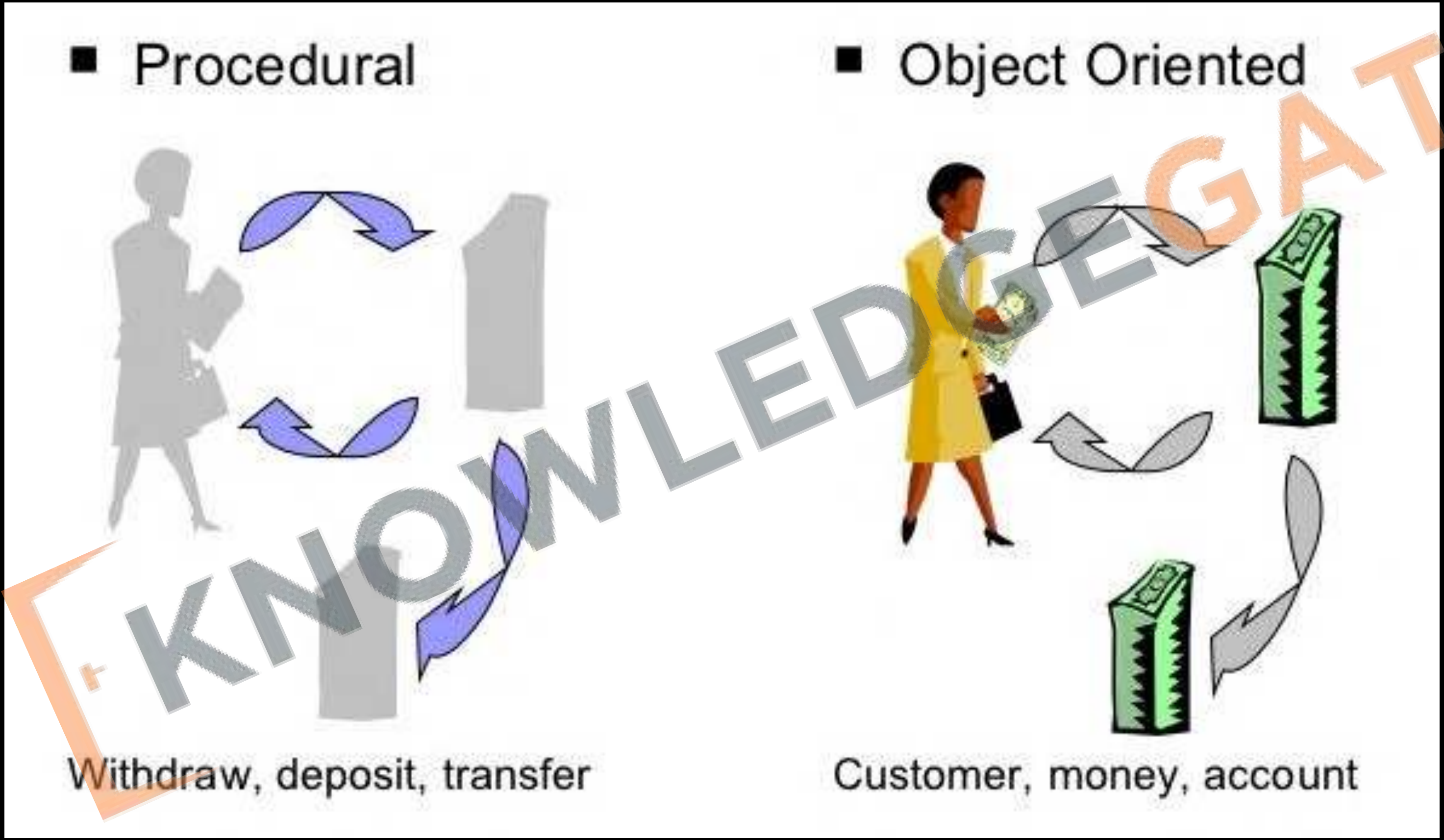
6 Classes & Objects

1. Process vs Object Oriented
2. Instance Variables and Methods
3. Declaring and Using Objects
4. Class vs Object
5. This & Static Keyword
6. Constructors & Code Blocks
7. Stack vs Heap Memory
8. Primitive vs Reference Types
9. Variable Scopes
10. Garbage Collection & Finalize

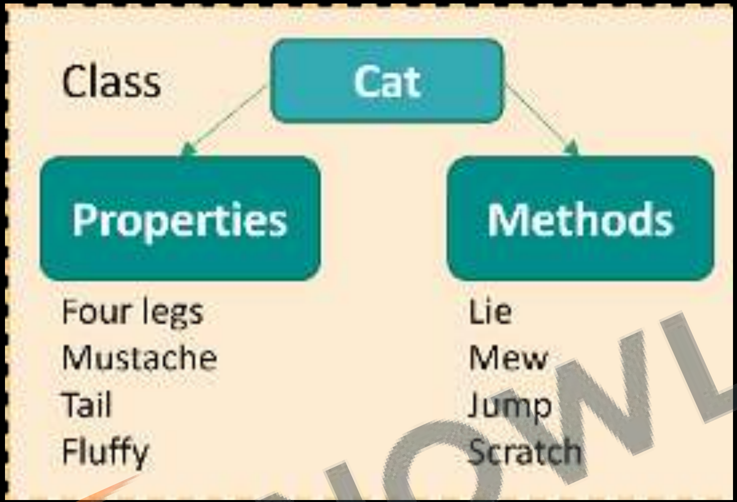
6.1 Process vs Object Oriented

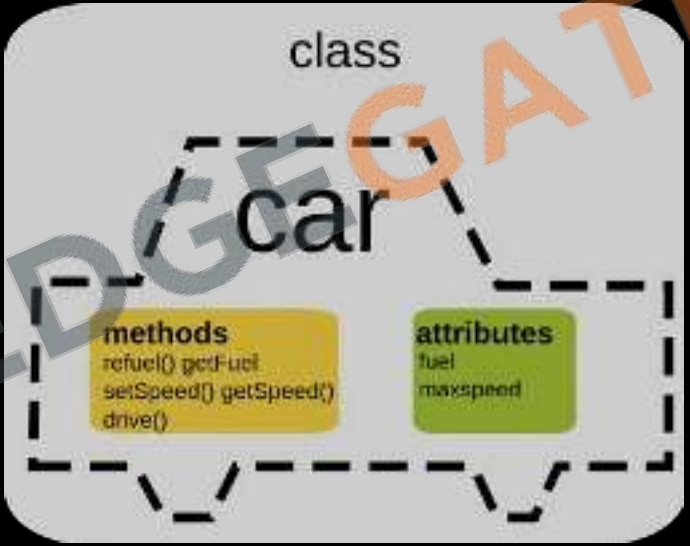


6.1 Process vs Object Oriented

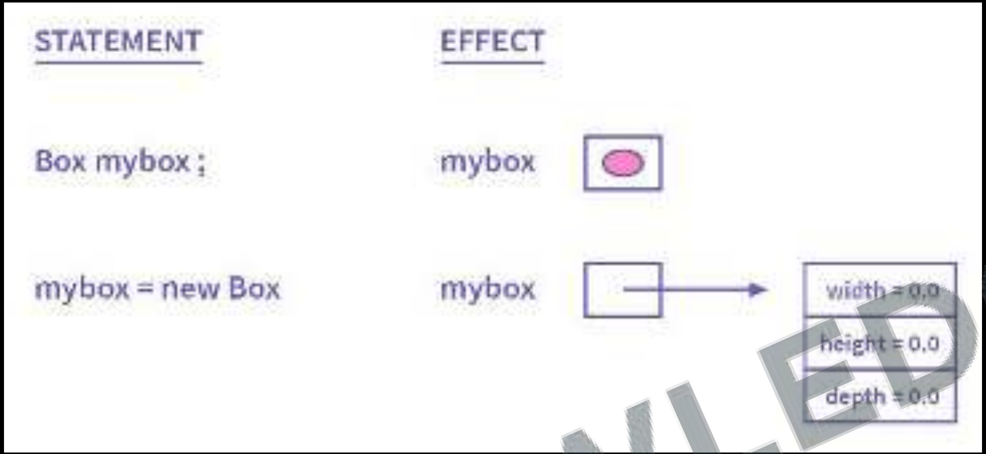


6.2 Instance Variables and Methods



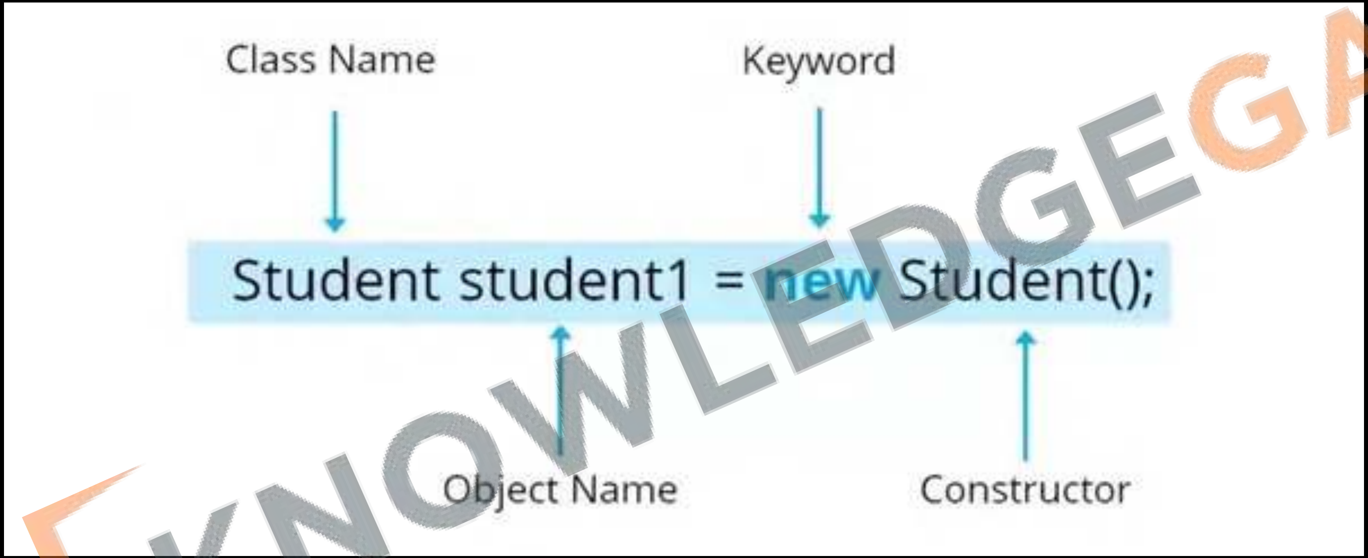


6.3 Declaring Objects

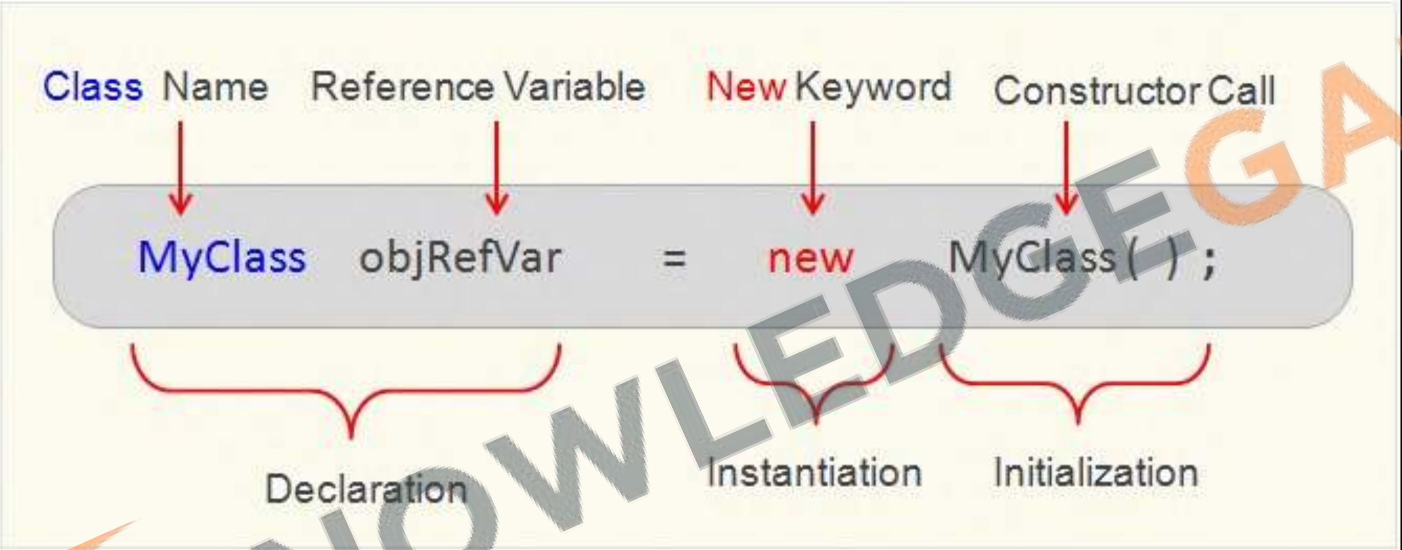


1. Object Creation: new instantiates a new object of a class.
2. Memory Allocation: Allocates memory for the object in the heap.
3. Constructor Invocation: Calls the class constructor to initialize the object.
4. Reference Return: Returns a reference to the created object.
5. Array Creation: Also used for creating arrays, like int[] arr = new int[5];.
6. Dynamic Allocation: Unlike static allocation, new allows for dynamic memory allocation, allocating memory at runtime.

6.3 Declaring Objects Syntax



6.3 Declaring Objects Syntax



objRefVar is a Reference Variable of MyClass Type .

new is a Java Keyword used To Instantiate a Class .

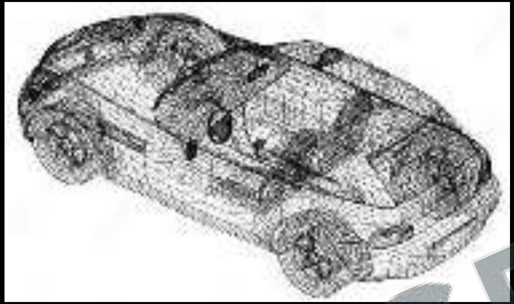
6.3 Using Objects

Access properties using . Operator like product.price

6.4 Class vs Object

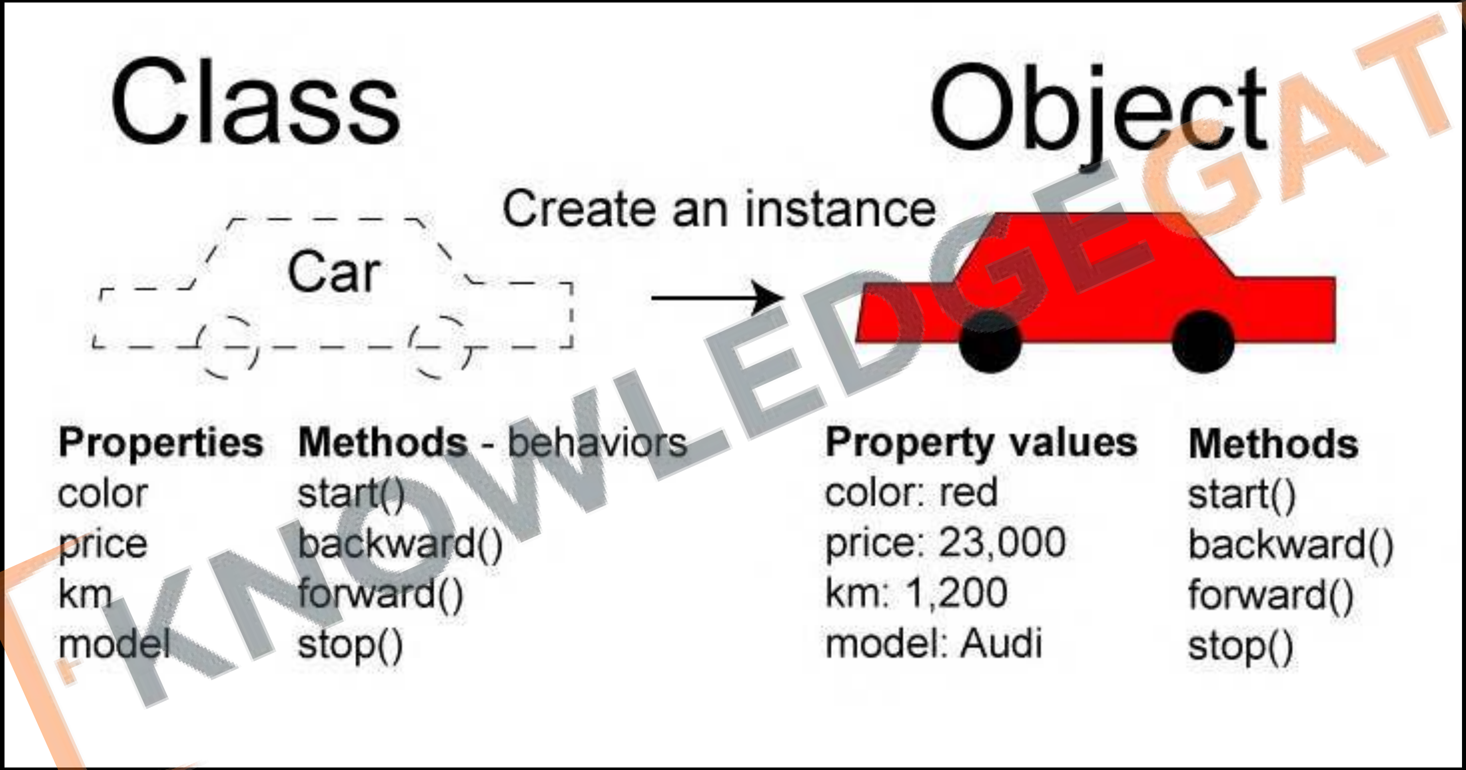
Class is a blueprint; Objects are real values in memory.

6.4 Class vs Object



Class is a blueprint; Objects are real values in memory.

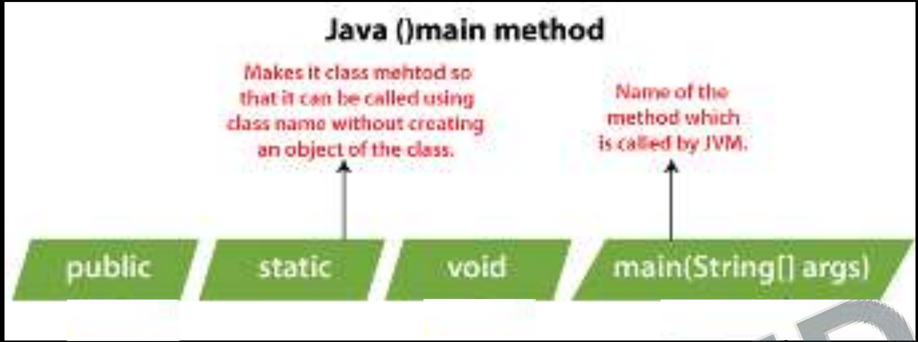
6.4 Class vs Object



6.5 This Keyword

1. Current Instance: Refers to the current class instance variable.
2. Constructor Call: Can be used to invoke a constructor of the same class (this()).
3. Method Call: Invokes a method of the current object.
4. Pass as Argument: Can be passed as an argument in the method call.
5. Return the Current Class Instance: Can return the current class instance from the method.

6.5 Static Keyword



1. Static Variables: Belong to the class, not individual instances. Shared among all instances of the class.
2. Static Methods: Can be called without creating an object of the class. Can only directly access static variables and other static methods.
3. No Access to Non-static Members: Static methods and blocks cannot directly access non-static members (variables and methods) of the class.

6.6 Constructors

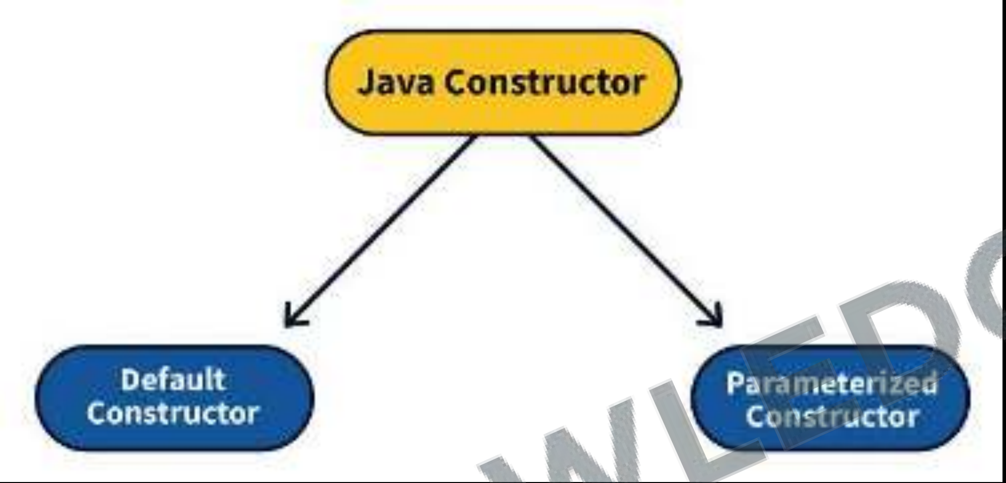
```
public class Car {
    1 usage
    String color;
    1 usage
    float price;

    1 usage   new *
    Car() { // Default constructor
        color = "Black";
        price = 50000;
    }

    new *
    public static void main(String[] args) {
        Car swift = new Car();
    }
}
```

1. Purpose: Constructors initialize new objects and set initial states for the object's attributes.
2. Naming: A constructor must have the same name as the class in which it is declared.
3. No Return Type: Constructors do not have a return type, not even void.
4. Automatic Calls: A constructor is automatically called when an instance of a class is created.

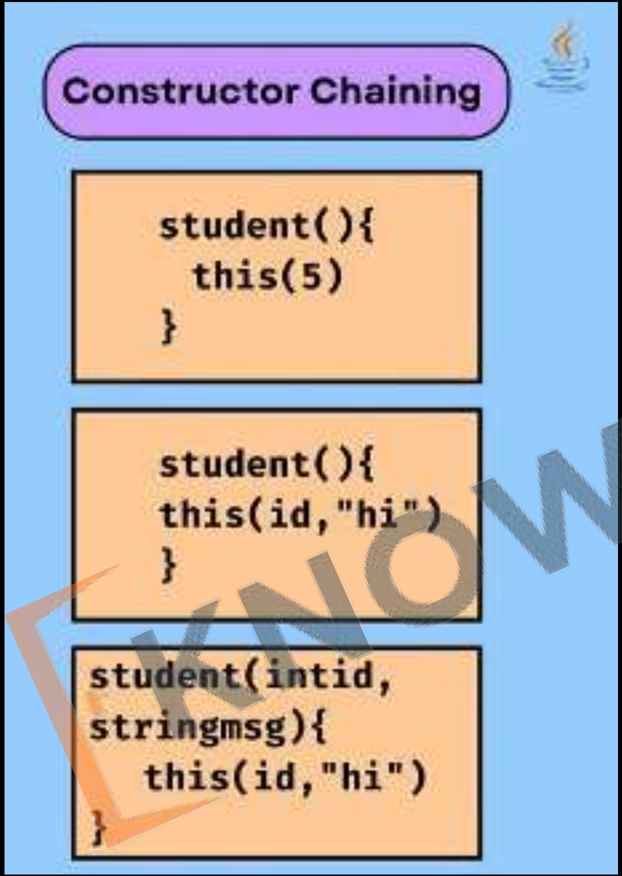
6.6 Constructors (Types)




```
Car(String carColor, float currPrice) {
    color = carColor;
    price = currPrice;
}
```

1. Default Constructor: If no constructor is explicitly defined, Java provides a default constructor that initializes all member variables to default values.
2. Parameterized Constructors: Constructors can have parameters to pass values when creating an object, allowing for different initializations.

6.6 Constructors (Chaining)



1. Within Same Class: Using this() to call another constructor in the same class.
2. First Statement: this() must be the first statement in a constructor.
3. No Loop: Constructor chaining can't form a loop; it must have a termination point.

6.6 Code Blocks

```java
{
    System.out.println("This is a Initialization Block");
    color = "Black";
    price = 58000;
}
```

```java
if (true) { // code block
    System.out.println("Code Block");
}
```

```java
static {
    System.out.println("This is a Static Block");
}
```

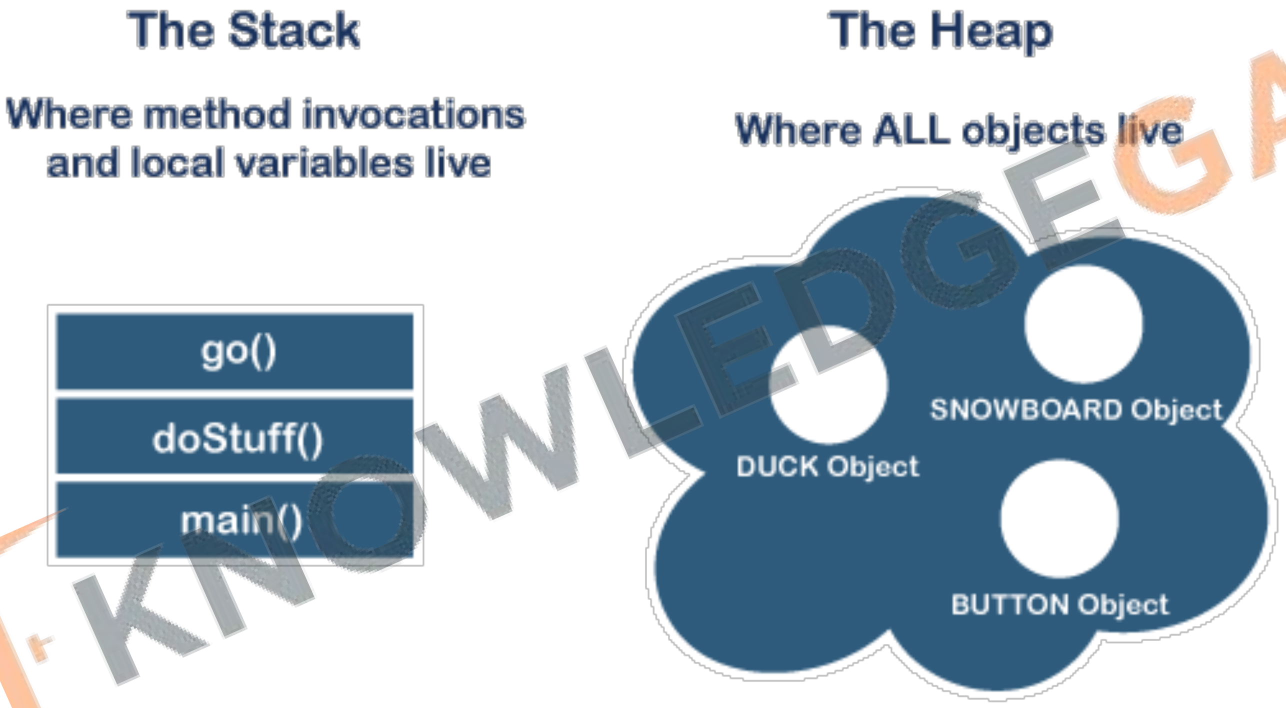
1. Scope: Code blocks {} determine the scope of variables.
2. Local Variables: Variables inside a block are not accessible outside it.
3. Initialization Block: Blocks without static run each time an instance is created.
4. Static Block: Blocks with static run once when the class is loaded.

CHALLENGE

51. Create a Book class for a library system.
    - Instance variables: title, author, isbn.
    - Static variable: totalBooks, a counter for the total number of book instances.
    - Instance methods: borrowBook(), returnBook().
    - Static method: getTotalBooks(), to get the total number of books in the library.
52. Design a Course class.
    - Instance variables: courseName, enrolledStudents.
    - Static variable: maxCapacity, the maximum number of students for any course.
    - Instance methods: enrollStudent(String studentName), unenrollStudent(String studentName).
    - Static method: setMaxCapacity(int capacity), to set the maximum capacity for courses.

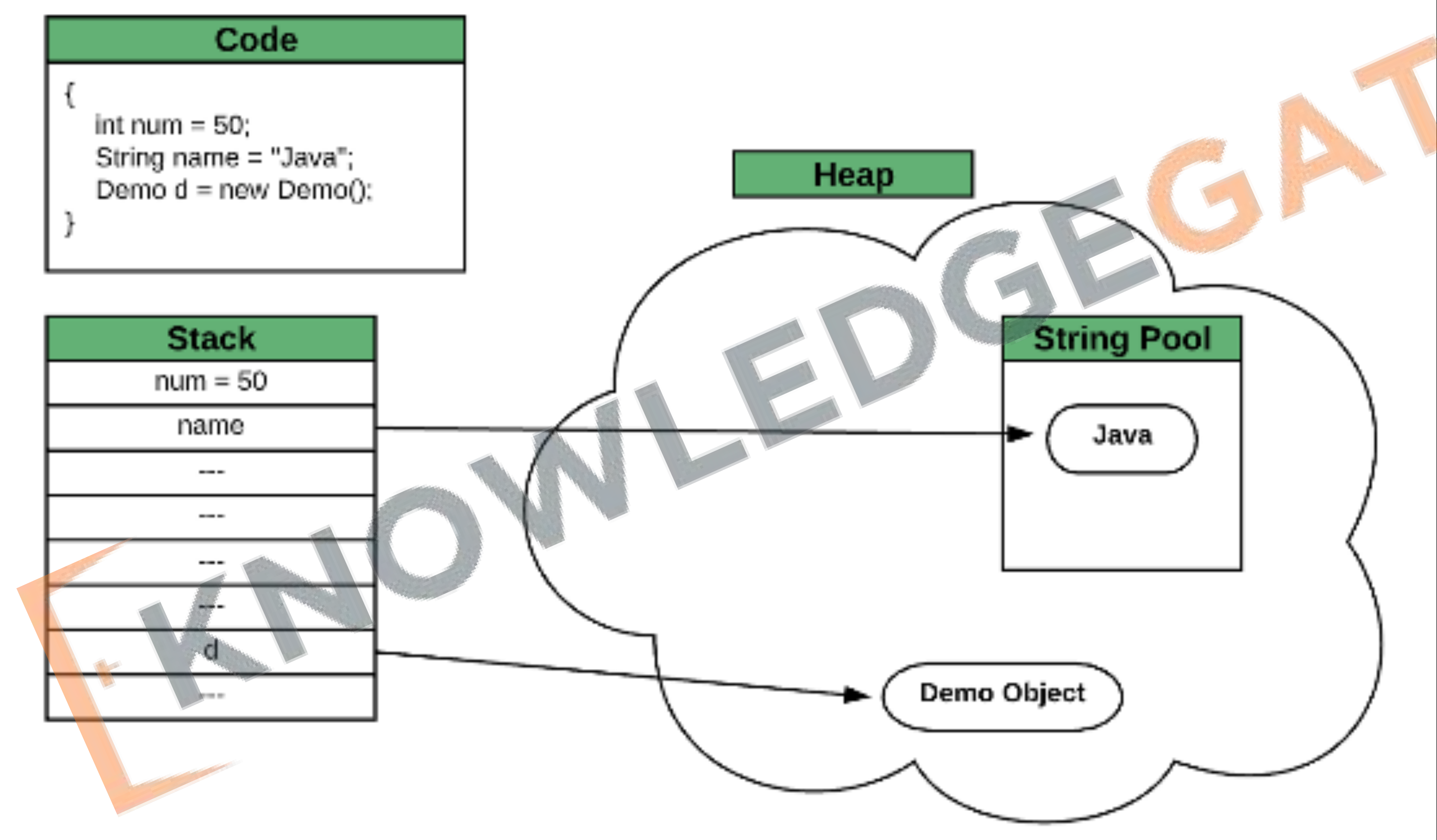
6.7 Stack vs Heap Memory

6.7 Stack vs Heap Memory

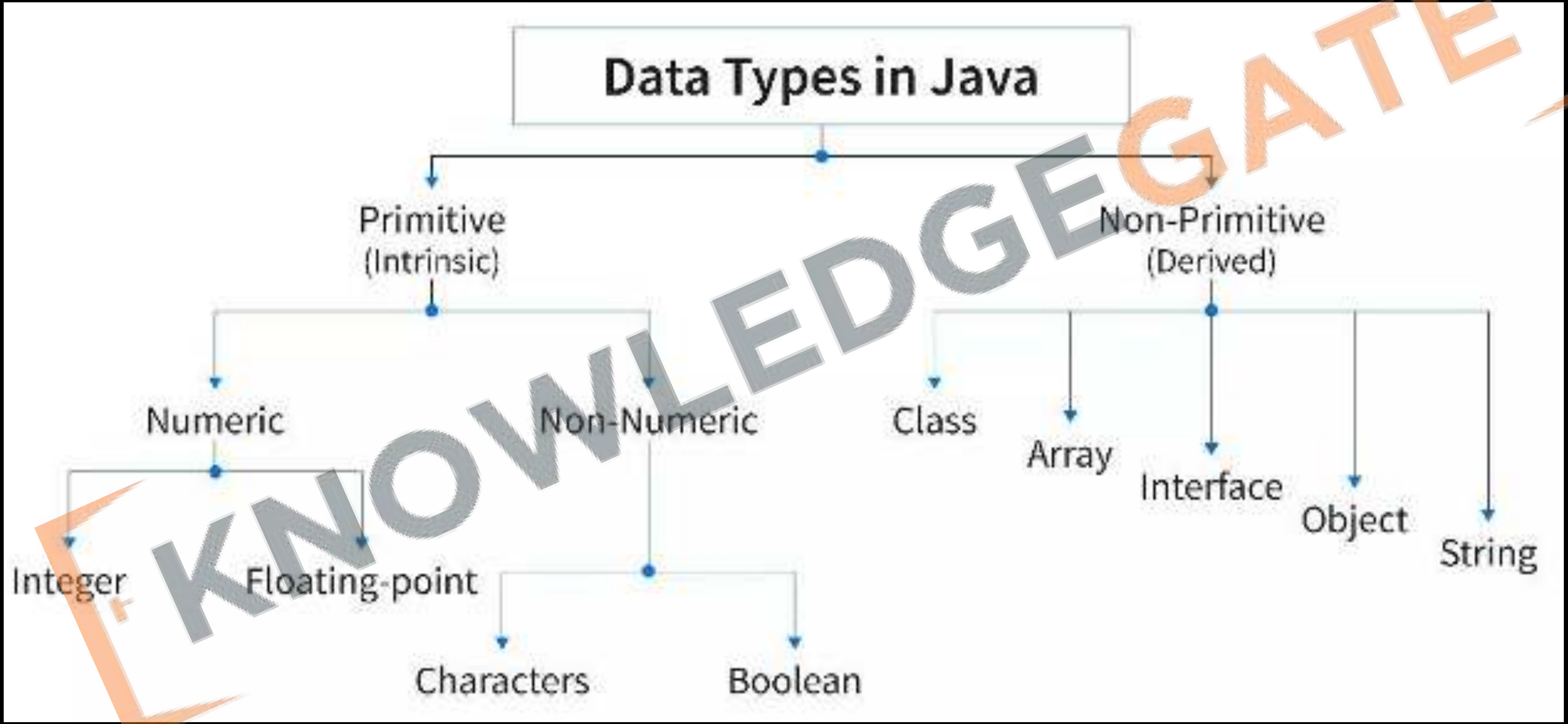


Stack Vs Heap

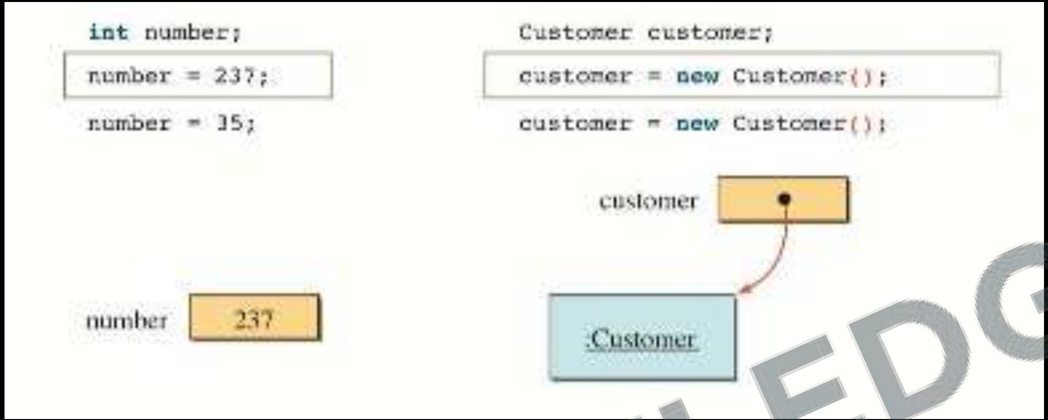
6.7 Stack vs Heap Memory



6.8 Primitive vs Reference Types

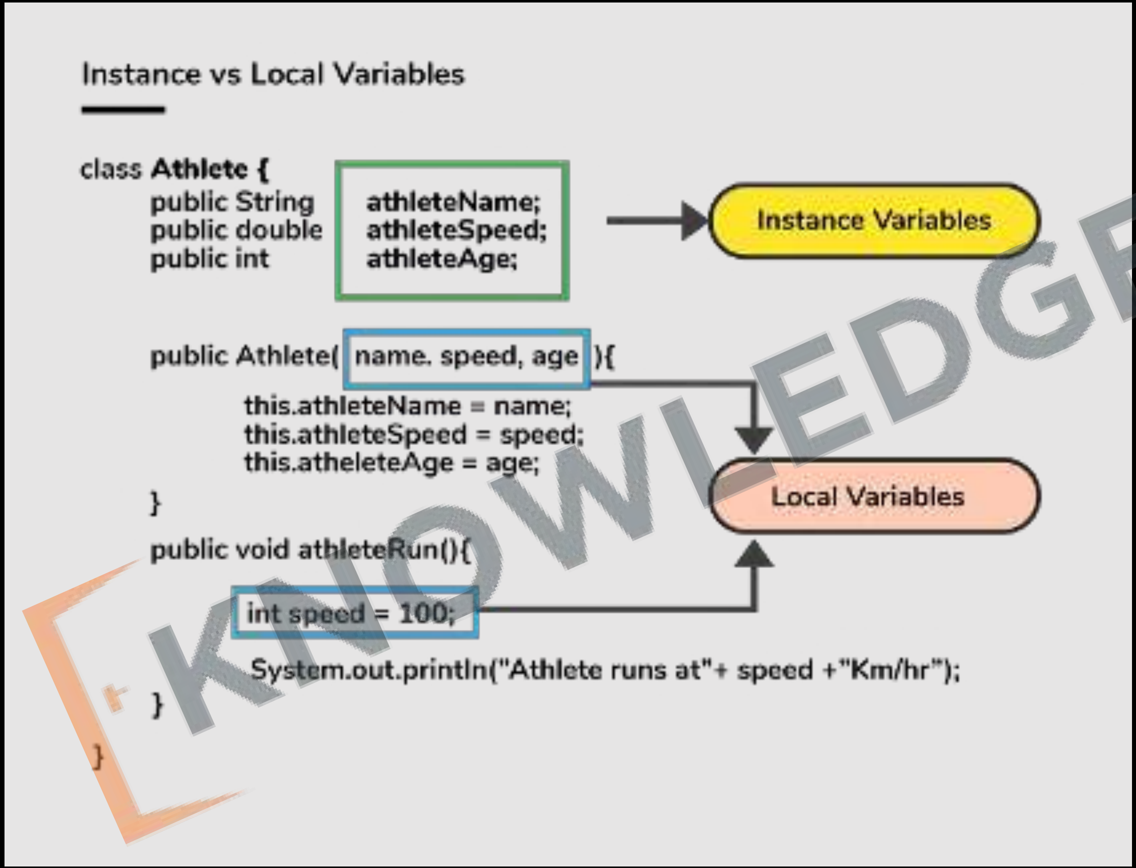


6.8 Primitive vs Reference Types

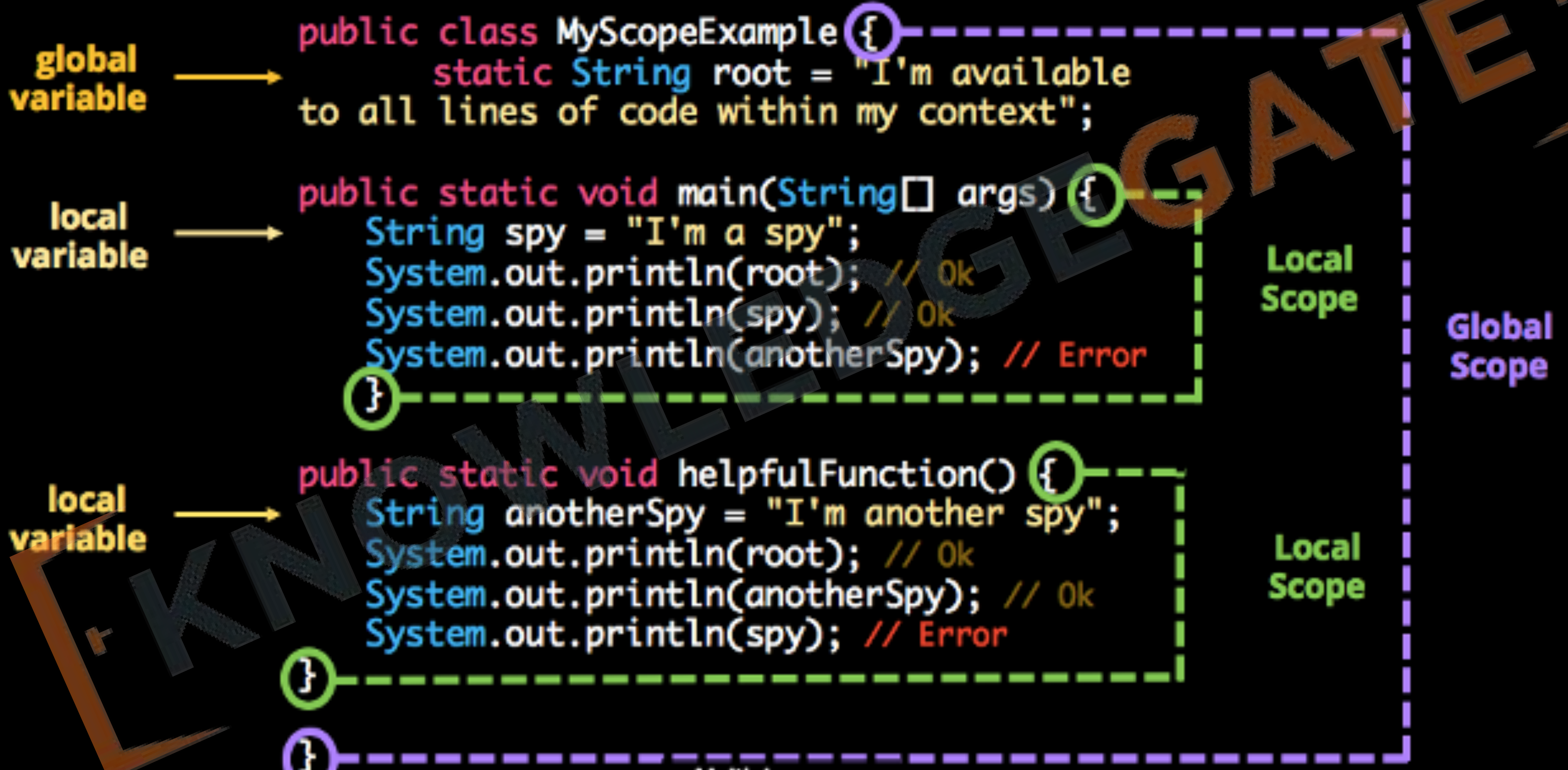


1. Memory: Primitives store actual values; reference types store addresses to objects.
2. Default Values: Primitives have specific defaults like 0 or false; reference types default to null.
3. Speed: Access to primitives is generally faster.
4. Storage Location: Primitives are stored in the stack; reference types are stored in the heap.
5. Comparison: Primitives compared by value; reference types compared by reference.

6.9 Variable Scopes



6.9 Variable Scopes

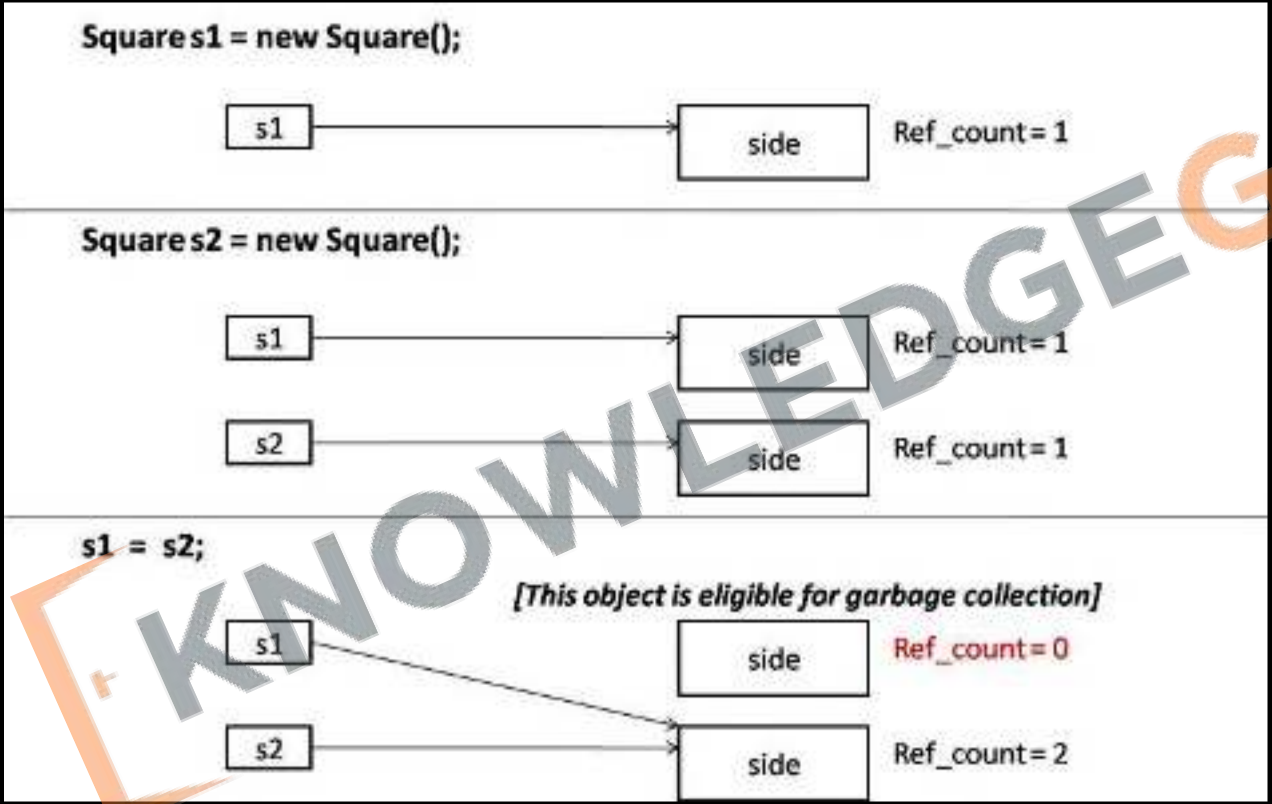


Multiple scopes

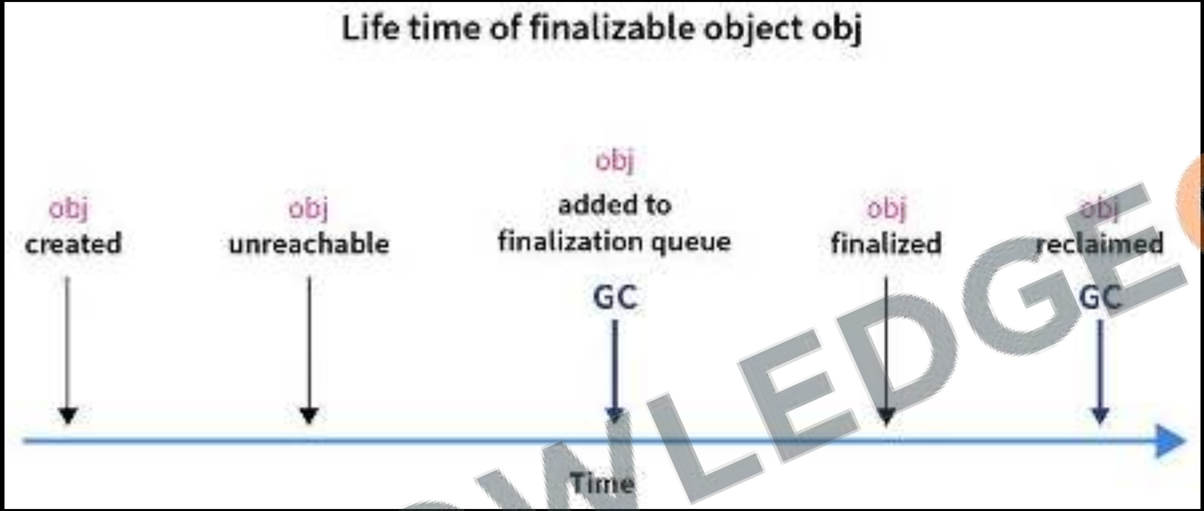
6.10 Garbage Collection & Finalize

1. Automatic Process: Garbage collection is managed by the Java Virtual Machine (JVM), running in the background.
2. Object Eligibility: Objects that are no longer reachable, meaning no active references to them, are eligible for garbage collection.
3. No Manual Control: Unlike languages like C++, Java developers cannot explicitly deallocate memory. Garbage collection is automatic and non-deterministic.
4. Generational Collection: Java uses a generational garbage collection strategy, which divides memory into different regions (young, old, and permanent generations) based on object ages.
5. Heap Memory: Garbage collection occurs in the heap memory, where all Java objects reside.
6. Performance Impact: Garbage collection can affect application performance, particularly if it runs frequently or takes a long time to complete.

6.10 Garbage Collection & Finalize



6.10 Garbage Collection & Finalize



1. Finalization: Before an object is garbage collected, the finalize() method may be called, giving the object a chance to clean up resources. However, it's not guaranteed to run, and its usage is generally discouraged.
2. Optimization: Developers can optimize the process indirectly through code practices, like setting unnecessary object references to null.
3. System.gc() Call: While System.gc() suggests that the JVM performs garbage collection, it's not a guarantee.

Revision

1. Process vs Object Oriented
2. Instance Variables and Methods
3. Declaring and Using Objects
4. Class vs Object
5. This & Static Keyword
6. Constructors & Code Blocks
7. Stack vs Heap Memory
8. Primitive vs Reference Types
9. Variable Scopes
10. Garbage Collection & Finalize

Practice Exercise

Classes, Objects

Answer in True/False:

1. Object-oriented programming is mainly concerned with data rather than logic.
2. Instance methods in Java can be called without creating an object of the class.
3. A class in Java is a template that can be used to create objects.
4. Static methods belong to the class and can be called without an object.
5. The new keyword is used to declare an array in Java.
6. Variables declared within a method are called instance variables.
7. In Java, you can access instance variables through static methods directly.
8. Every object in Java has its own unique set of instance variables.
9. The static keyword in Java means that a particular member belongs to a type itself, rather than to instances of that type.
10. Variable scope in Java is determined at runtime.

Practice Exercise

Classes, Objects

Answer in True/False:

1. Object-oriented programming is mainly concerned with data rather than logic. — True
2. Instance methods in Java can be called without creating an object of the class. — False
3. A class in Java is a template that can be used to create objects. — True
4. Static methods belong to the class and can be called without an object. — True
5. The new keyword is used to declare an array in Java. — True
6. Variables declared within a method are called instance variables. — False
7. In Java, you can access instance variables through static methods directly. — False
8. Every object in Java has its own unique set of instance variables. — True
9. The static keyword in Java means that a particular member belongs to a type itself, rather than to instances of that type. — True
10. Variable scope in Java is determined at runtime. — False

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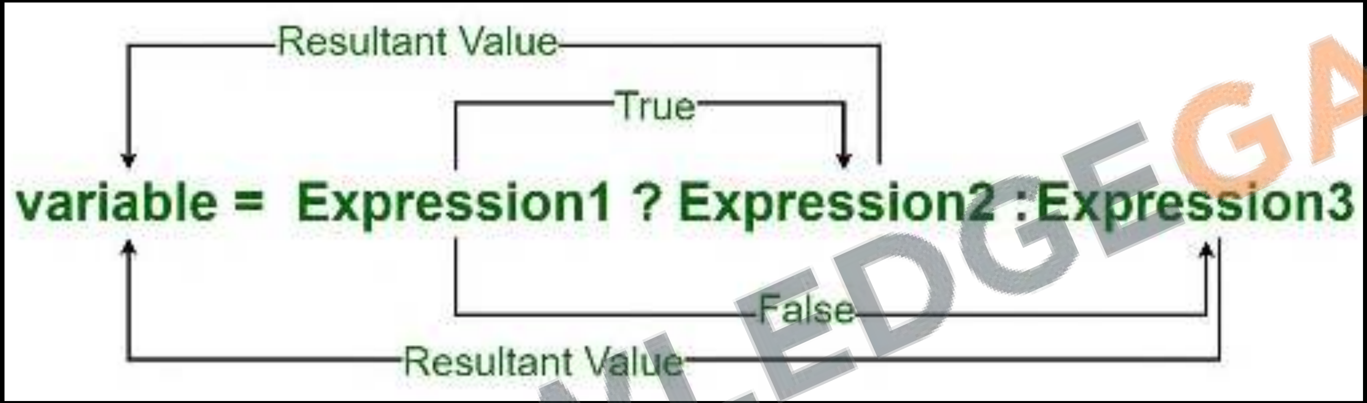
7. Control Statements, Math & String

1. Ternary operator
2. Switch
3. Loops (Do-while, For, For each)
4. Using break & continue
5. Recursion
6. Random Numbers & Math class
7. Don't Learn Syntax
8. toString Method
9. String class
10. StringBuffer vs StringBuilder
11. Final keyword

7.1 Ternary operator



1. Syntax: condition ? expression1 : expression2
2. Condition: Boolean expression, evaluates to true or false.
3. Expressions: Both expressions must return compatible types.
4. Use Case: Suitable for simple conditional assignments.
5. Readability: Good for simple conditions, but can reduce clarity if overused.

7.2 Switch

```
switch (day) {
    case 1: System.out.println("Monday");
            break;
    case 2: System.out.println("Tuesday");
            break;
    case 3: System.out.println("Wednesday");
            break;
    case 4: System.out.println("Thursday");
            break;
    case 5: System.out.println("Friday");
            break;
    case 6: System.out.println("Saturday");
            break;
    case 7: System.out.println("Sunday");
            break;
    default: System.out.println("Invalid day.");
}
```

1. Multiple Cases: Handles multiple values for an expression efficiently.
2. Supported Types: Accepts byte, short, char, int, String, enums, and from Java 14, long, float, double.
3. Case Labels: Each case ends with a colon (:) and is followed by code.
4. Break Statement: Typically used to prevent fall-through between cases.
5. Default Case: Executes if no case matches; optional and doesn't require break.
6. Type Safety: Case label types must match the switch expression's type.

7.2 Switch

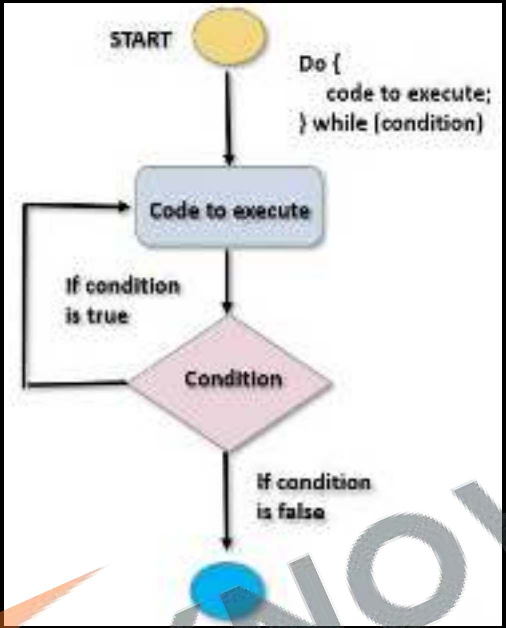
```java
String output = switch (day) {
        case 1 -> "Monday";
        case 2 -> "Tuesday";
        case 3 -> "Wednesday";
        case 4 -> "Thursday";
        case 5 -> "Friday";
        case 6 -> "Saturday";
        case 7 -> "Sunday";
        default -> "Invalid";
};
System.out.println(output);
```

1. Enhanced Switch: Java 12 introduced enhancements like yield and multiple constants per case.
2. Switch Expression: From Java 14, switch can return a value using yield.

CHALLENGE

53. Create a program to find the minimum of two numbers.
54. Create a program to find if the given number is even or odd.
55. Create a program to calculate the absolute value of a given integer.
56. Create a program to Based on a student's score, categorize as "High", "Moderate", or "Low" using the ternary operator (e.g., High for scores > 80, Moderate for 50-80, Low for < 50).
57. Create a program to print the month of the year based on a number (1-12) input by the user.
58. Create a program to create a simple calculator that uses a switch statement to perform basic arithmetic operations like addition, subtraction, multiplication, and division.

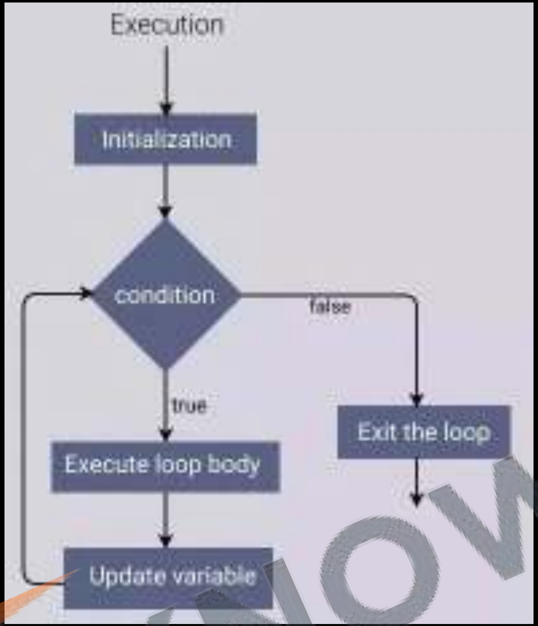
7.3 Loops (Do-while)




```
do {
  // Body of the loop
}
while (condition);
```

1. Executes block first, then checks condition.
2. Guaranteed to run at least one iteration.
3. Unlike while, first iteration is unconditional.
4. Don't forget to update condition to avoid infinite loops.

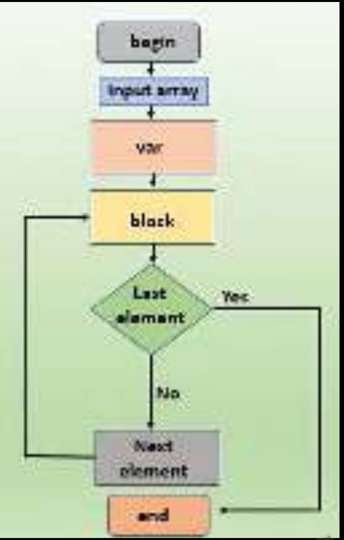
7.3 Loops (For)




```
for (initialisation; condition; update) {
  // Body of the loop
}
```

1. Standard loop for running code multiple times.
2. Generally preferred for counting iterations.

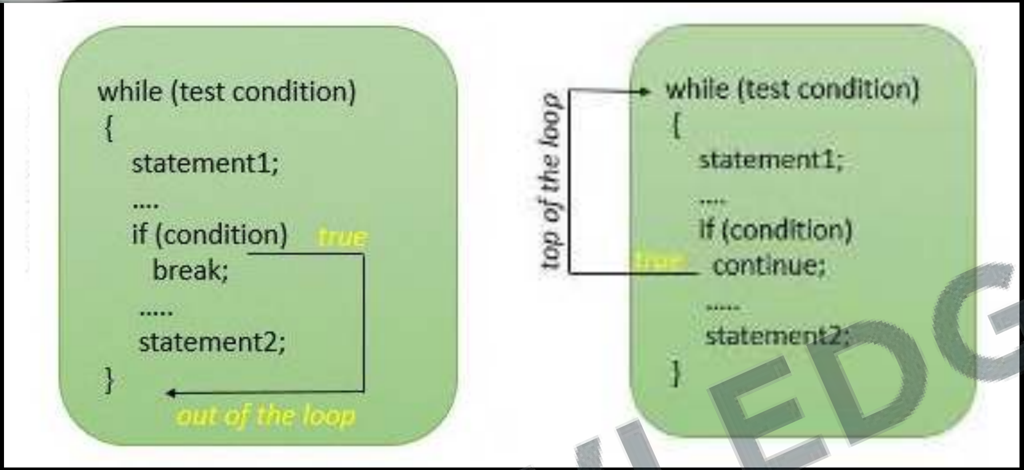
7.3 Loops (For each)




```java
String[] names = new String[] {
    "Ram", "Shyam", "Mohan", "Geeta",
    "Sita", "Sohan"
};
for (String name: names) {
    System.out.println(name);
}
```

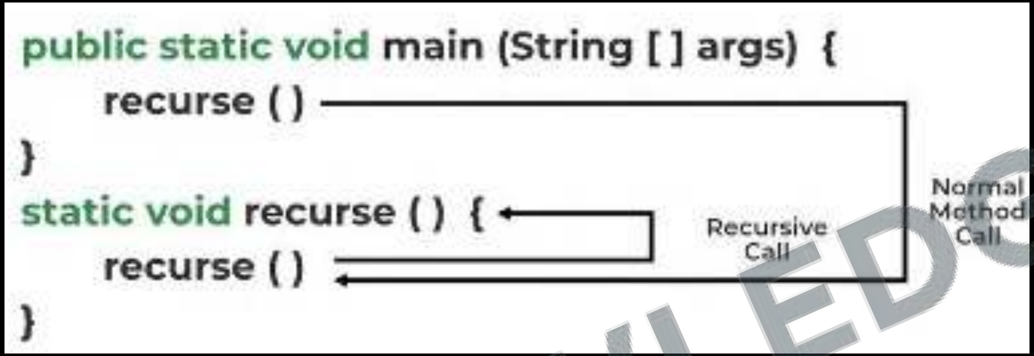
1. A method for array iteration, often preferred for readability.
2. Parameters: One for item, optional second for index.
3. Using return is similar to continue in traditional loops.
4. Not straightforward to break out of a forEach loop.
5. When you need to perform an action on each array element and don't need to break early.

7.4 Using break & continue



1. Break lets you stop a loop early, or break out of a loop
2. Continue is used to skip one iteration or the current iteration
3. In while loop remember to do the increment manually before using continue

7.5 Recursion

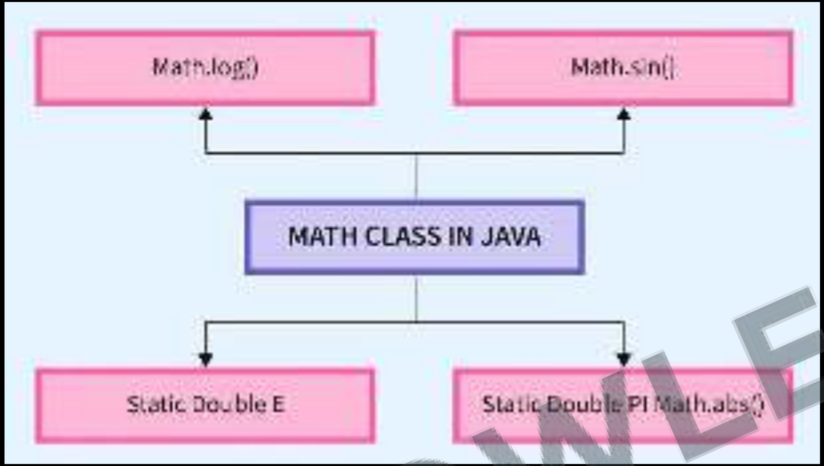


1. Self-Calling Function: Recursion is when a function calls itself.
2. Base Case: Essential to stop recursion and prevent infinite loops.
3. Recursive Case: The part where the function makes a recursive call.
4. Stack Overflow Risk: Excessive recursion can cause stack overflow errors.
5. Problem Solving: Ideal for problems divisible into similar, smaller problems.

CHALLENGE

59. Create a program using do-while to find password checker until a valid password is entered.
60. Create a program using do-while to implement a number guessing game.
61. Create a program using for loop multiplication table for a number.
62. Create a program using for to display if a number is prime or not.
63. Create a program using for-each to find the maximum value in an integer array.
64. Create a program using for-each to the occurrences of a specific element in an array.
65. Create a program using break to read inputs from the user in a loop and break the loop if a specific keyword (like "exit") is entered.
66. Create a program using continue to sum all positive numbers entered by the user; skip any negative numbers.
67. Create a program using continue to print only even numbers using continue for odd numbers.
68. Create a program using recursion to display the Fibonacci series upto a certain number.
69. Create a program using recursion to check if a string is a palindrome using recursion.

7.6 Random Numbers & Math class



Key Methods:

1. abs(): Absolute value.
2. ceil(): Rounds up.
3. floor(): Rounds down.
4. round(): Rounds to nearest integer.
5. max(), min(): Maximum and minimum of two numbers.
6. pow(): Power calculation.
7. sqrt(): Square root.
8. random(): Random number generation.
9. exp(), log(): Exponential and logarithmic functions.
10. Trigonometric functions: sin(), cos(), tan().

1. Static Class: Math methods are static and accessed directly.
2. Constants: Includes PI and E for π and the base of natural logarithms.

7.7 Don't Learn Syntax

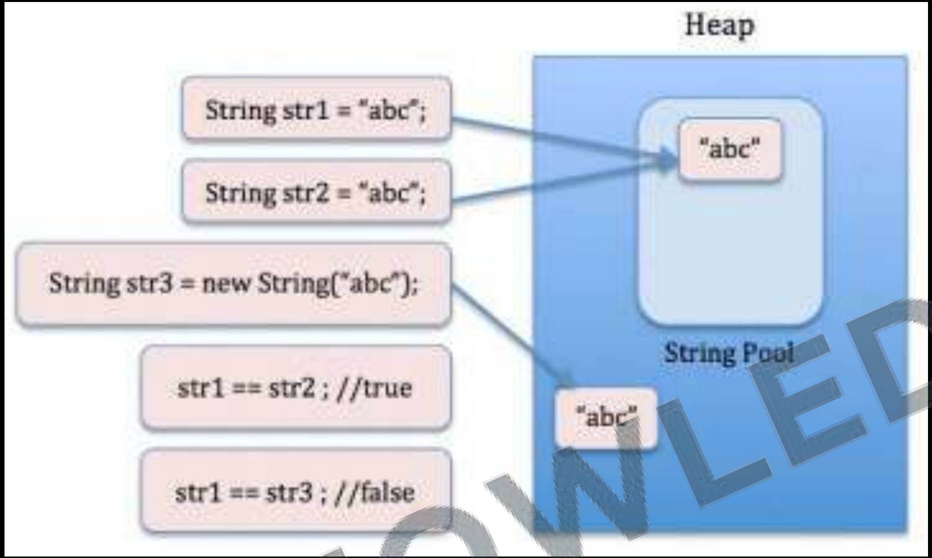
1. Google: Quick answers to coding problems.
2. Oracle: In-depth guides and documentation. https://docs.oracle.com/
3. ChatGPT: Real-time assistance for coding queries.
4. Focus: Understand concepts, not just syntax.

7.8 toString Method

```java
@Override
public String toString() {
    return "Car{" +
            "noOfWheels=" + noOfWheels +
            ", color='" + color + '\'' +
            ", maxSpeed=" + maxSpeed +
            ", currentFuelInLiters=" + currentFuelInLiters +
            ", noOfSeats=" + noOfSeats +
            '}';
}
```

1. Function: toString() provides a string representation of an object.
2. Inheritance: It's inherited from the Object class.
3. Default Format: By default, returns class name, "@", and hashcode.
4. Overriding: Commonly overridden in custom classes for meaningful output.
5. Implicit Call: Automatically called in string concatenation.

7.9 String class



1. Immutability: Once created, a String object's value cannot be changed. Modifications create new String objects.
2. String Pool: Java maintains a pool of strings for efficiency. When a new string is created, it's checked against the pool for a match to reuse.
3. Comparing: equals() method for value comparison, == operator checks reference equality.

7.9 String class

```
"Car{" +
 "noOfWheels=" + noOfWheels +
 ", color='" + color + '\'' +
 ", maxSpeed=" + maxSpeed +
 ", currentFuelInLiters=" + currentFuelInLiters +
 ", noOfSeats=" + noOfSeats +
 '}';
```

1. Concatenation: Strings can be concatenated using the + operator, but each concatenation creates a new String.
2. Methods: Provides methods like length(), substring(), equals(), compareTo(), indexOf(), for various operations.
3. Memory: Being immutable, strings can use more memory when frequently modified.

7.9 String class

Printf specifier	Data type
%s	String of text
%f	floating point value (float or double)
%e	Exponential, scientific notation of a float or double
%b	boolean true or false value
%c	Single character char
%d	Base 10 integer, such as a Java int, long, short or byte
%o	Octal number
%x	Hexadecimal number
%%	Percentage sign
%n	New line, aka carriage-return
%tY	Year in four digits
%tT	Time in format of HH:MM:SS (ie 21:46:30)

printf flag	Purpose
–	Aligns the formatted *printf* output to the left
+	The output includes a negative or positive sign
(	Places negative numbers in parenthesis
0	The formatted *printf* output is zero padded
,	The formatted output includes grouping separators
<space>	A blank space adds a minus sign for negative numbers and a leading space when positive

```
% [flags] [width] [.precision] specifier-character
```

7.9 String class

Pattern	Data	*Printf* Output
'%s'	Java	'Java'
'%15s'	Java	' Java'
'%-15s '	Java	'Java '
'%-15S '	Java	'JAVA '

7.9 String class

Pattern	Data	*Printf* output
'%d'	123,457,890	'123457890'
'%,15d'	123,457,890	' 123,457,890'
'%+,15d'	123457890	' +123,457,890'
'%-+,15d'	123457890	'+123,457,890 '
'%0,15d'	123457890	'0000123,457,890'

7.10 StringBuffer vs StringBuilder

```java
StringBuilder sentence = new StringBuilder("This is a sentence.");
sentence.append("Added word.");
System.out.println(sentence.toString()); //This is a sentence.Added word.
```

Parameter	String	StringBuilder	StringBuffer
mutability	immutable	mutable	mutable
Storage	String constant pool	heap	heap
Thread safety	Not used in the threaded environment as it is immutable	Not thread-safe so it is used in a single-threaded environment	Thread-safe so it is used in a multi-threaded environment
Speed	Comparably slowest	Comparably fastest	Faster than String but Slower than StringBuilder

7.11 Final keyword

```java
public class Main {
    public final String name = "John";

    public void setName(String name) {
        this.name = name; // cannot reassign final variables
    }
}
```

1. Variable: When applied to a variable, it becomes a constant, meaning its value cannot be changed once initialized.
2. Efficiency: Using final can lead to performance optimization, as the compiler can make certain assumptions about final elements.
3. Null Safety: A final variable must be initialized before the constructor completes, reducing null pointer errors.
4. Immutable Objects: Helps in creating immutable objects in combination with private fields and no setter methods.

CHALLENGE

70. Define a Student class with fields like name and age, and use toString to print student details.
71. Concatenate and Convert: Take two strings, concatenate them, and convert the result to uppercase.
72. Calculate the area and circumference of a circle for a given radius using Math.PI
73. Simulate a dice roll using Math.random() and display the outcome (1 to 6).
74. Create a number guessing game where the program selects a random number, and the user has to guess it.
75. Take an array of words and concatenate them into a single string using StringBuilder.
76. Create an object with final fields and a constructor to initialize them.

Revision

1. Ternary operator
2. Switch
3. Loops (Do-while, For, For each)
4. Using break & continue
5. Recursion
6. Random Numbers & Math class
7. Don't Learn Syntax
8. toString Method
9. String class
10. StringBuffer vs StringBuilder
11. Final keyword

Practice Exercise

Control Statements, Math & String

Answer in True/False:

1. The ternary operator in Java can replace certain types of if-else statements.
2. A switch statement in Java can only test for equality with constants.
3. The do-while loop will execute at least once even if the condition is false.
4. A for-each loop in Java is used to iterate over arrays.
5. The break statement skips the current iteration.
6. The continue statement exits the loop immediately, regardless of the condition.
7. In Java, recursion is a process where a method can call itself directly.
8. Random numbers generated by Math.random() are inclusive of 0 and 1.
9. The String class is mutable, meaning it can be changed after it's created.
10. StringBuilder is faster than StringBuffer since it is not synchronized.
11. A final variable in Java can be assigned a value once and only once.

Practice Exercise

Control Statements, Math & String

Answer in True/False:

1. The ternary operator in Java can replace certain types of if-else statements. — True
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3. The do-while loop will execute at least once even if the condition is false. — True
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5. The break statement skips the current iteration. — False
6. The continue statement exits the loop immediately, regardless of the condition. — False
7. In Java, recursion is a process where a method can call itself directly. — True
8. Random numbers generated by Math.random() are inclusive of 0 and 1. — False
9. The String class is mutable, meaning it can be changed after it's created. — False
10. StringBuilder is faster than StringBuffer since it is not synchronized. — True
11. A final variable in Java can be assigned a value once and only once. — True

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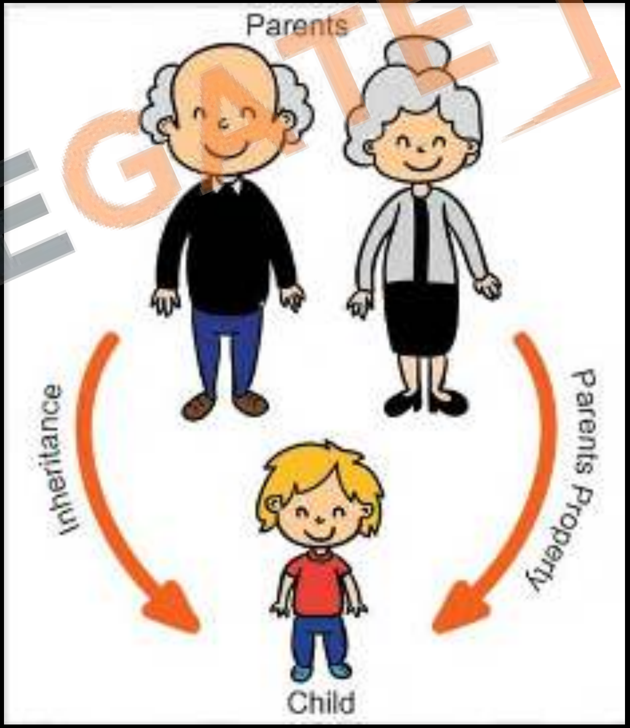
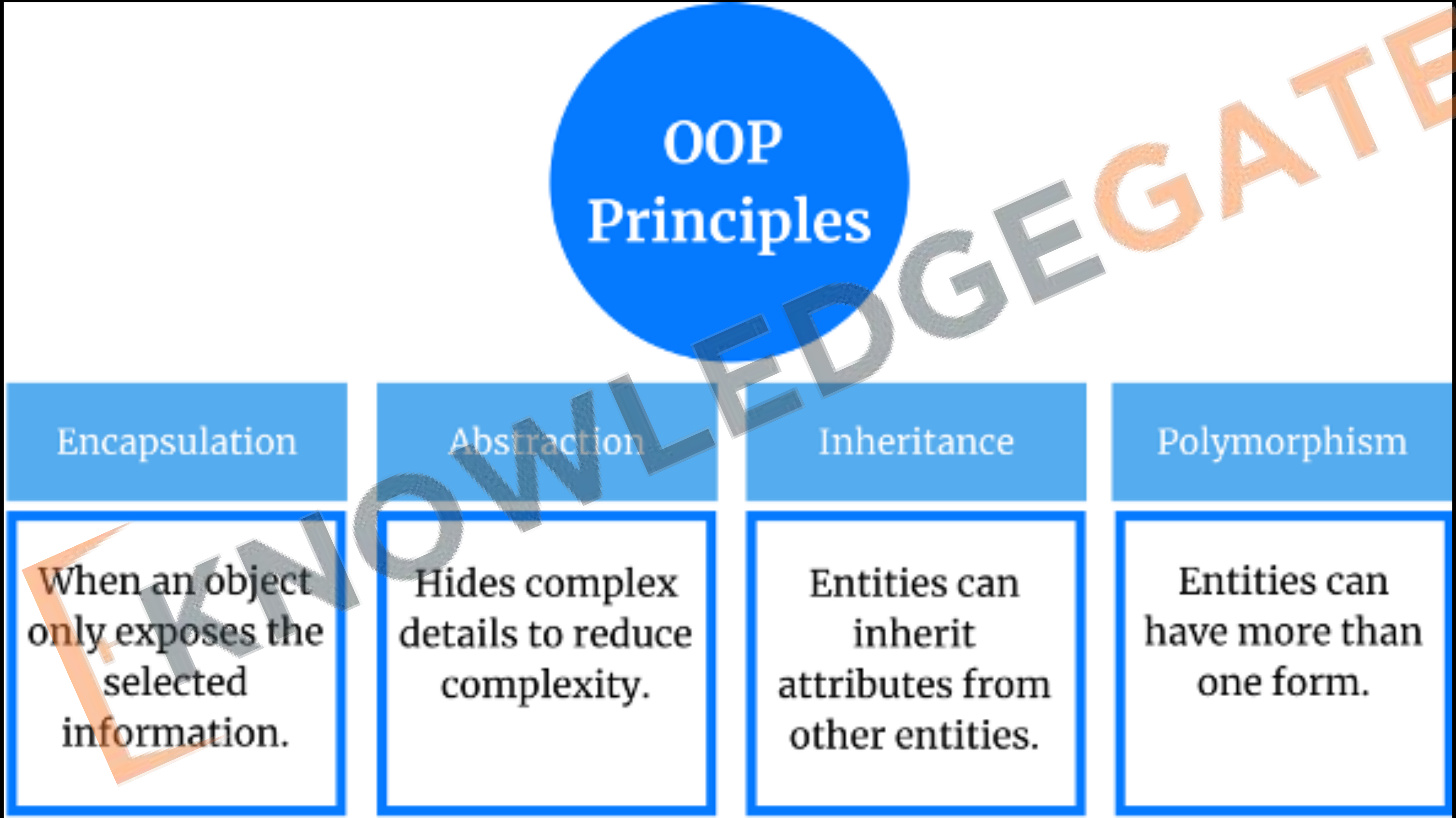
8 Encapsulation & Inheritance

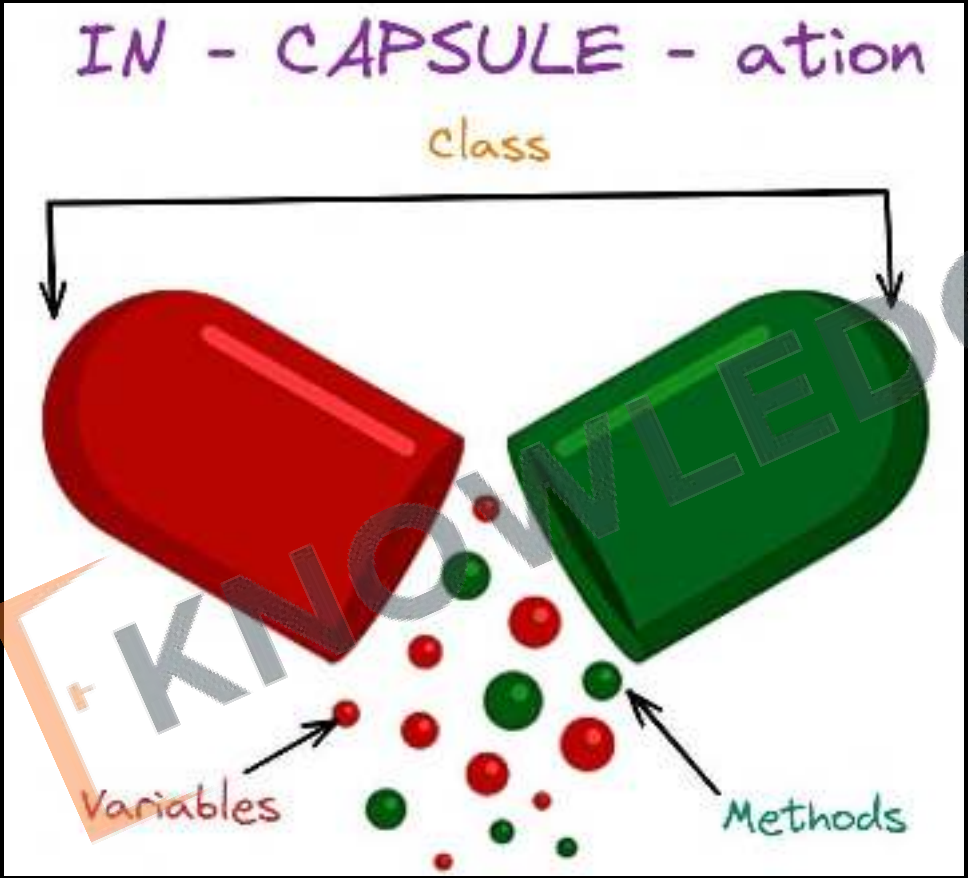
1. Intro to OOPs Principle
2. What is Encapsulation
3. Import & Packages
4. Access Modifiers
5. Getter and Setter
6. What is Inheritance
7. Types of Inheritance
8. Object class
9. Equals and Hash Code
10. Nested and Inner Classes



8.1 Intro to OOPs Principle



8.2 What is Encapsulation



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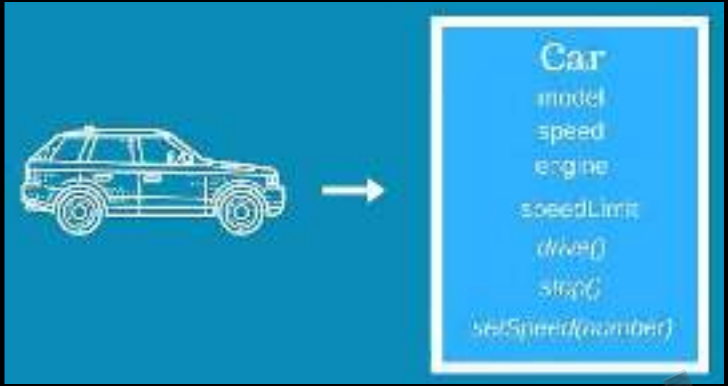
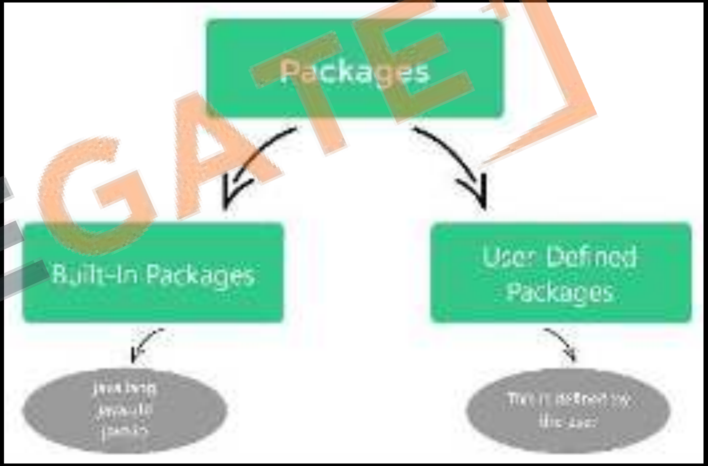


1. Data Hiding: Encapsulation hides internal data, allowing access only through methods.
2. Access Modifiers: Uses private, public, protected to control access to class members.
3. Getter/Setter: Provides public methods for controlled property access.
4. Maintains Integrity: Protects object state from external interference.
5. Enhances Modularity: Keeps classes separate and reduces coupling.

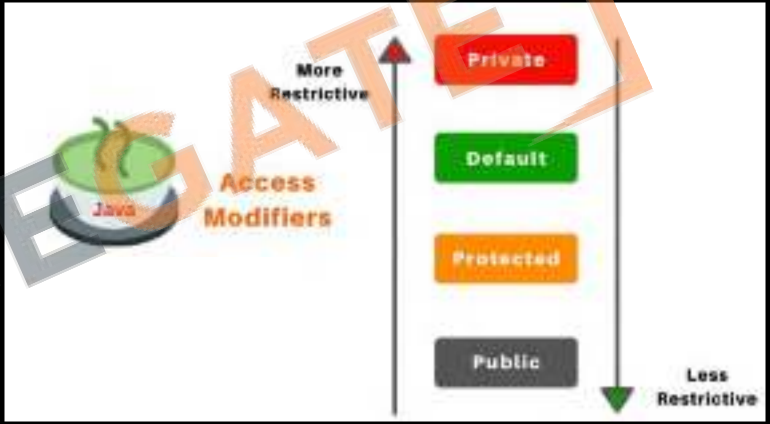
8.3 Import & Packages

1. Package Definition: A package in Java is a namespace that organizes classes and interfaces, preventing naming conflicts.
2. Package Declaration: Packages are declared at the beginning of a Java source file using the package keyword followed by the package name.
3. Import Statement: An import statement in Java is used to bring in classes or interfaces from other packages into the current file, making them accessible without using a fully qualified name.
4. Types of Import:
    - Single-Type Import: Imports a single class or interface from a package (e.g., import java.util.List;).
    - On-Demand Import: Imports all classes and interfaces from a package (e.g., import java.util.*;).
5. Avoiding Collisions: Packages help in avoiding name collisions by categorizing similar classes together.
6. Built-in Packages: Java comes with built-in packages like java.lang (automatically imported), java.util, java.io, etc.



8.4 Access Modifiers

1. Types: The four access modifiers are public, protected, default (no modifier), and private.
2. Public: The public modifier allows access from any other class. For classes, it means they can be accessed from any other package.
3. Protected: The protected modifier allows access within the same package and subclasses.
4. Default: If no access modifier is specified, it's the default, allowing access only within the same package.
5. Private: The private modifier restricts access to the defining class only.
6. Class-Level Access: Only public and default (no modifier) access modifiers are applicable for top-level classes.
7. Member-Level Access: Methods, constructors, and variables can use all four access modifiers to control visibility.



8.4 Access Modifiers

Access Specifiers in Java

		public	private	protected	default
Same Package	Class	YES	YES	YES	YES
	Sub class	YES	NO	YES	YES
	Non sub class	YES	NO	YES	YES
Different Package	Sub class	YES	NO	YES	NO
	Non sub class	YES	NO	NO	NO

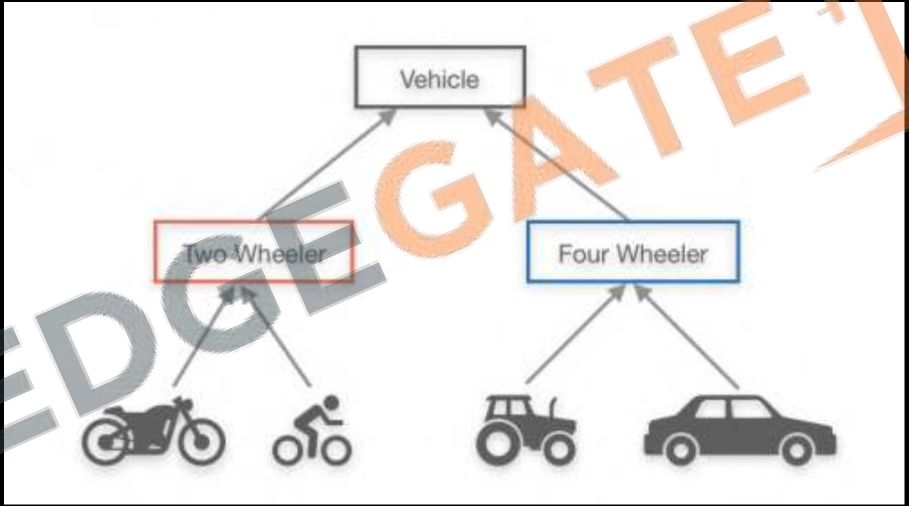
8.5 Getter and Setter

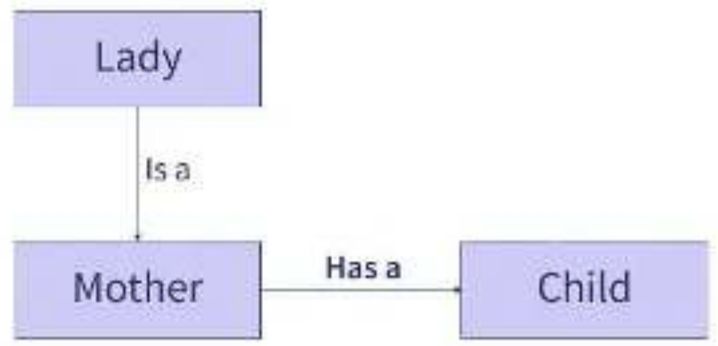
1. Getters: Retrieve private field values, typically named get<FieldName>.
2. Setters: Set or update private field values, usually named set<FieldName>.
3. Control and Validation: Offer controlled access and allow for validation logic.
4. Encapsulation: Facilitate read-only or write-only access to class fields.
5. Flexibility: Allow for internal changes without affecting external interfaces.

CHALLENGE

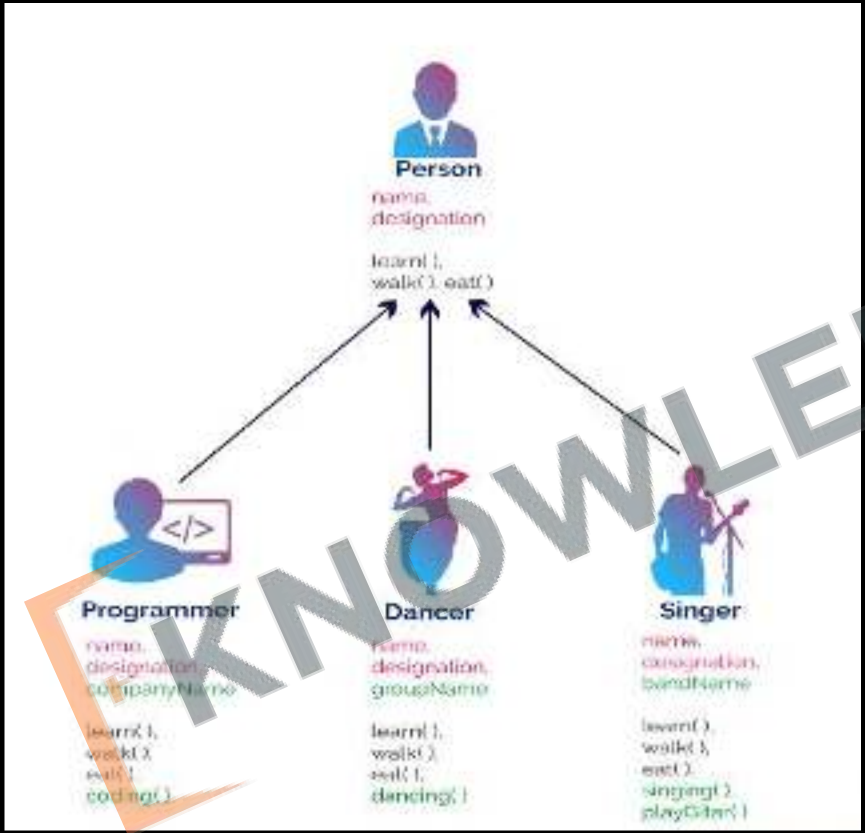
77. Create a simple application with at least two packages: com.example.geometry and com.example.utils. In the geometry package, define classes like Circle and Rectangle. In the utils package, create a Calculator class that can compute areas of these shapes.
78. Define a BankAccount class with private attributes like accountNumber, accountHolderName, and balance. Provide public methods to deposit and withdraw money, ensuring that these methods don't allow illegal operations like withdrawing more money than the current balance.
79. Define a class Employee with private attributes (like name, age, and salary), public methods to get and set these attributes, and a package-private method to displayEmployeeDetails. Create another class in the same package to test access to the displayEmployeeDetails method.

8.6 What is Inheritance



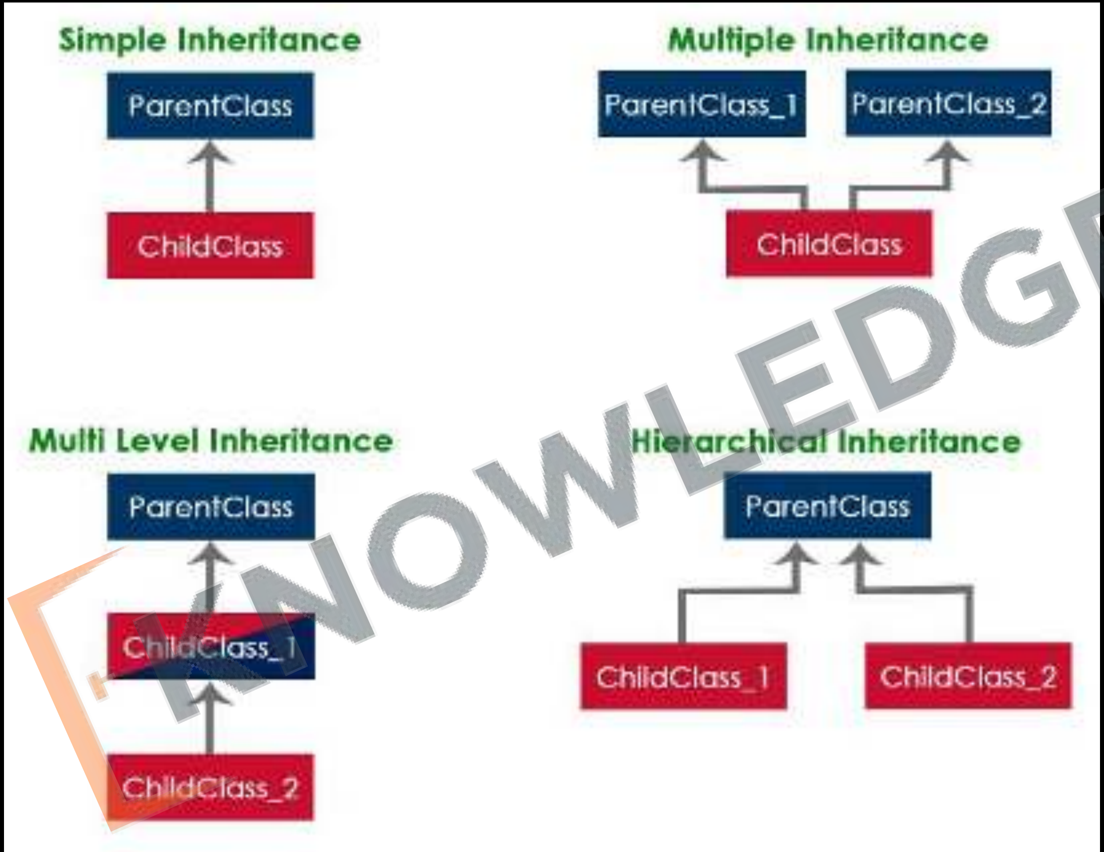


8.6 What is Inheritance

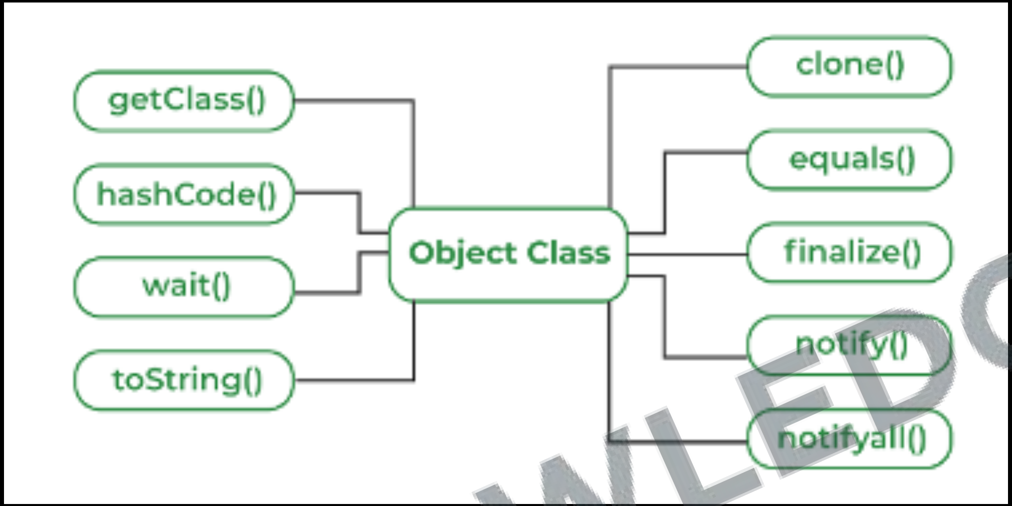


1. Inheritance allows a new class (subclass) to inherit features from an existing class (superclass).
2. Code Reusability: It enables subclasses to use methods and variables of the superclass, reducing code duplication.
3. Access Control: The protected access modifier is often used in inheritance to allow subclass access to superclass members.

8.7 Types of Inheritance

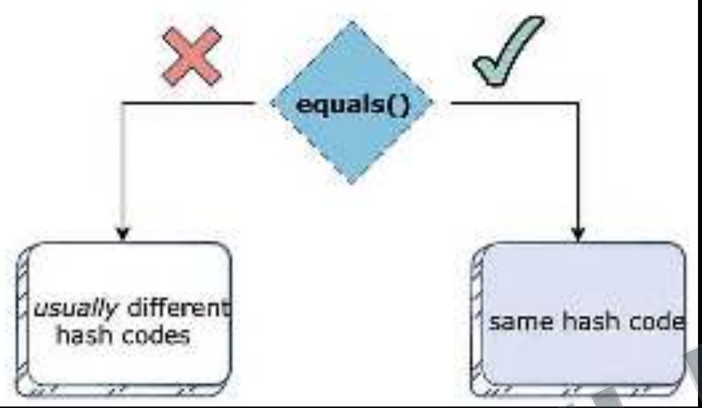


8.8 Object class



1. Root Class: The Object class is the parent class of all classes in Java, forming the top of the class hierarchy.
2. Default Methods: It provides fundamental methods like equals(), hashCode(), toString(), and getClass() that can be overridden by subclasses.
3. String Representation: The toString() method returns a string representation of the object, often overridden for more informative output.

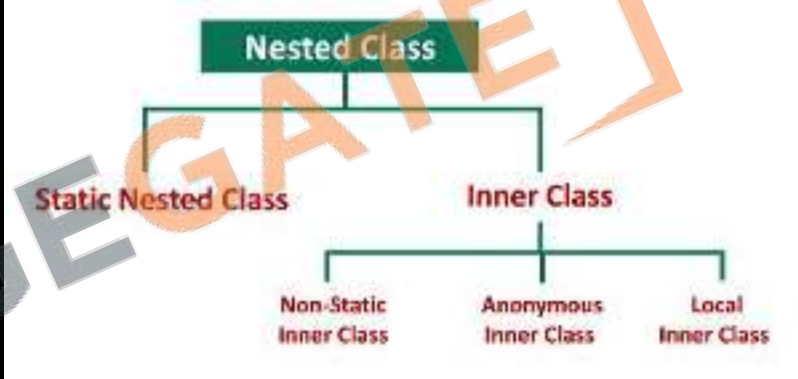
8.9 Equals and HashCode



1. equals() Method: Used for logical equality checks between objects. By default, it compares object references, but it's commonly overridden to compare object states.
2. hashCode() Method: Generates an integer hash code representing an object. It's crucial for the performance of hash-based collections like HashMap.
3. Equals-Hashcode Contract: If two objects are equal according to equals(), they must have the same hash code. However, two objects with the same hash code aren't necessarily equal.
4. Overriding Both: If equals() is overridden, hashCode() should also be overridden to maintain consistency between these methods.

8.10 Nested and Inner Classes

1. Nested Classes: Classes defined within another class, divided into static (static nested classes) and non-static (inner classes).
2. Static Nested Classes: Act as static members of the outer class; can access outer class's static members but not its non-static members.
3. Inner Classes: Associated with an instance of the outer class; can access all members of the outer class, including private ones.
4. Local and Anonymous Inner Classes: Local inner classes are defined within a block (like a method) and are not visible outside it. Anonymous inner classes are nameless and used for single-use implementations.
5. Use Cases: Useful for logically grouping classes, improving encapsulation, and enhancing code readability.



CHALLENGE

80. Start with a base class LibraryItem that includes common attributes like itemID, title, and author, and methods like checkout() and returnItem(). Create subclasses such as Book, Magazine, and DVD, each inheriting from LibraryItem. Add unique attributes to each subclass, like ISBN for Book, issueNumber for Magazine, and duration for DVD.

81. Create a class Person with attributes name and age. Override equals() to compare Person objects based on their attributes. Override hashCode() consistent with the definition of equals().

82. Create a class ArrayOperations with a static nested class Statistics. Statistics could have methods like mean(), median(), which operate on an array.

Revision

1. Intro to OOPs Principle
2. What is Encapsulation
3. Import & Packages
4. Access Modifiers
5. Getter and Setter
6. What is Inheritance
7. Types of Inheritance
8. Object class
9. Equals and Hash Code
10. Nested and Inner Classes

Practice Exercise

Encapsulation & Inheritance

Answer in True/False

1. Encapsulation is the OOP principle that ensures an object's data can be changed by any method, without restrictions.
2. The import statement in Java can be used to bring a single class or an entire package into visibility.
3. The private access modifier means that the member is only accessible within its own class.
4. Setters are used to retrieve the value of a private variable from outside the class.
5. Inheritance in OOP can be used to create a general class that defines traits to be inherited by more specific subclasses.
6. The Object class in Java has protected access by default.
7. If two objects are equal according to their equals() method, then their hashCode() methods must return different integers.
8. Inner classes are defined within the scope of a method and cannot exist independently of the method.
9. A static nested class can access the instance variables of its enclosing outer class.
10. Overriding the equals() method requires also overriding the hashCode() method to maintain consistency.

Practice Exercise

Encapsulation & Inheritance

Answer in True/False

1. Encapsulation is the OOP principle that ensures an object's data can be changed by any method, without restrictions. False
2. The import statement in Java can be used to bring a single class or an entire package into visibility. True
3. The private access modifier means that the member is only accessible within its own class. True
4. Setters are used to retrieve the value of a private variable from outside the class. False
5. Inheritance in OOP can be used to create a general class that defines traits to be inherited by more specific subclasses. True
6. The Object class in Java has protected access by default. False
7. If two objects are equal according to their equals() method, then their hashCode() methods must return different integers. False
8. Inner classes are defined within the scope of a method and cannot exist independently of the method. False
9. A static nested class can access the instance variables of its enclosing outer class. False
10. Overriding the equals() method requires also overriding the hashCode() method to maintain consistency. True

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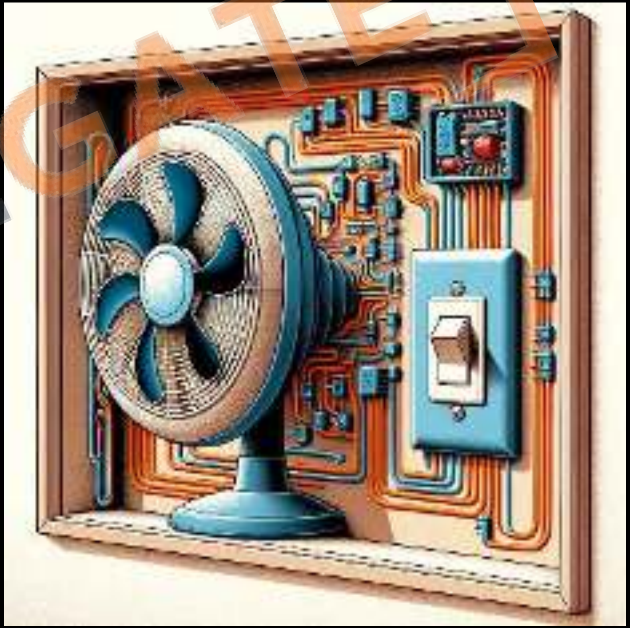
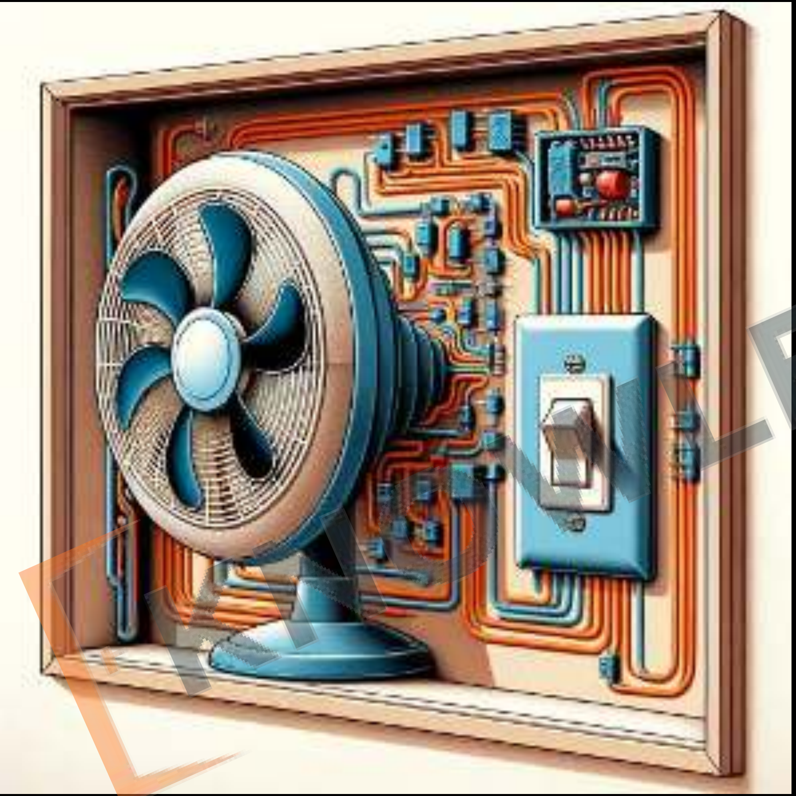
9 Abstraction and Polymorphism

1. What is Abstraction
2. Abstract Keyword
3. Interfaces
4. What is Polymorphism
5. References and Objects
6. Method / Constructor Overloading
7. Super Keyword
8. Method / Constructor Overriding
9. Final keyword revisited
10. Pass by Value vs Pass by reference.

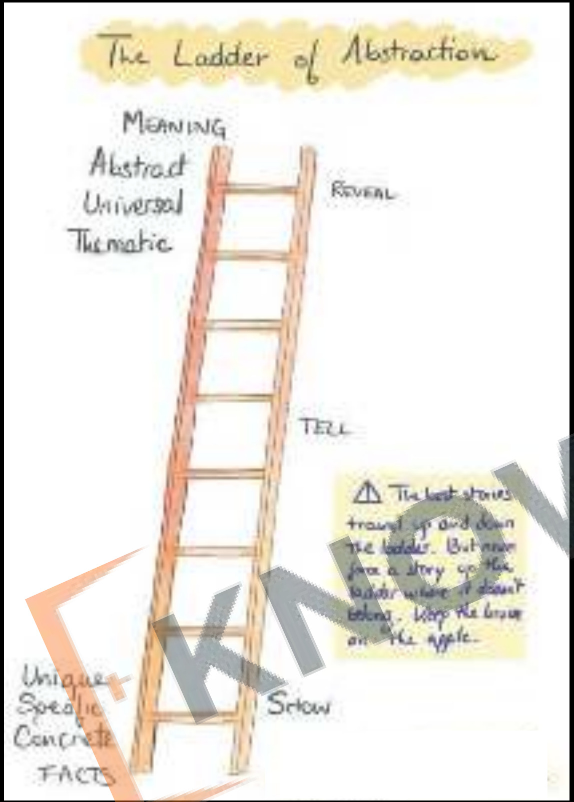


9.1 What is Abstraction



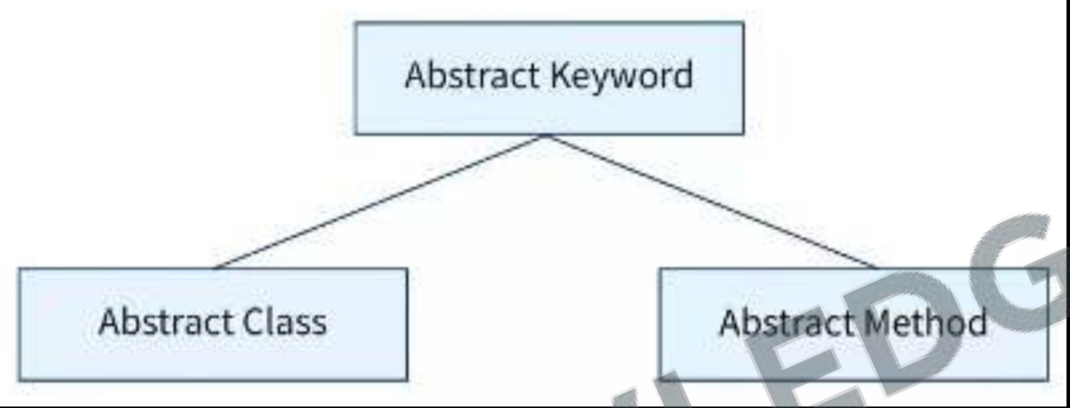
1. Core Principle: Abstraction hides complex implementation details, focusing only on essential features.
2. Focus on Functionality: Emphasizes what an object does, not how it does it, through clear interfaces.

9.1 What is Abstraction



1. **Simplifies Complexity:** It reduces complexity by showing only relevant information in class design.
2. **Real-World Modelling:** Abstraction allows creating objects that represent real-life entities with key attributes and behaviours.

9.2 Abstract Keyword



1. Abstract Class: Used to declare non-instantiable abstract classes that serve as base classes.
2. Abstract Method: Defines methods without implementations, requiring subclasses to provide specific functionality.
3. Mandatory Implementation: Subclasses must implement all abstract methods of an abstract class.
4. Design Flexibility: Allows for flexible class design by defining a contract for subclasses.

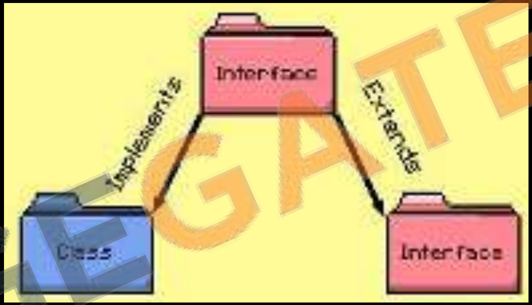
9.3 Interfaces

```
interface Pet {
    public void play();
}

class Dog extends Animal implements Pet{
    // Dog has its own implementation of run method
    public void run() { }

    public void bark() { }

    // Define the methods of the interface
    public void play() { }
}
```



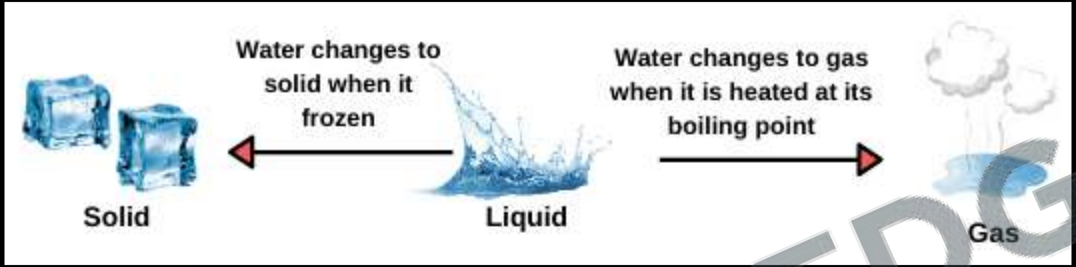
1. Interfaces primarily declare abstract methods for implementation by classes.
2. A class can implement multiple interfaces, allowing for more flexible designs.
3. Interfaces can have default methods with implementation and static methods.
4. Interface methods are inherently public and abstract, except for default and static methods.

CHALLENGE

83. Create an abstract class Shape with an abstract method calculateArea(). Implement two subclasses: Circle and Square. Each subclass should have relevant attributes (like radius for Circle, side for Square) and their own implementation of the calculateArea() method.

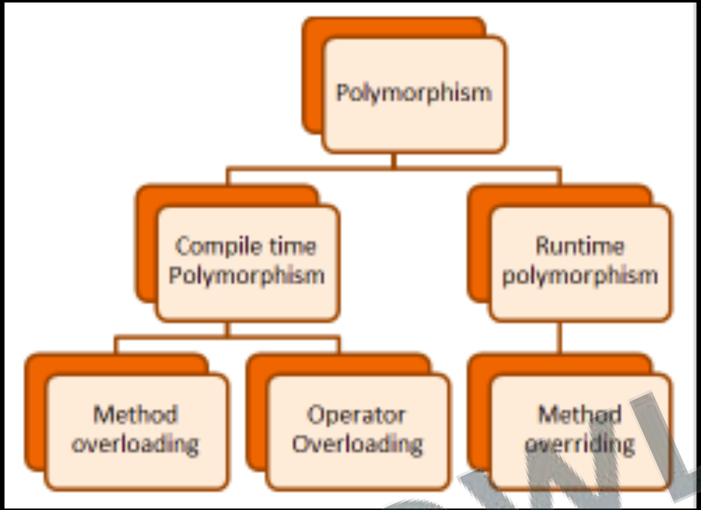
84. Create an interface Flyable with an abstract method fly(). Create an abstract class Bird that implements Flyable. Implement a subclass Eagle that extends Bird. Provide an implementation for the fly() method.

9.4 What is Polymorphism



1. Polymorphism is the ability of objects of different classes to respond to the same message (method call) in different ways.
2. Flexibility: Allows for writing more flexible and reusable code.
3. Simplicity: Enables developers to write more simple and readable code by using the same interface for different underlying data types.

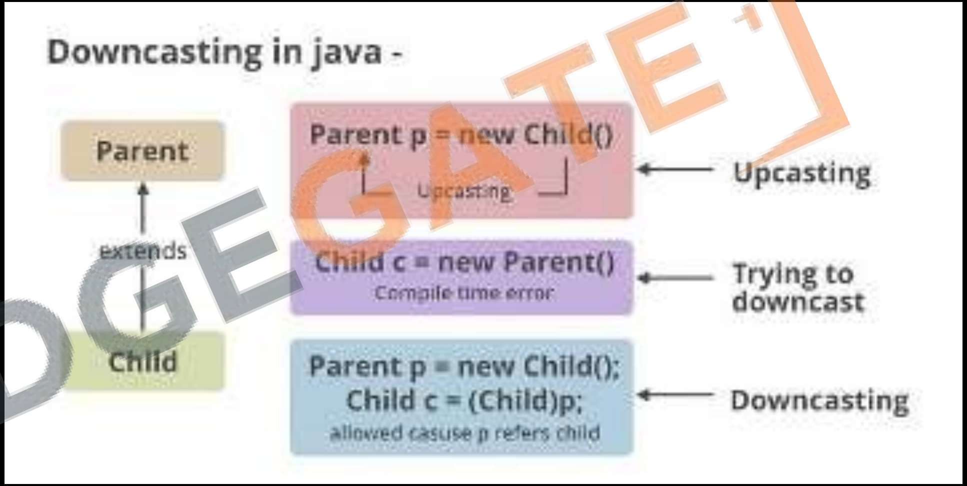
9.4 What is Polymorphism



1. Compile-Time Polymorphism: Achieved through method overloading or operator overloading.
2. Run-Time Polymorphism: Achieved through method overriding, where a subclass provides a specific implementation of a method already defined in its superclass.

9.5 References and Objects

1. Upcasting:
   - Converts subclass to superclass reference.
   - Automatic and safe.
   - Access only to superclass methods.
2. Downcasting:
   - Converts superclass to subclass reference.
   - Manual and risky, needs instanceof check.
   - Access to subclass-specific methods.
3. Usage:
   - Upcasting for generalization in methods.
   - Downcasting for specific subclass behaviors.



9.6 Method / Constructor Overloading

```java
// add() method is overloaded
public static int add(int a,int b) {
    return a + b; //adds integers
}

public static String add(String s1,String s2) {
    return s1.concat(s2); //concats strings
}
```

```java
class Dog{
    public void bark(){
        System.out.println("woof ");
    }
    //overloading method
    public void bark(int num){
        for(int i=0; i<num; i++)
            System.out.println("woof ");
    }
}
```

Same Method Name,
Different Parameter

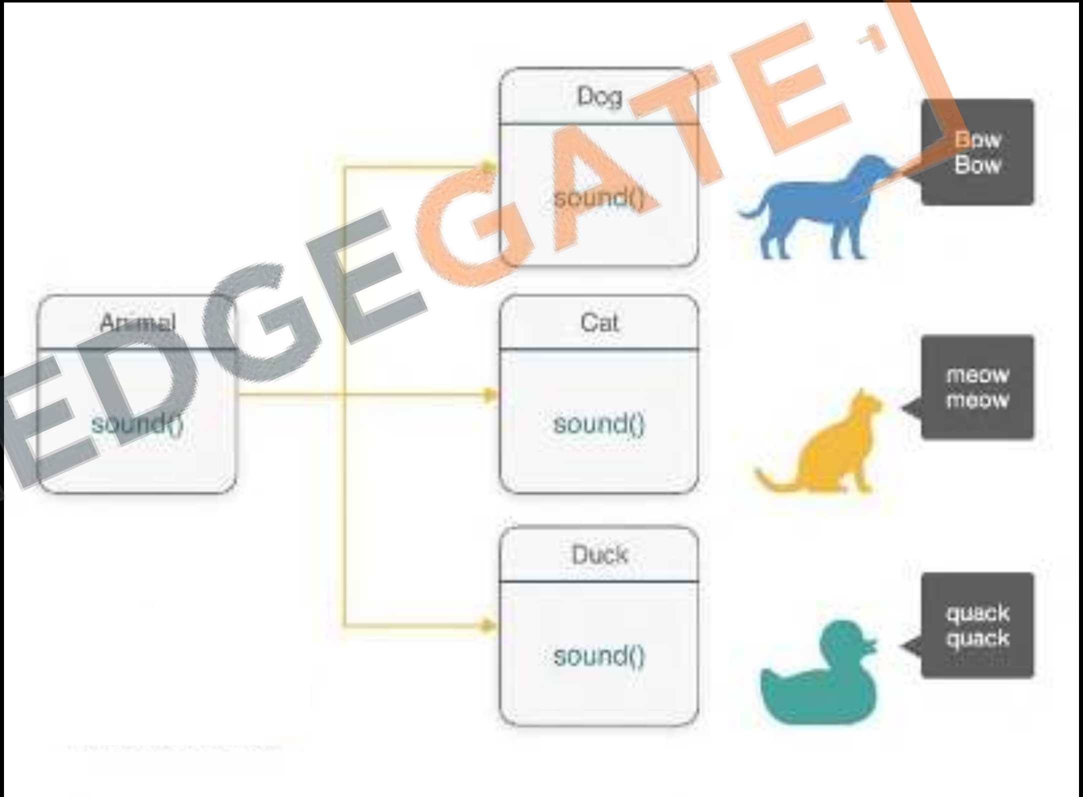
1. Method overloading occurs when multiple methods in the same class have the same name but different parameter lists.
2. Parameter Difference: Overloaded methods must differ in the number, type, or sequence of their parameters.
3. Return Type: Can vary between overloaded methods, but the return type alone does not distinguish them.
4. Compile-Time Polymorphism: It's a form of polymorphism that is resolved during compile time.

9.7 Super Keyword

1. Super can be used to refer immediate parent class instance varible.
2. Super can be used to invoke immediate parent class method.
3. Super () can be used to invoke immediate parent class constructor.

9.8 Method / Constructor Overriding

1. Method overriding occurs when a subclass provides a specific implementation for a method already defined in its superclass.
2. Run-Time Polymorphism: Overriding is a basis for runtime polymorphism, where the method call is determined by the object's type at runtime.
3. Superclass Reference: An overridden method can be called through a superclass reference holding a subclass object.



9.8 Method / Constructor Overriding

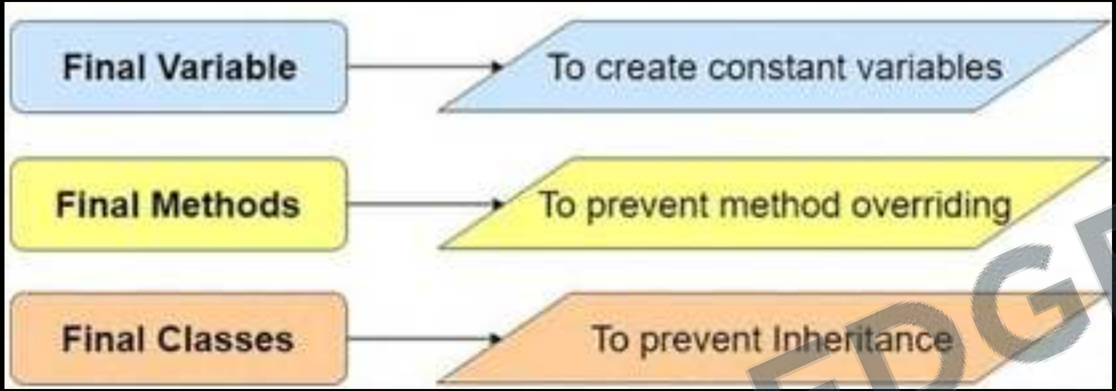
```java
class Dog{
    public void bark(){
        System.out.println("woof ");
    }
}
class Hound extends Dog{
    public void sniff(){
        System.out.println("sniff ");
    }

    public void bark(){
        System.out.println("bowl");
    }
}
```

Same Method Name.
Same parameter

1. Same Signature: Overridden methods must have the same name, return type, and parameters as the method in the parent class.
2. Access Level: The access level cannot be more restrictive than the overridden method's access level.
3. **@Override Annotation:** This annotation is optional but helps to ensure that the method is correctly overridden.

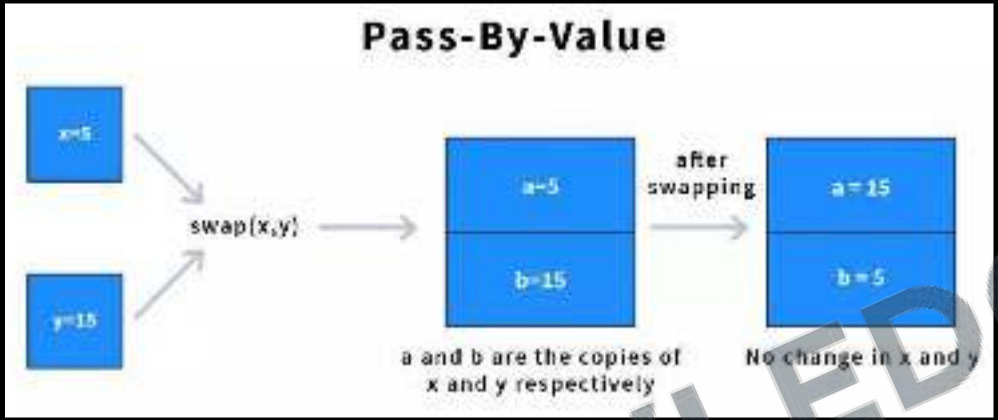
9.9 Final keyword revisited

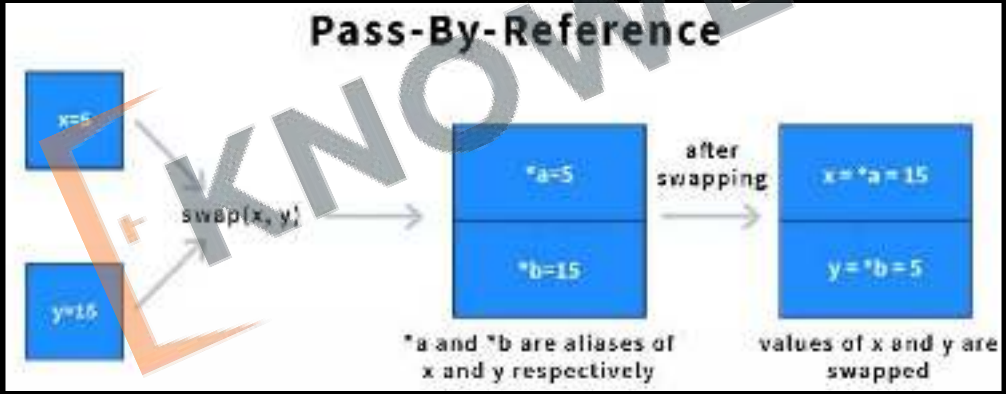


1. Variable becomes a constant, meaning its value cannot be changed once initialized.
2. Method: A final method cannot be overridden by subclasses.
3. Class: A final class cannot be subclassed, securing the class from being extended.
4. Efficiency: Using final can lead to performance optimization, as the compiler can make certain assumptions about final elements.
5. Null Safety: A final variable must be initialized before the constructor completes, reducing null pointer errors.
6. Immutable Objects: Helps in creating immutable objects in combination with private fields and no setter methods.

9.10 Pass by Value vs Pass by reference

9.10 Pass by Value vs Pass by reference





1. Variable Pass by Value:
   - Java's default method.
   - Copies argument's value to function's parameter.
   - Changes in function don't affect original variable.
2. Objects and References:
   - Java passes the reference's value for objects.
   - Modifications to objects in methods affect originals.
3. Primitive Types:
   - Always passed by value.
   - In-function changes don't impact originals.

CHALLENGE

85. In a class Calculator, create multiple add() methods that overload each other and can sum two integers, three integers, or two doubles. Demonstrate how each can be called with different numbers of parameters.

86. Define a base class Vehicle with a method service() and a subclass Car that overrides service(). In Car's service(), provide a specific implementation that calls super.service() as well, to show how overriding works.

Revision

1. What is Abstraction
2. Abstract Keyword
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4. What is Polymorphism
5. References and Objects
6. Method / Constructor Overloading
7. Super Keyword
8. Method / Constructor Overriding
9. Final keyword revisited
10. Pass by Value vs Pass by reference.

Practice Exercise

Abstraction and Polymorphism

Answer in True/False

1. Abstraction hides the internal implementation details and shows only the functionality to the users.
2. An abstract class must contain at least one abstract method.
3. When you pass an object to a method, Java copies the reference to the object.
4. Overloaded methods must have different return types.
5. The 'super' keyword can be used to access methods of the superclass that are hidden by the subclass.
6. Method overriding is used to change the default behaviour of a method in the superclass.
7. The final keyword in Java is used to define constants.
8. Java supports pass by reference for primitive types.
9. An interface in Java can contain instance fields (variables).
10. A subclass can override a protected method from its superclass with a public access modifier.

Practice Exercise

Abstraction and Polymorphism

Answer in True/False

1. Abstraction hides the internal implementation details and shows only the functionality to the users.	True
2. An abstract class must contain at least one abstract method.	False
3. When you pass an object to a method, Java copies the reference to the object.	True
4. Overloaded methods must have different return types.	False
5. The 'super' keyword can be used to access methods of the superclass that are hidden by the subclass.	True
6. Method overriding is used to change the default behaviour of a method in the superclass.	True
7. The final keyword in Java is used to define constants.	True
8. Java supports pass by reference for primitive types.	False
9. An interface in Java can contain instance fields (variables).	False
10. A subclass can override a protected method from its superclass with a public access modifier.	True

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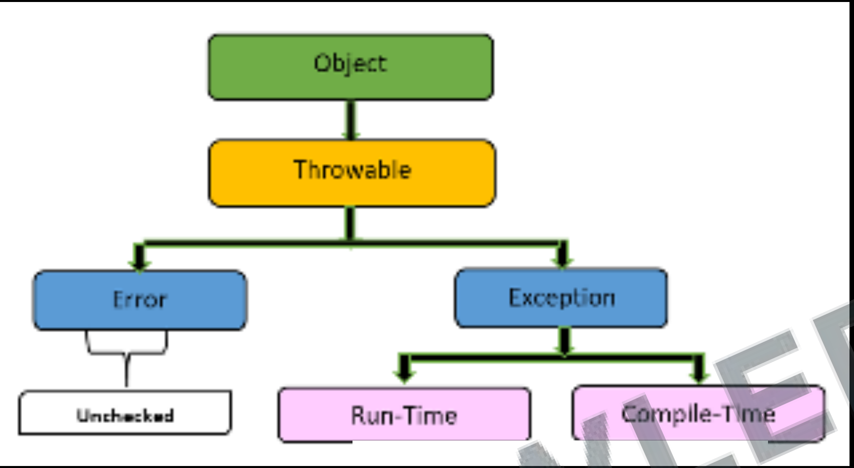
10. Exception & File Handling

1. What is an Exception
2. Try-Catch
3. Types of Exception
4. Throw and Throws
5. Finally Block
6. Custom Exceptions
7. FileWriter class
8. FileReader class

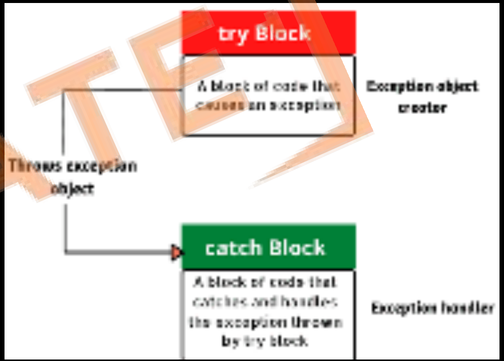
10.1 What is an Exception



1. In Java, an exception is a disruptive event that occurs during the execution of a program, interrupting its normal flow. It's an instance of a problem that arises while the program is running, such as arithmetic errors, null pointer accesses, or resource overflows.
2. Exceptions are objects in Java that encapsulate information about an error event, including its type and the state of the program when the error occurred.

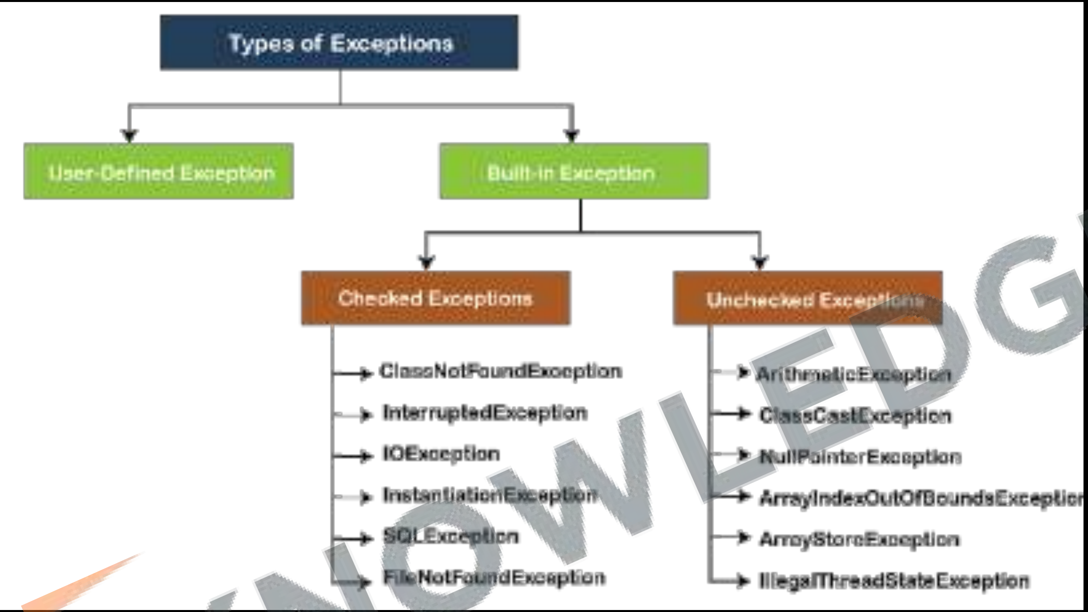
10.2 Try-Catch

```java
try{
    int num = 60/0;
    System.out.println(num);
} catch (ArrayIndexOutOfBoundsException e) {
    System.out.println("Array index exception: " + e.getStackTrace());
} catch (NumberFormatException | ArithmeticException e) {
    System.out.println("Multiple exceptions");
} catch (Exception e) {
    System.out.println("Last exception");
}
```



1. Try Block: Contains code that is susceptible to exceptions.
2. Catch Block: Follows the try block and handles the exceptions thrown by the try block.
3. When an exception occurs in the try block, the control is transferred to the catch block, where the exception is handled.

10.3 Types of Exceptions



1. Checked Exceptions: These are exceptions that must be either caught or declared in the method.
2. Unchecked Exceptions: These are exceptions that do not need to be explicitly handled.

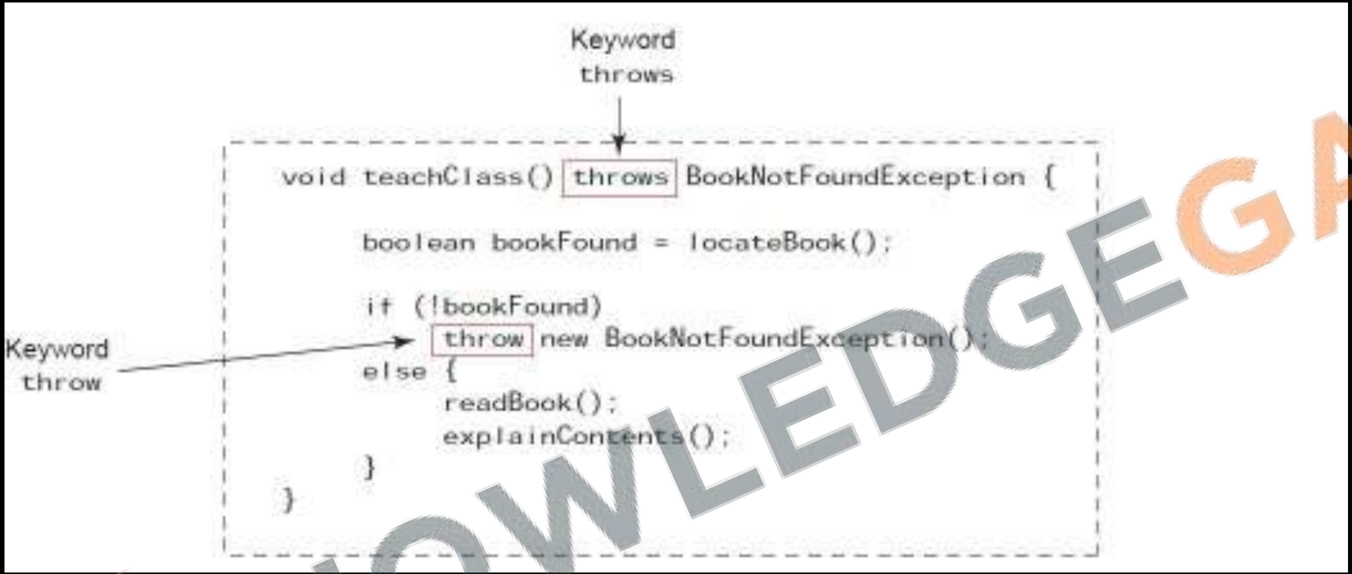
10.4 Throw and Throws

```
public void printName(String name)
        throws IllegalArgumentException {
    if (name.contains("-")) {
        throw new IllegalArgumentException("Name contains -");
    }
    System.out.println(name);
}
```

throws Keyword:

1. Declares that a method may throw one or more exceptions.
2. Used in the method signature to indicate that the method might throw exceptions of specified types.
3. A method declared with throws requires the calling method to handle or further declare the exception.

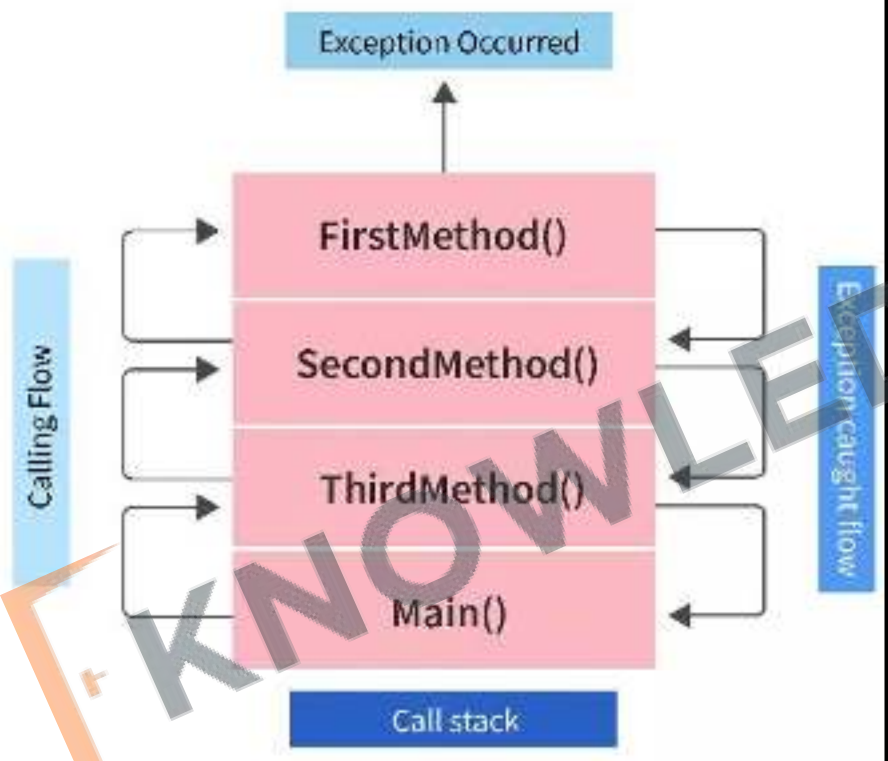
10.4 Throw and Throws



throw Keyword:

1. Used to explicitly throw an exception from any method or block of code.
2. You can throw either a new instance of an exception or an existing exception object using throw.
3. Example: throw new ArithmeticException("Division by zero");

10.4 Throw and Throws



10.4 Throw and Throws

throw	throws
Java **throw** keyword is used to explicitly throw an exception.	Java **throws** keyword is used to declare an exception.
Checked exception cannot be propagated using **throw**.	Checked exception can be propagated with **throws**.
If we see syntax wise, **throw** is followed by an instanceof Exception class Example : throw new NumberFormatException("The month entered, is invalid.");	If we see syntax wise, **throws** is followed by exception class names. Example : throws IOException,SQLException
The keyword **throw** is used inside method body.	**throws clause** is used in method declaration (signature).
By using **throw keyword** in java you cannot throw more than one exception. Example: throw new IOException("Connection failed!!")	By using **throws** you can declare multiple exceptions. Example: public void method() throws IOException,SQLException.

10.5 Finally Block

```
try {
    try {
        int result = 1 / 0;
    } catch (SomeException e) {
        System.out.println("Something caught");
    } finally {
        System.out.println("Not quite finally");
    }
} catch (ArithmeticException e) {
    System.out.println("ArithmeticException caught");
} finally {
    System.out.println("Finally");
}
```

1. Executes code after the try-catch blocks, used mainly for cleanup operations.
2. Always runs regardless of whether an exception is thrown or caught in the try-catch blocks.
3. Ideal for closing resources like files or database connections to prevent resource leaks.

10.6 Custom Exceptions

```java
public class TemperatureException extends Exception {
    private double degrees;

    public TemperatureException(double degrees) {
        this.degrees = degrees;
    }

    public String getMessage() {
        return "The temperature (" + degrees
            + "C) isn't in the normal range.";
    }

    public double getDegrees() {
        return degrees;
    }
}
```

1. Custom exceptions are user-defined exception classes that extend either Exception (for checked exceptions) or RuntimeException (for unchecked exceptions).
2. They are created to represent specific error conditions relevant to an application.

CHALLENGE

87. Arithmetic Exception Handling

Write a program that asks the user to enter two integers and then divides the first by the second. The program should handle any arithmetic exceptions that may occur (like division by zero) and display an appropriate message.

Key Points:

- Use Scanner to read user input.
- Implement a try-catch block to handle ArithmeticException.
- Display a user-friendly message if division by zero occurs.

10.7 FileWriter class

1. FileWriter is used for writing streams of characters to files.
2. It's a character-based stream, which means it's best used for writing text rather than binary data.
3. Constructors:
   - FileWriter(String fileName): Creates a FileWriter object given the name of the file to write to.
   - FileWriter(File file): Creates a FileWriter object given a File object.
4. Common Methods:
   - write(int c): Writes a single character.
   - write(char[] cbuf): Writes an array of characters.
   - write(String str): Writes a string.
   - flush(): Flushes the stream, ensuring all data is written out.
   - close(): Closes the stream, releasing any associated system resources.

10.7 FileWriter class

```java
public class FileWriterExample {
    new *
    public static void main(String[] args) {
        // Define the filename
        String fileName = "example.txt";

        // Create a FileWriter object
        try (FileWriter writer = new FileWriter(fileName)) {
            // Write a string to the file
            writer.write( str: "Hello, this is a test.");

            // Optionally, you can flush the writer
            writer.flush();

            System.out.println("Successfully written to the file.");
        } catch (IOException e) {
            System.out.println("An error occurred.");
            e.printStackTrace();
        }
    }
}
```

10.8 FileReader class

1. The FileReader class is used for reading streams of characters from files.
2. It's a character-based stream, meaning it reads characters (as opposed to bytes). This makes it suitable for reading text files.
3. Constructors:
   - FileReader(String fileName): Creates a FileReader object to read from a file with the specified name.
   - FileReader(File file): Creates a FileReader object to read from the specified File object.
4. Common Methods:
   - read(): Reads a single character and returns it as an integer. Returns -1 if the end of the stream is reached.
   - read(char[] cbuf): Reads characters into an array and returns the number of characters read.

```java
public class FileReaderExample {
    new *
    public static void main(String[] args) {
        // Define the filename
        String fileName = "example.txt";

        // Create a FileReader object
        try (FileReader reader = new FileReader(fileName)) {
            int character;

            // Read and display characters one by one
            while ((character = reader.read()) != -1) {
                System.out.print((char) character);
            }
        } catch (IOException e) {
            System.out.println("An error occurred.");
            e.printStackTrace();
        }
    }
}
```

CHALLENGE

88. File Not Found Exception Handling

Write a program to read a filename from the user and display its content. The program should handle the situation where the file does not exist.

Key Points:

- Use Scanner to read the filename from the user.
- Use FileReader to read the file content.
- Implement a try-catch block to handle FileNotFoundException.
- Display a message informing the user if the file is not found.

Revision

1. What is an Exception
2. Try-Catch
3. Types of Exception
4. Throw and Throws
5. Finally Block
6. Custom Exceptions
7. FileWriter class
8. FileReader class

Practice Exercise

Exception & File Handling

Answer in True/False

1. A method that throws a checked exception must declare the exception using the throws keyword in its signature.
2. It is possible to handle more than one type of exception in a single catch block.
3. The finally block is always executed, even if a return statement is encountered in the try or catch block.
4. If both try and catch blocks have return statements, the finally block will not execute.
5. You can throw an exception of any type, including custom exceptions, using the throw keyword.
6. If a try block does not throw any exception, the catch block is executed for cleanup purposes.
7. Unchecked exceptions are a direct subclass of Throwable.
8. A catch block can exist independently of a try block.
9. You can nest try blocks within another try block.
10. The throw keyword is used to propagate an exception up the call stack.

Practice Exercise

Exception & File Handling

Answer in True/False

1. A method that throws a checked exception must declare the exception using the throws keyword in its signature.	True
2. It is possible to handle more than one type of exception in a single catch block.	True
3. The finally block is always executed, even if a return statement is encountered in the try or catch block.	True
4. If both try and catch blocks have return statements, the finally block will not execute.	False
5. You can throw an exception of any type, including custom exceptions, using the throw keyword.	True
6. If a try block does not throw any exception, the catch block is executed for cleanup purposes.	False
7. Unchecked exceptions are a direct subclass of Throwable.	False
8. A catch block can exist independently of a try block.	False
9. You can nest try blocks within another try block.	True
10. The throw keyword is used to propagate an exception up the call stack.	True

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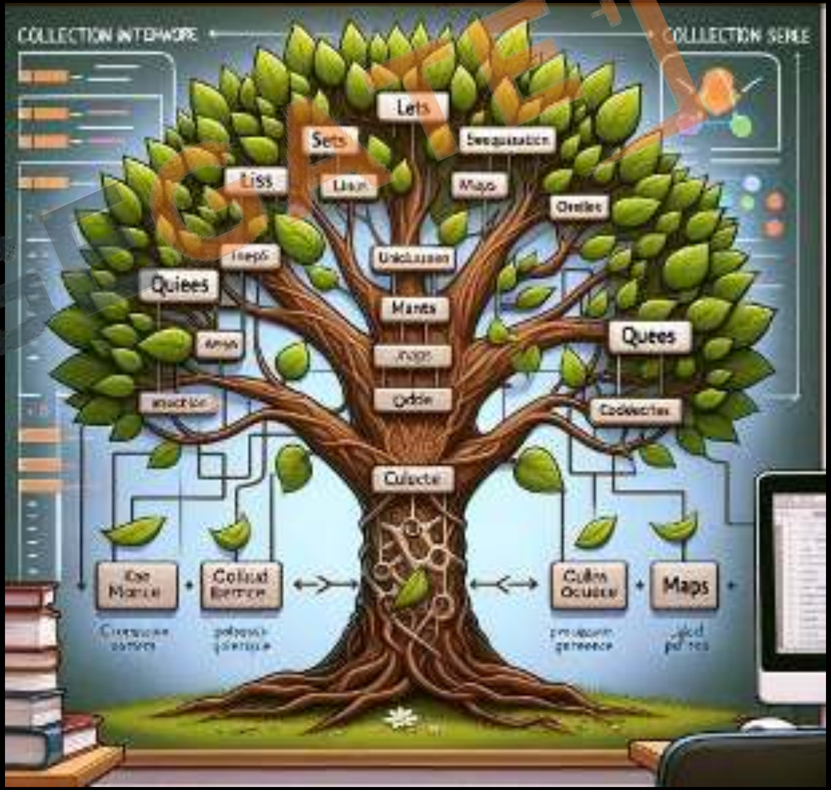
11 Collections & Generics

1. Variable Arguments
2. Wrapper Classes & Autoboxing
3. Collections Library
4. List Interface
5. Queue Interface
6. Set Interface
7. Collections Class
8. Map Interface
9. Enums
10. Generics & Diamond Operators



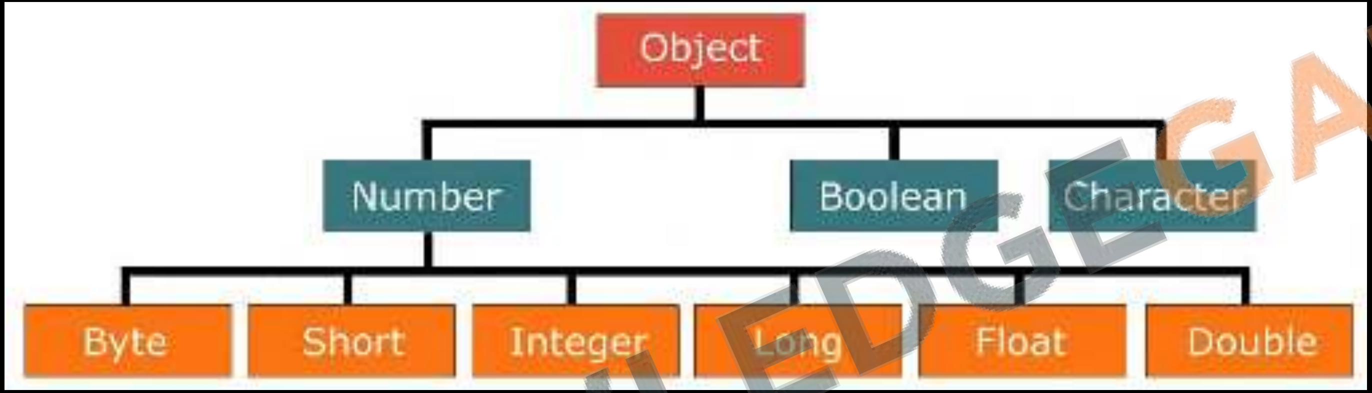
11.1 Variable Arguments

```java
public class MyClass {
    4 usages   new *
    static void printMany(String ...elements) {
        for (String element : elements) {
            System.out.printf(element);
        }
    }

    new *
    public static void main(String[] args) {
        printMany( ...elements: "one", "two", "three");
        printMany( ...elements: "one", "two");
        printMany();
        printMany(new String[]{"one", "two", "three"});
    }
}
```

1. Java's varargs allow methods to accept any number of arguments.
2. Declared using an ellipsis (...), e.g., void method(int... nums).
3. Internal Handling: Treated as arrays, e.g., int... nums is int[] nums.
4. Placement: Must be the last in the method's parameters.
5. Usage: Call with varying argument counts, e.g., method(1, 2) or method().
6. Introduced in: Java 5.

11.2 Wrapper Classes

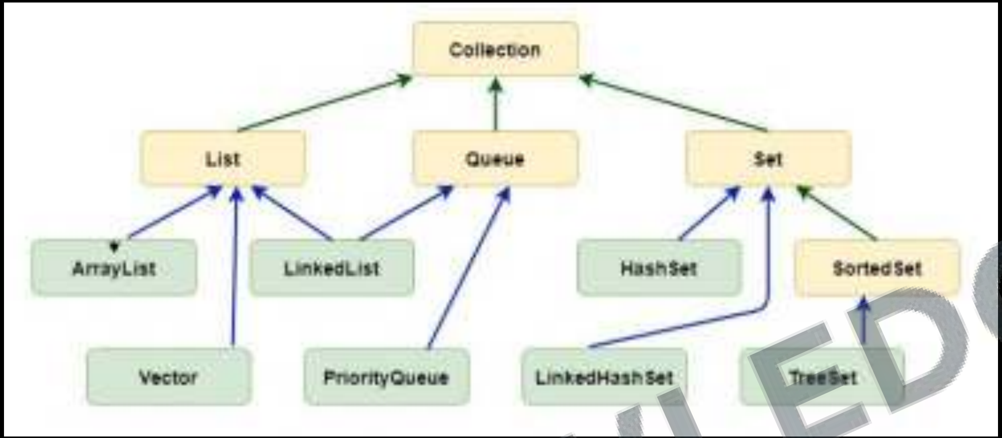


1. Provide a way to use primitive data types (int, char, boolean, etc.) as objects.
2. Automatic conversion between the primitive types and their corresponding wrapper classes.
3. Once created, the value of a wrapper object cannot be changed.
4. Utility Methods: Each wrapper class provides useful methods, like compareTo, valueOf, and parseXxx (e.g., parseInt for Integer).
5. Required for storing primitives in collection objects like ArrayList, HashMap, etc.
6. Allows assignment of null to primitive values when needed.

11.2 Autoboxing

1. Autoboxing: Automatic conversion of primitive types to their corresponding wrapper class objects.
2. Unboxing: Automatic conversion of wrapper class objects back to their respective primitive types.

11.3 Collections Library



1. Collection Interface: The root interface of the collection hierarchy. It declares basic operations like add, remove, clear, and size.
2. List Interface: An ordered collection. Lists can contain duplicate elements.
3. Set Interface: A collection that cannot contain duplicate elements.
4. Queue Interface: A collection used for holding elements in FIFO prior to processing.
5. Map Interface: Not a true Collection, but part of the Collections Framework. Maps store key-value pairs. Keys are unique, but different keys can map to the same value.

11.4 List Interface

1. An ordered collection (also known as a sequence).
2. Allows duplicate elements.
3. Elements can be accessed by their integer index.
4. Maintains the insertion order of elements.
5. Performance: Offers fast random access and quick iteration.
6. Capacity: Grows automatically as elements are added.
7. Preferred over arrays when the size is dynamic or unknown.

11.4 List Interface

1. add(E e): Appends the specified element
2. add(int index, E element): Inserts at specified position
3. remove(Object o): Removes the first occurrence of the specified element
4. remove(int index): Removes the element at the specified position
5. get(int index): Returns the element at the specified position
6. set(int index, E element): Replaces the element at the specified position
7. size(): Returns the number of elements
8. clear(): Removes all of the elements
9. contains(Object o): Returns `true` if the list contains the specified element.
10. indexOf(Object o): Returns the index of the first occurrence, or -1 if the list does not contain the element.

```java
//List Creation
List<Integer> list = new ArrayList<Integer>();
//List Addition
list.add(1);
list.add(2);
//Printing the List
System.out.println(list);
```

11.5 Queue Interface

1. It's a collection designed for holding elements prior to processing.
2. Ordering: Typically, it orders elements in a FIFO (First-In-First-Out) manner.
3. End Points: Offers two ends - one for insertion (tail) and the other for removal (head).

11.5 Queue Interface

1. add(E e): Inserts the specified element into the queue. Throws an exception if the element cannot be added.
2. offer(E e): Inserts the specified element into the queue. Returns false if the element cannot be added.
3. remove(): Retrieves and removes the head of the queue. Throws an exception if the queue is empty.
4. poll(): Retrieves and removes the head of the queue, or returns null if the queue is empty.
5. element(): Retrieves, but does not remove, the head of the queue. Throws an exception if the queue is empty.
6. peek(): Retrieves, but does not remove, the head of the queue, or returns null if the queue is empty.

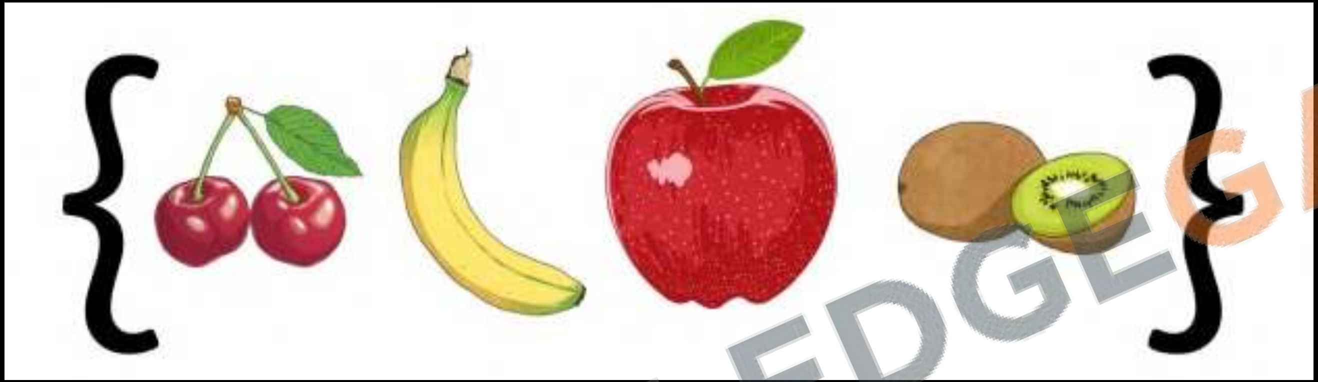
```java
// Creating a Queue using LinkedList
Queue<String> queue = new LinkedList<>();

// Adding elements (offer method can also be used)
queue.add("First");
queue.add("Second");

// Displaying the head of the queue
System.out.println("Head of queue: " + queue.peek());

// Removing elements from the queue
while (!queue.isEmpty()) {
    System.out.println("Removed: " + queue.poll());
}
```

11.6 Set Interface



1. Unique Elements: Does not allow duplicate elements.
2. Unordered Collection: it does not guarantee any specific ordering of elements.
3. No Positional Access: Unlike lists, it doesn't support indexing-based access to elements.
4. Implementation: Common implementations include HashSet, LinkedHashSet, and TreeSet.

11.6 Set Interface

```java
// Creating a Set
Set<String> set = new HashSet<>();

// Adding elements
set.add("Apple");

// Attempting to add a duplicate element
boolean isAdded = set.add("Apple"); // This will be false
System.out.println("Apple added again: " + isAdded);

// Checking if a specific element is in the set
boolean containsOrange = set.contains("Orange");
System.out.println("Contains Orange: " + containsOrange);
```

1. add(E e): Adds the specified element to the set
2. remove(Object o): Removes the specified element from the set
3. contains(Object o): Checks if the set contains the specified element.
4. size(): Returns the number of elements in the set.
5. isEmpty(): Checks if the set is empty.

11.7 Collections Class

```java
// Creating a list
List<Integer> list = new ArrayList<>();
Collections.addAll(list, ...elements: 5, 1, 8, 3, 2);

// Sorting the list
Collections.sort(list);
System.out.println("Sorted list: " + list);

// Finding max and min in the list
int max = Collections.max(list);
int min = Collections.min(list);
System.out.println("Max: " + max + ", Min: " + min);
```

```java
// Reversing the list
Collections.reverse(list);
System.out.println("Reversed list: " + list);

// Searching in the list (List must be sorted)
Collections.sort(list);
int index = Collections.binarySearch(list, key: 3);
System.out.println("Index of 3: " + index);

// Creating an immutable list
List<Integer> unmodifiableList = Collections.unmodifiableList(list);
System.out.println("Unmodifiable list: " + unmodifiableList);
```

1. Offers methods like sort to sort lists.
2. Provides methods like binarySearch for searching sorted lists.
3. Allows reversing the order of elements in a list with reverse.
4. Can shuffle the elements of a list randomly using shuffle.
5. Creates unmodifiable collections using methods like unmodifiableList, etc.
6. Methods like singletonList, create immutable collections with a single element.
7. The copy method is used to copy all elements from one list to another.

CHALLENGE

89. Write a method concatenate Strings that takes variable arguments of String type and concatenates them into a single string.
90. Write a program that sorts a list of String objects in descending order using a custom Comparator.
91. Use the Collections class to count the frequency of a particular element in an ArrayList.
92. Write a method that swaps two elements in an ArrayList, given their indices.
93. Create a program that reverses the elements of a List and prints the reversed list.
94. Create a PriorityQueue of a custom class Student with attributes name and grade. Use a comparator to order by grade.
95. Write a program that takes a string and returns the number of unique characters using a Set.

11.8 Map Interface

1. Stores data as key-value pairs.
2. Each key can map to at most one value.
3. Keys are unique, but multiple keys can map to the same value.
4. It is part of the Collections Framework but does not extend the Collection interface.

11.8 Map Interface

```java
// Creating a Map
Map<String, Integer> map = new HashMap<>();

// Adding key-value pairs to the Map
map.put("Apple", 10);

// Accessing a value
Integer appleCount = map.get("Apple");
System.out.println("Apples count: " + appleCount);

// Checking if a key exists
if (map.containsKey("Banana")) {
    System.out.println("Banana is in the map");
}

// Removing a key-value pair
map.remove( key: "Orange");
```

1. put(K key, V value): Associates the specified value with the specified key in the map.
2. get(Object key): Returns the value to which the specified key is mapped, or null if the map contains no mapping for the key.
3. remove(Object key): Removes the mapping for a key from the map if it is present.
4. containsKey(Object key): Checks if the map contains a mapping for the specified key.
5. keySet(): Returns a Set view of the keys contained in the map.
6. values(): Returns a Collection view of the values contained in the map.

11.9 Enums

```java
enum TrafficSignal{
    RED("stop"), GREEN("start"), ORANGE("slow down");

    private String action;

    public String getAction(){
        return this.action;
    }
    private TrafficSignal(String action){
        this.action = action;
    }
}

// String to Enum using valueOf
TrafficSignal signal = TrafficSignal.valueOf("RED");
signal = TrafficSignal.valueOf("GREEN"); //OK
signal = TrafficSignal.valueOf("Green"); //Not OK
```

1. Enums in Java: Special types for fixed sets of constants like days, colors.
2. Declaration: Use enum keyword, e.g., enum Color { RED, GREEN, BLUE; }.
3. Access: Access constants with dot syntax, e.g., Color.RED.
4. Features: Type-safe, readable, can have methods and fields.
5. Usage: Useful in switch statements and iterating with values() method.

11.10 Generics & Diamond Operators

```java
class SpecificClass {
    2 usages
    private String thing;
    no usages   new *
    public String getThing() {
        return thing;
    }
    no usages   new *
    public void setThing (String thing) {
        this.thing = thing;
    }
}
```

```java
class GenericClass<T> {
    2 usages
    private T thing;
    no usages   new *
    public T getThing() {
        return thing;
    }
    no usages   new *
    public void setThing(T thing) {
        this.thing = thing;
    }
}
```


1. They allow you to write flexible and reusable code by enabling types (classes and interfaces) to be parameters when defining classes, interfaces, and methods.
2. Generics provide compile-time type safety by allowing you to enforce that certain objects are of a specific type.
3. With generics, you don't need to cast objects because the type is known.
4. Generics are denoted by angle brackets <>, e.g., List<String> means a list of strings.
5. Diamond Operator: Introduced in Java 7, the diamond operator <> allows you to infer the type parameter from the context, simplifying instantiation of generic classes.

CHALLENGE

96. Create an enum called Day that represents the days of the week. Write a program that prints out all the days of the week from this enum.
97. Enhance the Day enum by adding an attribute that indicates whether it is a weekday or weekend. Add a method in the enum that returns whether it's a weekday or weekend, and write a program to print out each day along with its type.
98. Create a Map where the keys are country names (as String) and the values are their capitals (also String). Populate the map with at least five countries and their capitals. Write a program that prompts the user to enter a country name and then displays the corresponding capital, if it exists in the map.

Revision

1. Variable Arguments
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7. Collections Class
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9. Enums
10. Generics & Diamond Operators

Practice Exercise

Collections & Generics

Answer in True/False

1. Variable arguments can be used to pass an arbitrary number of values to a method.
2. Autoboxing is the process where primitive types are automatically converted into their corresponding wrapper class objects by the Java compiler.
3. Every class in the Collections Library implements the Collection interface.
4. Lists guarantee the order of insertion for the elements they contain.
5. Sets in Java inherently maintain the elements in sorted order.
6. In a Map, you can have multiple entries with the same value but not with the same key.
7. Enums in Java can have constructors, methods, variables, and can implement interfaces.
8. With the diamond operator, you do not need to specify the type on the right-hand side of a statement when initializing an object.
9. The List interface provides methods for inserting elements at a specific position in the list.
10. Wrapper classes in Java, such as Integer and Double, can be extended like any other class.

Practice Exercise

Collections & Generics

Answer in True/False

1. Variable arguments can be used to pass an arbitrary number of values to a method. True
2. Autoboxing is the process where primitive types are automatically converted into their corresponding wrapper class objects by the Java compiler. True
3. Every class in the Collections Library implements the Collection interface. False
4. Lists guarantee the order of insertion for the elements they contain. True
5. Sets in Java inherently maintain the elements in sorted order. False
6. In a Map, you can have multiple entries with the same value but not with the same key. True
7. Enums in Java can have constructors, methods, variables, and can implement interfaces. True
8. With the diamond operator, you do not need to specify the type on the right-hand side of a statement when initializing an object. True
9. The List interface provides methods for inserting elements at a specific position in the list. True
10. Wrapper classes in Java, such as Integer and Double, can be extended like any other class. False

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12 Multi threading & Executor Service

1. Intro to Multi-threading
2. Creating a Thread
3. States of a Thread
4. Thread Priority
5. Join Method
6. Synchronize keyword
7. Thread Communication
8. Intro to Executor Service
9. Multiple Threads with Executor
10. Returning Futures

12.1 What is a Thread

1. What is a Thread: A thread in Java is a small part of a program that can run at the same time as other parts.
2. Purpose: Threads help a program do many things at once, like handling many users or doing different tasks simultaneously.
3. Creating Threads: You can make a thread by using the Thread class or the Runnable interface.
4. Using Threads: Use threads for tasks that can happen at the same time, like managing many requests or splitting up a big job.
5. Thread Talk: Threads can talk to each other using wait(), notify(), and notifyAll() to coordinate their work.

12.1 Need of Multi-threading

```
// First Task
for (int i = 1; i <= 1000; i++) {
    System.out.printf("%d:* ", i);
}
System.out.println("\nFirst Task Done");

// Second Task
for (int i = 1; i <= 1000; i++) {
    System.out.printf("%d:& ", i);
}
System.out.println("\nSecond Task Done");

// Third Task
for (int i = 1; i <= 1000; i++) {
    System.out.printf("%d:$ ", i);
}
System.out.println("\nThird Task Done");
```

1. Tasks might be very important
2. Tasks are independent of each other
3. A Multi-core CPU is sitting idle most of the time
4. A big task can be divided into smaller parts
5. Making your code responsive

12.2 Creating a Thread

(Extending Thread Class)

```java
// Step 1: Define a Class that Extends Thread
4 usages
public class PrintTask extends Thread {
    // Step 2: Override the run() Method
    no usages
    public void run() {
        // First Task
        for (int i = 1; i <= 1000; i++) {
            System.out.printf("%d:%c ", i, targetChar);
        }
        System.out.printf("\n%c Task Done\n", targetChar);
    }
    3 usages
    private final char targetChar;
    2 usages
    public PrintTask(char targetChar) {
        this.targetChar = targetChar;
    }
}
```

```java
public static void main(String... args) {
    // Step 3: Create an Instance of Your Class
    PrintTask t1 = new PrintTask( targetChar: '*');
    t1.start(); // Start the first thread

    PrintTask t2 = new PrintTask( targetChar: '$');
    t2.start(); // Start the second thread
}
```

In the main method, two threads (t1 and t2) are created and started. They will execute independently and print their values.

12.2 Creating a Thread

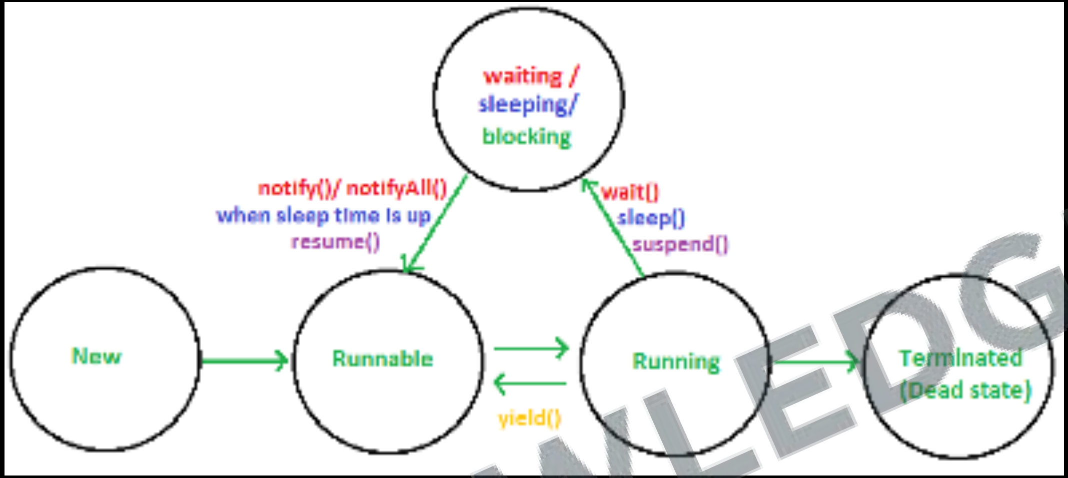
(Creating Runnables)

```java
// Step 1: Define a Class that implements Runnable
4 usages
public class PrintRunnable implements Runnable {
    // Step 2: Override the run() Method
    no usages
    @Override
    public void run() {
        // First Task
        for (int i = 1; i <= 1000; i++) {
            System.out.printf("%d:%c ", i, targetChar);
        }
        System.out.printf("\n%c Task Done\n", targetChar);
    }
    3 usages
    private final char targetChar;
    2 usages
    public PrintRunnable(char targetChar) {
        this.targetChar = targetChar;
    }
}
```

```java
public static void main(String... args) {
    // Step 3: Create an Instance of Your Class
    PrintRunnable t1 = new PrintRunnable( targetChar: '*');
    // Step4: Wrap your class with a thread
    new Thread(t1).start(); // Start the first thread

    PrintRunnable t2 = new PrintRunnable( targetChar: '$');
    new Thread(t1).start(); // Start the second thread
}
```

In the main method, two threads (t1 and t2) are created and started. They will execute independently and print their values.

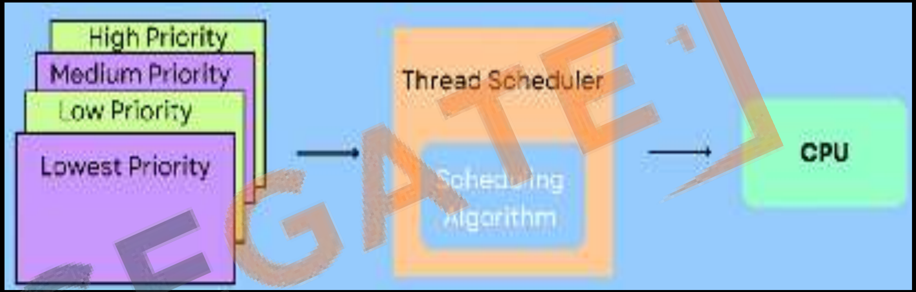
12.3 States of a Thread



1. New: Thread is created but not started.
2. Runnable: Thread is ready or running.
3. Running: Thread is actively executing tasks.
4. Blocked/Waiting: Thread is alive but not active because it's waiting for resources or other threads.
5. Terminated: Thread has finished or stopped running.

12.4 Thread Priority

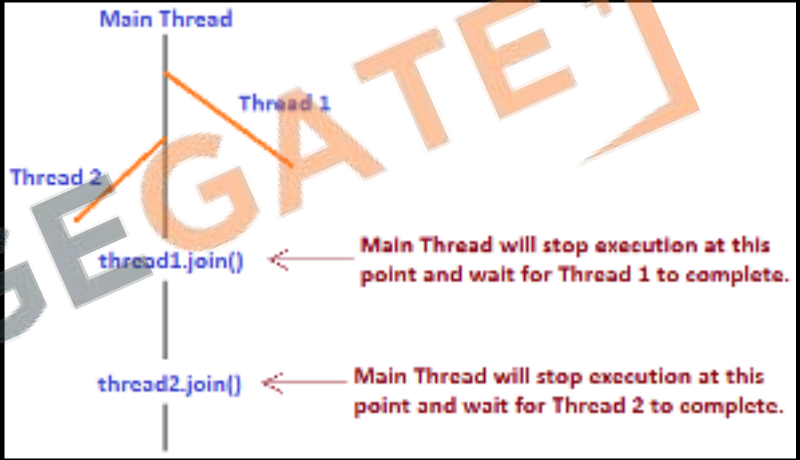
```java
class MyThread extends Thread {
    no usages
    public void run() {
        Thread current = Thread.currentThread();
        System.out.printf("Running thread name: %s\n",
                current.getName());
        System.out.printf("Running thread priority: %s\n",
                current.getPriority());
    }
}

public class ThreadPriority {
    public static void main(String args[]) {
        MyThread t1 = new MyThread();
        MyThread t2 = new MyThread();
        t1.setPriority(Thread.MIN_PRIORITY); // Setting priority to 1
        t2.setPriority(Thread.MAX_PRIORITY); // Setting priority to 10

        t1.setName("Thread-1");
        t2.setName("Thread-2");
        t1.start();
        t2.start();
    }
}
```



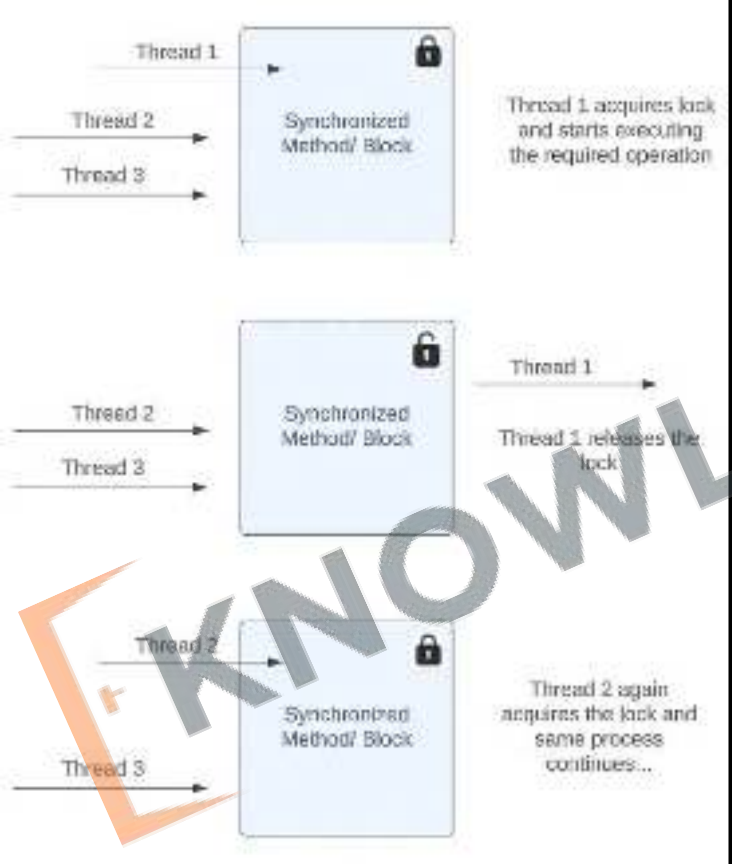
1. Priority Levels: Java threads have priority levels from 1 (lowest) to 10 (highest), with a default value of 5.
2. Influence on Execution: A thread's priority suggests the importance of a thread to the scheduler, though it doesn't guarantee the order of execution.
3. Set and Get Priority: Use setPriority(int) to change a thread's priority and getPriority() to retrieve it.

12.5 Join Method

1. Purpose of join: The join method is used to make the calling thread wait until the thread on which join has been called completes its execution.
2. Synchronization of Threads: join helps in synchronizing multiple threads, ensuring that a thread completes its execution before the next steps in the calling thread proceed.
3. Overloaded Versions: join comes in three versions:
   - join(): Waits indefinitely until the thread on which it's called finishes.
   - join(long millis): Waits for the thread to die for the specified milliseconds.
   - join(long millis, int nanos): Waits for the thread to die for the specified milliseconds plus nanoseconds.



12.6 Synchronize keyword



1. Mutual Exclusion: The synchronize keyword in Java ensures that only one thread can execute a block of code at a time, providing mutual exclusion and preventing race conditions.
2. Object Lock: When a thread enters a synchronized block or method, it acquires a lock on the object or class, depending on whether the method is an instance method or a static method.
3. Visibility: It ensures that changes made by one thread to shared data are visible to other threads.

12.6 Synchronize keyword

```java
class Counter {
    2 usages
    private int count = 0;
    // Synchronized method to increment the counter
    1 usage
    public synchronized void increment() {
        count++;
    }
    // Method to get the current count
    1 usage
    public int getCount() {
        return count;
    }
}
4 usages
class SynchronizedThread extends Thread {
    2 usages
    private Counter counter;
    2 usages
    public SynchronizedThread(Counter counter) {
        this.counter = counter;
    }
    no usages
    public void run() {
        for (int i = 0; i < 1000; i++) {
            counter.increment();
        }
    }
}
```

```java
public class SynchronizedExample {
    public static void main(String[] args) throws InterruptedException {
        Counter counter = new Counter();
        SynchronizedThread t1 = new SynchronizedThread(counter);
        SynchronizedThread t2 = new SynchronizedThread(counter);

        t1.start();
        t2.start();

        t1.join();
        t2.join();
        System.out.println("Final count is: " + counter.getCount());
    }
}
```

12.7 Thread Communication

1. sleep(long millis): Causes the currently executing thread to sleep (temporarily cease execution) for the specified number of milliseconds.
2. yield(): Causes the currently executing thread to pause and allow other threads to execute. It's a way of suggesting that other threads of the same priority can run.
3. wait(): Causes the current thread to wait until another thread invokes the notify() or notifyAll() method for this object. It releases the lock held by this thread.
4. notify(): Wakes up a single thread that is waiting on the object's monitor. If any threads are waiting, one is chosen to be awakened.
5. notifyAll(): Wakes up all threads that are waiting on the object's monitor.

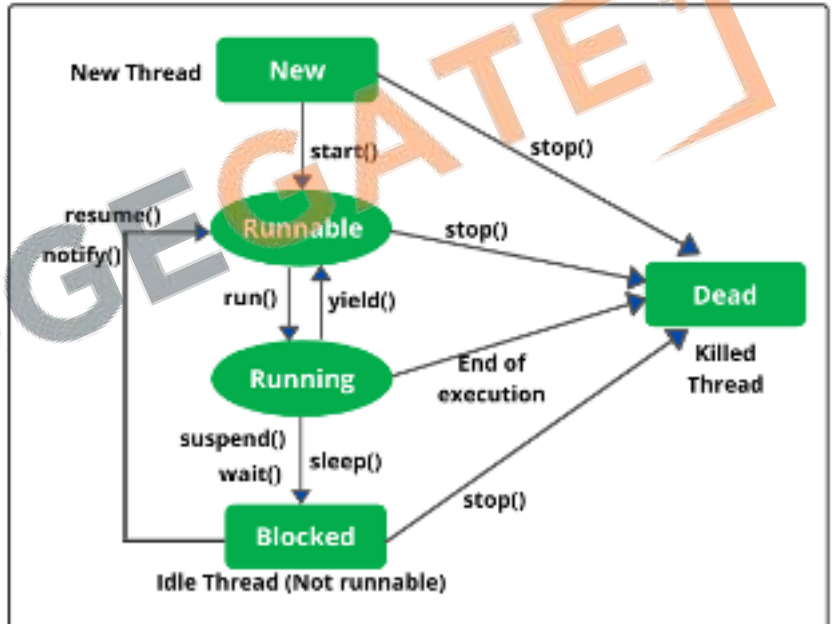


Fig: State Transition Diagram of a Thread

CHALLENGE

99. Write a program that creates two threads. Each thread should print "Hello from Thread X", where X is the number of the thread (1 or 2), ten times, then terminate.

100. Write a program that starts a thread and prints its state after each significant event (creation, starting, and termination). Use Thread.sleep() to simulate long-running tasks and Thread.getState() to print the thread's state.

101. Create three threads. Ensure that the second thread starts only after the first thread ends and the third thread starts only after the second thread ends using the join method. Each thread should print its start and end along with its name.

102. Simulate a traffic signal using threads. Create three threads representing three signals: RED, YELLOW, and GREEN. Each signal should be on for a certain time, then switch to the next signal in order. Use sleep for timing and synchronize to make sure only one signal is active at a time.

12.8 Intro to Executor Service

1. Purpose: ExecutorService is a framework provided by the Java Concurrency API to manage and execute submitted tasks without the need to manually manage thread life cycles.
2. Thread Pool Management: ExecutorService efficiently reuses a fixed pool of threads to execute tasks, thereby improving performance by reducing the overhead of thread creation, especially for short-lived asynchronous tasks.

12.8 Intro to Executor Service

```java
// Step 1: Define a Class that implements Runnable
5 usages
public class PrintRunnable implements Runnable {
    // Step 2: Override the run() Method
    no usages
    @Override
    public void run() {
        // First Task
        for (int i = 1; i <= 1000; i++) {
            System.out.printf("%d:%c ", i, targetChar);
        }
        System.out.printf("\n%c Task Done\n", targetChar);
    }
    3 usages
    private final char targetChar;
    3 usages
    public PrintRunnable(char targetChar) {
        this.targetChar = targetChar;
    }
}
```

```java
public class SingleThreadExecutorExample {
    public static void main(String[] args) {
        // Create a single-threaded executor
        ExecutorService executor = Executors.newSingleThreadExecutor();
        // Define a task (a Runnable)
        Runnable task = new PrintRunnable( targetChar: 'E');

        // Submit the task to the executor
        executor.submit(task);
        executor.shutdown();
    }
}
```

12.9 Multiple Threads with Executor

```java
public class PrintRunnable implements Runnable {
    // Step 2: Override the run() Method
    no usages
    @Override
    public void run() {
        // Task
        String threadName = Thread.currentThread().getName();
        System.out.printf("Executing task with char %c on" +
                "Thread: %s \n", targetChar, threadName);
        for (int i = 1; i <= 1000; i++) {
            System.out.printf("%d:%c ", i, targetChar);
        }
        System.out.printf("\n%c Task Done\n", targetChar);
    }
    4 usages
    private final char targetChar;
    3 usages
    public PrintRunnable(char targetChar) {
        this.targetChar = targetChar;
    }
}
```

```java
public class MultiThreadExecutorExample {
    public static void main(String[] args) throws InterruptedException {
        // Create a single-threaded executor
        ExecutorService executor = Executors.newFixedThreadPool(nThreads: 3);

        // Define a task (a Runnable)
        for (int i = 0; i < 5; i++) {
            int taskNumber = i + 1;
            Runnable task = new PrintRunnable((char)taskNumber);
            // Submit the task to the executor
            executor.submit(task);
        }

        executor.shutdown();

        // Wait for all tasks to finish executing or timeout after 10 seconds
        if (!executor.awaitTermination( 10, TimeUnit.SECONDS)) {
            // Try to stop all actively executing tasks
            executor.shutdownNow();
        }
    }
}
```

12.10 Returning Futures

```java
// Create an executor with a fixed thread pool of 2 threads
ExecutorService executor = Executors.newFixedThreadPool( nThreads: 2);
// Submit the task to the executor and get a Future object
Future<String> future = executor.submit(task);

try {
    // Get the result from the Future object.
    String result = future.get();
    System.out.println("Result from future: " + result);
} catch (ExecutionException | InterruptedException e) {
    // Handle the interruption during the get
    Thread.currentThread().interrupt();
    System.out.println("Task was interrupted");
}

// Shut down the executor
executor.shutdown();
```

```java
// Define a callable task that returns a result
Callable<String> task = new Callable<String>() {
    @Override
    public String call() throws Exception {
        // Simulate task execution time
        TimeUnit.SECONDS.sleep( timeout: 1);
        return "Result from task";
    }
};
```

CHALLENGE

103. Write a program that creates a single-threaded executor service. Define and submit a simple Runnable task that prints numbers from 1 to 10. After submission, shut down the executor.
104. Create a fixed thread pool with a specified number of threads using Executors.newFixedThreadPool(int). Submit multiple tasks to this executor, where each task should print the current thread's name and sleep for a random time between 1 and 5 seconds. Finally, shut down the executor and handle proper termination using awaitTermination.
105. Write a program that uses an executor service to execute multiple Callable tasks. Each task should calculate and return the factorial of a number provided to it. Use Future objects to receive the results of the calculations. After all tasks are submitted, retrieve the results from the futures, print them, and ensure the executor service is shut down correctly.

Revision

1. Intro to Multi-threading
2. Creating a Thread
3. States of a Thread
4. Thread Priority
5. Join Method
6. Synchronize keyword
7. Thread Communication
8. Intro to Executor Service
9. Multiple Threads with Executor
10. Returning Futures

Practice Exercise

Multi threading & Executor Service

Answer in True/False

1. In Java, the main thread is the only user thread that is created automatically when a program starts.
2. The run() method is automatically called when a thread is started using the start() method.
3. A thread can be in both the Runnable and Running state at the same time.
4. Lower priority threads are executed first to ensure equal processing time for all threads.
5. The join() method causes the calling thread to immediately stop executing until the thread it joins with stops running.
6. Using the synchronized keyword can prevent two threads from executing a method simultaneously.
7. The notify() method wakes up all threads that are waiting on the object's monitor.
8. The ExecutorService must be explicitly shut down to terminate the threads it manages.
9. A Callable task cannot be submitted to an ExecutorService.
10. The Thread.yield() method forces a thread to stop its execution permanently.

Practice Exercise

Multi threading & Executor Service

Answer in True/False

1. In Java, the main thread is the only user thread that is created automatically when a program starts. True
2. The run() method is automatically called when a thread is started using the start() method. True
3. A thread can be in both the Runnable and Running state at the same time. False
4. Lower priority threads are executed first to ensure equal processing time for all threads. False
5. The join() method causes the calling thread to immediately stop executing until the thread it joins with stops running. True
6. Using the synchronized keyword can prevent two threads from executing a method simultaneously. True
7. The notify() method wakes up all threads that are waiting on the object's monitor. False
8. The ExecutorService must be explicitly shut down to terminate the threads it manages. True
9. A Callable task cannot be submitted to an ExecutorService. False
10. The Thread.yield() method forces a thread to stop its execution permanently. False

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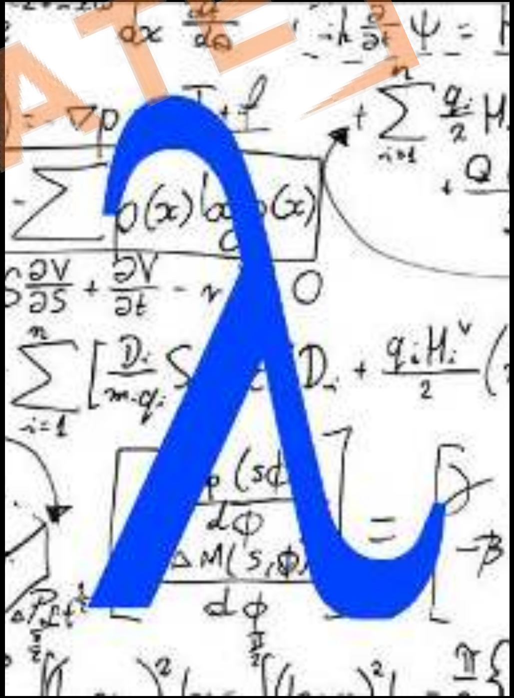
Sanchit Socket

13 Functional Programming

1. What is Functional Programming
2. Lambda Expression
3. What is a Stream
4. Filtering & Reducing
5. Functional Interfaces
6. Method References
7. Functional vs Structural Programming
8. Optional Class
9. Intermediate vs Terminal Operations
10. Max, Min, Collect to List
11. Sort, Distinct, Map

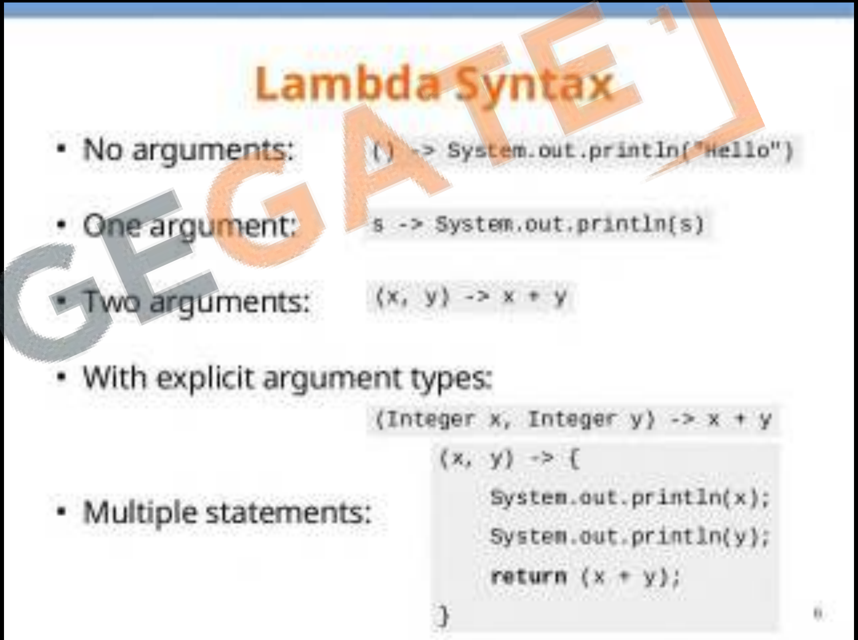


13.1 What is Functional Programming

1. Functional Programming: It's a way of writing programs where you use functions like small building blocks.
2. Functions as First-Class: Functions can be passed as arguments, returned from other functions, and assigned to variables.
3. Immutable Data: Once you create a piece of data, you don't change it.
4. Pure Functions: These are special functions that always give the same result for the same input with no Side-Effects.
5. Functional Interfaces: These are like templates for functions, making it easier to use them in different parts of your program.

13.2 Lambda Expression

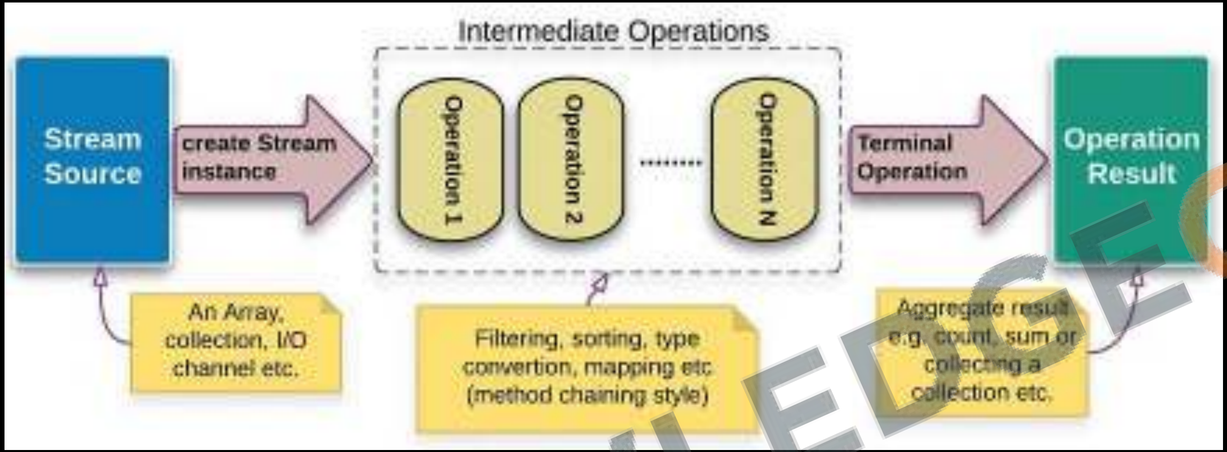
1. Shortcuts: Lambda expressions are quick, nameless functions for small tasks.
2. Syntax: Written as (parameters) -> {body}, linking inputs to actions.
3. Functional Interfaces: They work with interfaces that have only one method, making code concise.
4. Readability: They make code shorter and clearer, especially with collections.
5. Useful with Collections: Great for managing lists and sets, like filtering or sorting.



13.2 Lambda Expression

```
// Simple Addition:
(int a, int b) -> a + b;

// Check if a Number is Even:
(int number) -> number % 2 == 0;

// Print a Message:
(String message) -> System.out.println(message);

// Sort a List of Strings by Length:
(String s1, String s2) -> s1.length() - s2.length();

// Runnable with Lambda (No Parameters):
() -> System.out.println("Hello, World!");
```

13.3 What is a Stream



1. Element Sequence: Streams represent a sequence of elements
2. Functional Operations: Operations like map, filter, and reduce.
3. No Storage: Streams don't store data; they process it on-the-fly from sources like collections or arrays.
4. Efficiency: Stream operations can be lazy, processing elements only as needed, which is efficient for large data.
5. One-Time Use: Streams are consumable; once processed, they cannot be.
6. Parallel Capable: They support parallel processing, making operations faster by utilizing multiple threads.

13.4 Filtering & Reducing (Filter)

```
List<String> myList = List.of("apple", "banana", "cherry", "date");
myList.stream()
    .filter(s -> s.endsWith("e"))
    .forEach(s -> System.out.println(s));
```

1. Purpose: Used to filter elements of a stream based on a given predicate (a condition). Only elements that satisfy the condition are included in the resulting stream.
2. Lazy Operation: It's a lazy operation, meaning it's not executed until a terminal operation (like collect or forEach) is invoked on the stream.
3. Returns a Stream: filter itself returns a new stream with elements that match the predicate.

13.4 Filtering & Reducing (Reduce)

```java
List<Integer> numbers = Arrays.asList(1, 2, 3, 4, 5);
int sum = numbers.stream()
                 .reduce(0, (a, b) -> a + b);
// sum: 15
```

1. Purpose: Used to reduce the elements of a stream to a single value. It takes a binary operator as a parameter and applies it repeatedly, combining the elements of the stream.
2. Versatile: Can be used for summing, finding min or max, and combining elements in a myriad of ways.
3. Optional or Default Value: Without an identity value, reduce returns an Optional. With an identity value, it returns a default value if the stream is empty.

CHALLENGE

106. Write a lambda expression that takes two integers and returns their multiplication. Then, apply this lambda to a pair of numbers.
107. Convert an array of strings into a stream. Then, use the stream to print each string to the console.
108. Given a list of strings, use stream operations to filter out strings that have length of 10 or more and then concatenate the remaining strings.
109. Given a list of integers, use stream operations to filter odd numbers and print them.

13.5 Functional Interfaces

1. Single Abstract Method (SAM): A functional interface has only one abstract method. However, it can have multiple default or static methods.
2. Lambda Compatibility: They are intended to be used with lambda expressions, providing a target type for lambdas and method references.
3. @FunctionalInterface Annotation: While not mandatory, this annotation helps the compiler to identify the intention of making an interface functional and to generate an error if the annotated interface does not satisfy the conditions.
4. Common Examples: Predicate, Consumer, BinaryOperator, Runnable, Callable, Comparator, and user-defined interfaces can be functional if they have only one abstract method.

```
Predicate<Integer> isPositive = x -> x > 0;
// Output: true
System.out.println(isPositive.test(5));
// Output: false
System.out.println(isPositive.test(-5));

Consumer<String> print =
        message -> System.out.println(message);
// Output: Hello, World!
print.accept("Hello, World!");

BinaryOperator<Integer> multiply = (a, b) -> a * b;
// Output: 15
System.out.println(multiply.apply(5, 3));
```

13.6 Method References

Lambda Expression	Method Reference
`s -> s.toLowerCase()`	`String::toLowerCase`
`s.toLowerCase()`	`String::toLowerCase`
`(a, b) -> a.compareTo(b)`	For Integers it will be `Integer::compareTo` and for strings it is `String::compareTo`
`(a, b) -> Person.compareByAge(a, b)`	`Person::compareByAge`

Syntax:

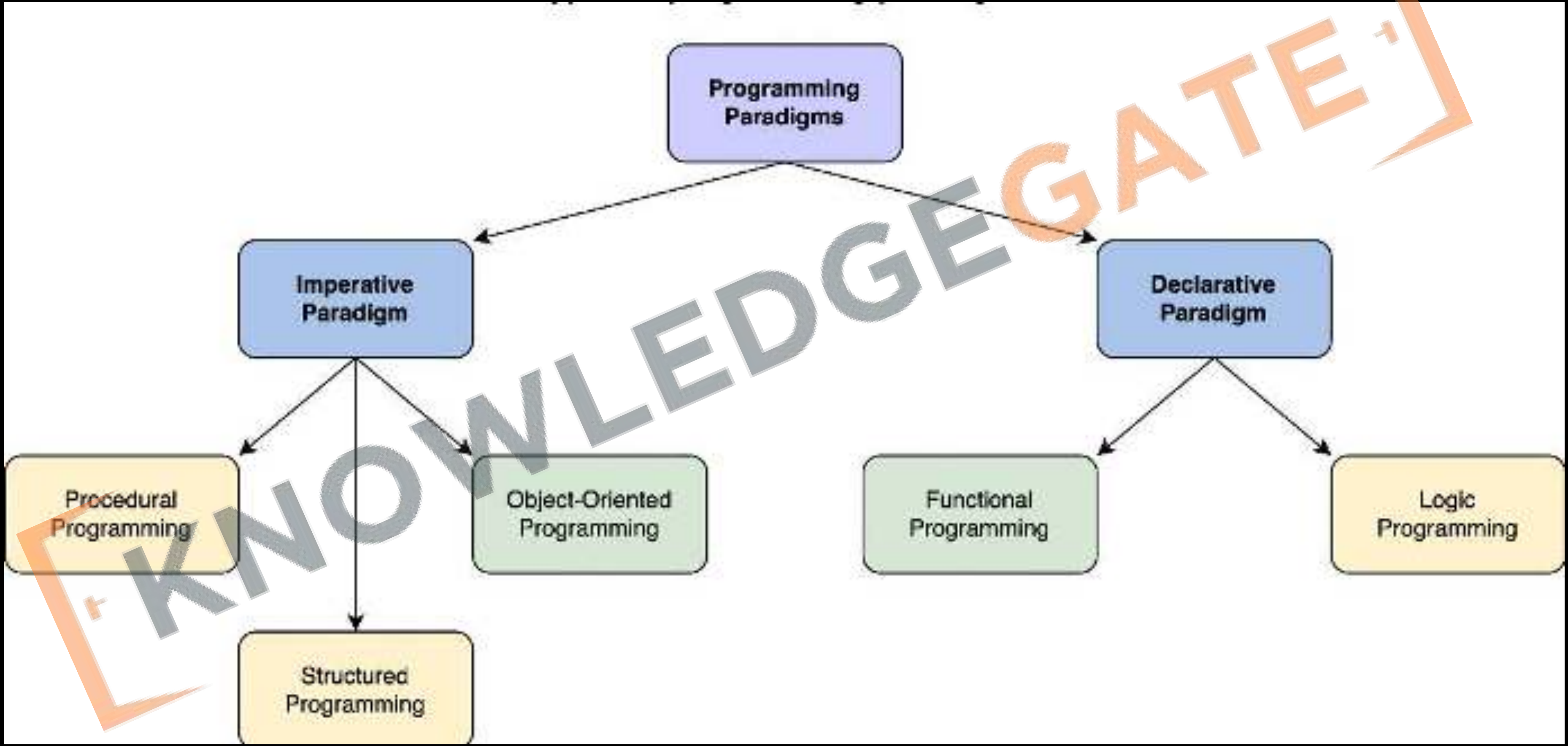
- Static Method References: ClassName::staticMethodName
- Instance Method: instance::instanceMethodName
- Instance Method Particular Class: ClassName::methodName
- Constructor References: ClassName::new.

1. Purpose Syntax & Usage: A method reference is described using :: (double colon) syntax. For example, System.out::println refers to the println method of the System.out object.
2. Functional Interfaces: They are used with functional interfaces.
3. Benefit: They make your code more readable and concise
4. Limitation: They can only be used for methods that fit the parameters and return type.

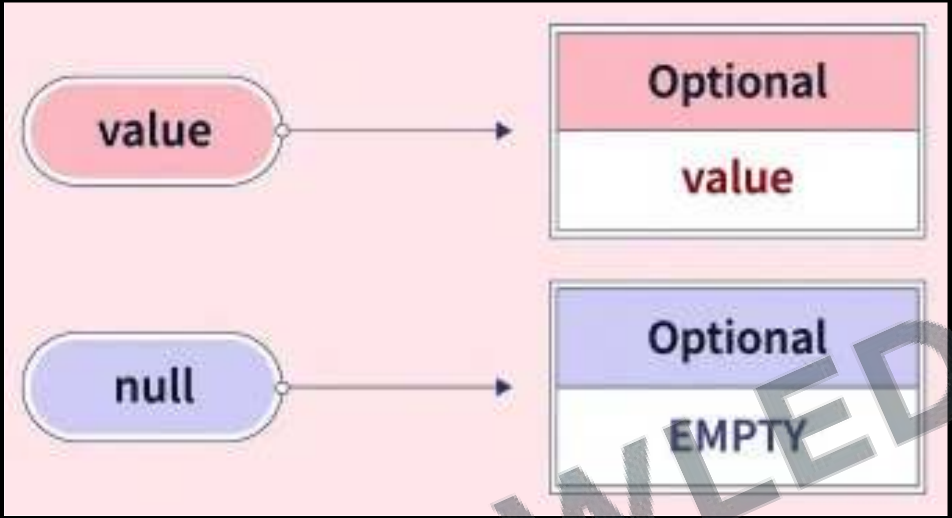
13.7 Functional vs Structural Programming

Imperative Programming	Declarative Programming
1. Computation	
You describe the step-by-step instructions for how an executed program achieves the desired results.	You set the conditions that trigger the program execution to produce the desired results.
2. Readability and complexity	
With the emphasis on the control flow, you can often follow the step-by-step process fairly easily. However, as you add more code, it can become longer and more complex	Step-by-step processes are eschewed. You'll discover that this paradigm is less complex and requires less code, making it easier to read.
3. Customization	
A straightforward way to customize and edit code and structure is offered. You have complete control and can easily adapt the structure of your program to your needs.	Customizing the source code is more difficult because of complicated syntax and the paradigm's dependence on implementing a pre-configured algorithm.
4. Optimization	
Adding extensions and making upgrades are supported, but doing so is significantly more challenging than with declarative programming, making it harder to optimize.	You can easily optimize code because an algorithm controls the implementation. Furthermore, you can add extensions and make upgrades.
5. Structure	
The code structure can be long and complex. The code itself specifies how it should run and in what order.	The code structure is concise and precise, and it lacks detail.

13.7 Functional vs Structural Programming



13.8 Optional Class



1. Creating Optional Objects: Optional.empty(), Optional.of(), Optional.ofNullable()
2. Checking Value Presence: isPresent() and ifPresent()
3. Default Values: orElse() and orElseGet()
4. Value Transformation: map()
5. Throwing Exception: orElseThrow()

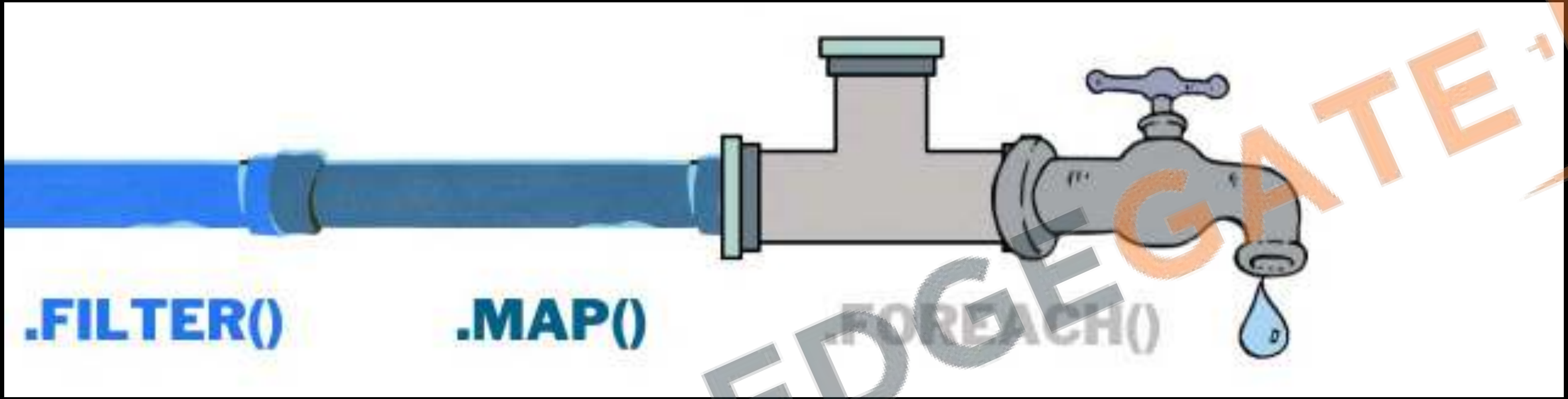
13.8 Optional Class

```java
// Creating Optional objects
Optional<String> optionalEmpty = Optional.empty();
Optional<String> optionalOf = Optional.of("Java");
Optional<String> optionalNullable = Optional.ofNullable(null);

// Checking presence of value
if (optionalOf.isPresent()) {
    System.out.println("Value is present: " + optionalOf.get());
}

// Using orElse to provide a fallback
String orElseExample = optionalEmpty.orElse("Default Value");
System.out.println("Using orElse: " + orElseExample);

// Using ifPresent to perform an action if value is present
optionalOf.ifPresent(System::out::println);
```

CHALLENGE

110. Create your own functional interface with a single abstract method that accepts an integer and returns a boolean. Implement it using a lambda that checks if the number is prime.
111. Write two versions of a program that calculates the factorial of a number: one using structural (procedural) programming, and the other using functional programming.
112. Write a function that accepts a string and returns an Optional<String>. If the string is empty or null, return an empty Optional, otherwise, return an Optional containing the uppercase version of the string.

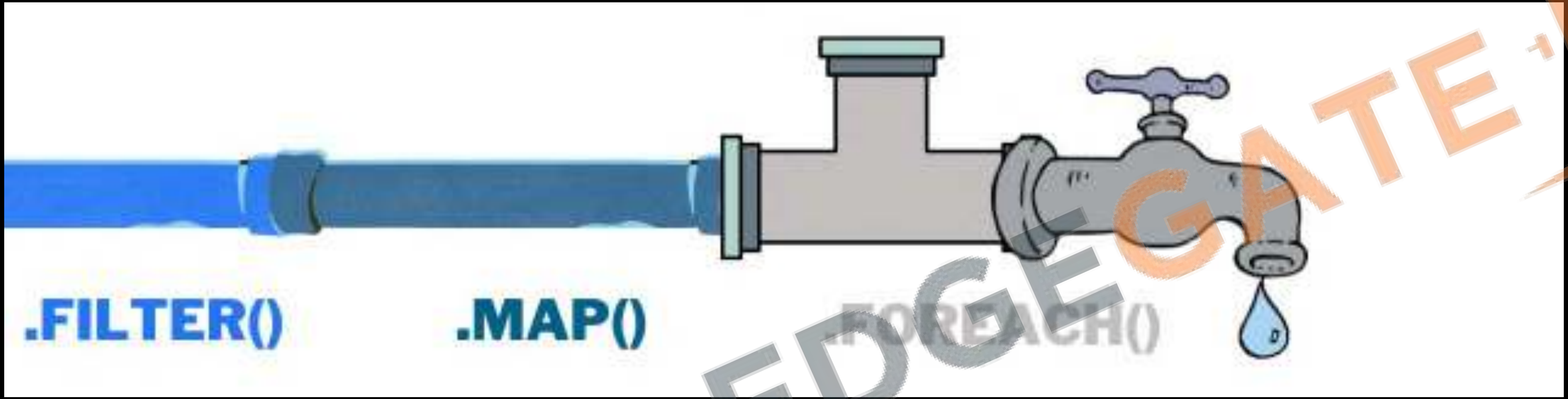
13.9 Intermediate vs Terminal Operations



Intermediate Operations

1. Laziness: Executed only when a terminal operation is invoked, setting up a pipeline without processing data.
2. Stream Transformation: Transform one stream into another, e.g., filter, map. They're chainable, allowing multiple transformations.
3. State Handling: Can be stateless (like map) or stateful(like sorted), affecting processing.

13.9 Intermediate vs Terminal Operations



Terminal Operations

1. Computation Trigger: Initiates the stream processing and closes the stream. After this, the stream can't be reused.
2. Final Outcome: Produces a result (like a sum or list) or a side-effect (like printing each element). Not chainable.
3. Examples: Operations like collect, forEach, reduce, sum, max, min, and count are terminal.

13.10 Max, Min, Collect to List

```java
List<Integer> numbers = List.of(4, 2, 5, 1, 3);
Optional<Integer> maxNumber = numbers.stream()
    .max(Integer::compareTo);
// Output: 5
maxNumber.ifPresent(System.out::println);

List<Integer> numbers = List.of(4, 2, 5, 1, 3);
Optional<Integer> minNumber = numbers.stream()
    .min(Integer::compareTo);
// Output: 1
minNumber.ifPresent(System.out::println);

List<String> words = Arrays.asList("Stream",
    "Operations", "Java");
List<String> collectedWords = words.stream()
    .collect(Collectors.toList());
// Output: [Stream, Operations, Java]
System.out.println(collectedWords);
```

1. max() finds the largest element in the stream according to a given comparator or natural ordering.
2. min() identifies the smallest element in the stream based on a provided comparator or natural ordering.
3. collect(Collectors.toList()) gathers all the elements of the stream into a new List.

13.11 Sort, Distinct, Map

```java
List<Integer> numbers = List.of(4, 2, 5, 1, 3);
List<Integer> sortedNumbers = numbers.stream()
    .sorted()
    .collect(Collectors.toList());
// Output: [1, 2, 3, 4, 5]
System.out.println(sortedNumbers);


List<String> items = List.of("apple",
    "banana", "apple", "orange", "banana");
List<String> distinctItems = items.stream()
    .distinct()
    .collect(Collectors.toList());
// Output: [apple, banana, orange]
System.out.println(distinctItems);


List<String> words = List.of("Stream",
    "Operations", "Java");
List<String> uppercaseWords = words.stream()
    .map(String::toUpperCase)
    .collect(Collectors.toList());
// Output: [STREAM, OPERATIONS, JAVA]
System.out.println(uppercaseWords);
```

1. sorted() orders the elements of a stream based on their natural order or a provided comparator.
2. distinct() filters out duplicate elements, ensuring that every element in the resulting stream is unique.
3. map() applies a function to each element of a stream, transforming them into a new stream of results based on the function logic.

CHALLENGE

113. Given an array of integers, create a stream, use the distinct operation to remove duplicates, and collect the result into a new list.
114. Create a list of employees with name and salary fields. Write a comparator that sorts the employees by salary. Then, use this comparator to sort your list using the sort stream operation.
115. Create a list of strings representing numbers ("1", "2", ...). Convert each string to an integer, then again calculating squares of each number using the map operation and sum up the resulting integers.

Revision

1. What is Functional Programming
2. Lambda Expression
3. What is a Stream
4. Filtering & Reducing
5. Functional Interfaces
6. Method References
7. Functional vs Structural Programming
8. Optional Class
9. Intermediate vs Terminal Operations
10. Max, Min, Collect to List
11. Sort, Distinct, Map

Practice Exercise

Functional Programming

Answer in True/False

1. Functions can't be assigned to variables, passed as arguments, or returned from other functions.
2. A lambda expression in Java can be used to implement any interface, regardless of the number of abstract methods.
3. A Java stream represents a sequence of elements and supports various methods which can be pipelined to produce the desired result.
4. The filter method in streams is a terminal operation that returns a boolean value.
5. A functional interface in Java is an interface with exactly one abstract method.
6. Method references in Java can only refer to static methods.
7. The Optional class in Java is used to avoid NullPointerException.
8. Intermediate operations on streams are executed immediately and are always followed by terminal operations.
9. The max and min operations on streams return an Optional describing the maximum or minimum element.
10. The sorted operation in streams is a terminal operation and it sorts the elements of the stream in their natural order.

Practice Exercise

Functional Programming

Answer in True/False

1. Functions can't be assigned to variables, passed as arguments, or returned from other functions. False
2. A lambda expression in Java can be used to implement any interface, regardless of the number of abstract methods. False
3. A Java stream represents a sequence of elements and supports various methods which can be pipelined to produce the desired result. True
4. The filter method in streams is a terminal operation that returns a boolean value. False
5. A functional interface in Java is an interface with exactly one abstract method. True
6. Method references in Java can only refer to static methods. False
7. The Optional class in Java is used to avoid NullPointerException. True
8. Intermediate operations on streams are executed immediately and are always followed by terminal operations. False
9. The max and min operations on streams return an Optional describing the maximum or minimum element. True
10. The sorted operation in streams is a terminal operation and it sorts the elements of the stream in their natural order. False

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Sanchit Socket

Module

Algorithm Concepts and Problems (Easy/Medium/Hard)

Topic

DSA by Prashant sir

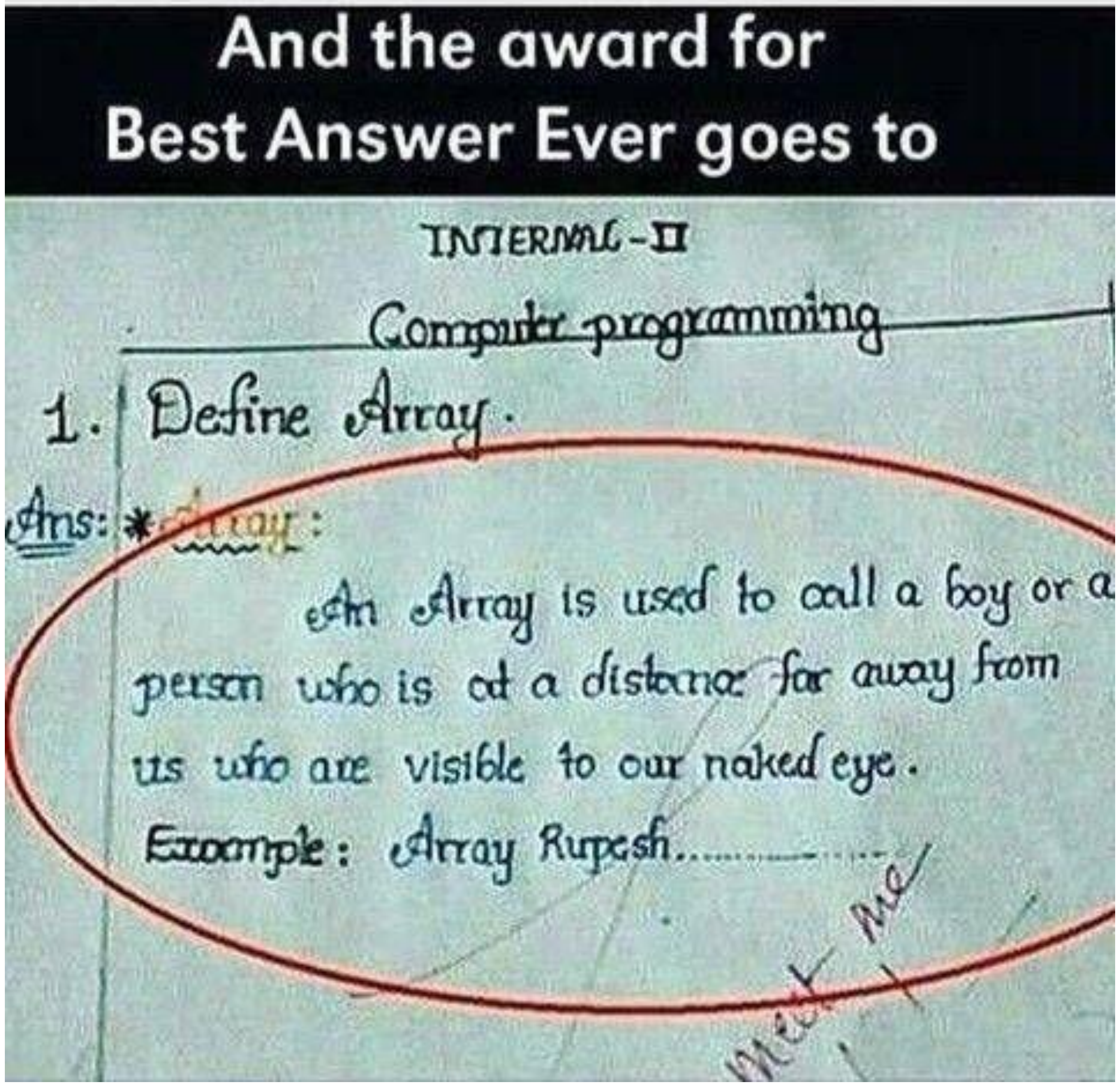
Subtopic

Arrays

Lesson

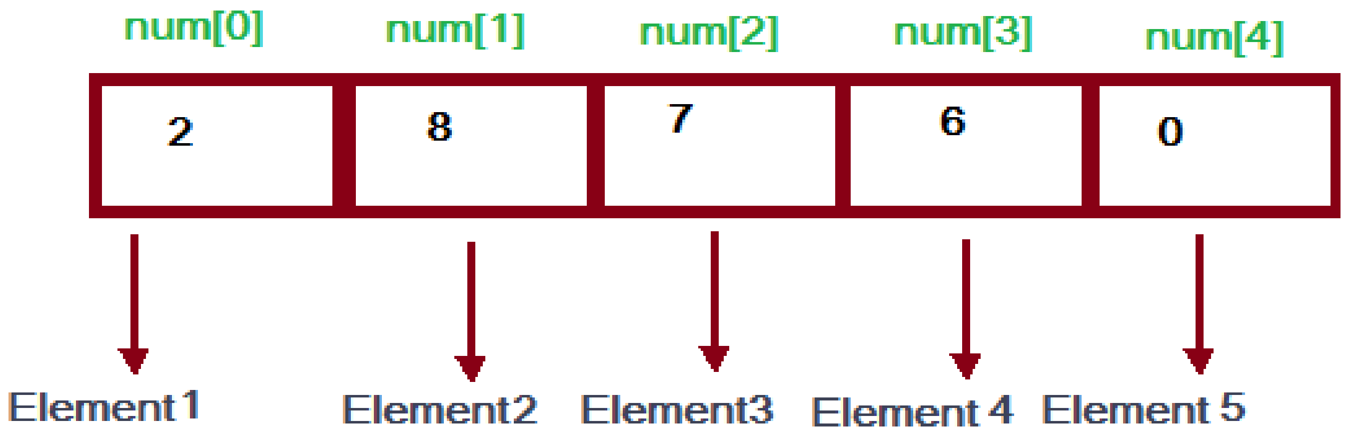
ARRAY

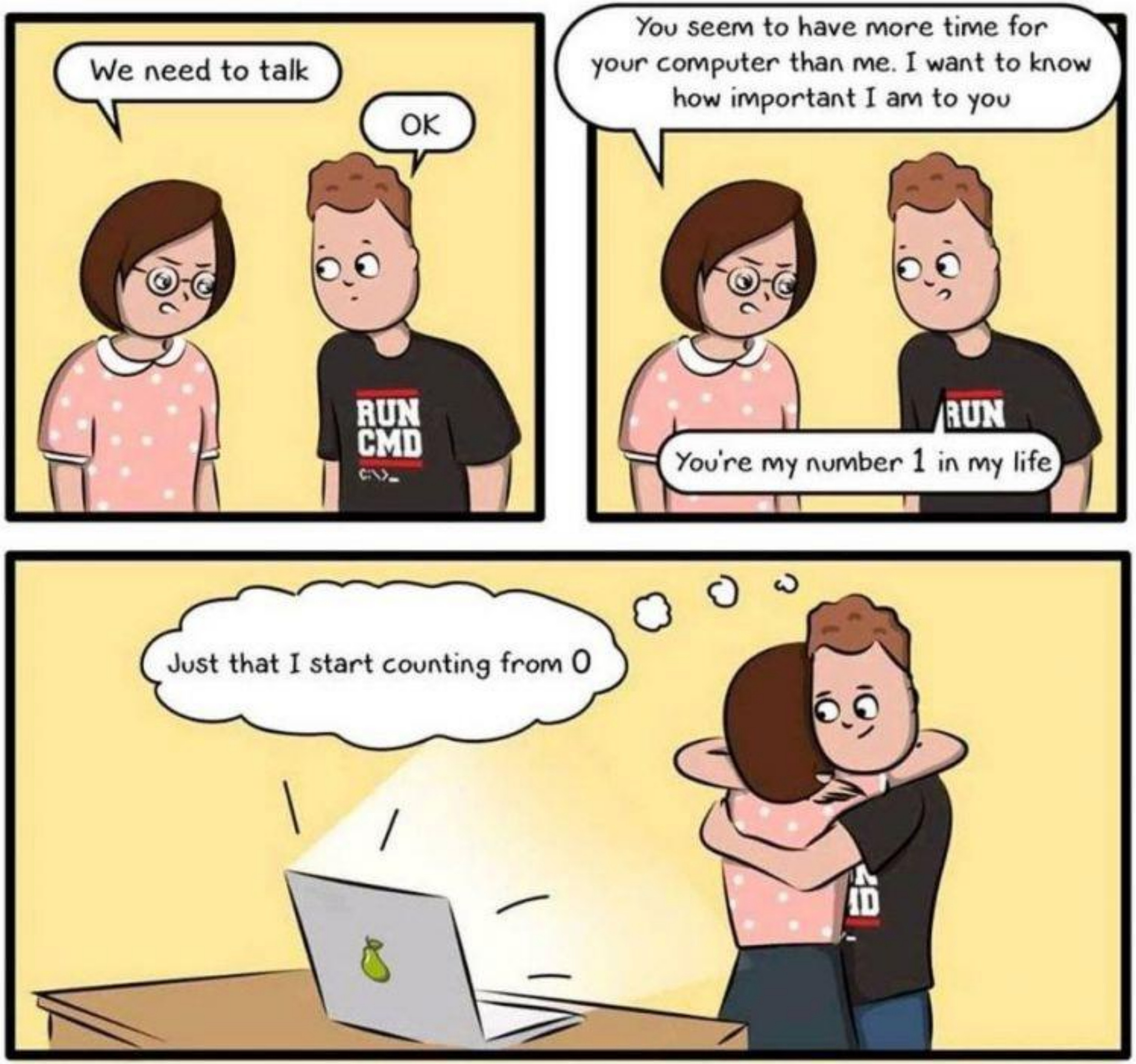
Array


And the award for
Best Answer Ever goes to
INTERNAL-II
Computer programming
1. Define Array.
Ans: * Array:
An Array is used to call a boy or a person who is at a distance far away from us who are visible to our naked eye.
Example: Array Rupesh..........
Meet me

Array

- An array is a collection of items stored at contiguous memory locations.
- The idea is to store multiple items of the same type together. This makes it easier to calculate the position of each element by simply adding an offset to a base value, i.e., the memory location of the first element of the array (generally denoted by the name of the array).
- Each element can be uniquely identified by their index in the array.



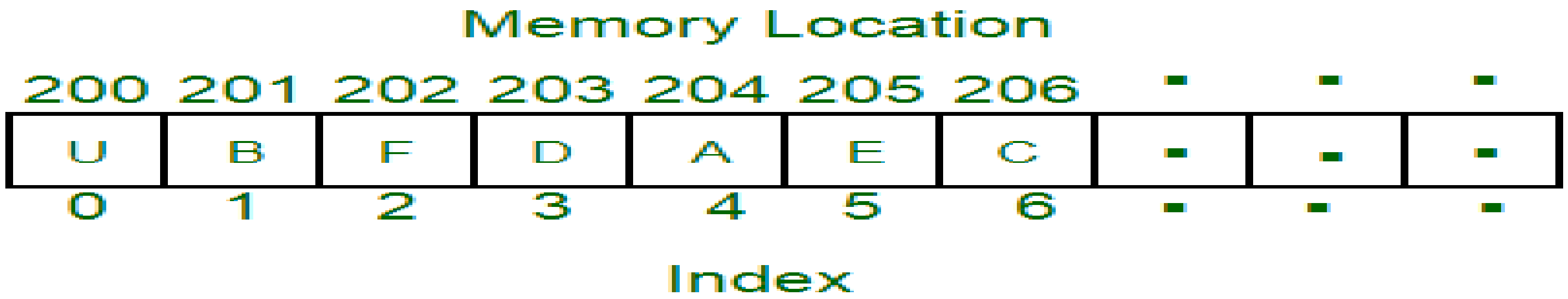

We need to talk
OK
RUN CMD
You seem to have more time for your computer than me. I want to know how important I am to you
RUN
You're my number 1 in my life
Just that I start counting from 0

Size of an array

- Number of elements = (Upper bound – Lower Bound) + 1
  - Lower bound index of the first element of the array
  - Upper bound index of the last element of the array


Array index starts from 1

- Size = number of elements * Size of each elements in bytes



Break


HOW DO I CREATE A MULTIDIMENSIONAL ARRAY FROM A SINGLE ARRAY?
imgflip.com

Two-Dimensional array

- The two-dimensional array can be defined as an array of arrays. The 2D array is organized as matrices which can be represented as the collection of rows and columns.

- However, 2D arrays are created to implement a relational database look a like data structure. It provides ease of holding the bulk of data at once which can be passed to any number of functions wherever required.

	Column 0	Column 1	Column 2
Row 0	x[0][0]	x[0][1]	x[0][2]
Row 1	x[1][0]	x[1][1]	x[1][2]
Row 2	x[2][0]	x[2][1]	x[2][2]

Two-Dimensional array

Module

Algorithm Concepts and Problems (Easy/Medium/Hard)

Topic

DSA by Prashant sir

Subtopic

Time and Space Complexity

Lesson

Time and Space Complexity

Time complexity
&
Space complexity

How to calculate running time of a program ?

```java
package in.knowledgegate.dsa.complexity;

public class Sum {
    public static void main(String[] args) {
        System.out.println("Sum:" + sum(a: 5, b: 6));
        System.out.println("Sum:" + sum(a: 8, b: 10));
    }

    2 usages
    private static int sum (int a, int b) {
        return a + b;
    }
}
```

How to calculate running time of a program ?

```java
package in.knowledgegate.dsa.complexity;

public class ArraySum {
    public static void main(String[] args) {
        System.out.println("Sum:" + sum(new int[]{5, 6}));
        System.out.println("Sum:" + sum(new int[]{1, 2, 3, 4, 5}));
    }

    2 usages
    private static int sum (int []array) {
        int result = 0;
        for (int i = 0; i < array.length; i++) {
            result += array[i];
        }
        return result;
    }
}
```

How to calculate space consumed by a program ?

```java
package in.knowledgegate.dsa.complexity;

public class ArraySum {
    public static void main(String[] args) {
        System.out.println("Sum:" + sum(new int[]{5, 6}));
        System.out.println("Sum:" + sum(new int[]{1, 2, 3, 4, 5}));
    }

    2 usages
    private static int sum (int []array) {
        int result = 0;
        for (int i = 0; i < array.length; i++) {
            result += array[i];
        }
        return result;
    }
}
```

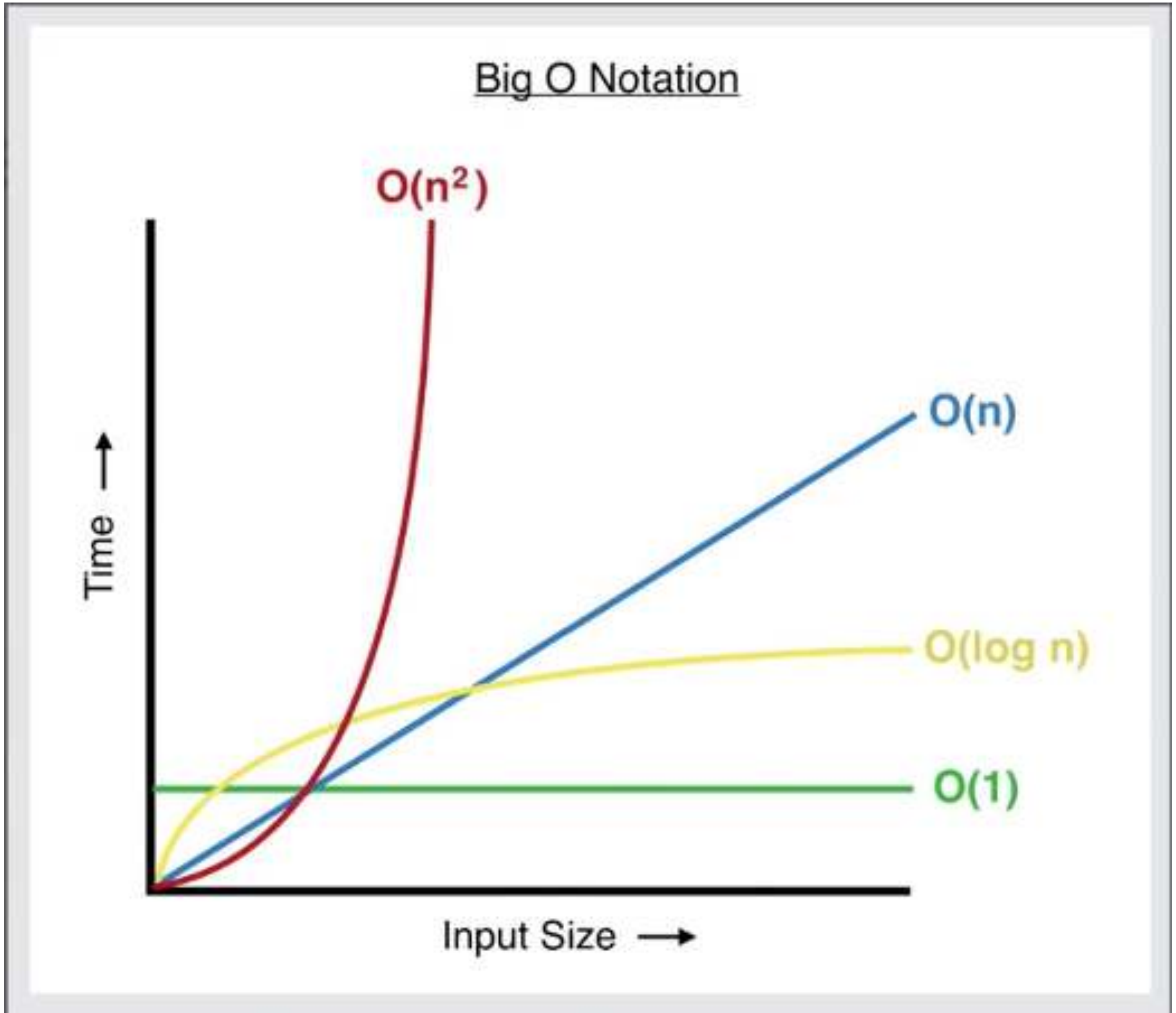
How to calculate space consumed by a program ?

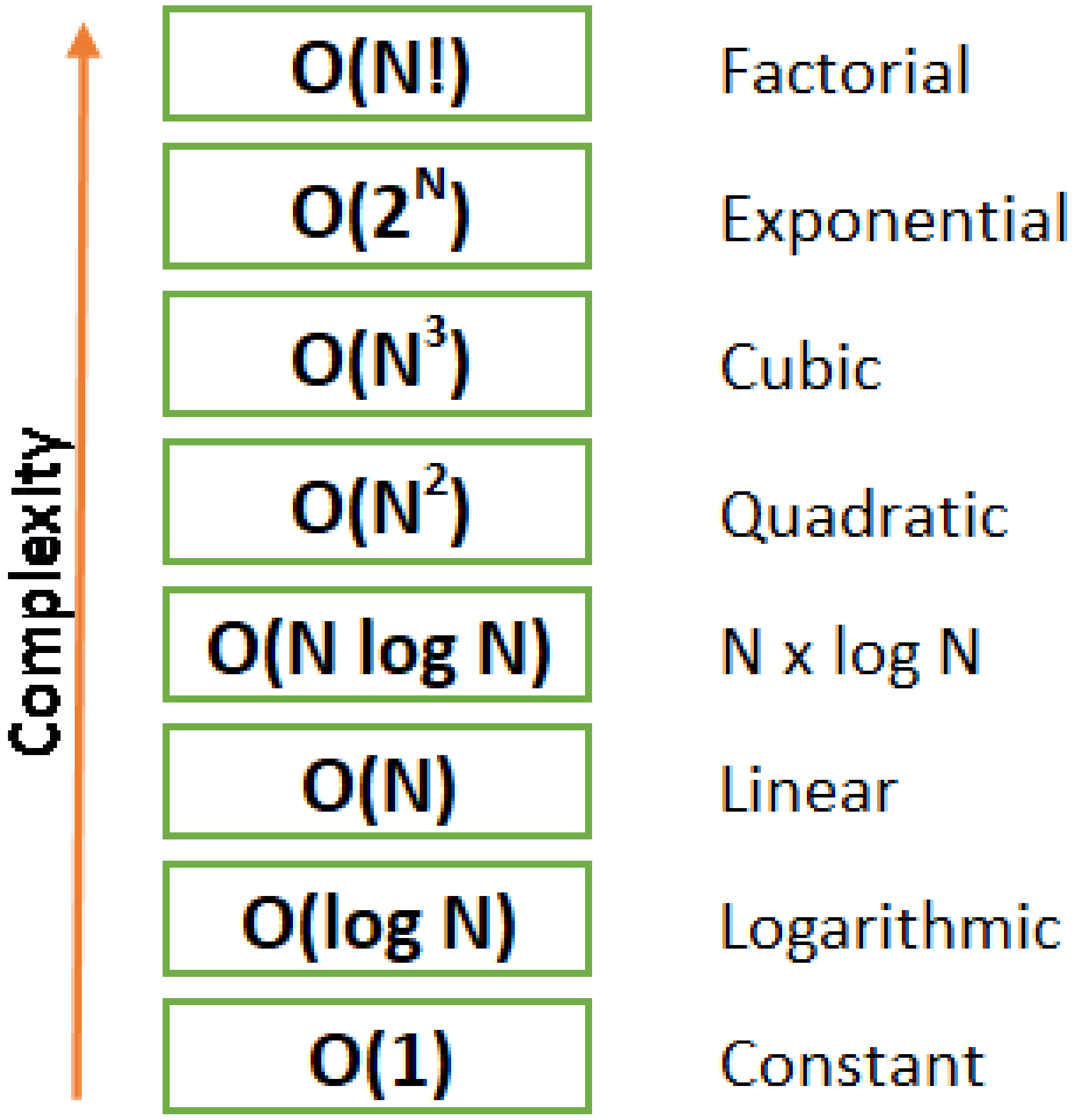
```java
package in.knowledgegate.dsa.complexity;

public class ArrayCopy {

    public static void main(String[] args) {
        int[] newCopy = copy(new int[]{1, 2, 3 , 4, 5});
    }

    1 usage
    private static int[] copy (int []array) {
        int newCopy[] = new int[array.length];
        for (int i = 0; i < array.length; i++) {
            newCopy[i] = array[i];
        }
        return newCopy;
    }
}
```

What are Asymptotic notations?

What is Big-O Notation

Understanding Cost




Complexity
O(N!) Factorial
O(2^N) Exponential
O(N^3) Cubic
O(N^2) Quadratic
O(N log N) N x log N
O(N) Linear
O(log N) Logarithmic
O(1) Constant

Tradeoff between Space and Time complexity

Need of Time Space Tradeoff:

1.) Less time by using more memory.

2.) By solving a problem in very little space by spending a long time.

Example involving the concept of Time Space Tradeoff:

1. If data is stored uncompressed, it takes more space but less time.
2. Usage of Maps
3. Dynamic Programming

Module

Algorithm Concepts and Problems (Easy/Medium/Hard)

Topic

DSA by Prashant sir

Subtopic

Strings

Lesson

Strings

Java Strings

String Class

- An object of the String class represents a string of characters.

- The String class belongs to the java.lang package, which does not require an import statement.

- Like other classes, String has constructors and methods.

Strings

- **string**: An object storing a sequence of text characters.
  - Two ways to create a string
    - As a new object through the String class

```
String name = new String("text");
```

    - Convenient shorthand method

```
String name = "text";
```

BREAK

Immutability

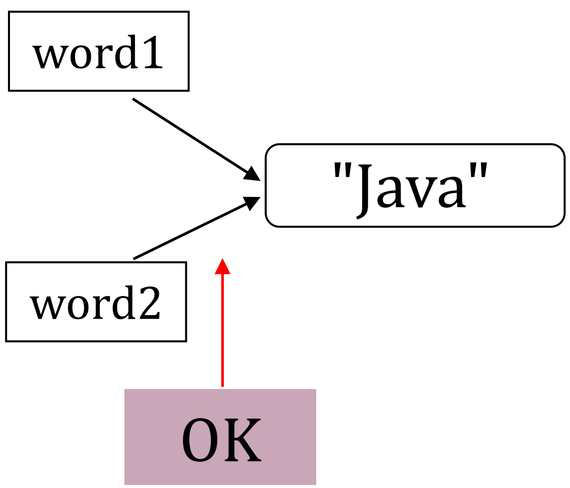
- *immutable*.
  - `String name = "text";`
  - Once created in memory, a string cannot be changed: none of its methods changes the string
  - If a string variable is reassigned, the memory address is deleted and a new memory address is created

- Immutable objects are convenient because several references can point to the same object safely: there is no danger of changing an object through one reference without the others being aware of the change.

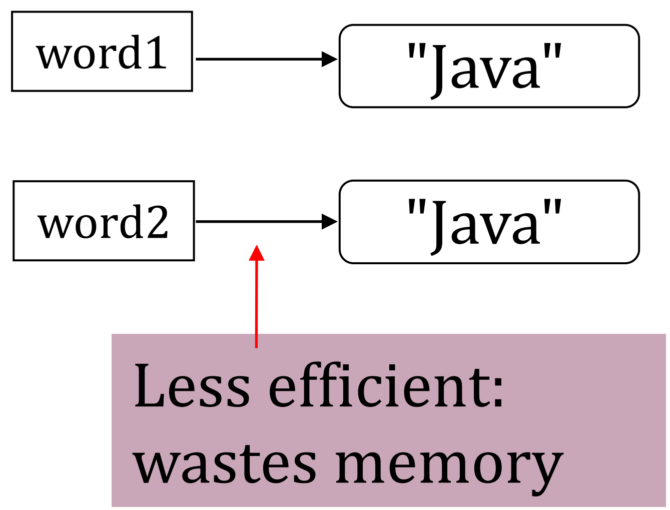
Advantages Of Immutability

Uses less memory.

```
String word1 = "Java";
String word2 = word1;
```



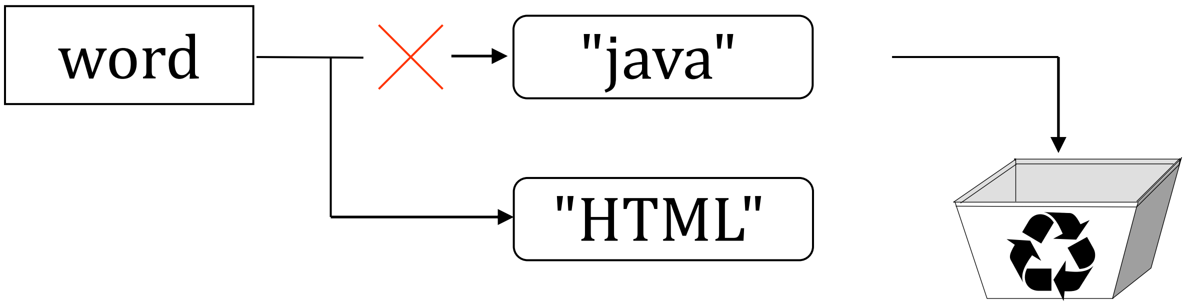

```
String word1 = “Java";
String word2 = new String(word1);
```



Disadvantages of Immutability

Less efficient — you need to create a new string and throw away the old one even for small changes.

```
String word = "java";
word = "HTML";
```



BREAK

Manipulating Characters

- `Character` class
  - Contains standard methods for testing values of characters
  - Methods that begin with “is”
    - Such as `isUpperCase()`
    - Return Boolean value
    - Can be used in comparison statements
  - Methods that begin with “to”
    - Such as `toUpperCase()`
    - Return character that has been converted to stated format

Indexes

- Characters of a string are numbered with 0-based *indexes*:

```
String name = "Java fun";
```

index	0	1	2	3	4	5	6	7
character	J	a	v	a		f	u	n

  - First character's index : 0
  - Last character's index : 1 less than the string's length

  - The individual characters are values of type `char`

BREAK

String Methods

Method name	Description
indexOf(str)	index where the start of the given string appears in this string (-1 if not found)
length()	number of characters in this string
substring(index1, index2) or substring(index1)	the characters in this string from index1 (inclusive) to index2 (exclusive); if index2 is omitted, grabs till end of string
toLowerCase()	a new string with all lowercase letters
toUpperCase()	a new string with all uppercase letters
charAt(index)	Returns the character at the specified index

- These methods are called using the dot notation:

```
String name = "Hello World";
System.out.println(name.length());
```

Length of a String

- It is sometimes useful to determine the length of a string
- Use the length( ) methods

```
String s = "Java is Fun";
int sLength = s.length();
System.out.println(sLength);    // 11
```

- The method counts the number of characters in the string variable.   It produces an integer result.

Modifying String

- Java is case sensitive

```
String s1 = "Java";
String s2 = "java";
```

  - s1 and s2 represent two different strings

- To get around this possible obstacle, Java has methods that display a string in all upper case letters or all lower case letters
  - toUpperCase( )
  - toLowerCase( )

Modifying String

- These methods build and return a new string, rather than modifying the current string.

- To modify a variable's value, you must reassign it:

```
String s = "Hello World";
s1 = s.toUpperCase();
s2 = s.toLowerCase();
System.out.println(s1 + "  " + s2);
      // HELLO WORLD  hello world
```

BREAK

Comparing Strings

- One thing we will be testing often in our programs is whether one string is the same as the other

- Two basic commands:
  - equals( )
  - compareTo( )

- **Do not** compare two `Strings` using the == operator
  - Not comparing values
  - Comparing computer memory locations

Comparing `Strings`

- `equals()` method
  - Evaluates contents of two `String` objects to determine if they are equivalent
  - Returns `true` if objects have identical contents
- `equalsIgnoreCase()` method
  - Ignores case when determining if two `Strings` equivalent
  - Useful when users type responses to prompts in programs

```
s1 = "Java";
s2 = "java";
boolean eq = s1.equals(s2);
System.out.println(eq);      // false
```

Comparing `Strings`

- `compareTo()` method
  - Compares two `Strings` and returns:
    - Zero
      - Only if two `Strings` refer to same value
    - Negative number
      - If calling object “less than” argument
    - Positive number
      - If calling object “more than” argument

Comparing Strings

Method	Description
equals(str)	whether two strings contain the same characters
equalsIgnoreCase(str)	whether two strings contain the same characters, ignoring upper vs. lower case
startsWith(str)	whether one contains other's characters at start
endsWith(str)	whether one contains other's characters at end
contains(str)	whether the given string is found within this one

These methods return a boolean value (true or false)

BREAK

String Concatenation

- Concatenation
  - Join multiple variables together
  - Two ways to concatenate
    - concat method
      - `String s3 = s1.concat(s2);`
    - Convenient shorthand
      - `String s3 = s1 + s2;`

```
s1 = "Hello";
s2 = "World";
s3 = s1 + " " + s2;
```

Numbers to Strings

Three ways to convert a number into a string:

1. String s = "" + num;

```
s = "" + 123;          //"123"
```

2. String s = Integer.toString (i);
   String s = Double.toString (d);

```
s = Integer.toString(123);      //"123"
s = Double.toString(3.14);      //"3.14"
```

3. String s = String.valueOf (num);

```
s = String.valueOf(123);        //"123"
```

Integer and **Double** are “wrapper” classes from **java.lang** that represent numbers as objects. They also provide useful static methods.

`Strings` to Numbers

- `Integer` and `Double` class
  - Automatically imported into programs (java.lang)
  - Convert `String` to integer
    - `valueOf()` method
      - Convert `String` to `Integer` class object
    - `intValue()` method
      - Extract simple integer from wrapper class
    - `parseInt()` method
      - Returns its integer value
  - Convert `String` to double
    - `parseDouble()` method
      - Returns its double value

BREAK

`StringBuilder`

The `StringBuilder/StringBuffer` class is an alternative to the `String` class.

- Can be used wherever a string is used.
- More flexible than String.
  - You can add, insert, or append new contents into a string buffer, whereas the value of a String object is fixed once the string is created.

`StringBuilder`

- Using `StringBuilder` objects
  - Provides improved computer performance over `String` objects
  - Can insert or append new contents into `StringBuilder`
- `StringBuilder` constructors

```
public StringBuilder ()
public StringBuilder (int capacity)
public StringBuilder (String s)
```

```
StringBuilder name = new
 StringBuilder("Hello");
```

StringBuilder

Method	Description
append(str)	Adds a string to the end of the StringBuilder object
delete(start, end)	Deletes characters at the specified index
insert(index, str)	Inserts string into the builder at the position offset
replace(start, end, str)	Replaces the characters in this builder at the specified index with the specified string
setCharAt(index, char)	Changes a character at the specified index
CharAt(index)	Returns character at the specified index

Module

Algorithm Concepts and Problems (Easy/Medium/Hard)

Topic

DSA by Prashant sir

Subtopic

Math

Lesson

Number System

NUMBER SYSTEM

Number System

When we type some letters or words, the computer translates them in binary numbers as computers can understand only binary numbers.

Decimal number system has base 10 as it uses 10 digits from 0 to 9. In decimal number system, the successive positions to the left of the decimal point represent units, tens, hundreds, thousands and so on.

A value of each digit in a number can be determined using

- The digit
- Symbol value (is the digit value 0 to 9)
- The position of the digit in the number
- Increasing Power of the base (i.e. 10) occupying successive positions moving to the left

Example

Decimal number (592):

Number	Symbol Value	Position from the right end	Positional Value	Decimal Equivalent
5 9 2				
	2	0	10^0	$2*10^0 = 2$
	9	1	10^1	$9*10^1 = 90$
	5	2	10^2	$5*10^2 = 500$
				592

BREAK

Binary number system

- Uses two digits, 0 and 1.
- Also called base 2 number system

$(110011)_2 = (51)_{10}$

Number	Symbol Value	Position from the right end	Positional Value	Decimal Equivalent
1 1 0 0 1 1				
	1	0	2^0	1*0 = 1
	1	1	2^1	1*2 = 2
	0	2	2^2	0*4 = 0
	0	3	2^3	0*8 = 0
	1	4	2^4	1*16= 16
	1	5	2^5	1*32= 32
				51

BREAK

Convert decimal 17 into binary number

Step			Remainder
1	Divide 17 by 2	$2\,\lfloor\underline{\,17\,}$ 8	1
2	Divide 8 by 2	$2\,\lfloor\underline{\,8\,}$ 4	0
3	Divide 4 by 2	$2\,\lfloor\underline{\,4\,}$ 2	0
4	Divide 2 by 2	$2\,\lfloor\underline{\,2\,}$ 1	0
5	Divide 1 by 2	$2\,\lfloor\underline{\,1\,}$ 0	1

Convert Binary to Decimal

Binary Number	Positional value	operation	
1 1 1 0 0 1 0 1			
	1	$1*2^0$	1
	0	$0*2^1$	0
	1	$1*2^2$	4
	0	$0*2^3$	0
	0	$0*2^4$	0
	1	$1*2^5$	32
	1	$1*2^6$	64
	1	$1*2^7$	128 = 229

BREAK

Octal number notation

- Octal is base 8 counting system having digit values 0 through 7
- The octal system groups three binary bits together into one digit symbol.

Octal	Binary
0	000
1	001
2	010
3	011
4	100
5	101
6	110
7	111

Hexadecimal

- Hexadecimal number system is a base 16 counting system
- It uses 16 Symbols: 0 to 9 and the capital letter A,B…F.
- Each Hexadecimal is equivalent to a group of 4 binary bits.

Hexadecimal	Binary	Hexadecimal	Binary
0	0000	8	1000
1	0001	9	1001
2	0010	A	1010
3	0011	B	1011
4	0100	C	1100
5	0101	D	1101
6	0110	E	1110
7	0111	F	1111

BREAK

Convert binary to Hexadecimal

- Divide the given binary number into groups of 4 bits each(from right to left).
- Replace each group by its hexadecimal Equivalent.

Questions:

1. Convert $(101111100001)_2$ into its hexadecimal.

   Ans: $(BEI)_{16.}$

2. Convert $(10101111.0010111)_2$ into its hexadecimal.

   Ans: $(AF.2E)_{16}$

Convert Decimal to Hexadecimal

- Divide the number by 16.
- Write the dividend under the number. This become the new number.
- Write the remainder at the right in a column.
- Repeat steps 1 to 3 until a ‘0’ is produced as a new number.

Question: Convert the Decimal 87 to hexadecimal number.

$(87)_{10} = (57)_{16}$

Convert hexadecimal to Decimal

- Write out the Hexadecimal digits as power of 16.
- Convert each power of 16 into its decimal equivalent term.
- Add these terms to produce the required decimal number.

Question: $(A2D)_{16}=(2605)_{10}$

Module

Algorithm Concepts and Problems (Easy/Medium/Hard)

Topic

DSA by Prashant sir

Subtopic

Sorting

Lesson

Sorting

Sorting

Outline

- How to use standard sorting methods in the Java API
- Understand these sorting algorithms:
  - Selection sort
  - Bubble sort
  - Insertion sort
  - Merge sort
  - Heapsort
  - Quicksort

Using Java API Sorting Methods

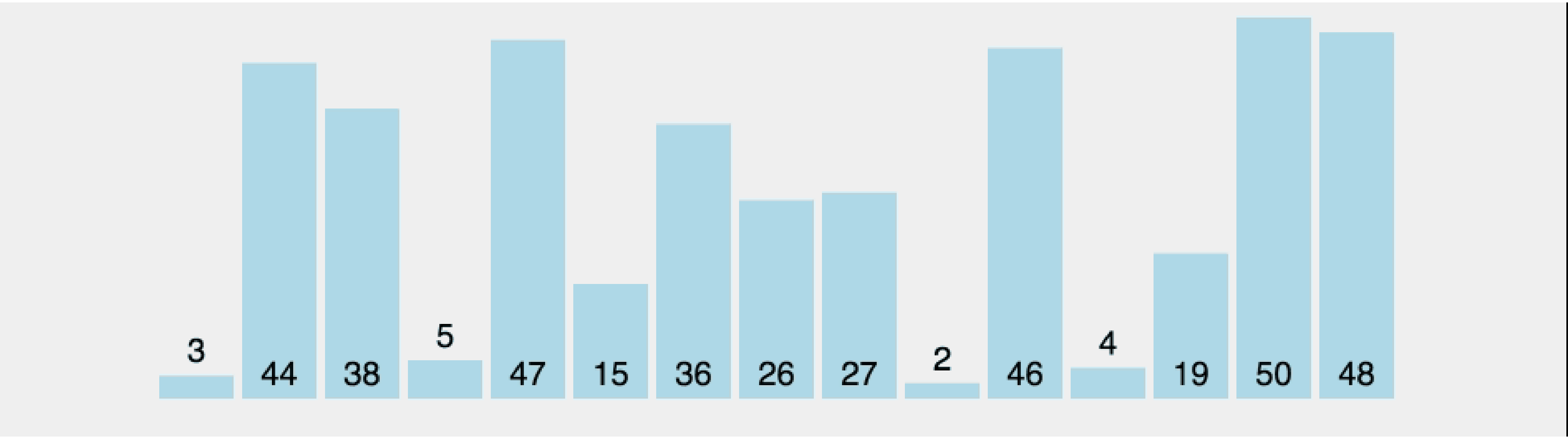
- Java API provides a class **`Arrays`** with several overloaded sort methods for different array types
- Class **`Collections`** provides similar sorting methods
- Sorting methods for arrays of primitive types:
    - Based on the *<u>Quicksort</u>* algorithm
- Method of sorting for arrays of objects (and **`List`**):
    - Based on *<u>Mergesort</u>*
- In practice you would tend to use these
    - In this class, you will implement some yourself

BREAK

Selection Sort

- A relatively easy to understand algorithm
- Sorts an array in *passes*
  - Each pass selects the next smallest element
  - At the end of the pass, places it where it belongs
- Efficiency is $O(n^2)$, hence called a *quadratic* sort
- Performs:
  - $O(n^2)$ comparisons
  - $O(n)$ exchanges (swaps)

Selection Sort



Selection Sort Algorithm

1. for **fill** = 0 to n-2 do // steps 2-6 form a <u>*pass*</u>
2. set **posMin** to **fill**
3. for **next** = **fill**+1 to n-1 do
4. if item at **next** < item at **posMin**
5. set **posMin** to **next**
6. Exchange item at **posMin** with one at **fill**

BREAK

Bubble Sort

- *Compares adjacent* array elements
  - *Exchanges* their values if they are *out of order*

- Smaller values *bubble up* to the top of the array
  - Larger values sink to the bottom

Bubble Sort

8 5 3 1 4 7 9

Bubble Sort Algorithm

1. do
2. for each pair of adjacent array elements
3. if values are out of order
4. Exchange the values
5. while the array is not sorted

Analysis of Bubble Sort

- Excellent performance *in some cases*
  - But *very poor* performance in others!
- Works ***best*** when array is nearly sorted to begin with
- Worst case number of comparisons: $O(n^2)$
- Worst case number of exchanges: $O(n^2)$
- *Best case* occurs when the array is already sorted:
  - $O(n)$ comparisons
  - $O(1)$ exchanges (none actually)

BREAK

Insertion Sort

- Based on technique of card players to arrange a hand
  - Player keeps cards picked up so far in *sorted order*
  - When the player picks up a new card
    - Makes room for the new card
    - Then inserts it in its proper place

FIGURE 10.3
Picking Up a Hand of Cards

Insertion Sort

6 5 3 1 8 7 2 4

Insertion Sort Algorithm

- For each element from 2nd (nextPos = 1) to last:
  - Insert element at nextPos where it belongs
  - Increases sorted subarray size by 1

- To make room:
  - Hold nextPos value in a variable
  - Shuffle elements to the right until gap at right place

Analysis of Insertion Sort

- Maximum number of comparisons: $O(n^2)$
- In the best case, number of comparisons: $O(n)$
- # shifts for an insertion = # comparisons - 1
  - When new value smallest so far, # comparisons
- A shift in insertion sort moves only <u>*one item*</u>
  - Bubble or selection sort exchange: <u>*3 assignments*</u>

BREAK

Comparison of Quadratic Sorts

- None good for large arrays!

TABLE 10.2
Comparison of Quadratic Sorts

	Number of Comparisons		Number of Exchanges	
	Best	Worst	Best	Worst
Selection sort	$O(n^2)$	$O(n^2)$	$O(n)$	$O(n)$
Bubble sort	$O(n)$	$O(n^2)$	$O(1)$	$O(n^2)$
Insertion sort	$O(n)$	$O(n^2)$	$O(n)$	$O(n^2)$

TABLE 10.3
Comparison of Rates of Growth

n	n^2	$n \log n$
8	64	24
16	256	64
32	1,024	160
64	4,096	384
128	16,384	896
256	65,536	2,048
512	262,144	4,608

BREAK

Merge Sort

- A *merge* is a common data processing operation:
  - Performed on *two sequences* of data
    - Items in both sequences use same **`compareTo`**
    - Both sequences in ordered of this **`compareTo`**
- **Goal:** Combine the two sorted sequences in one larger sorted sequence

- *Merge sort* merges longer and longer sequences

Merge Algorithm (Two Sequences)

Merging two sequences:

1. Access the first item from both sequences
2. While neither sequence is finished
   1. Compare the current items of both
   2. Copy smaller current item to the output
   3. Access next item from that input sequence
3. Copy any remaining from first sequence to output
4. Copy any remaining from second to output

Picture of Merge

6 5 3 1 8 7 2 4

Analysis of Merge

- Two input sequences, total length n elements
  - Must move each element to the output
  - Merge time is O(n)
- Must store *both input* and *output* sequences
  - An array cannot be merged *in place*
  - Additional space needed: O(n)

BREAK

Quicksort

- Developed in 1962 by C. A. R. Hoare
- Given a *pivot value:*
  - Rearranges array into two parts:
    - Left part $\leq$ pivot value
    - Right part > pivot value
- Average case for Quicksort is O(n log n)
  - Worst case is $O(n^2)$

Quicksort

6 5 3 1 8 7 2 4

Algorithm for Quicksort

first and **last** are end points of region to sort

- if **first** < **last**
- Partition using **pivot**, which ends in **pivIndex**
- Apply Quicksort recursively to left subarray
- Apply Quicksort recursively to right subarray

Performance: O(n log n) provide **pivIndex** not always too close to the end

Performance $O(n^2)$ when **pivIndex** always near end

Algorithm for Partitioning

1. Set pivot value to a[fst]
2. Set **up** to fst and **down** to lst
3. do
4. Increment **up** until a[**up**] > pivot or **up** = lst
5. Decrement **down** until a[**down**] <= pivot or **down** = fst
6. if **up** < **down**, swap a[**up**] and a[**down**]
7. while **up** is to the left of **down**
8. swap a[fst] and a[**down**]
9. return **down** as pivIndex

Trace of Algorithm for Partitioning

FIGURE 10.14
Locating First Values to Exchange

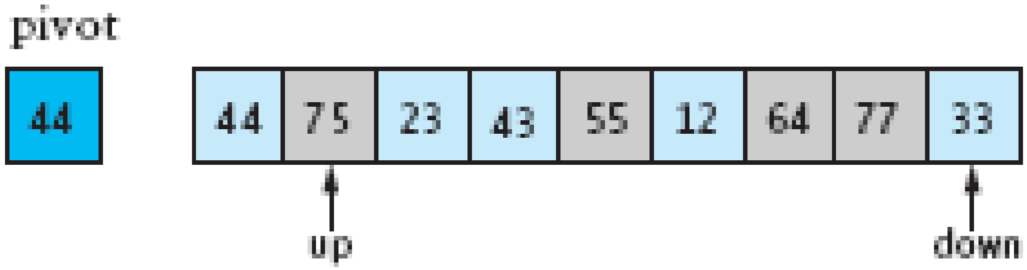


FIGURE 10.15
Array After the First Exchange

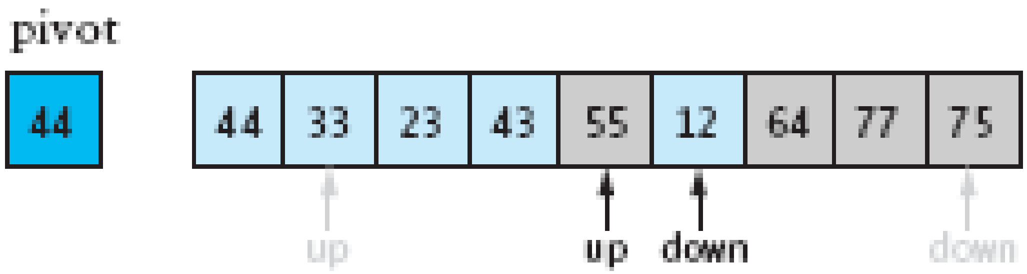
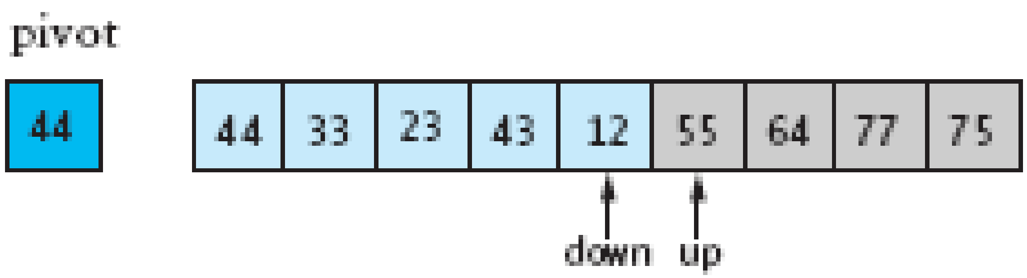


FIGURE 10.16
Array After the Second Exchange



BREAK

Summary

- Three quadratic sorting algorithms:
  - Selection sort, bubble sort, insertion sort
- Quicksort: average-case O(n log n)
  - If the pivot is picked poorly, get worst case: $O(n^2)$
- Merge sort and heapsort: guaranteed O(n log n)
  - Merge sort: space overhead is O(n)
- Java API has good implementations

BREAK

Module

Algorithm Concepts and Problems (Easy/Medium/Hard)

Topic

DSA by Prashant sir

Subtopic

Binary Search

Lesson

Binary Search

Binary Search

Binary search. Given `value` and sorted array `a[]`, find index `i` such that `a[i] = value`, or report that no such index exists.

Invariant. Algorithm maintains `a[lo]` $\leq$ `value` $\leq$ `a[hi]`.

Ex. Binary search for 33.

6	13	14	25	33	43	51	53	64	72	84	93	95	96	97
0	1	2	3	4	5	6	7	8	9	10	11	12	13	14
lo														hi

Binary Search

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Ex. Binary search for 33.

6	13	14	25	33	43	51	53	64	72	84	93	95	96	97
0	1	2	3	4	5	6	7	8	9	10	11	12	13	14

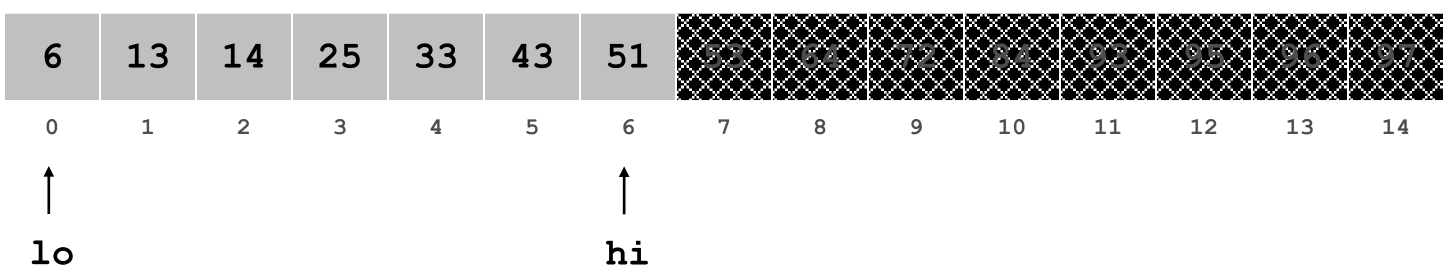
lo (0) **mid** (7) **hi** (14)

Binary Search

Binary search. Given `value` and sorted array `a[]`, find index `i` such that `a[i] = value`, or report that no such index exists.

Invariant. Algorithm maintains `a[lo]` $\leq$ `value` $\leq$ `a[hi]`.

Ex. Binary search for 33.



Binary Search

Binary search. Given `value` and sorted array `a[]`, find index `i` such that `a[i] = value`, or report that no such index exists.

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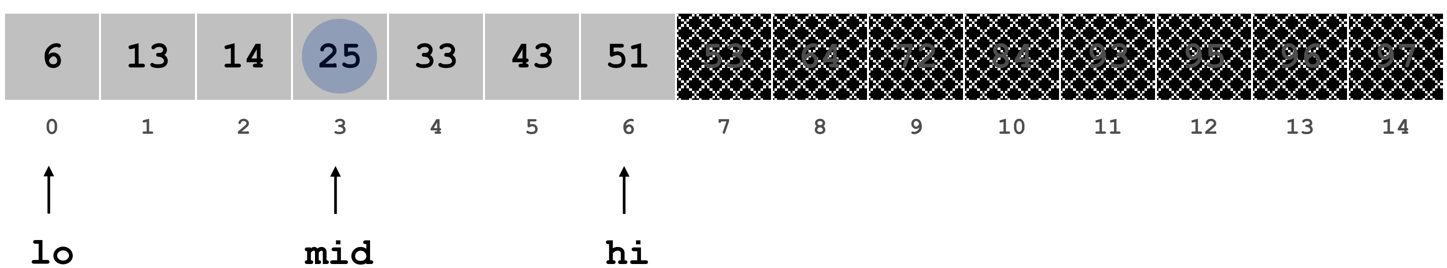
Ex. Binary search for 33.

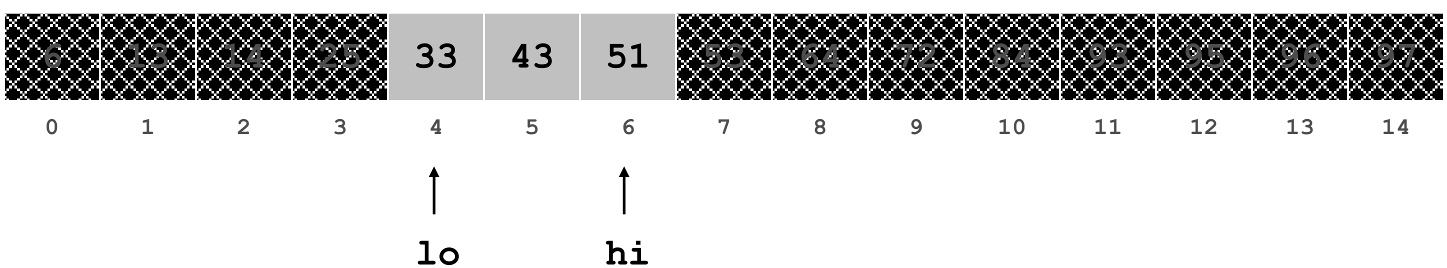


Binary Search

Binary search. Given `value` and sorted array `a[]`, find index `i` such that `a[i] = value`, or report that no such index exists.

Invariant. Algorithm maintains `a[lo]` ≤ `value` ≤ `a[hi]`.

Ex. Binary search for 33.



Binary Search

Binary search. Given `value` and sorted array `a[]`, find index `i` such that `a[i] = value`, or report that no such index exists.

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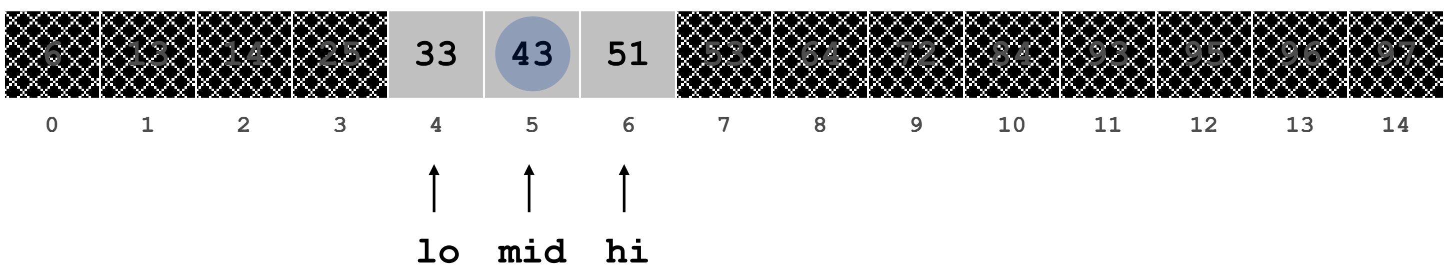

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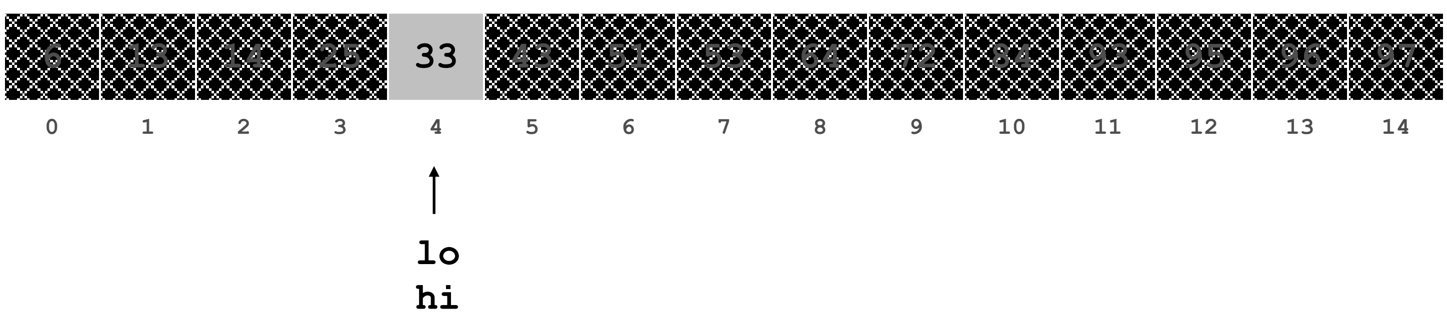

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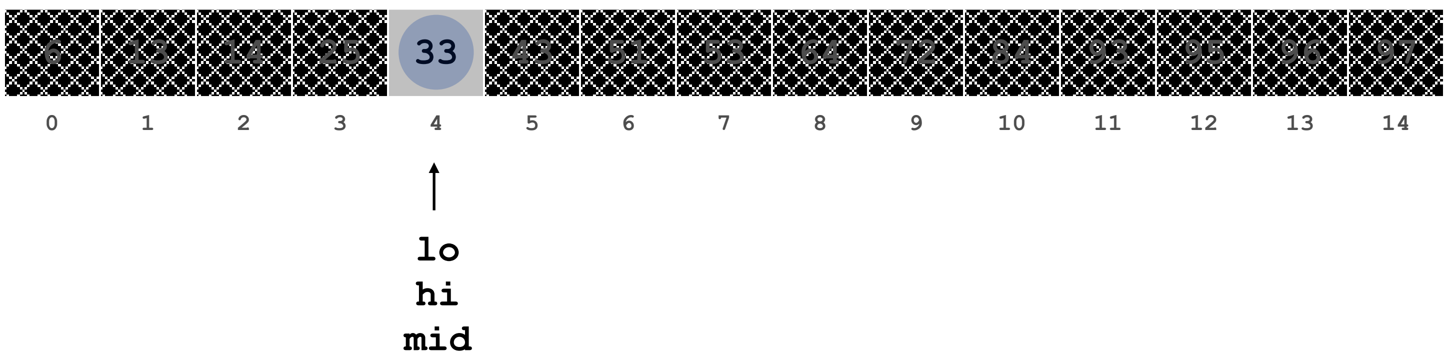

Binary Search

Binary search. Given `value` and sorted array `a[]`, find index `i` such that `a[i] = value`, or report that no such index exists.

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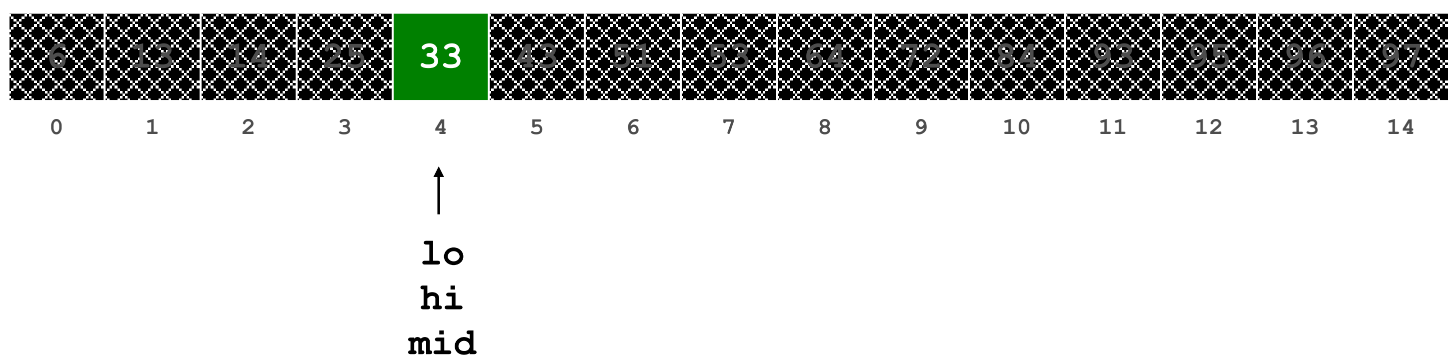


Module

Algorithm Concepts and Problems (Easy/Medium/Hard)

Topic

DSA by Prashant sir

Subtopic

Bit Manipulation

Lesson

Bit Manipulation

BIT Manipulation

Objectives

- To understand
  - Bitwise Logical Operations
    - OR
    - AND
    - XOR
  - Shift Operations
    - Shift Left
    - Shift Right

Logical AND Operator

- AND (truth table)

A	B	A AND B
0	0	0
0	1	0
1	0	0
1	1	1

- **AND :** bit-by-bit operation leaves a 1 in the result only if both bits of the operands are 1

Example : Logical AND Operator

- Assume that register A and B are 8-bit register
- Let A stores value 0011 1011
- Let B stores value 1001 0010
- Find A AND B

0	0	1	1	1	0	1	1	AND
1	0	0	1	0	0	1	0	
0	0	0	1	0	0	1	0	

Logical NOT Operator

- Single operand. Reverses the input
- **Example**

0	0	1	1	NOT

Logical OR Operator

- OR (truth table)

A	B	A OR B
0	0	0
0	1	1
1	0	1
1	1	1

- OR : bit-by-bit operation leaves a 1 in the result if *either* bit of the operands is 1

Example : Logical OR Operator

- Assume that register A and B are 8-bit register
- Let A = 0011 1011, and B = 1001 0010
- Find A OR B

0	0	1	1	1	0	1	1	OR
1	0	0	1	0	0	1	0	
1	0	1	1	1	0	1	1	

Logical XOR Operator

- XOR (truth table)

A	B	A XOR B
0	0	0
0	1	1
1	0	1
1	1	0

- XOR : bit-by-bit operation leaves a 1 in the result if bit of the operands are different

Example : Logical XOR Operator

- Assume that register A and B are 8-bit register
- Let A = 0011 1011, and B = 1001 0010
- Find A XOR B

0	0	1	1	1	0	1	1	XOR
1	0	0	1	0	0	1	0	
1	0	1	0	1	0	0	1	

Shift Operations

- Shift means move all the bits in a word to the **left** or **right** by number of bits

Shift Operations

- Move all the bits in a word to the left or right by a number of bits, filling the emptied bits with **0**s
- **Example :**

  Given A an 8-bit register which stores data 1100 1010

					1	1	0	0	1	0	1	0
Shift right by 4 bits					0	0	0	0	1	1	0	0	1	0	1	0
Shift left by 4 bits	1	1	0	0	1	0	1	0	0	0	0	0

Module

Algorithm Concepts and Problems (Easy/Medium/Hard)

Topic

DSA by Prashant sir

Subtopic

Linked List

Lesson

link list

Linked List

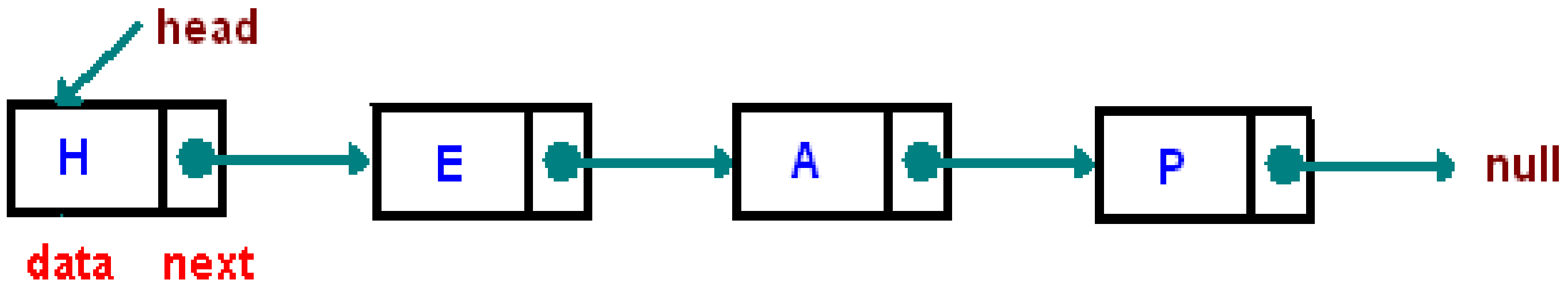
<u>Problem with array</u>

- One disadvantage of using arrays to store data is that, arrays are static structures and therefore cannot be easily extended or reduced to fit the data set (suffer from both internal and external fragmentation).

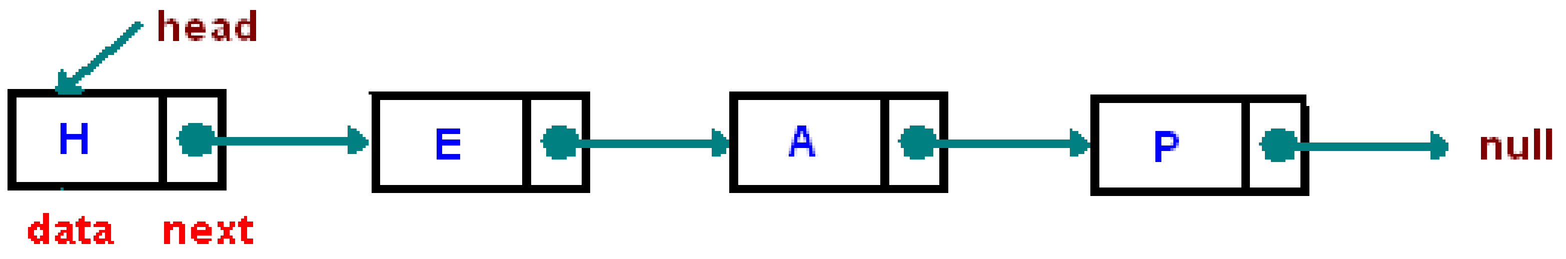
- Arrays are also expensive to maintain new insertions and deletions.

0	1	2	3	4	5	6	7

- Solution is link list
  - A linked list is a linear collection of data elements called nodes, where the linear order is given by the means of pointers.
  - Each node is divided into two parts the first part is the information part of the node, which contains data.
  - The second part called linked field or next pointer field, contains the address of the next node of the list.



- The pointer of the last node contains a null pointer, which is an invalid address (0 or negative value).
- The linked also contains a list pointer variable called start/first/head which contain the address of the first node in the list.
- A special case is the list that has no nodes, such a list is called null list or empty list and is denoted by a null pointer in the variable start/first/head.

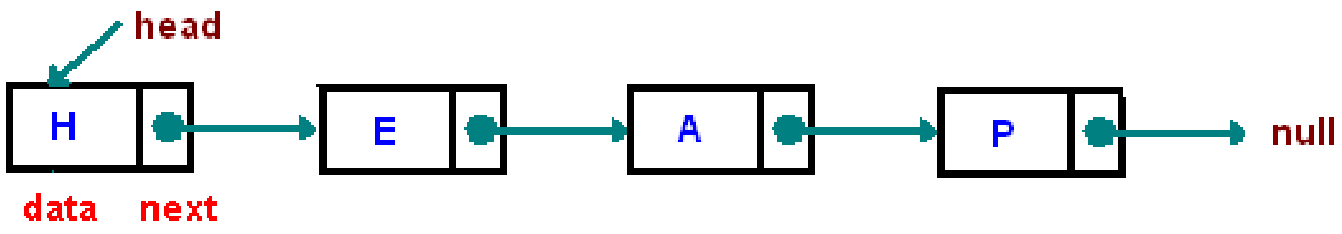


BREAK

Advantage of link list

- Do not suffer from internal and external fragmentation.

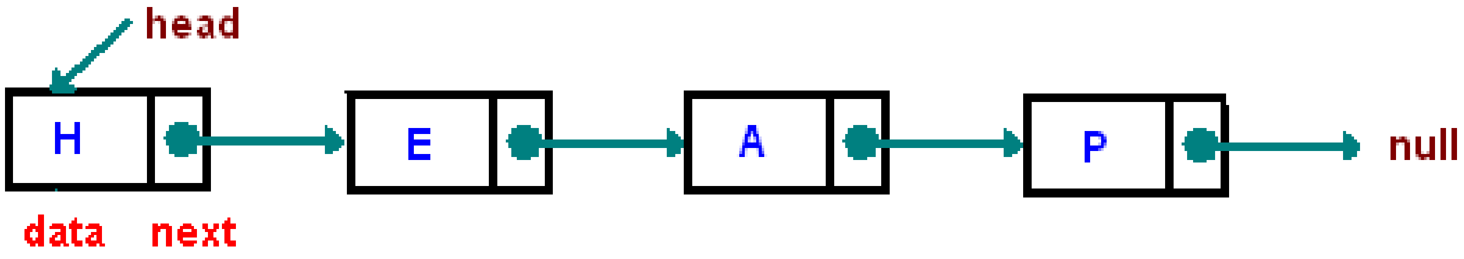
- A linked list is a dynamic data structure. The number of nodes in a list is not fixed and can grow and shrink on demand. Can easily support insertion or deletion.



Disadvantage of link list

- Slow access: - It does not allow direct search, supports only sequential search even if link list is sorted. i.e. If we want to access a particular node then we have to start from the first node.

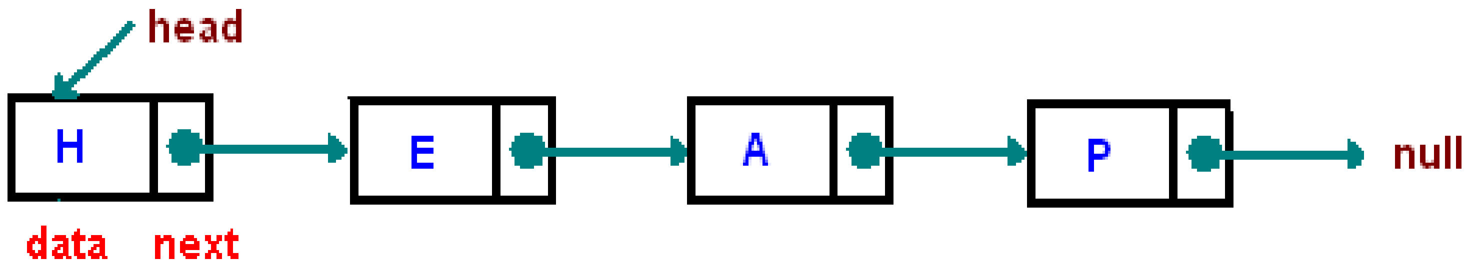
- A good amount of space is wasted in storing pointers. For e.g. 4 bytes (on 32-bit CPU) to store a reference to the next node.



BREAK

Java Implementation of link list

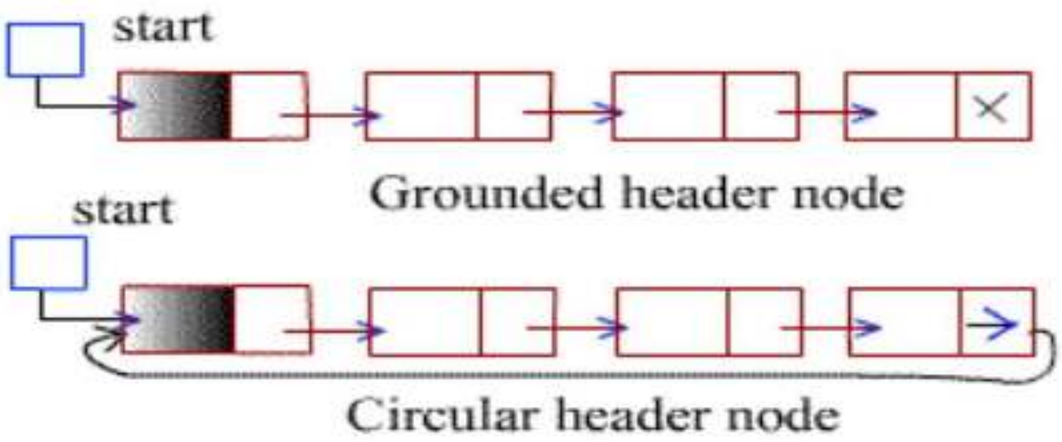
```
Class ListNode {
    int data;
    ListNode next;
};
```



BREAK

Header link list

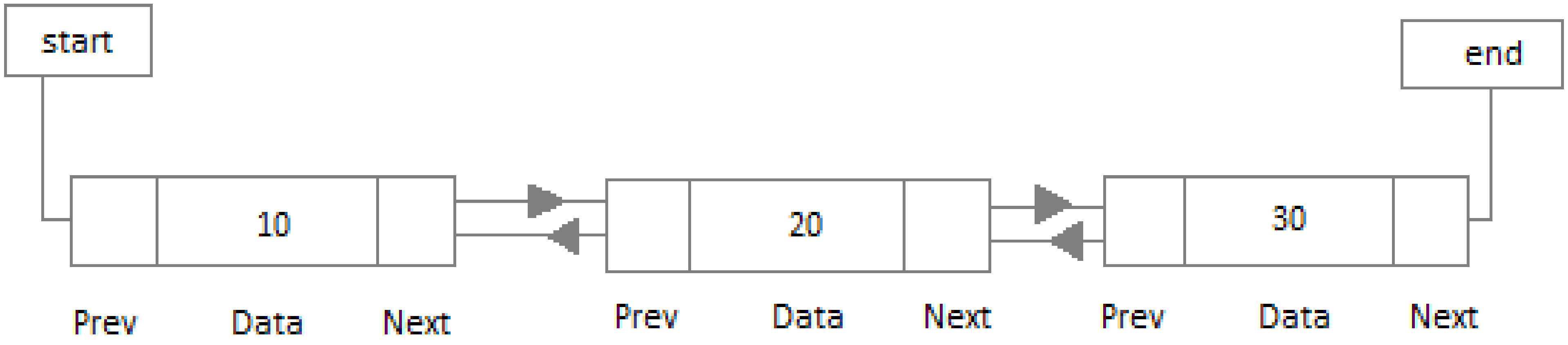
- A header linked list is a linked list which always contains a special node, called head node at the beginning of the list. This special node in general used to store number of nodes present in the linked list or some other meta data



- Header circular link list are frequently used instead of ordinary linked list, because many operations are much easier to state and implement using header list.
- Null pointer is not used and hence all pointer contains valid address.
  - loop condition while(ptr != start)

Doubly link list

- The list which can be traversed in two directions: in the unusual forward direction from the beginning of the list to the ends or in the end direction from the end of the list to the beginning of the list.
- Furthermore given the location LOC of a node N in the list, one now has immediate access to both the next node and the preceding node of the list. This means in particular that one is able to delete N from the list without traversing any part of the list.



BREAK

Module

Algorithm Concepts and Problems (Easy/Medium/Hard)

Topic

DSA by Prashant sir

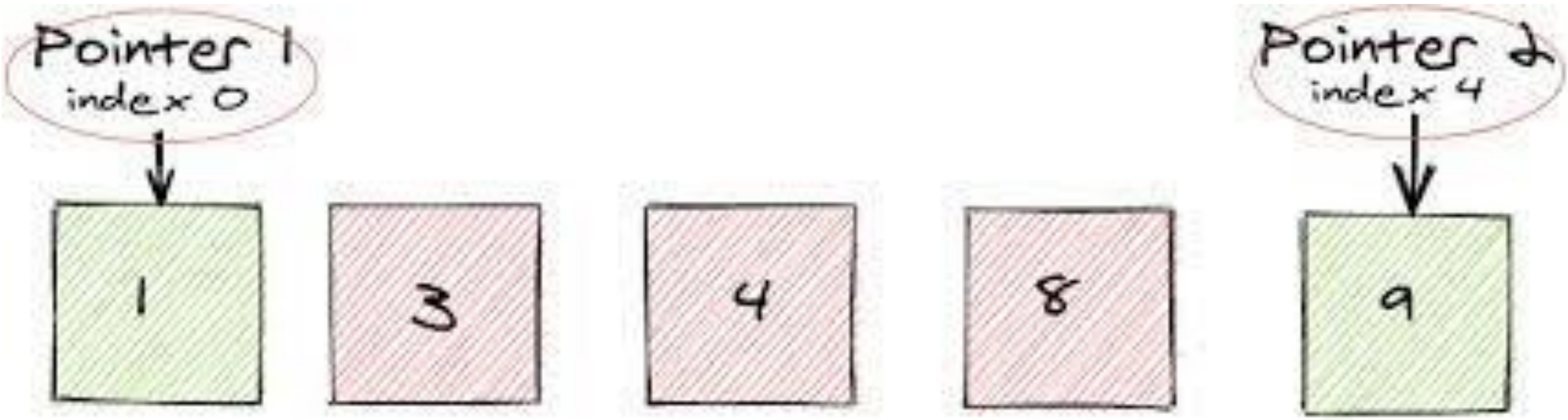
Subtopic

Two pointer

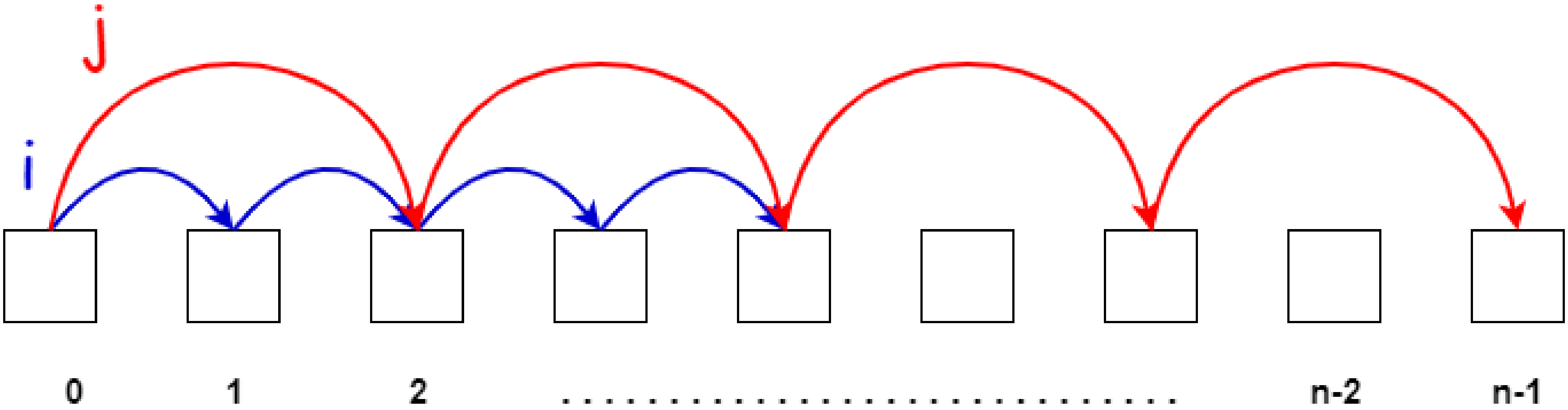
Lesson

Two Pointer

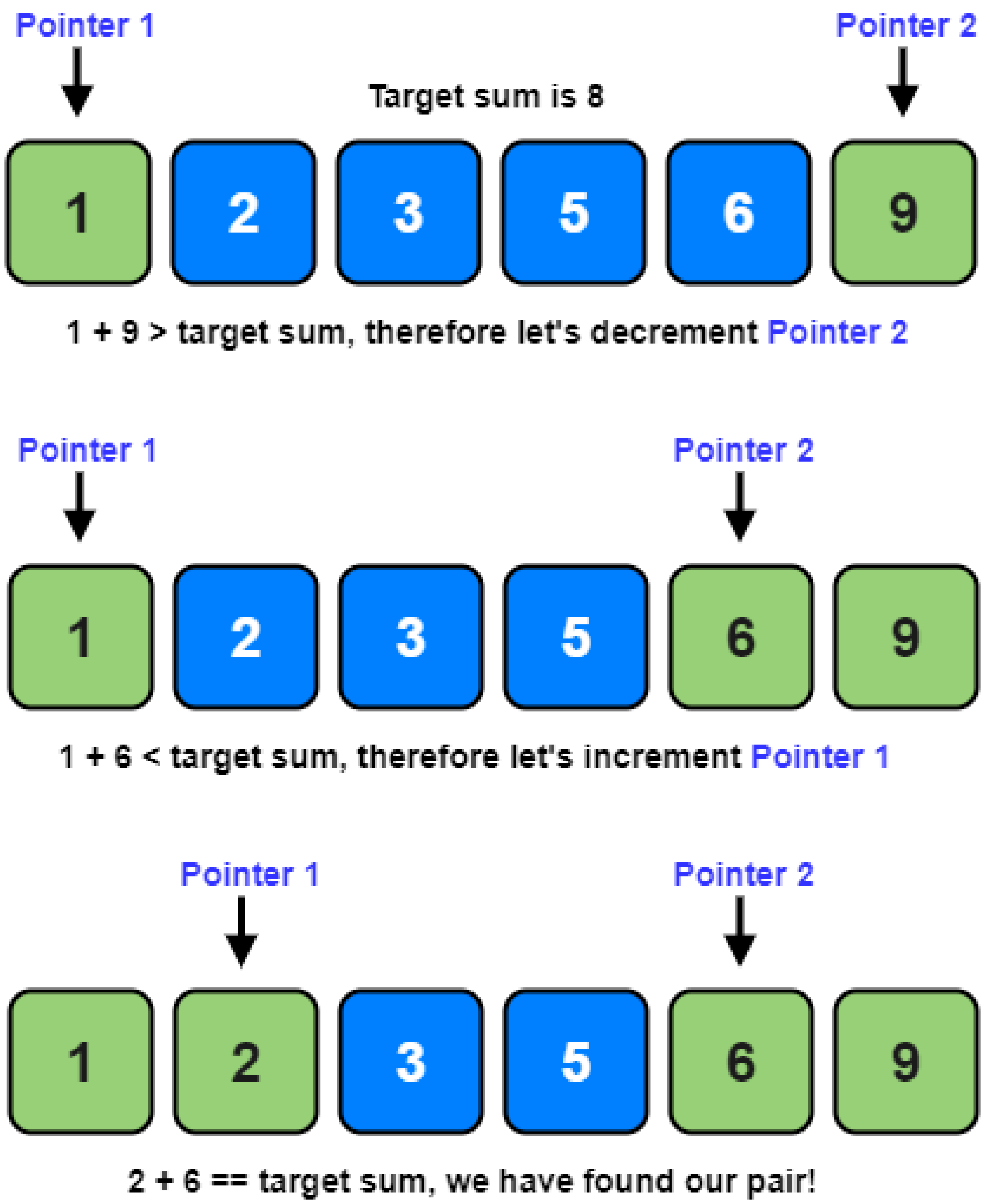
Two Pointer Approach



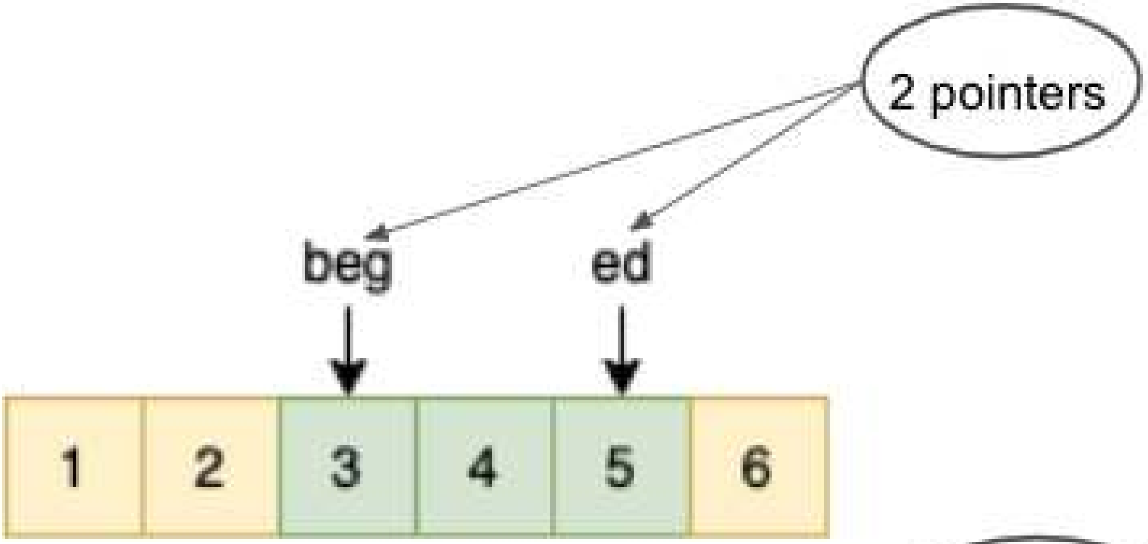
Pointer 1
index 0
Pointer 2
index 4
1
3
4
8
9



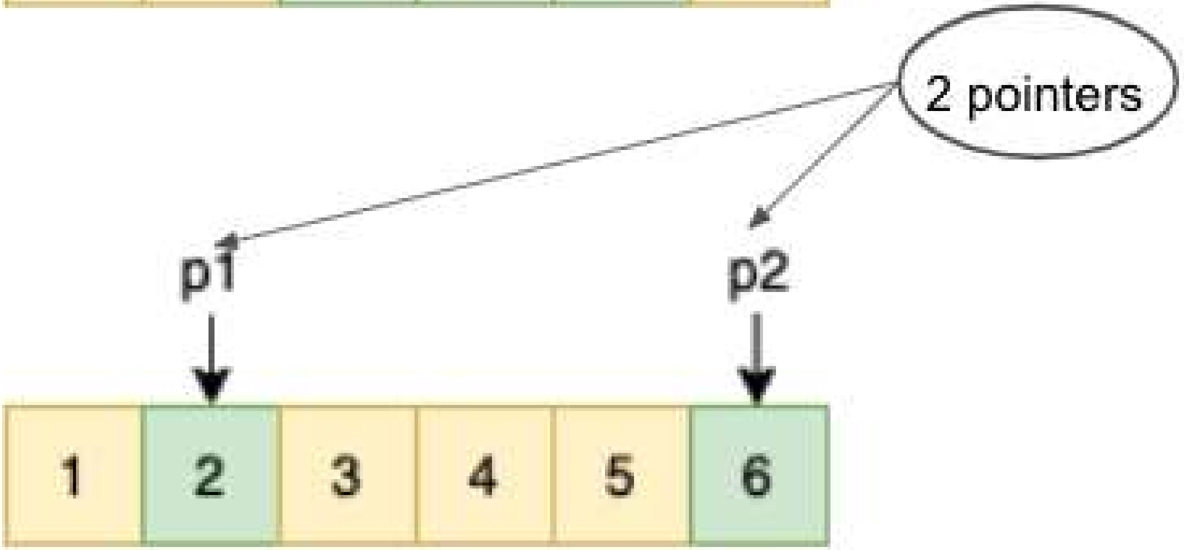
j
i
0
1
2
n-2
n-1


Pointer 1
Pointer 2
Target sum is 8
1
2
3
5
6
9
1 + 9 > target sum, therefore let's decrement Pointer 2
Pointer 1
Pointer 2
1
2
3
5
6
9
1 + 6 < target sum, therefore let's increment Pointer 1
Pointer 1
Pointer 2
1
2
3
5
6
9
2 + 6 == target sum, we have found our pair!

1. Given an **Array**
2. Example 1:
   Find a **subarray** with sum = 12



3. Example 2:
   Find a **pair** with difference = 4



Module

Algorithm Concepts and Problems (Easy/Medium/Hard)

Topic

DSA by Prashant sir

Subtopic

Recursion

Lesson

Recursion

Recursion

Objectives

- Become familiar with the idea of recursion
- Learn to use recursion as a programming tool

Overview

Recursion: a definition in terms of itself.

Recursion in algorithms:

- Natural approach to **some** (not all) problems
- A *recursive algorithm* uses itself to solve one or more smaller identical problems

Recursion in Java:

- Recursive methods implement recursive algorithms
- A *recursive method* includes a call to itself

Recursive Methods
Must Eventually Terminate

A recursive method must have
at least one base, or stopping, case.

- A base case does not execute a recursive call
  - stops the recursion

- Each successive call to itself must be a "smaller version of itself"
  - an argument that describes a smaller problem
  - a base case is eventually reached

Key Components of a Recursive Algorithm Design

1. What is a smaller ***identical*** problem(s)?
    - Decomposition
2. How are the answers to smaller problems combined to form the answer to the larger problem?
    - Composition
3. Which is the smallest problem that can be solved easily (without further decomposition)?
    - Base/stopping case

Break

Factorial (N!)

- $N! = (N\text{-}1)! \;\; * \; N \quad$ [for $N > 1$]
- 1! = 1
- 3!

  = 2! * 3

  = (1! * 2) * 3

  = 1 * 2 * 3
- Recursive design:
  - Decomposition: $(N\text{-}1)!$
  - Composition: * N
  - Base case: 1!

`factorial` Method

```
public static int factorial(int n)
{
  int fact;
  if (n > 1) // recursive case (decomposition)
    fact = factorial(n - 1) * n; // composition
  else // base case
    fact = 1;

  return fact;
}
```

```
public static int factorial(int 3)
{
  int fact;
  if (n > 1)
    fact = factorial(2) * 3;
  else
    fact = 1;
  return fact;
}
```

```
public static int factorial(int 3)
{
  int fact;
  if (n > 1)
    fact = factorial(2) * 3;
  else
    fact = 1;
  return fact;
}
```

```
public static int factorial(int 2)
{
  int fact;
  if (n > 1)
    fact = factorial(1) * 2;
  else
    fact = 1;
  return fact;
}
```

```
public static int factorial(int 3)
{
  int fact;
  if (n > 1)
    fact = factorial(2) * 3;
  else
    fact = 1;
  return fact;
}
```

```
public static int factorial(int 2)
{
  int fact;
  if (n > 1)
    fact = factorial(1) * 2;
  else
    fact = 1;
  return fact;
}
```

```
public static int factorial(int 1)
{
  int fact;
  if (n > 1)
    fact = factorial(n - 1) * n;
  else
    fact = 1;
  return fact;
}
```

```
public static int factorial(int 3)
{
  int fact;
  if (n > 1)
    fact = factorial(2) * 3;
  else
    fact = 1;
  return fact;
}
```

```
public static int factorial(int 2)
{
  int fact;
  if (n > 1)
    fact = factorial(1) * 2;
  else
    fact = 1;
  return fact;
}
```

```
public static int factorial(int 1)
{
  int fact;
  if (n > 1)
    fact = factorial(n - 1) * n;
  else
    fact = 1;
  return 1;
}
```

```
public static int factorial(int 3)
{
  int fact;
  if (n > 1)
    fact = factorial(2) * 3;
  else
    fact = 1;
  return fact;
}
```

```
public static int factorial(int 2)
{
  int fact;
  if (n > 1)
    fact = 1 * 2;
  else
    fact = 1;
  return fact;
}
```

```
public static int factorial(int 1)
{
  int fact;
  if (n > 1)
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  return 1;
}
```

```
public static int factorial(int 3)
{
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     fact = factorial(2) * 3;
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     fact = 1;
   return fact;
}
```

```
public static int factorial(int 2)
{
   int fact;
   if (n > 1)
     fact = 1 * 2;
   else
     fact = 1;
   return 2;
}
```

```
public static int factorial(int 3)
{
   int fact;
   if (n > 1)
      fact = 2 * 3;
   else
      fact = 1;
   return fact;
}
```

```
public static int factorial(int 2)
{
   int fact;
   if (n > 1)
      fact = 1 * 2;
   else
      fact = 1;
   return 2;
}
```

```
public static int factorial(int 3)
{
  int fact;
  if (n > 1)
    fact = 2 * 3;
  else
    fact = 1;
  return 6;
}
```

Execution Trace (decomposition)

```
public static int factorial(int n)
{
  int fact;
  if (n > 1) // recursive case (decomposition)
    fact = factorial(n – 1) * n; (composition)
  else // base case
    fact = 1;
  return fact;
}
```

factorial(4)

factorial(3)   4

Execution Trace (decomposition)

```
public static int factorial(int n)
{
  int fact;
  if (n > 1) // recursive case (decomposition)
    fact = factorial(n - 1) * n; (composition)
  else // base case
    fact = 1;
  return fact;
}
```

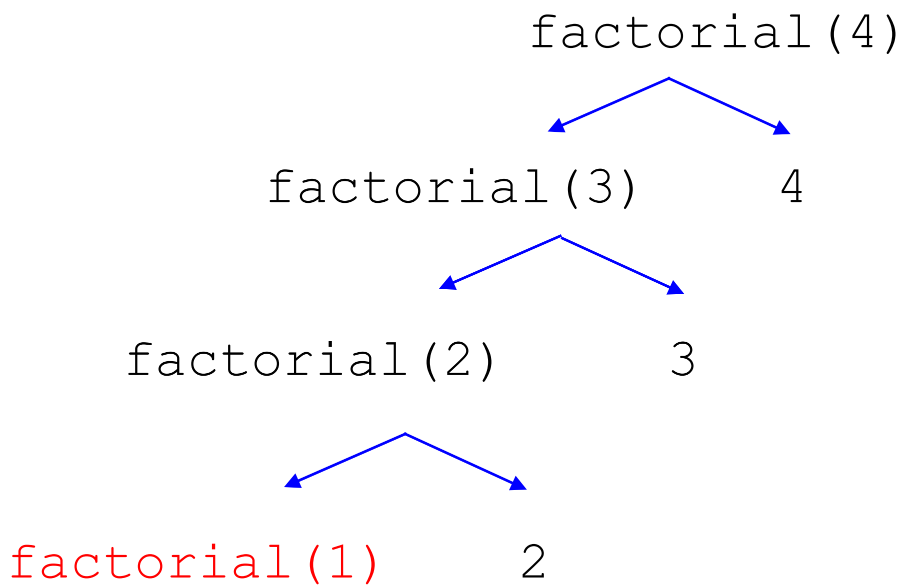
factorial(4)

factorial(3)   4

factorial(2)   3

Execution Trace (decomposition)

```java
public static int factorial(int n)
{
  int fact;
  if (n > 1) // recursive case (decomposition)
    fact = factorial(n - 1) * n; (composition)
  else // base case
    fact = 1;
  return fact;
}
```



Execution Trace (composition)

```
public static int factorial(int n)
{
  int fact;
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  else // base case
    fact = 1;
  return fact;
}
```
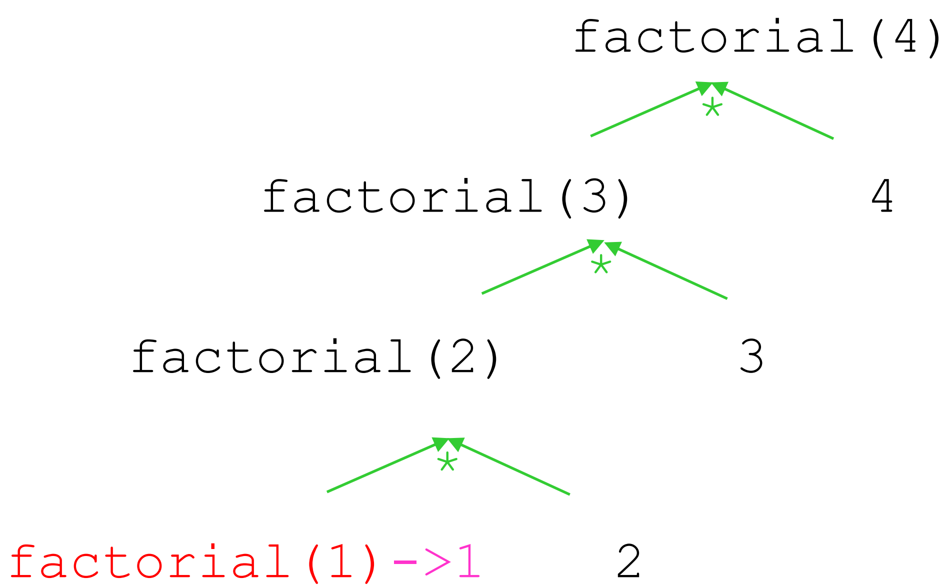



Execution Trace (composition)

```
public static int factorial(int n)
{
  int fact;
  if (n > 1) // recursive case (decomposition)
    fact = factorial(n - 1) * n; (composition)
  else // base case
    fact = 1;
  return fact;
}
```

```
              factorial(4)
                   *
       factorial(3)          4
             *
factorial(2)->2      3
```

Execution Trace (composition)

```
public static int factorial(int n)
{
  int fact;
  if (n > 1) // recursive case (decomposition)
    fact = factorial(n - 1) * n; (composition)
  else // base case
    fact = 1;
  return fact;
}
```

```
           factorial(4)
                *
factorial(3)->6      4
```

Execution Trace (composition)

```
public static int factorial(int n)
{
  int fact;
  if (n > 1) // recursive case (decomposition)
    fact = factorial(n - 1) * n; (composition)
  else // base case
    fact = 1;
  return fact;
}
```

```
factorial(4)->24
```

Break

Fibonacci Numbers

- The *N*th Fibonacci number is the sum of the previous two Fibonacci numbers
- 0, 1, 1, 2, 3, 5, 8, 13, …
- Recursive Design:
  - Decomposition & Composition
    - fibonacci(n) = fibonacci(n-1) + fibonacci(n-2)
  - Base case:
    - fibonacci(1) = 0
    - fibonacci(2) = 1

`fibonacci` Method

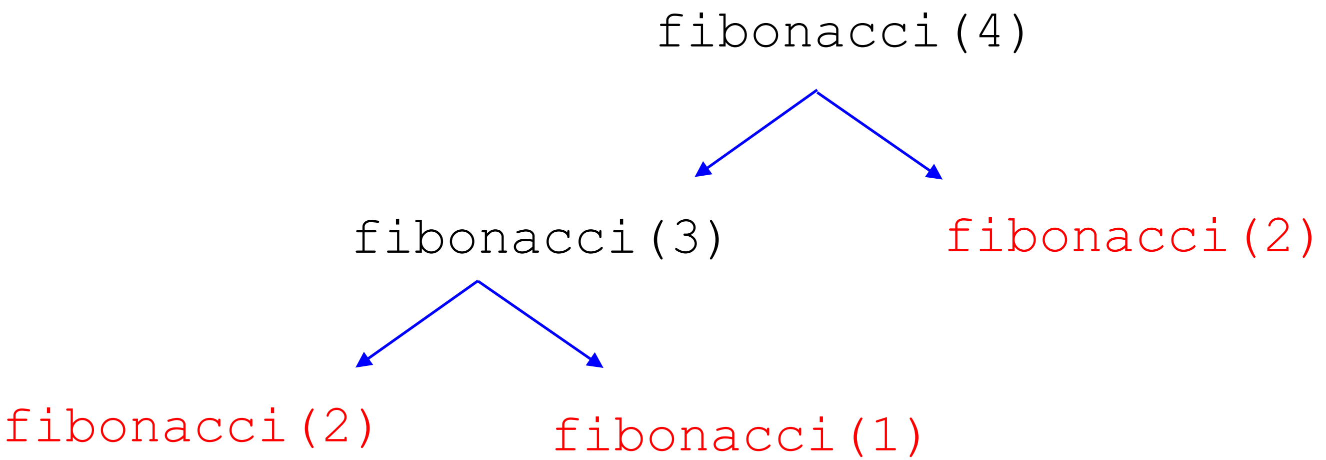
```
public static int fibonacci(int n)
{
   int fib;
   if (n > 2)
     fib = fibonacci(n-1) + fibonacci(n-2);
   else if (n == 2)
     fib = 1;
   else
     fib = 0;
   return fib;
}
```

Execution Trace (decomposition)

fibonacci(4)

fibonacci(3) fibonacci(2)

Execution Trace (decomposition)



Execution Trace (composition)

`fibonacci(4)`

`fibonacci(3)` `fibonacci(2)`

`fibonacci(2)->1` `fibonacci(1)->0`

Execution Trace (composition)

```
          fibonacci(4)
               +
fibonacci(3)->1    fibonacci(2)->1
```

Execution Trace (composition)

```
fibonacci(4)->2
```

Break

Remember:
Key to Successful Recursion

- if-else statement (or some other branching statement)
- Some branches: recursive call
  - "smaller" arguments or solve "smaller" versions of the same task (*decomposition*)
  - Combine the results (*composition*) [if necessary]
- Other branches: no recursive calls
  - stopping cases or base cases

Template

```
… method(…)
{
    if ( … )// base case
    {
    }
    else // decomposition & composition
    {
    }
    return … ; // if not void method
}
```

Break

What Happens Here?

```
public static int factorial(int n)
{
  int fact=1;

  if (n > 1)
    fact = factorial(n) * n;

  return fact;
}
```

What Happens Here?

```
public static int factorial(int n)
{
  return factorial(n - 1) * n;
}
```

Warning: Infinite Recursion May Cause a Stack Overflow Error

- Infinite Recursion
  - Problem not getting smaller (no/bad decomposition)
  - Base case exists, but not reachable (bad base case and/or decomposition)
  - No base case
- *Stack:* keeps track of recursive calls by JVM (OS)
  - Method begins: add data onto the stack
  - Method ends: remove data from the stack
- Recursion never stops; stack eventually runs out of space
  - **Stack overflow error**

Break

Summary

- *Recursive call*: a method that calls itself
- Powerful for algorithm design at times
- Recursive algorithm design:
    - Decomposition (smaller identical problems)
    - Composition (combine results)
    - Base case(s) (smallest problem, no recursive calls)
- Implementation
  - Conditional (e.g. if) statements to separate different cases
  - Avoid infinite recursion
    - Problem is getting smaller (decomposition)
    - Base case exists and reachable
  - Composition could be tricky

Module

Algorithm Concepts and Problems (Easy/Medium/Hard)

Topic

DSA by Prashant sir

Subtopic

Tree and Binary Tree

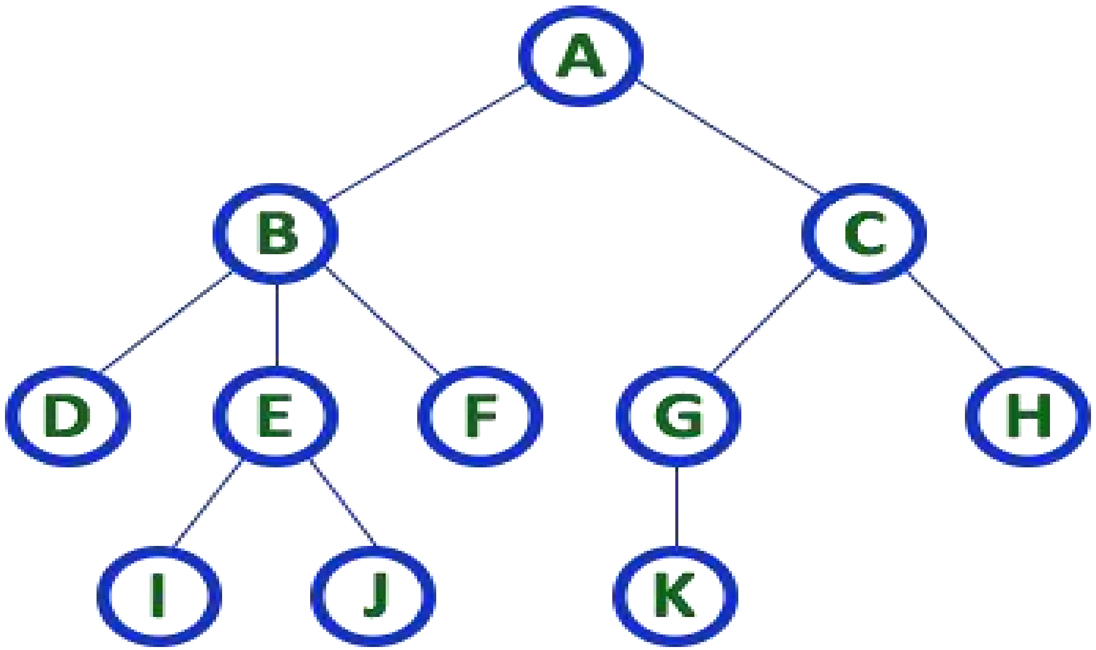
Lesson

Tree and Binary Tree

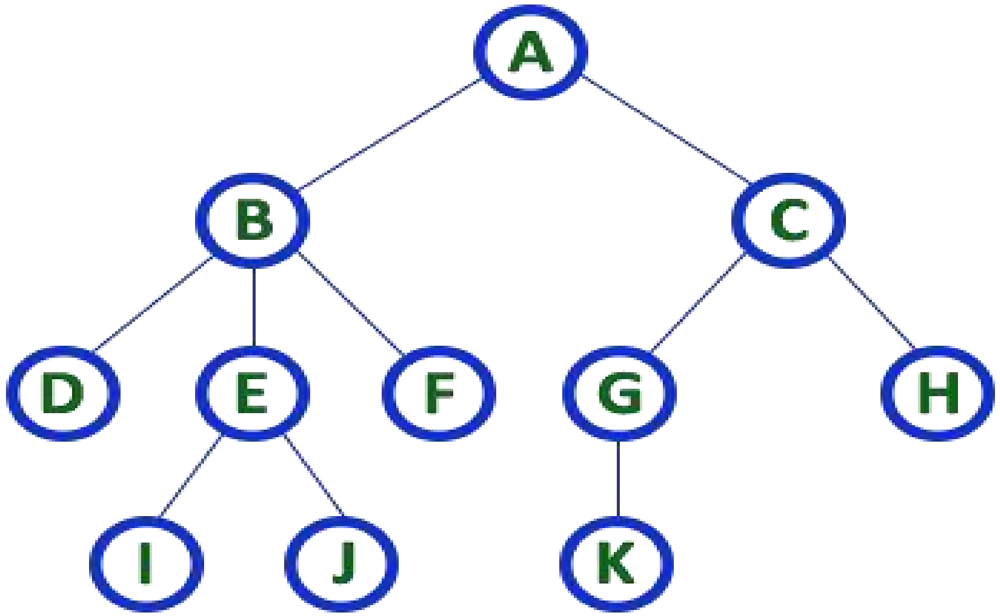
TREE

Tree

- The tree is one of the most powerful, flexible, versatile and nonlinear advanced data structures, it represents hierarchical relationship existing between several data items. it is used in wide range of applications.



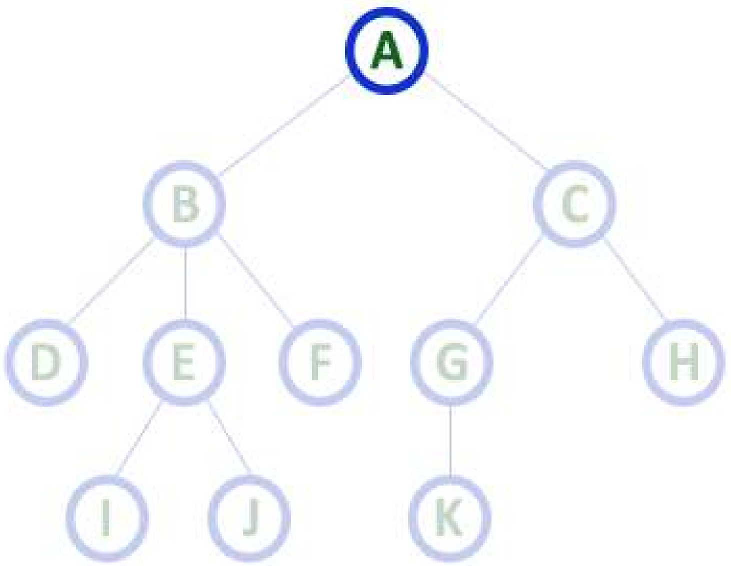
- A tree is a finite set of one or more data items(nodes) such that
  - There is a special data item called root of the tree
  - And its remaining data items are partitioned into number of mutually exclusive (disjoint) subsets, each of which is itself a tree and they are called subtree. i.e. Every node (exclude a root) is connected by a directed edge *from* exactly one other node; A direction is: *parent -> children*



BREAK

Root

- The first/Top most node is called as Root Node. We always have exactly one root node in every tree. We can say that root node is the origin of tree data structure.

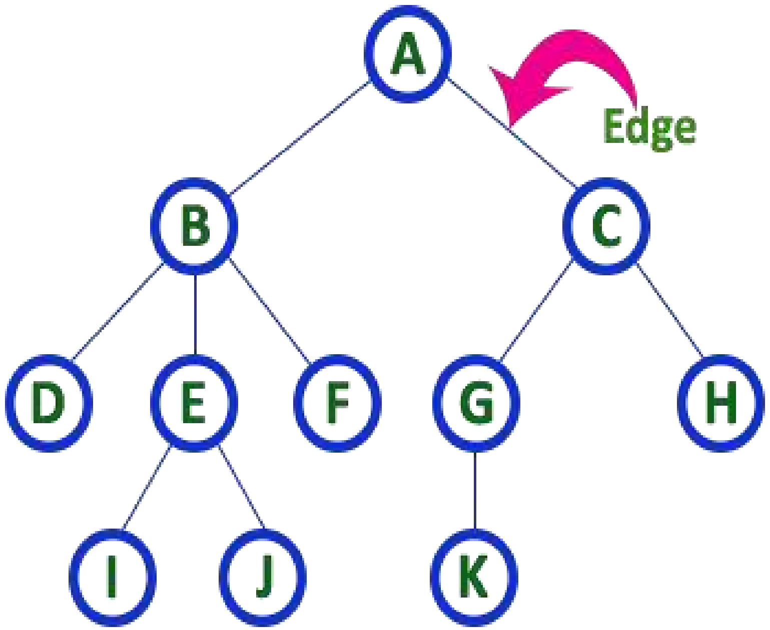


Here 'A' is the 'root' node

- In any tree the first node is called as ROOT node

Edge

- In a tree data structure, the connecting link between any two nodes is called as EDGE. In a tree with 'N' number of nodes there will be exactly of 'N-1' number of edges.

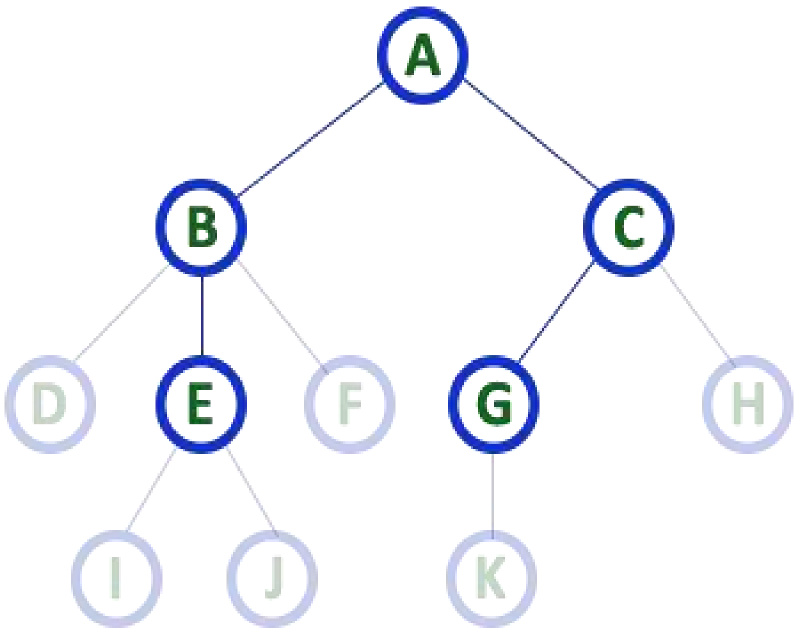


- In any tree, ‘Edge’ is a connecting link between two nodes.

Parent

- In a tree data structure, the node which is predecessor of any node is called as PARENT NODE.

- In simple words, the node which has branch from it to any other node is called as parent node. Parent node can also be defined as "The node which has child / children".



Here A, B, C, E & G are Parent nodes

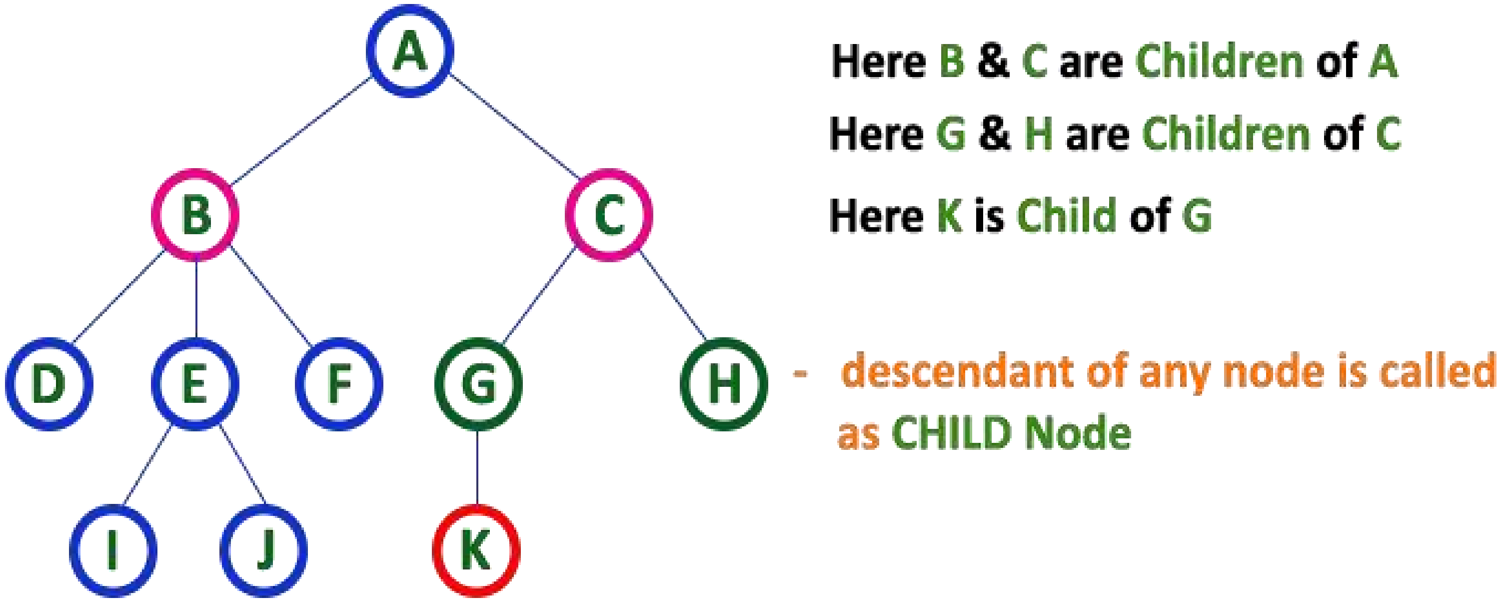
- In any tree the node which has child / children is called 'Parent'

- A node which is predecessor of any other node is called 'Parent'

Child

- In a tree data structure, the node which is descendant of any node is called as CHILD Node.

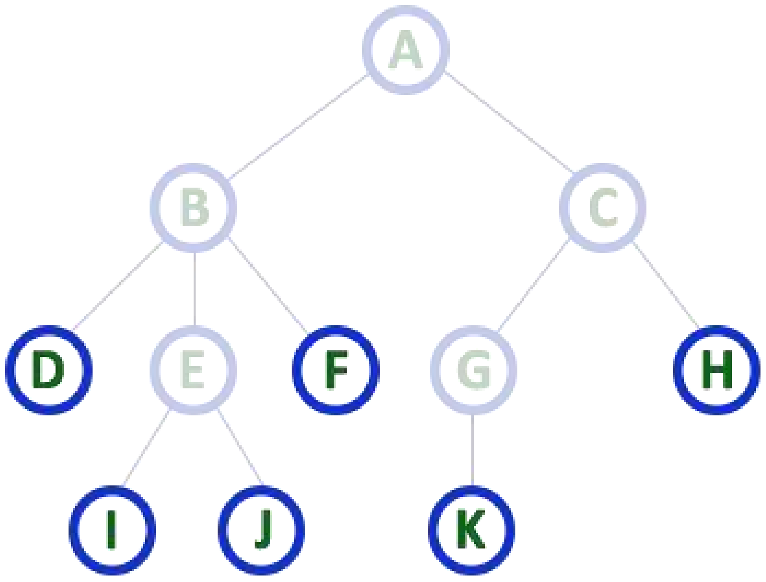
- In simple words, the node which has a link from its parent node is called as child node. In a tree, any parent node can have any number of child nodes. In a tree, all the nodes except root are child nodes.



Leaf / External

- In a tree data structure, the node which does not have a child is called as LEAF Node. In simple words, a leaf is a node with no child.

- In a tree data structure, the leaf nodes are also called as External Nodes. External node is also a node with no child. In a tree, leaf node is also called as 'Terminal' node.



Here D, I, J, F, K & H are Leaf nodes

- In any tree the node which does not have children is called 'Leaf'

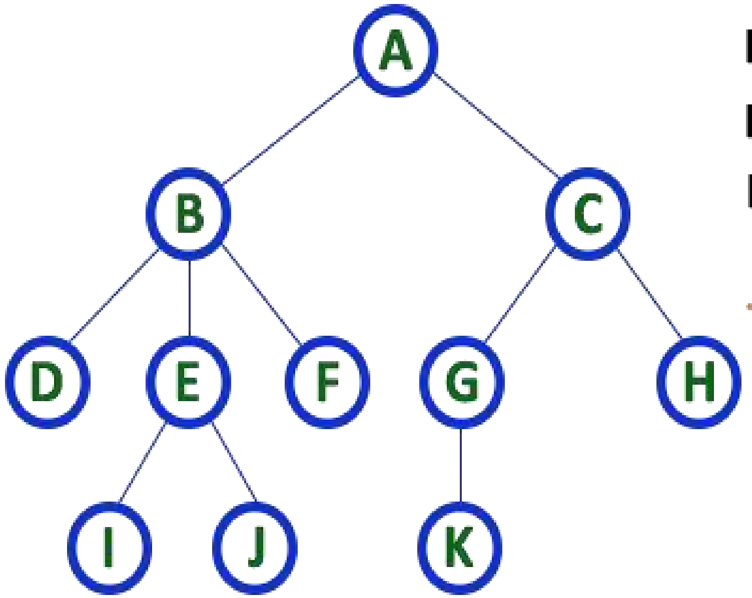
- A node without successors is called a 'leaf' node

BREAK

Degree

- In a tree data structure, the total number of children of a node is called as DEGREE of that Node. In simple words, the Degree of a node is total number of children it has.

- The highest degree allowed of a node in a tree is called as 'Degree of Tree'



Here Degree of B is 3

Here Degree of A is 2

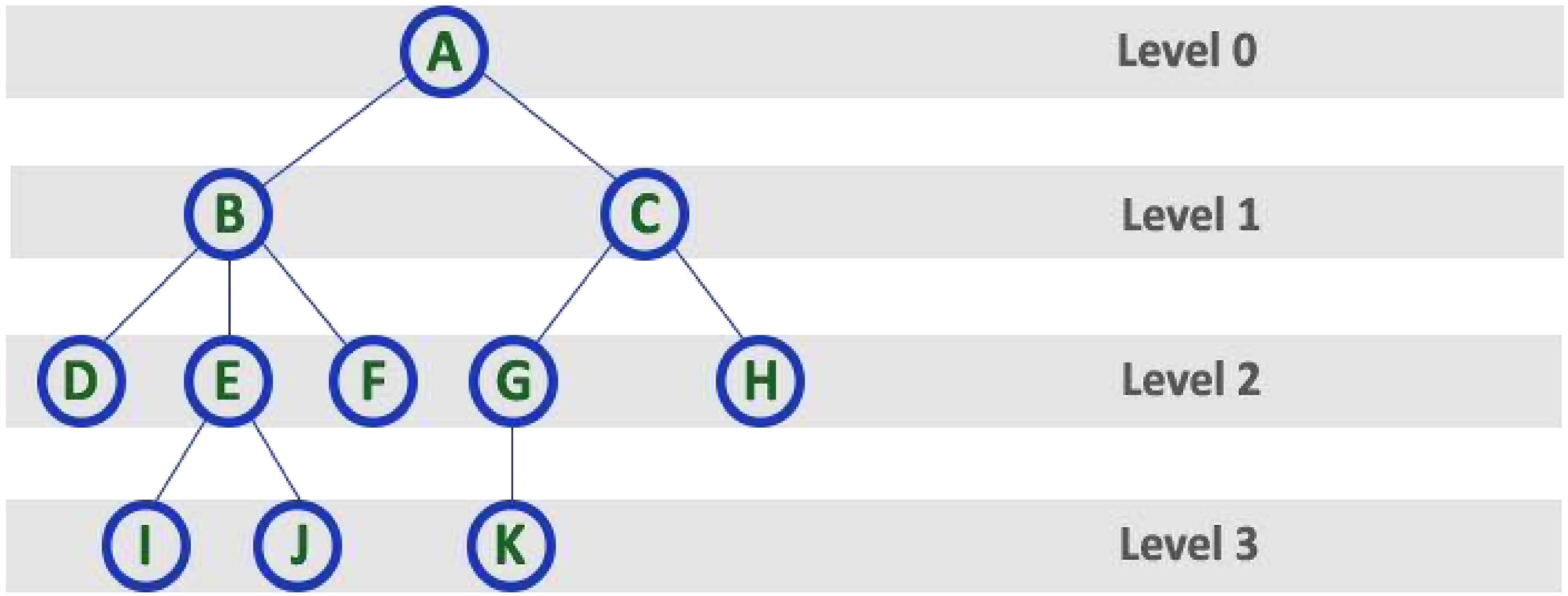
Here Degree of F is 0

- In any tree, 'Degree' a node is total number of children it has.

Level / Depth / Height

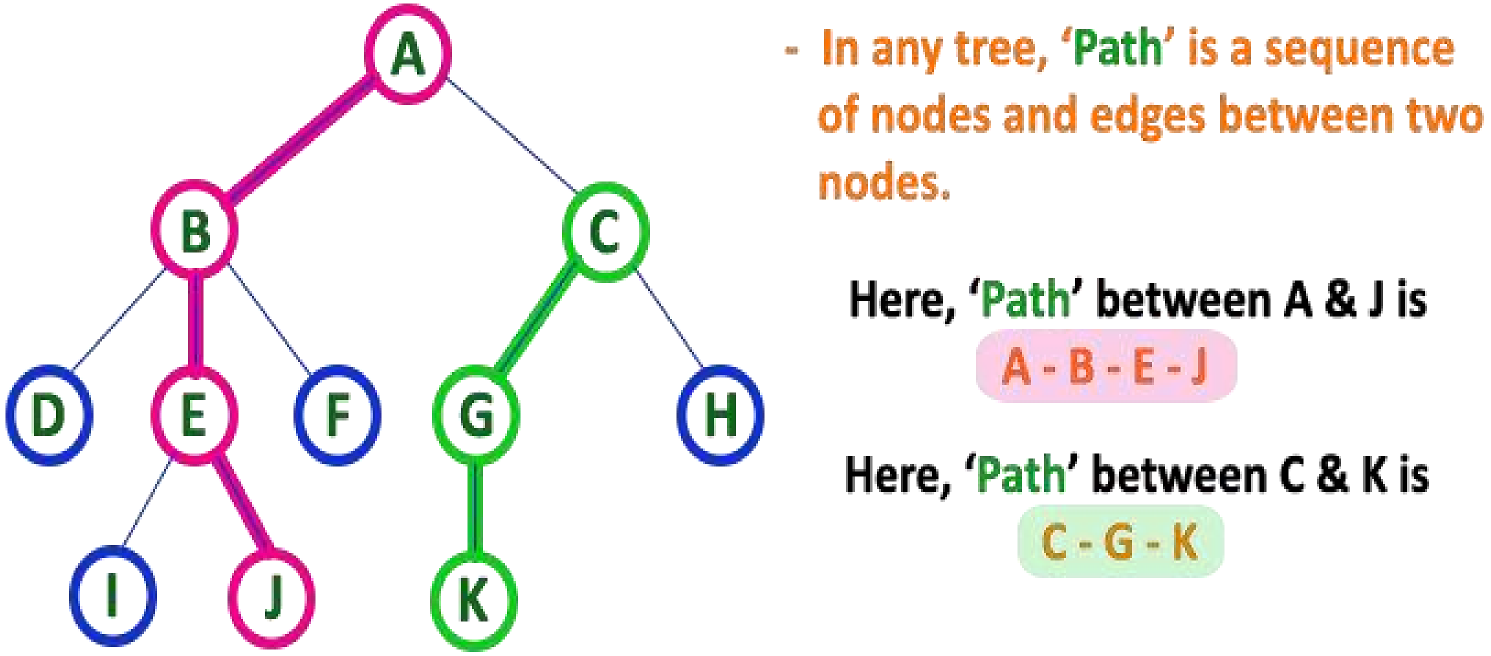
- In a tree data structure, the root node is said to be at Level 0 and the children of root node are at Level 1 and the children of the nodes which are at Level 1 will be at Level 2 and so on...

- In simple words, in a tree each step from top to bottom is called as a Level and the Level count starts with '0' and incremented by one at each level (Step).



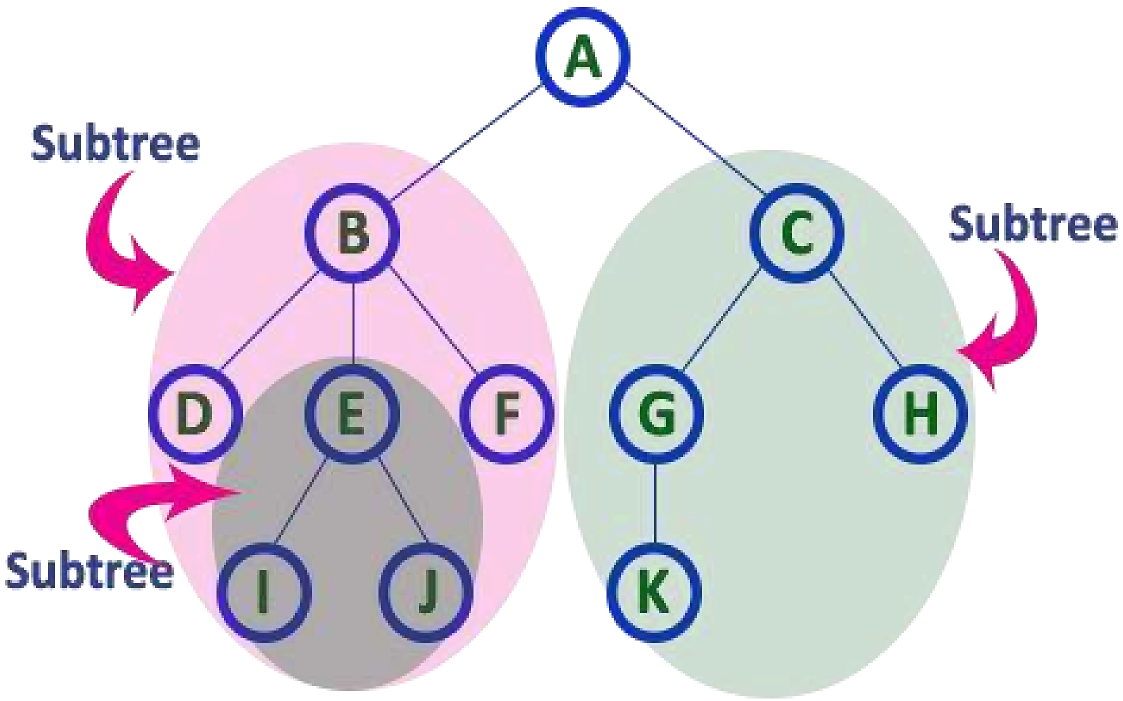
Path

- In a tree data structure, the sequence of Nodes and Edges from one node to another node is called as PATH between that two Nodes. Length of a Path is total number of nodes in that path. In below example the path A - B - E - J has length 4.



Sub Tree

- In a tree data structure, each child from a node forms a subtree recursively. Every child node will form a subtree on its parent node.



Break

Binary tree

- A binary tree T is defined as a finite set of elements called nodes such that,
  - T is empty (null tree)
  - T contain a distinguished node R, called the root of T, and the remaining nodes of T form an ordered pair of disjoint binary tree T_1 and T_2
- Direct: - A tree T in which any node can have maximum two children (left and right)
- Representation of tree in memory
  - Sequential representation – using an array info and left child and right child
  - Linked Representation – using self-referential structure node

```
struct node {
   int data;
   struct node* left;
   struct node* right;
}
```

Traversal of binary tree

- The process of visiting (checking and/or updating) each node in a tree data structure, exactly once in called tree traversal. Such traversals are classified by the order in which the nodes are visited.

- Unlike linked lists, one-dimensional arrays and other linear data structures, which are canonically traversed in linear order, trees may be traversed in multiple ways.

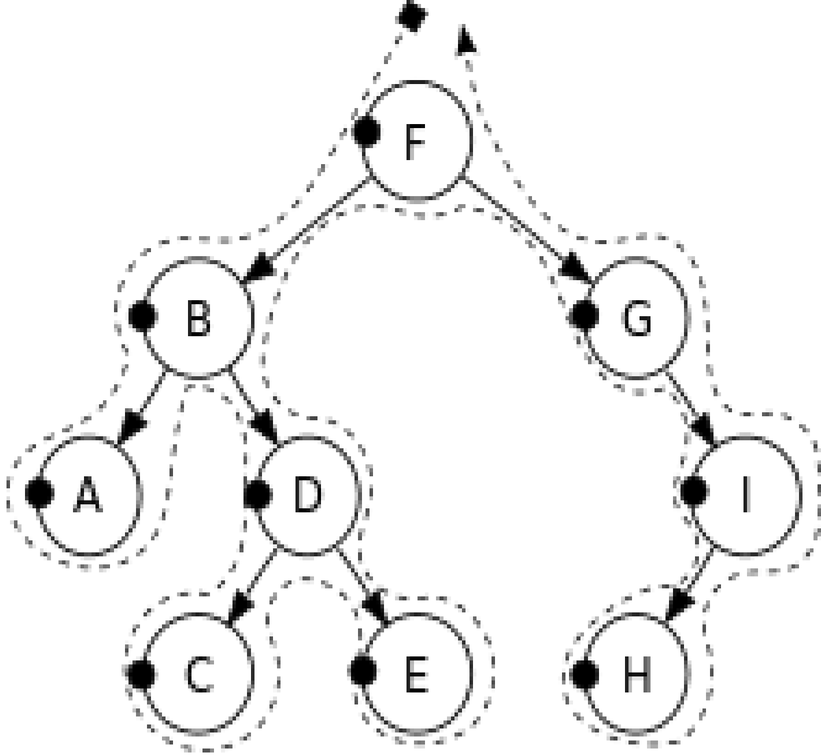
- They may be traversed in depth-first or breadth-first order. There are three common ways to traverse them in depth-first order: in-order, pre-order and post-order. Beyond these basic traversals, various more complex or hybrid schemes are possible, such as depth-limited searches like iterative deepening depth-first search.

- Some applications do not require that the nodes be visited in any particular order as long as each node is visited precisely once. For other applications, nodes must be visited in an order that preserves some relationship.

- These steps can be done *in any order*. If (L) is done before (R), the process is called left-to-right traversal, otherwise it is called right-to-left traversal. The following methods show left-to-right traversal:

Break

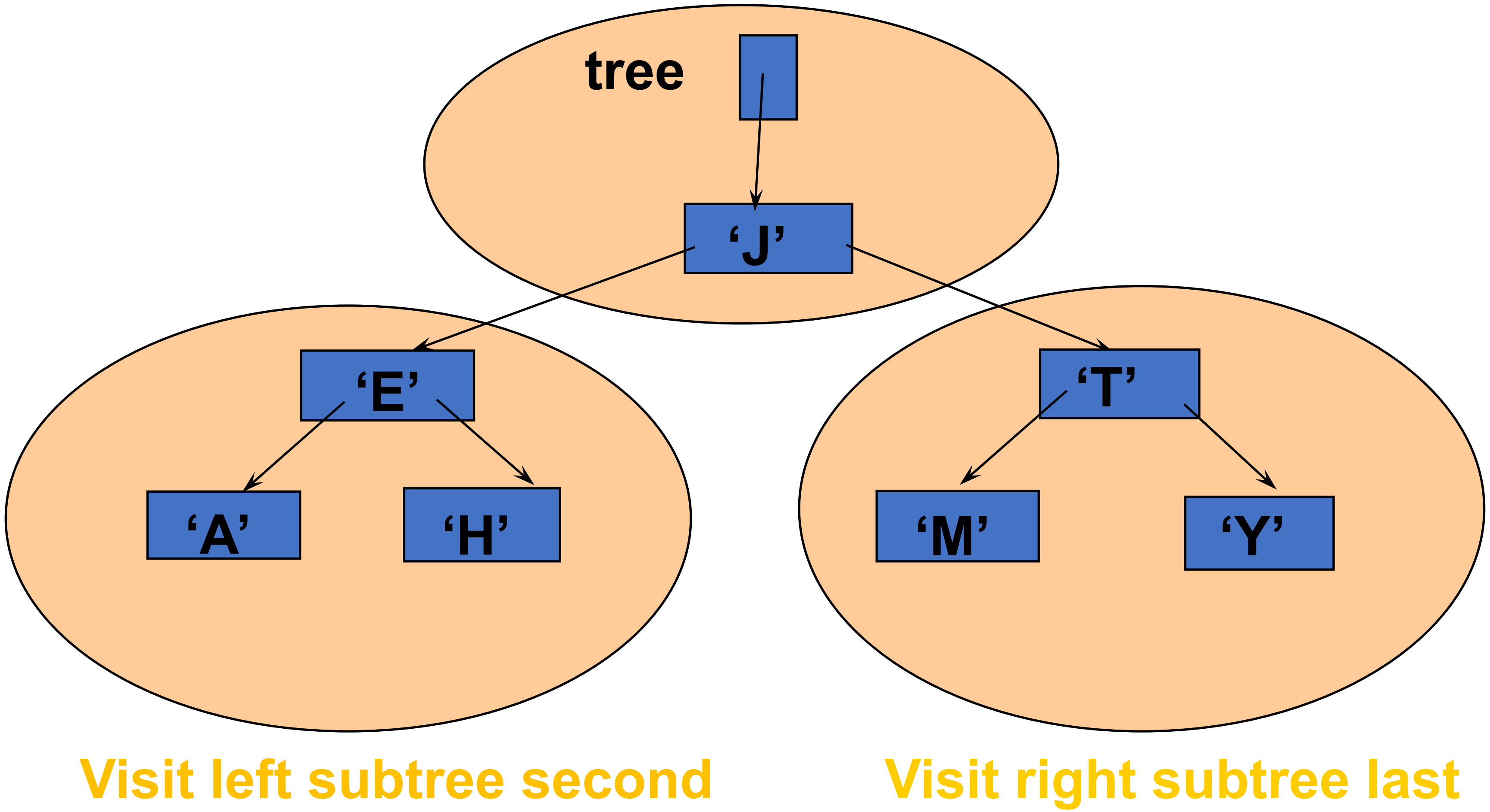
Pre-order (Root L R)



Pre-order: F, B, A, D, C, E, G, I, H.

- Check if the current node is empty or null.
- Display the data part of the root (or current node).
- Traverse the left subtree by recursively calling the pre-order function.
- Traverse the right subtree by recursively calling the pre-order function.

Preorder Traversal: J E A H T M Y

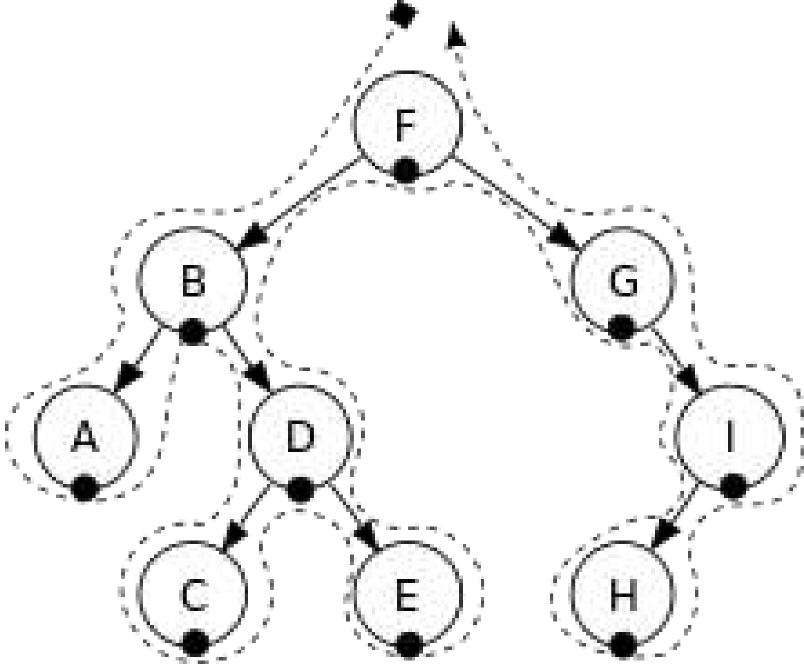


Break

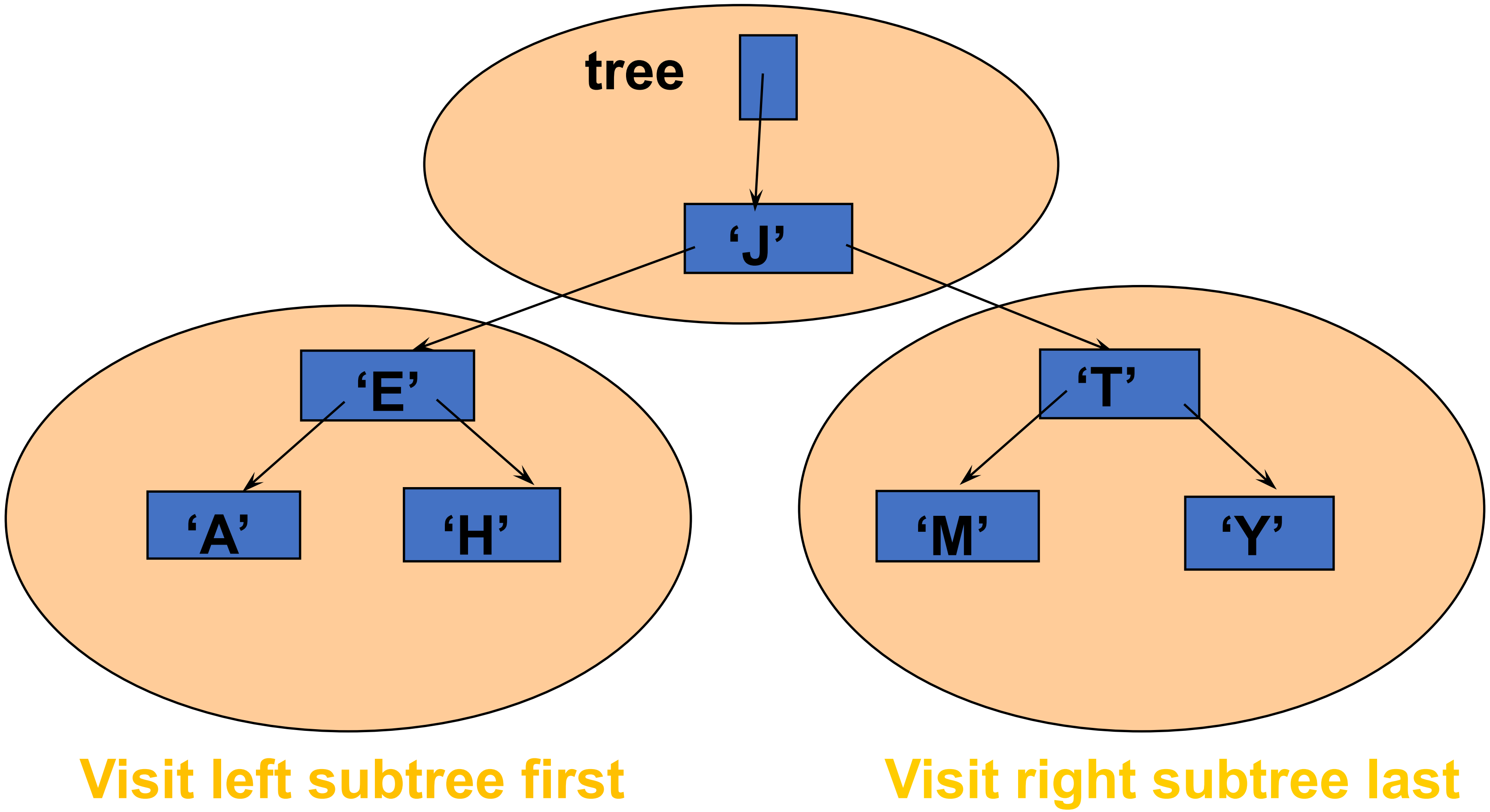
In-order (L root R)

In-order: A, B, C, D, E, F, G, H, I.

- Check if the current node is empty or null.
- Traverse the left subtree by recursively calling the in-order function.
- Display the data part of the root (or current node).
- Traverse the right subtree by recursively calling the in-order function.
- In a binary search tree, in-order traversal retrieves data in sorted order.

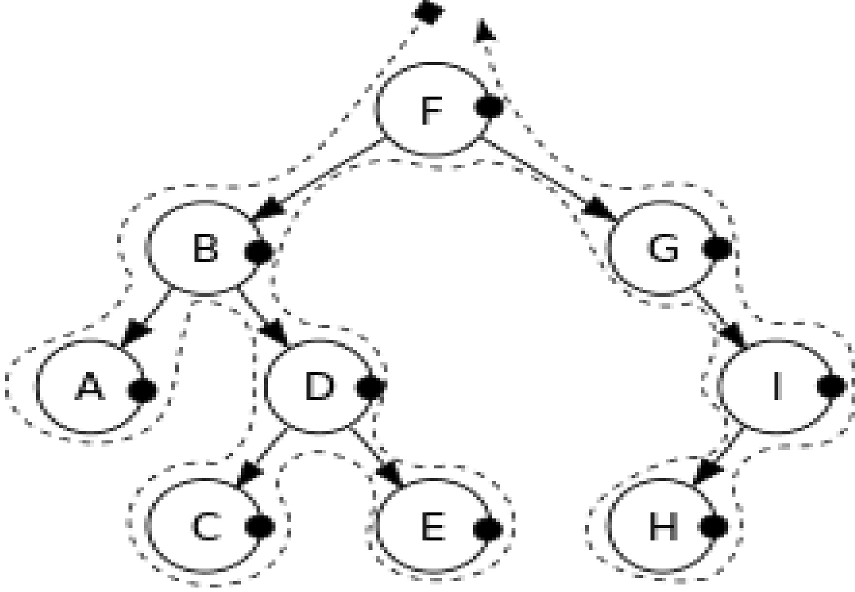


Inorder Traversal: A E H J M T Y



Break

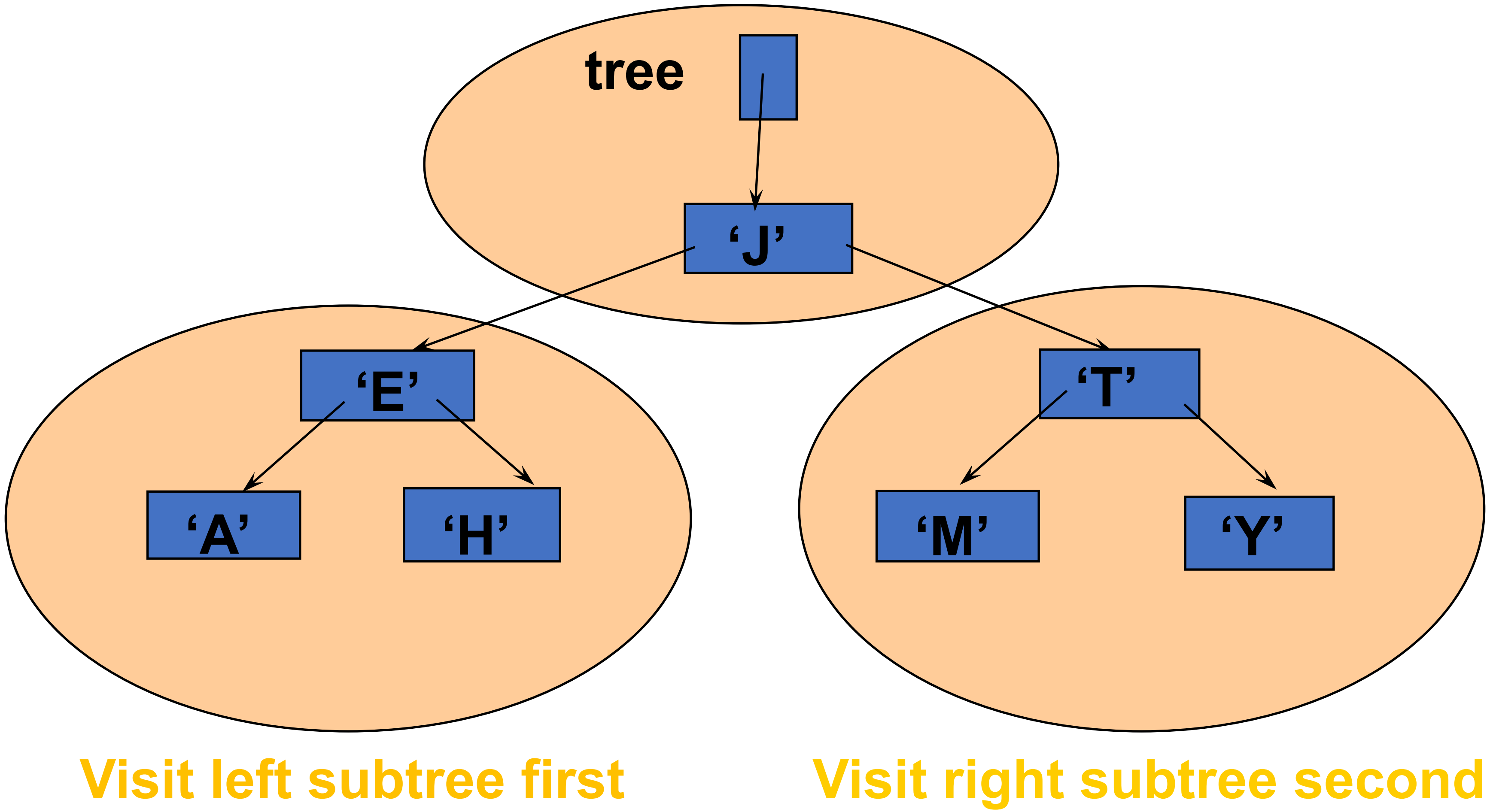
Post-order (L R Root)



Post-order: A, C, E, D, B, H, I, G, F.

- Check if the current node is empty or null.
- Traverse the left subtree by recursively calling the post-order function.
- Traverse the right subtree by recursively calling the post-order function.
- Display the data part of the root (or current node).

Postorder Traversal: A H E M Y T J



Module

Algorithm Concepts and Problems (Easy/Medium/Hard)

Topic

DSA by Prashant sir

Subtopic

Stack

Lesson

STACK

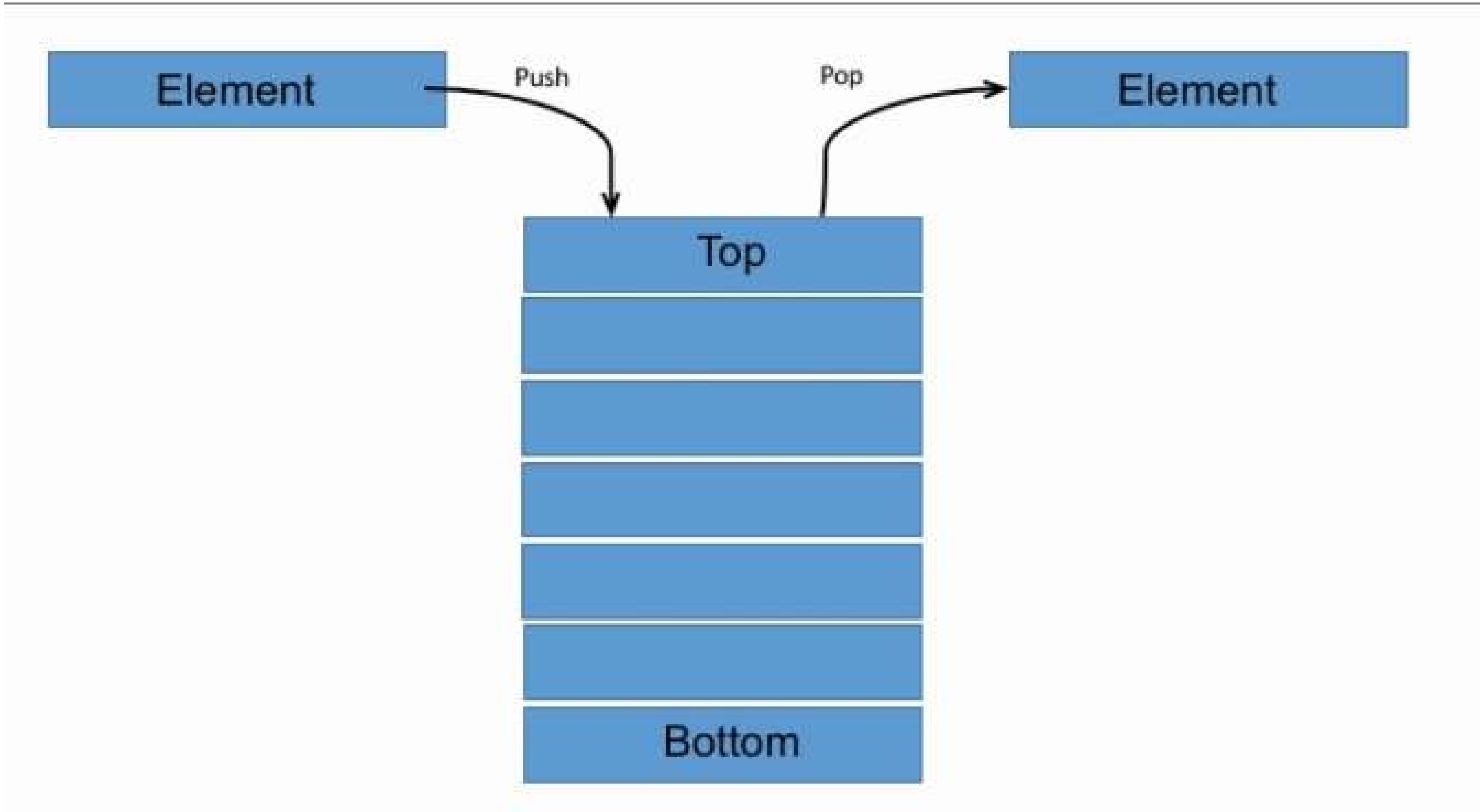
Stack

Basics of Stack

Basics of Stack

STACK

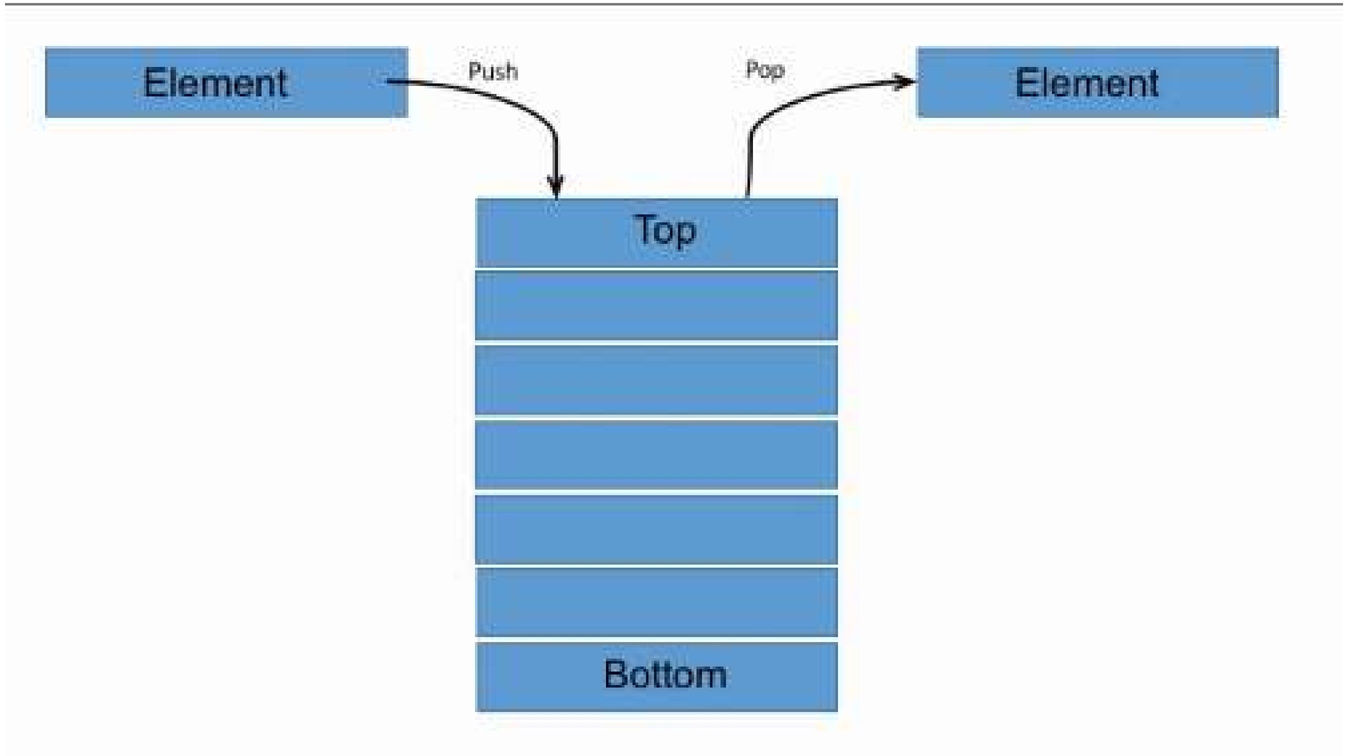
- A stack is a non-primitive linear data structure. it is an ordered list in which addition of a new data item and deletion of already existing data item is done from only one end known as top of stack (TOS).



- The element which is added in last will be first to be removed and the element which is inserted first will be removed in last.

- That is why it is called last in first out (LIFO) or first in last out (FILO) type of list.

- Most frequently accessible element in the stack is the top most element, whereas the least accessible element is the bottom of the stack.

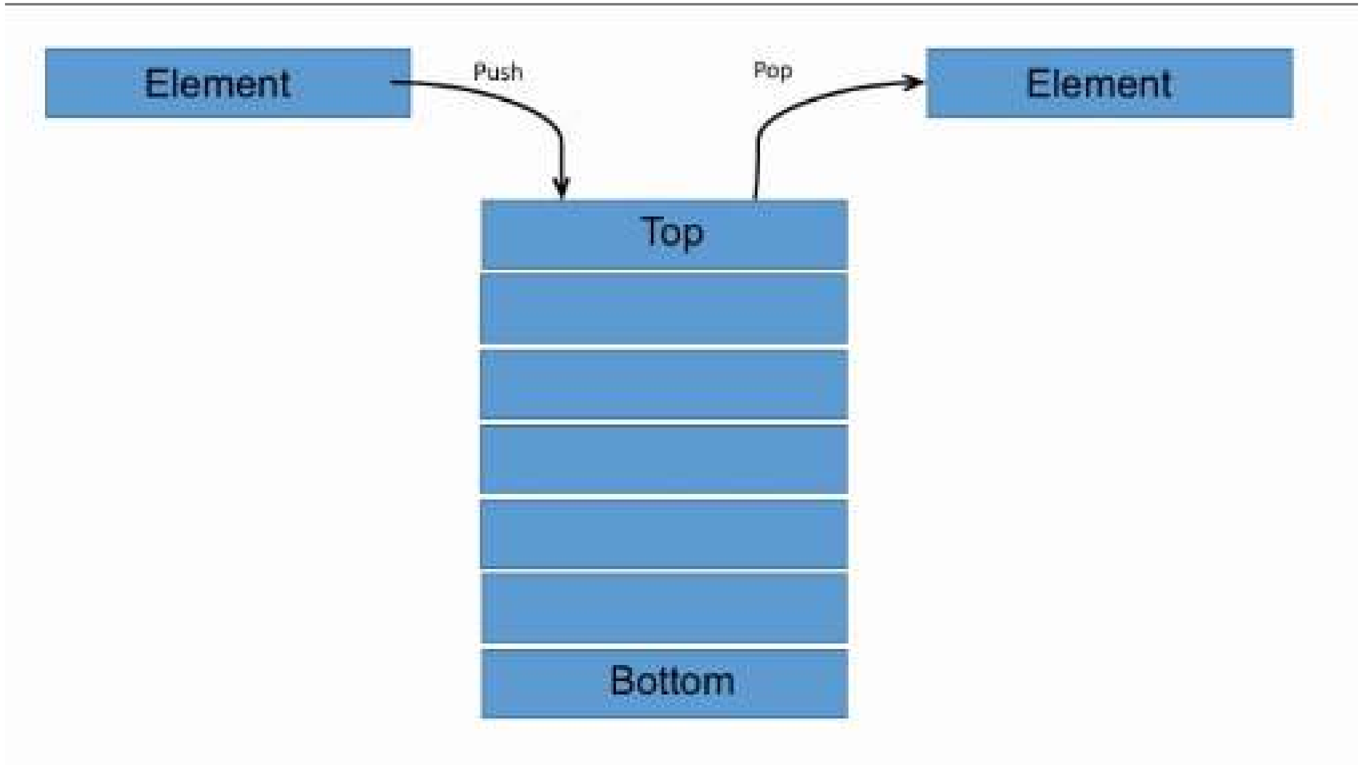


Break

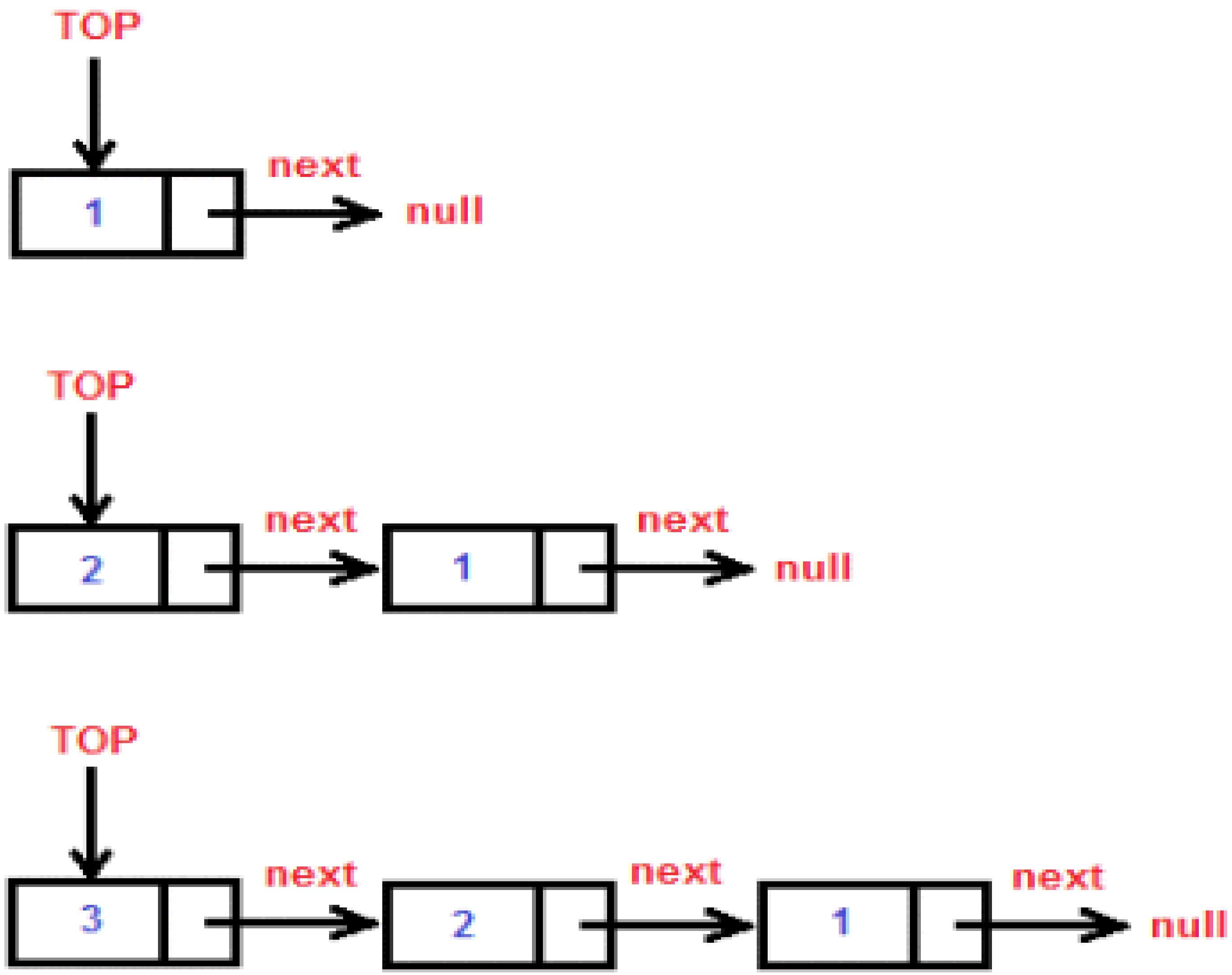
Stack Implementation

Stack is generally implemented in two ways.

- **Static Implementation**: - Here array is used to create stack. it is a simple technique but is not a flexible way of creation, as the size of stack has to be declared during program design, after that size implementation is not efficient with respect to memory utilization.



- **Dynamic implementation**: - It is also called linked list representation and uses pointer to implement the stack type of data structure.

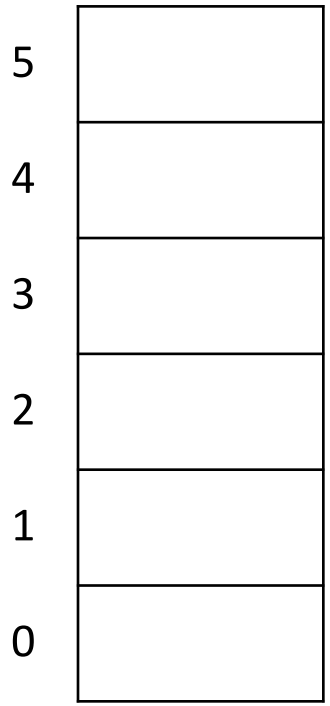


Break

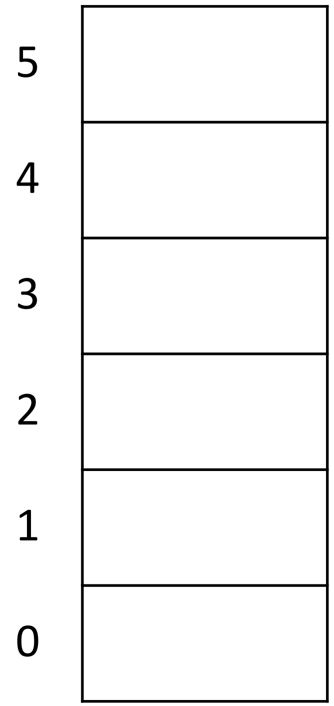
Basics operations on stack

- push - place an item on the stack
- peek - Look at the item on top of the stack, but do not remove it
- pop - Look at the item on top of the stack and remove it

<u>Push operation</u>: - The process of adding new element to the top of stack is called push operation. the new element will be inserted at the top after every push operation the top is incremented by one. in the case the array is full and no new element can be accommodated it is called over-flow condition.



Pop: - The process of deleting an element. from the top of stack is called POP operation, after every POP operation the stack is decremented by one if there is no element in the stack and the POP operation is requested then this will result into a stack underflow condition.



Break

Reversing a Word

- We can use a stack to reverse the letters in a word.
- How?

Reversing a Word

- Read each letter in the word and push it onto the stack
- When you reach the end of the word, pop the letters off the stack and print them out.

Break

Module

Algorithm Concepts and Problems (Easy/Medium/Hard)

Topic

DSA by Prashant sir

Subtopic

Queues

Lesson

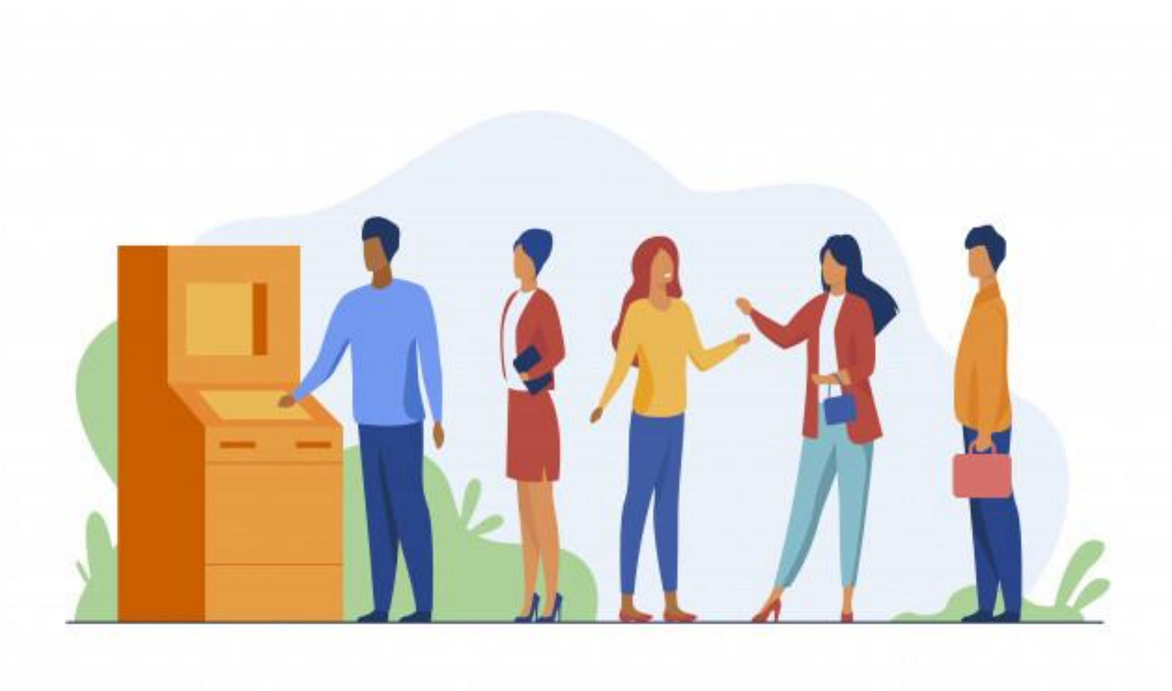
QUEUE

Queue

Queue

- A queue is a linear list of elements in which deletions can take place only at one end called the front, and insertions can take place only at the end called rear.

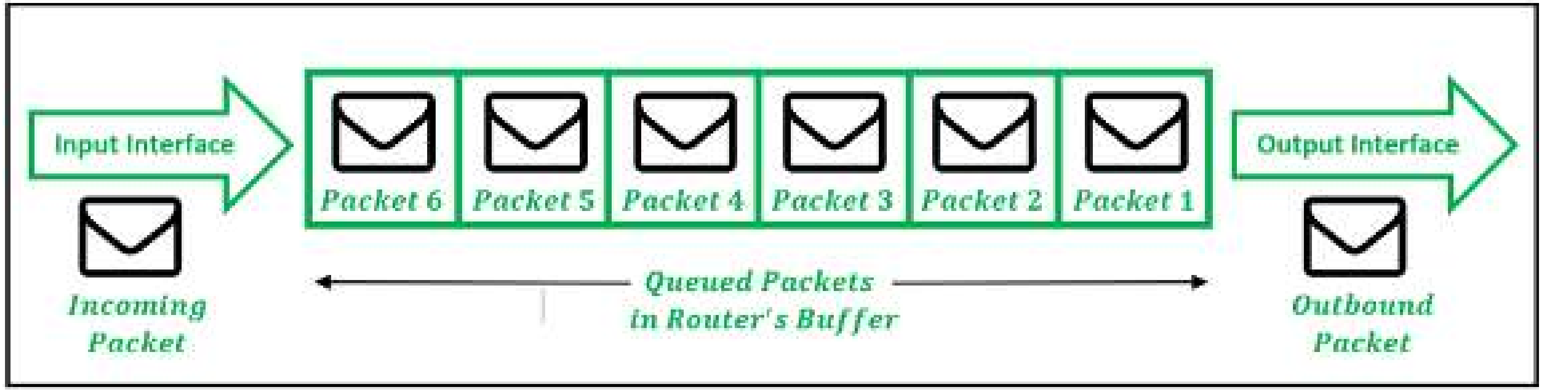
- Queue is a first in first out types of data structure(FIFO), the terms ‘Front’ and ‘Rear’ are used in describing a linear list only when it is implemented as a queue.

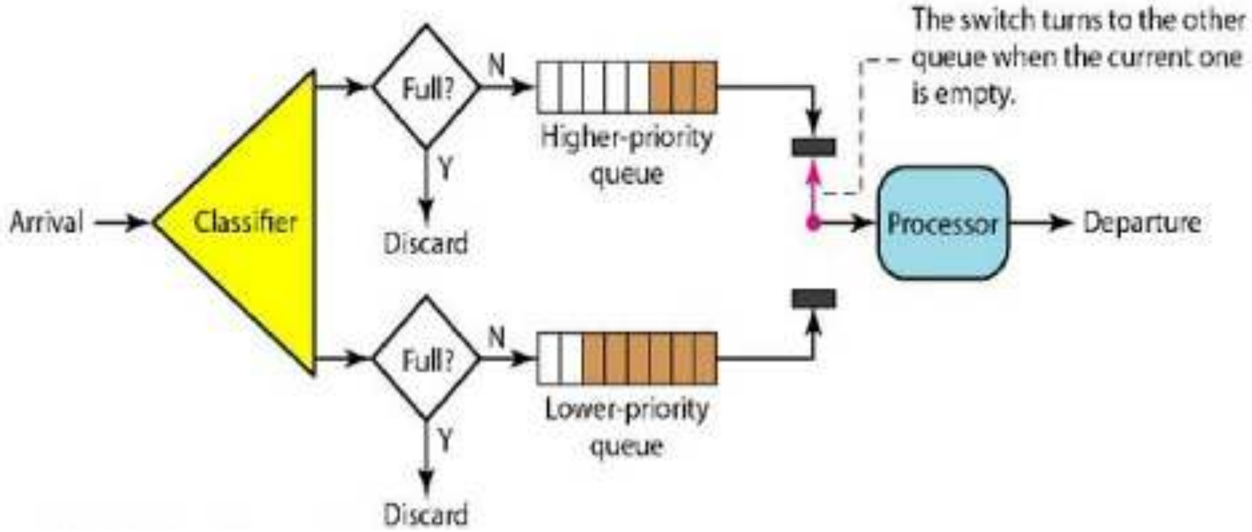





- In computer science queue are used in multiple places e.g. in time sharing system program with the same priority from a queue waiting to be executed.

- A queue is a non-primitive linear data structure. it is homogeneous collection of elements.





Break

Queues

- What can we do with a queue?
  - Enqueue - Add an item to the queue
  - Dequeue - Remove an item from the queue
- These ops are also called insert and getFront in order to simplify things.

Queues

- A queue is called a FIFO (First in-First out) data structure.
- What are some applications of queues?
  - Round-robin scheduling in processors
  - Input/Output processing
  - Queueing of packets for delivery in networks

Break

Representation of Queues

- Mostly each of our queues will be maintained by a linear array QUEUE and two pointer variables: FRONT containing the location of the Front element of the queue and REAR, containing the location of the rear element of the queue.

0	1	2	3	4	5	6	7

- The condition FRONT = null, will indicate that the queue is empty. Whenever an element is deleted from the queue, the value of FRONT is increased by 1
  - Front = Front + 1

0	1	2	3	4	5	6	7

- Whenever an element is added to the queue, the value of REAR is increased by 1
  - REAR = REAR + 1

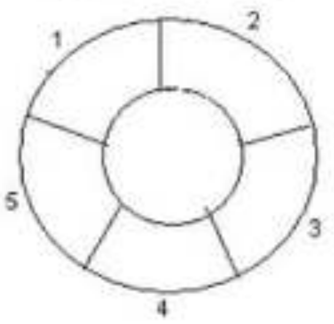
0	1	2	3	4	5	6	7

- This means that after N insertion the rear element of the queue will occupy QUEUE[N] or queue will occupy the last part of the array. This may occur even though the queue itself may not contain many elements.

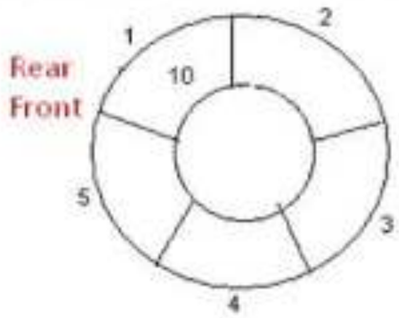
- Total number of elements in a queue
  - Rear – Front + 1

Break

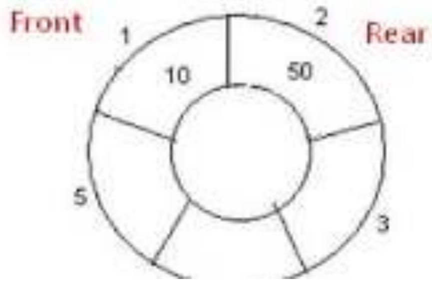
Circular Queue

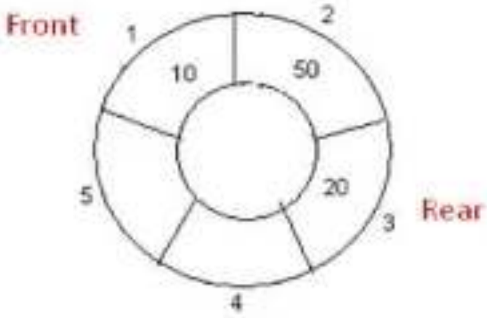


2. Insert 10, Rear = 1, Front = 1.



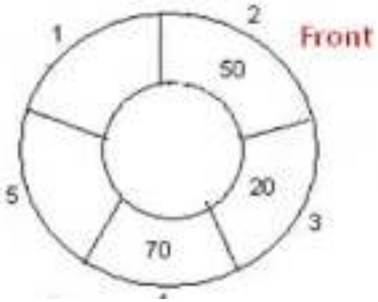
3. Insert 50, Rear = 2, Front = 1.





5. Insert 70, Rear = 4, Front = 1.

6. Delete front, Rear = 4, Front = 2.



Priority Queue

- A priority queue is a collection of elements such that each element has been assigned a priority and such that the order in which elements are deleted and processed comes from the following rules.
  - An element of higher priority is processed before any element of lower priority
  - Two element with the same priority are processed according to the order in which they were added to the queue.

Break

Module

Algorithm Concepts and Problems (Easy/Medium/Hard)

Topic

DSA by Prashant sir

Subtopic

Hashing and Maps

Lesson

Hashing and Map

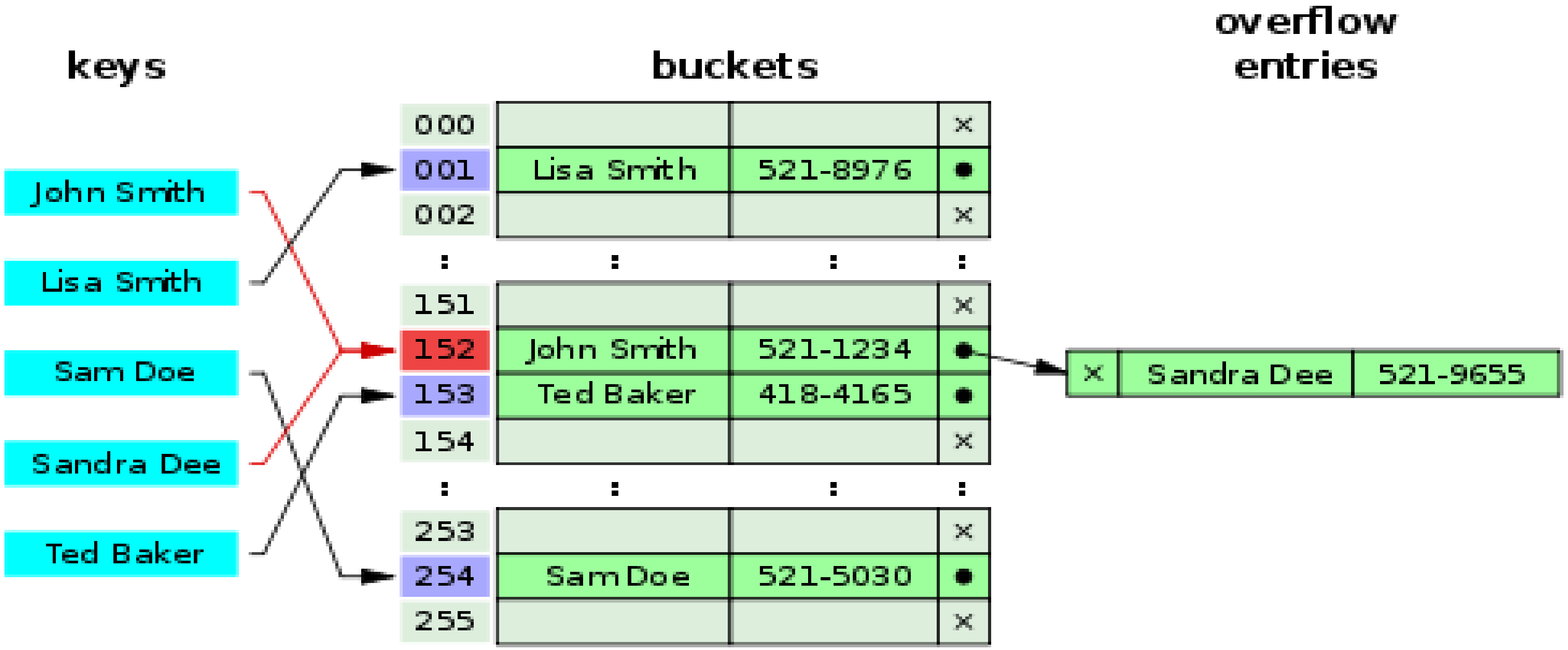
Hashing & Maps

Introduction to hashing

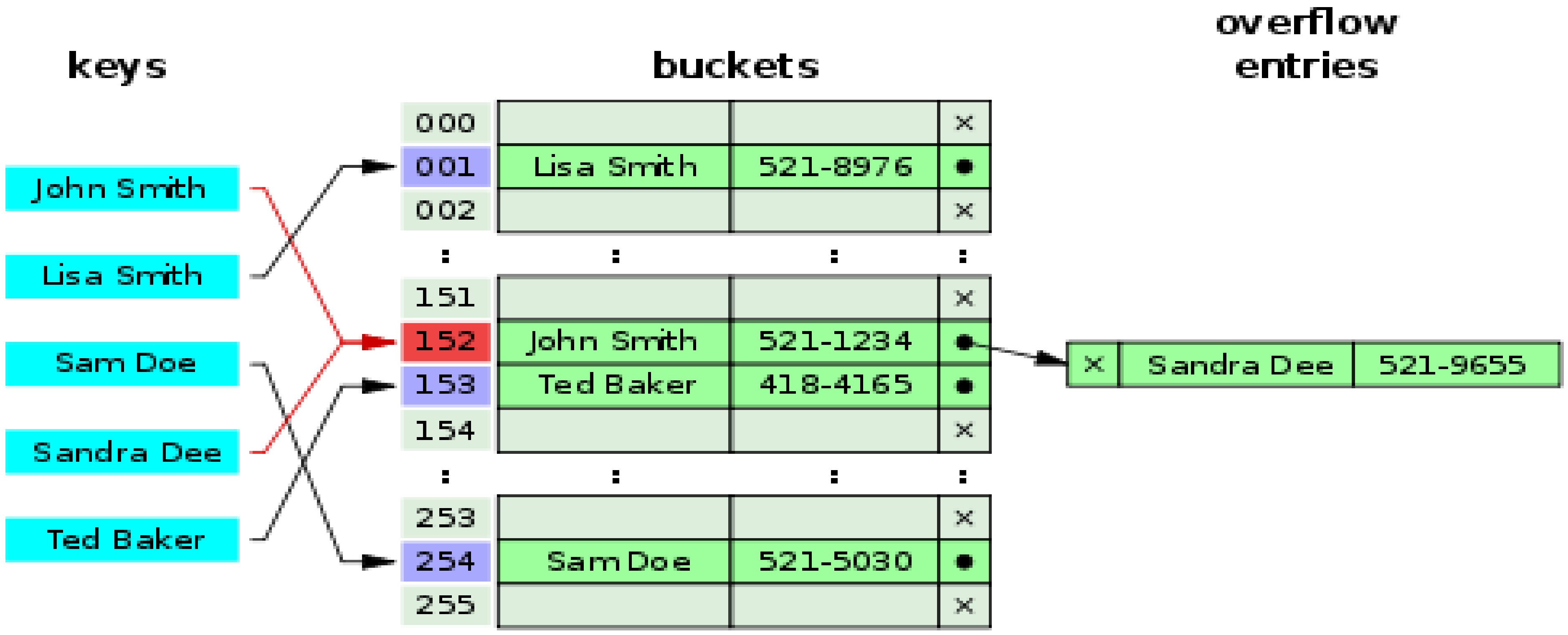
- Main idea of data structure is to help us store the data. But Most common operation on any data structure is not insert or delete but actually search, as even for insertion and deletion search is also required.

- In any of the data structure the search time first depends on the number of elements which data structure contains and then on type of structure. for e.g.
  - Unsorted array – O(n)
  - sorted array – O(logn)
  - link list – O(n)
  - BT – O(n)
  - BST – O(logn)

- So hashing is a technique where search time is independent of the number of items in which we are searching a data value.
- The basic idea is to use the key itself to find the address in the memory to make searching easy. For e.g. *to use phone number, roll no, Aadhar card, voter id or any other key and convert it into a smaller* practical *number (*but it must be modified so a great deal of space is not wasted*) and uses the small number as index in a table called hash table.*
- The values are then stored in **hash table**, By using that key you can access the element in **O(1)** time.



- This conversion called hash function which is from the set of K keys into the set of memory location L.
  - H: K→L
- In simple terms, a hash function maps a big number or string to a small integer that can be used as index in hash table. An array that stores pointers to records corresponding to our search key. The remaining entries can be nil.



BREAK

Most popular hash function

- **Division-remainder method***:* The size of the number of items in the table is estimated. That number is then used as a divisor into each original value or key to extract a quotient and a remainder.

- The remainder is the hashed value. (Since this method is liable to produce a number of collisions, any search mechanism would have to be able to recognize a collision and offer an alternate search mechanism.)
  - $H(K) = K(\mathrm{mod}\ m)$
  - $H(K) = K(\mathrm{mod}\ m) + 1$

BREAK

- Characteristics of good hash function
  - Easy to compute and understand
  - Efficiently computable- It must take less time to compute
  - Should uniformly distribute the keys (Each table position equally likely for each key) and should not result in clustering.
  - Must have low collision rate

BREAK

- **Collision**: - It is possible that two different set of keys K_1 and K_2 will yield the same hash address. This situation is called collision. The technique to resolve collision is called collision resolution.

- **Note**: Irrespective of how good a hash function is, collisions are bound to occur. Therefore, to maintain the performance of a hash table, it is important to manage collisions through various collision resolution techniques.

Collision resolution technique

- **Open Addressing/closed hashing** - In Open Addressing, all elements are stored in the hash table itself. i.e. collision is resolved by **probing** or searching through alternate locations in the Hash table itself in a particular *sequence.*

- When searching for an element, we one by one examine table slots until the desired element is found or it is clear that the element is not in the table. So, at any point, size of table must be greater than or equal to total number of keys.

- It is of three types linear probing, quadratic probing, double hashing

Linear probing

- Searches the table sequentially starting at the position given by the hash function, until finding a cell with a matching key or an empty cell.
- It takes constant expected time per search, insertion, or deletion when implemented using a random hash function.
- Using linear probing, dictionary operations can be implemented in constant expected time. In other words, insert, remove and search operations can be implemented in O(1), as long as the load factor of the hash table is a constant strictly less than one.
  - Insert(k): Keep probing until an empty slot is found. Once an empty slot is found, insert k.
  - Search(k): Keep probing until slot's key doesn't become equal to k or an empty slot is reached.
  - Delete(k): ***Delete operation is interesting***. If we simply delete a key, then search may fail. So slots of deleted keys are marked specially as "deleted". Insert can insert an item in a deleted slot, but search doesn't stop at a deleted slot.

- **Advantage**
  - The most popular implementation on standard hardware uses linear probing, which is both fast and simple.
  - Linear probing can provide high performance because of its good locality of reference.

- **Disadvantage**
    - Is more sensitive to the quality of its hash function than some other collision resolution schemes.
    - its performance degrades more quickly at high load factors because of primary clustering, a tendency for one collision to cause more nearby collisions. Basic operations takes more time. etc
    - Additionally, achieving good performance with this method requires a higher-quality hash function than for some other collision resolution schemes.

BREAK

Module

Algorithm Concepts and Problems (Easy/Medium/Hard)

Topic

DSA by Prashant sir

Subtopic

Binary Search Tree

Lesson

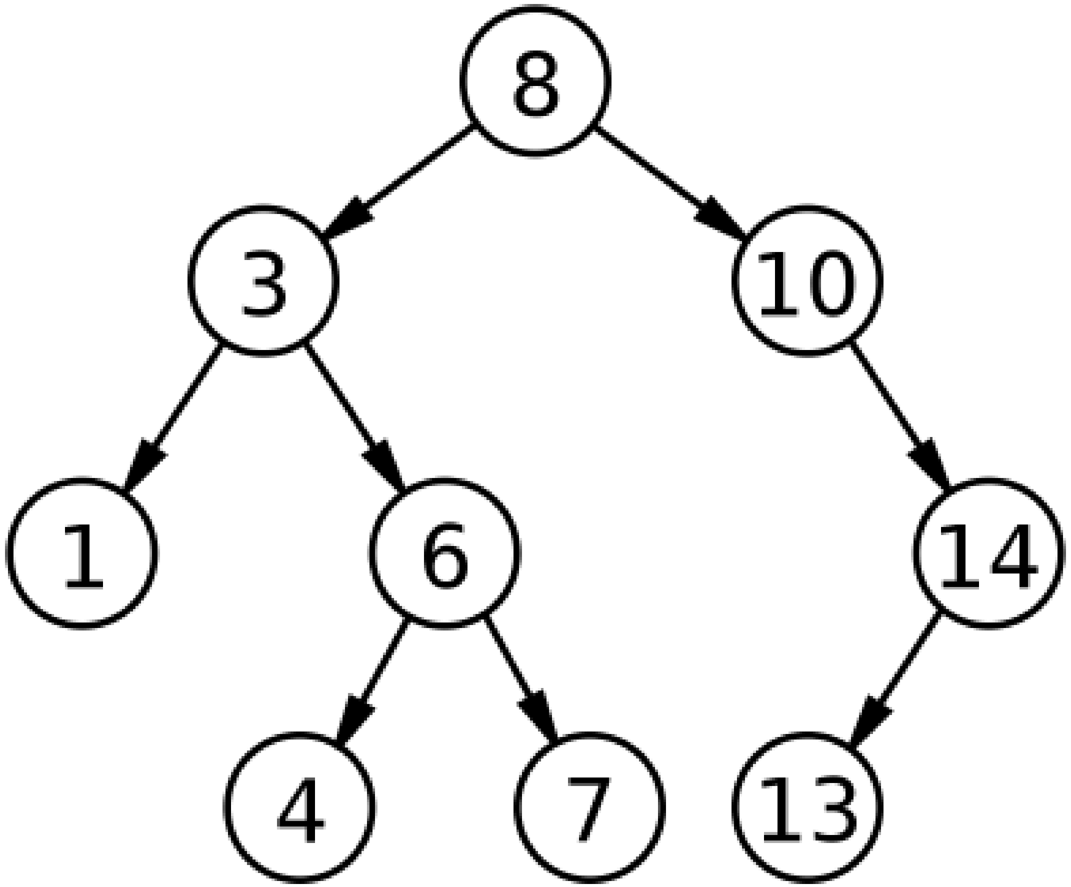
Binary Search Tree

BST

Binary Search Tree

Binary search tree / Ordered tree / Sorted binary tree

- A binary search tree (BST) is a binary tree in which left subtree of a node contains a key less than the node's key and right subtree of a node contains only the nodes with key greater than the node's key. Left and right sub tree must each also be a binary search tree.

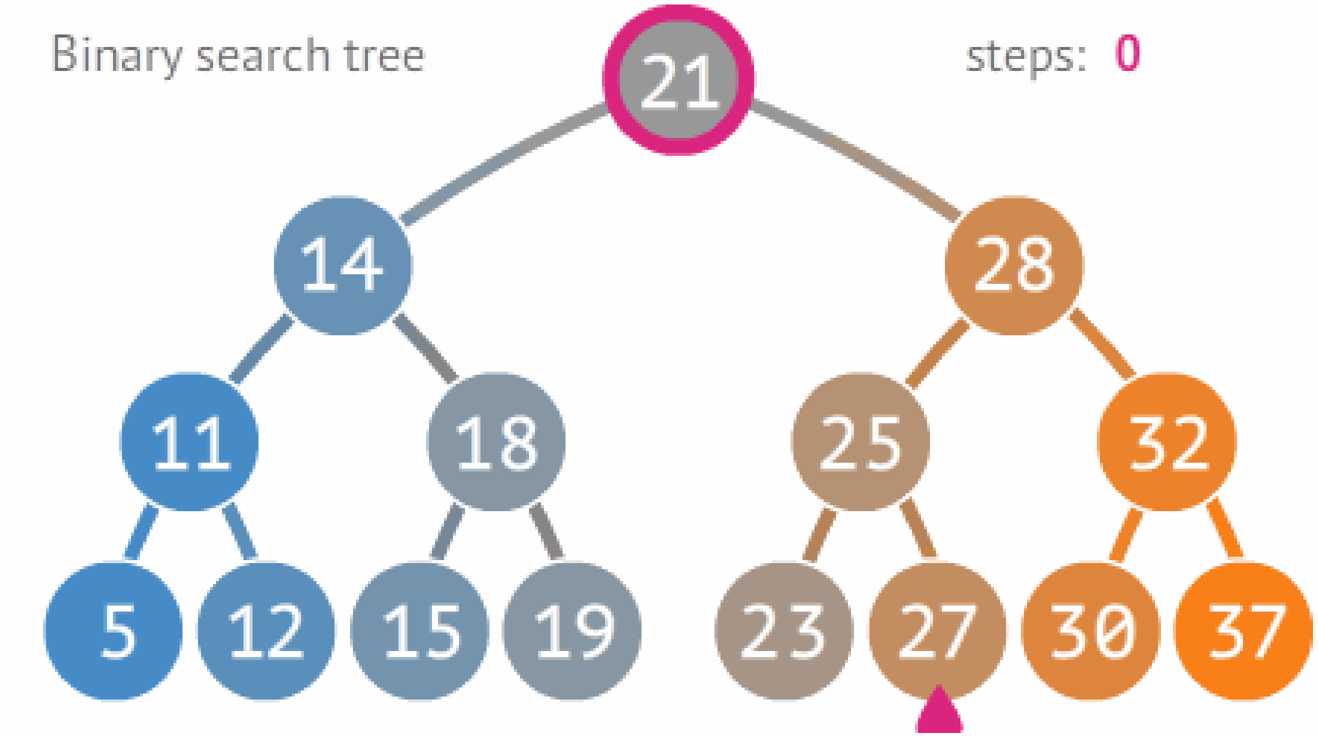


Searching

- We begin by examining the root node. If the tree is *null*, the key we are searching for does not exist in the tree. Otherwise, if the key equals that of the root, the search is successful and we return the node.

- If the key is less than that of the root, we search the left subtree. Similarly, if the key is greater than that of the root, we search the right subtree.

- This process is repeated until the key is found or the remaining subtree is *null*. If the searched key is not found after a *null* subtree is reached, then the key is not present in the tree.



Break

Operations

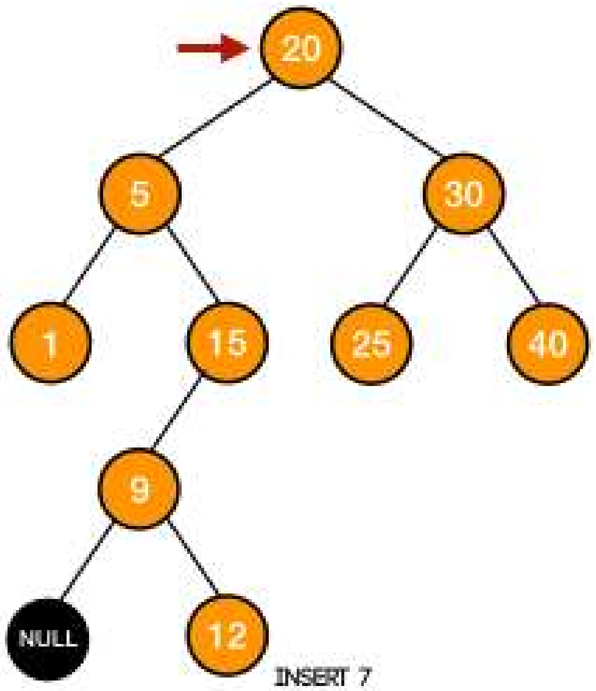
- Binary search trees support three main operations: insertion of elements, deletion of elements, and lookup (checking whether a key is present).

Insertion

- Insertion begins as a search would begin; if the key is not equal to that of the root, we search the left or right subtrees as before.

- Eventually, we will reach an external node and add the new key-value pair (here encoded as a record ‘new Node') as its right or left child, depending on the node's key.

- In other words, we examine the root and recursively insert the new node to the left subtree if its key is less than that of the root, or the right subtree if its key is greater than or equal to the root.

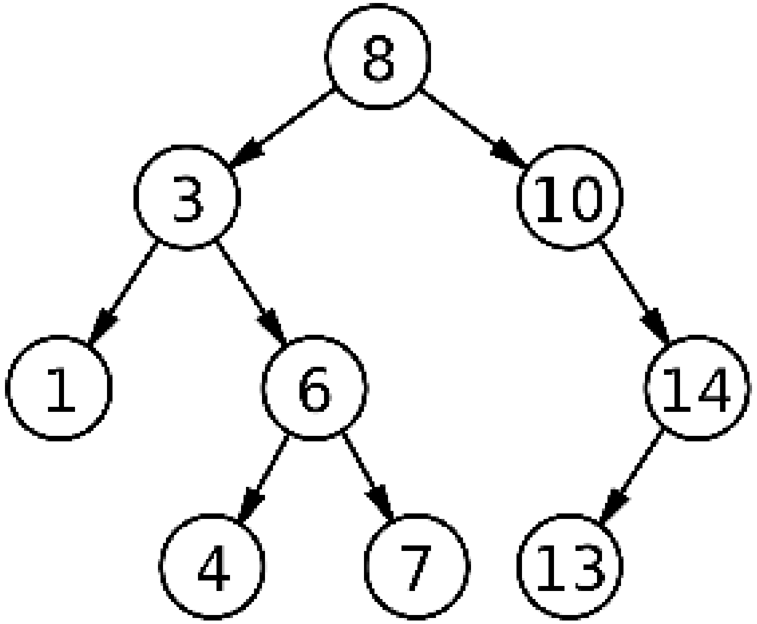


Deletion

- Deleting a node with no children: simply remove the node from the tree.
- Deleting a node with one child: remove the node and replace it with its child.
- Deleting a node with two children: call the node to be deleted *D*. Do not delete *D*.

- Instead, choose either its in-order predecessor node or its in-order successor node as replacement node *E* (s. figure). Copy the user values of *E* to *D*.

- If *E* does not have a child simply remove *E* from its previous parent *G*. If *E* has a child, say *F*, it is a right child. Replace *E* with *F* at *E*'s parent.



Algorithm	Average	Worst case
Space	O(n)	O(n)
Search	O(log n)	O(n)
Insert	O(log n)	O(n)
Delete	O(log n)	O(n)

- The major advantage of binary search trees over other data structures is that the related sorting algorithms and search algorithm such as in-order traversal can be very efficient; they are also easy to code

Binary search tree

Algorithm	Average	Worst case
Space	O(n)	O(n)
Search	O(log n)	O(n)
Insert	O(log n)	O(n)
Delete	O(log n)	O(n)

Break

Module

Algorithm Concepts and Problems (Easy/Medium/Hard)

Topic

DSA by Prashant sir

Subtopic

Graph

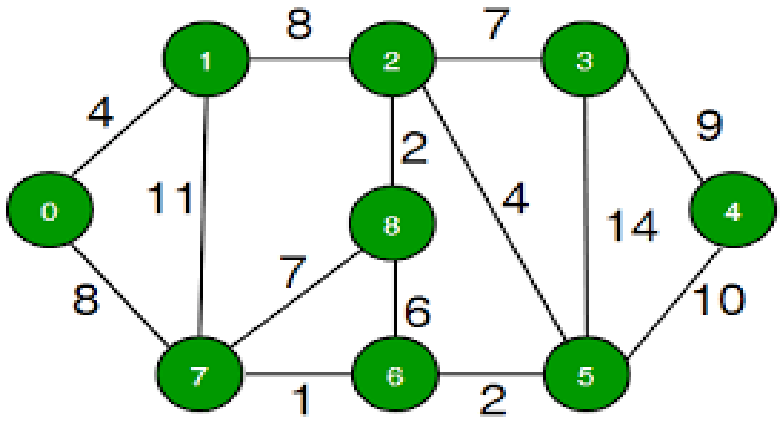
Lesson

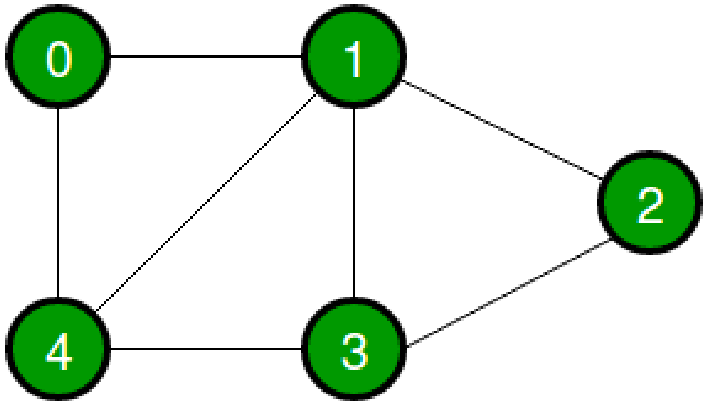
Graph

GRAPH

Graph

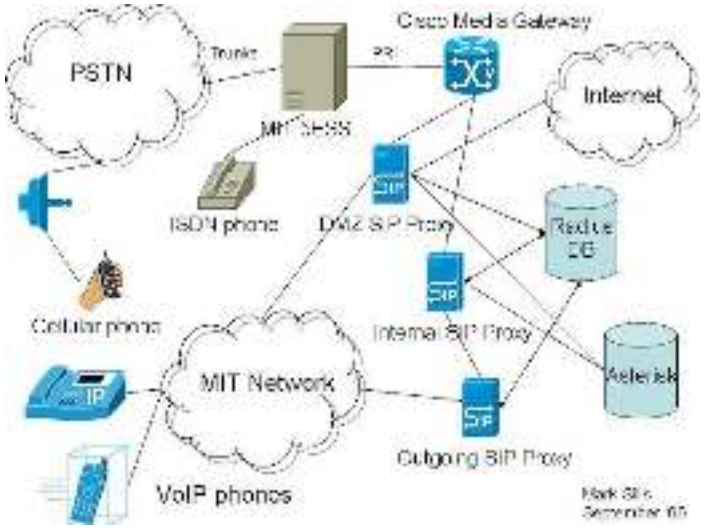
- Graph is a data structure that consists of following two components:
  - A finite set of vertices also called as nodes.
  - A finite set of ordered pair of the form (u, v) called as edge. The pair is ordered because (u, v) is not same as (v, u) in case of a directed graph(di-graph).
  - The pair of the form (u, v) indicates that there is an edge from vertex u to vertex v. The edges may contain weight/value/cost.





- Graphs are used to represent many real-life applications: Graphs are used to represent networks. The networks may include paths in a city or telephone network or circuit network.

- Graphs are also used in social networks like LinkedIn, Facebook. For example, in Facebook, each person is represented with a vertex (or node). Each node is a structure and contains information like person id, name, gender and locale.



Break

Representation of Graph in Memory

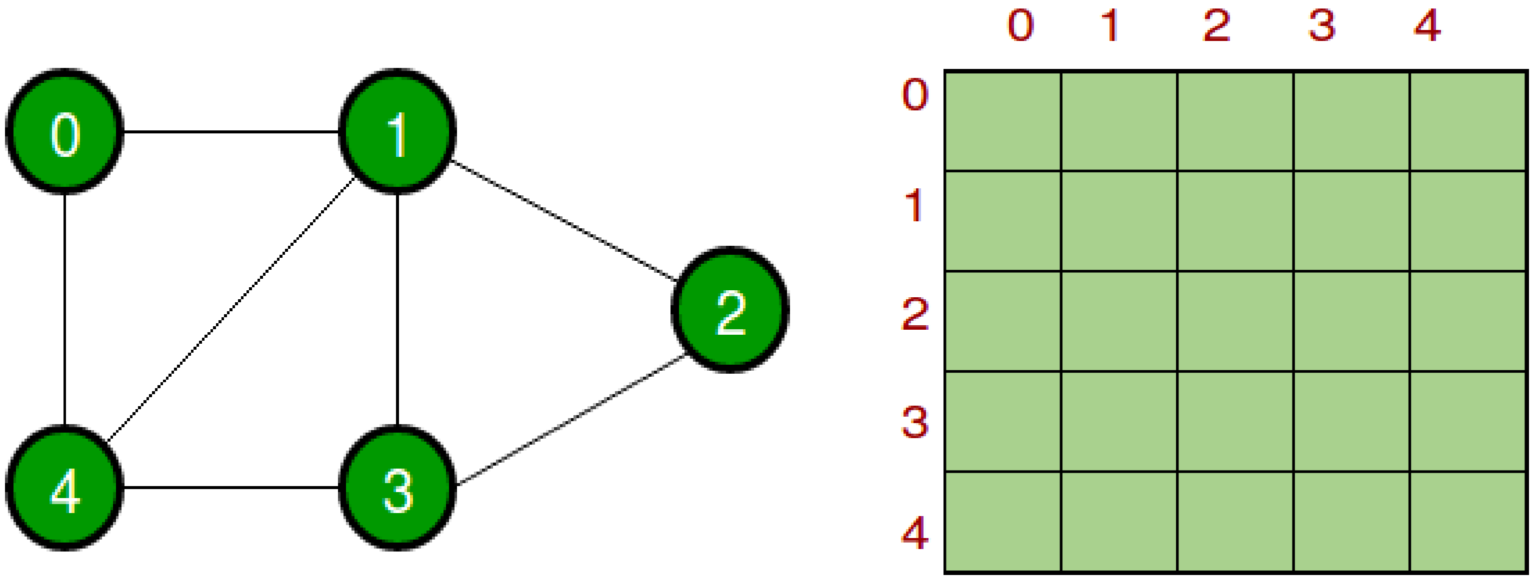
- Following two are the most commonly used representations of a graph.
  - Adjacency Matrix
  - Adjacency List

- There are other representations also like, Incidence Matrix and Incidence List. The choice of the graph representation is situation specific. It totally depends on the type of operations to be performed and ease of use.

- **Adjacency Matrix:** Adjacency Matrix is a 2D array of size V x V where V is the number of vertices in a graph. Let the 2D array be adj[][], a slot adj[i][j] = 1 indicates that there is an edge from vertex i to vertex j.

- Adjacency matrix for undirected graph is always symmetric.

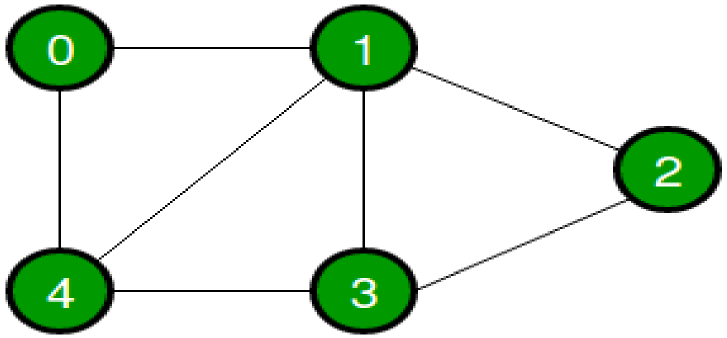
- Adjacency Matrix is also used to represent weighted graphs. If adj[i][j] = w, then there is an edge from vertex i to vertex j with weight w.

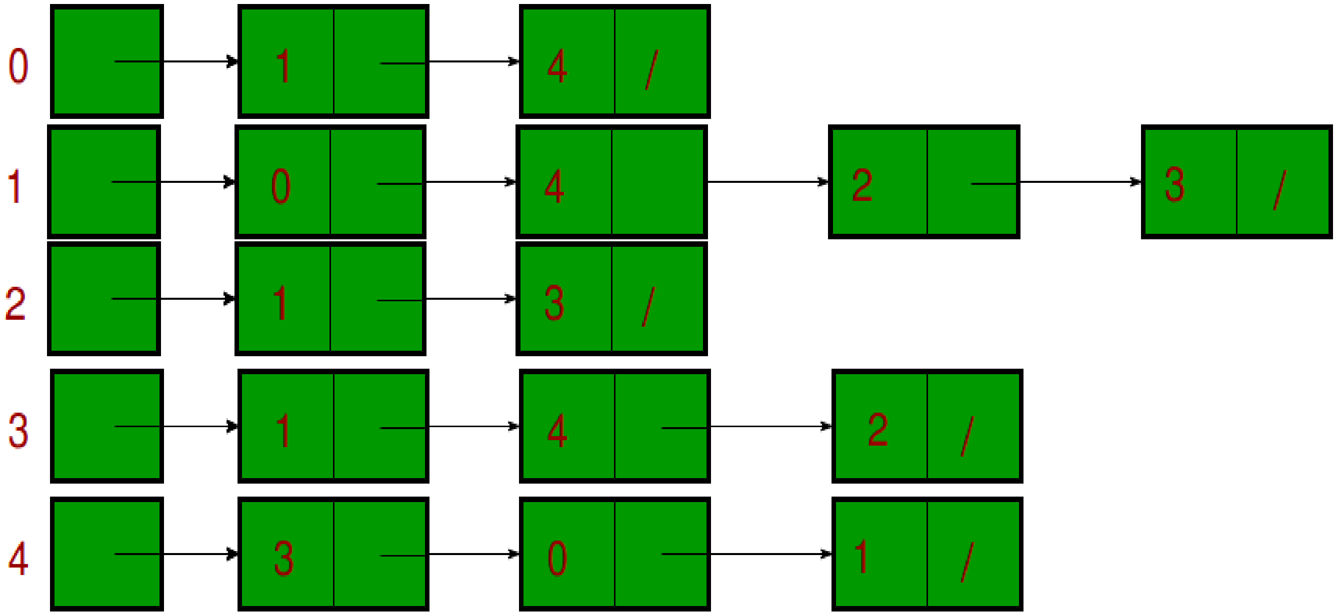


- **Pros:** Representation is easier to implement and follow. Removing an edge takes O(1) time. Queries like whether there is an edge from vertex ‘u’ to vertex ‘v’ are efficient and can be done O(1).

- **Cons:** Consumes more space $O(V^2)$. Even if the graph is sparse(contains less number of edges), it consumes the same space. Adding a vertex is $O(V^2)$ time.

- **Adjacency List:** An array of lists is used. Size of the array is equal to the number of vertices. Let the array be array[]. An entry array[i] represents the list of vertices adjacent to the ***i***th vertex. This representation can also be used to represent a weighted graph. The weights of edges can be represented as lists of pairs.

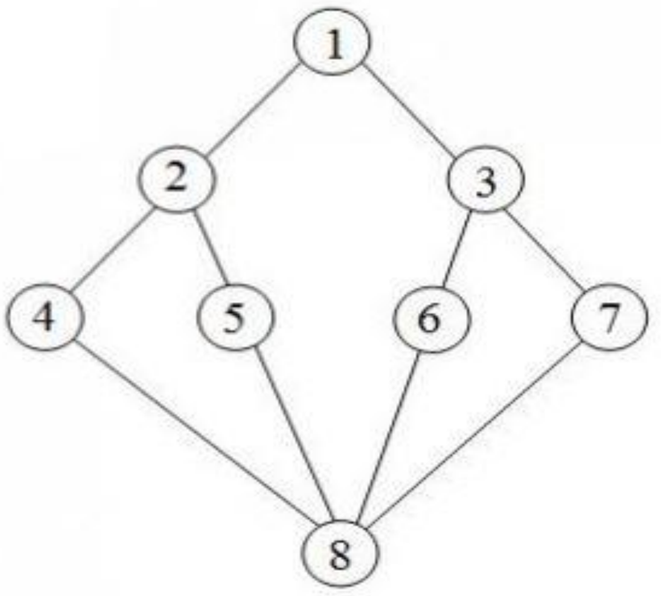




Break

Graph Traversal

- Traversal means visiting all the nodes of a graph.
- Depth First Traversal (or Search) for a graph is similar to Depth First Traversal of a tree.
- The only catch here is, unlike trees, graphs may contain cycles, so we may come to the same node again. To avoid processing a node more than once, we use a Boolean visited array.



Break

Module

Algorithm Concepts and Problems (Easy/Medium/Hard)

Topic

DSA by Prashant sir

Subtopic

DFS and BFS

Lesson

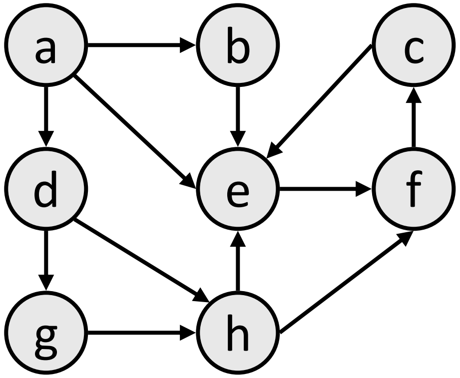
DFS and BFS

DFS & BFS

Depth-first search

- **depth-first search** (DFS): Finds a path between two vertices by exploring each possible path as far as possible before backtracking.
  - Often implemented recursively.
  - Many graph algorithms involve *visiting* or *marking* vertices.

- Depth-first paths from *a* to all vertices (assuming ABC edge order):
  - to b: {a, b}
  - to c: {a, b, e, f, c}
  - to d: {a, d}
  - to e: {a, b, e}
  - to f: {a, b, e, f}
  - to g: {a, d, g}
  - to h: {a, d, g, h}



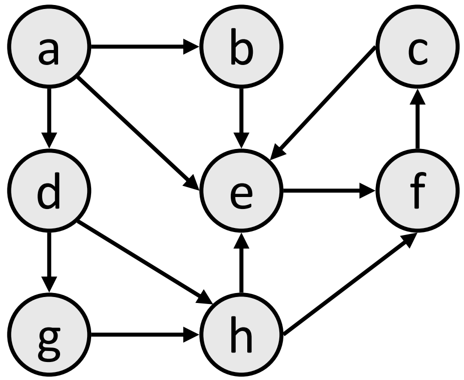
DFS pseudocode

function **dfs**(v_1, v_2):
  dfs(v_1, v_2, { }).

function **dfs**(v_1, v_2, *path*):
  path += v_1.
  mark v_1 as visited.
  if v_1 is v_2:
    a path is found!

  for each unvisited neighbor *n* of v_1:
    if dfs(*n*, v_2, *path*) finds a path: a path is found!

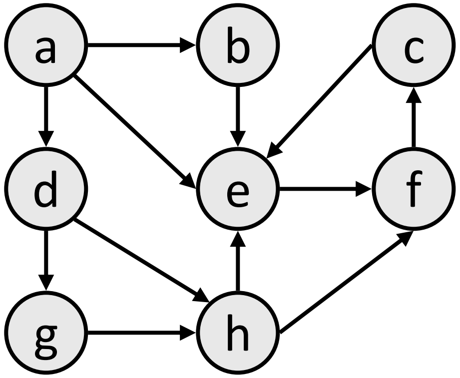
  path -= v_1. // path is not found.



- The *path* param above is used if you want to have the path available as a list once you are done.
  - Trace dfs(*a*, *f*) in the above graph.

DFS observations

- *discovery*: DFS is guaranteed to find <u>*a*</u> path if one exists.



- *retrieval*: It is easy to retrieve exactly what the path is (the sequence of edges taken) if we find it

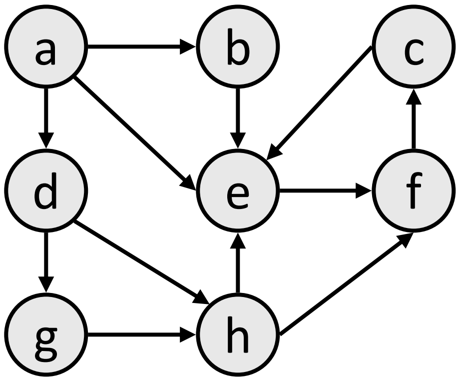
- *optimality*: not optimal. DFS is guaranteed to find <u>a</u> path, not necessarily the best/shortest path
  - Example: dfs(a, f) returns {a, d, c, f} rather than {a, d, f}.

BREAK

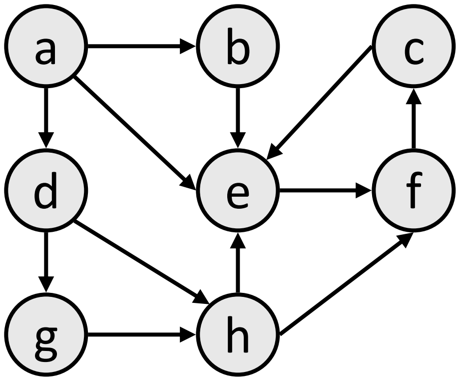
Breadth-first search

- **breadth-first search** (BFS): Finds a path between two nodes by taking one step down all paths and then immediately backtracking.
  - Often implemented by maintaining a queue of vertices to visit.

- BFS always returns the shortest path (the one with the fewest edges) between the start and the end vertices.
  - to b: {a, b}
  - to c: **{a, e, f, c}**
  - to d: {a, d}
  - to e: **{a, e}**
  - to f: **{a, e, f}**
  - to g: {a, d, g}
  - to h: **{a, d, h}**



BFS pseudocode



function **bfs**(v_1, v_2):
    queue := {v_1}.
    mark v_1 as visited.

    while *queue* is not empty:
        v := *queue*.removeFirst().
        if v is v_2:
            a path is found!

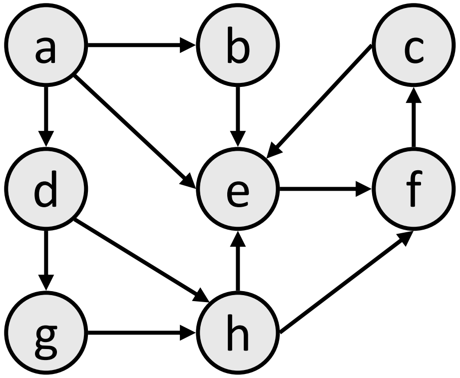
        for each unvisited neighbor n of v:
            mark n as visited.
            queue.addLast(n).

    // path is not found.

- Trace bfs(a, f) in the above graph.

BFS observations

- *optimality*:
  - always finds the shortest path (fewest edges).
  - in unweighted graphs, finds optimal cost path.
  - In weighted graphs, *not* always optimal cost.



- *retrieval*: harder to reconstruct the actual sequence of vertices or edges in the path once you find it
  - conceptually, BFS is exploring many possible paths in parallel, so it's not easy to store a path array/list in progress
  - solution: We can keep track of the path by storing predecessors for each vertex (each vertex can store a reference to a *previous* vertex).
- DFS uses less memory than BFS, easier to reconstruct the path once found; but DFS does not always find shortest path. BFS does.

BREAK

Module

Algorithm Concepts and Problems (Easy/Medium/Hard)

Topic

DSA by Prashant sir

Subtopic

Greedy Algo

Lesson

Greedy

Greedy Algorithm

- A **greedy algorithm** is any algorithm that follows the problem-solving heuristic of making the locally optimal choice at each stage with the intent of finding a global optimum.

- In many problems, a greedy strategy does not usually produce an optimal solution, but nonetheless a greedy heuristic may yield locally optimal solutions that approximate a globally optimal solution in a reasonable amount of time.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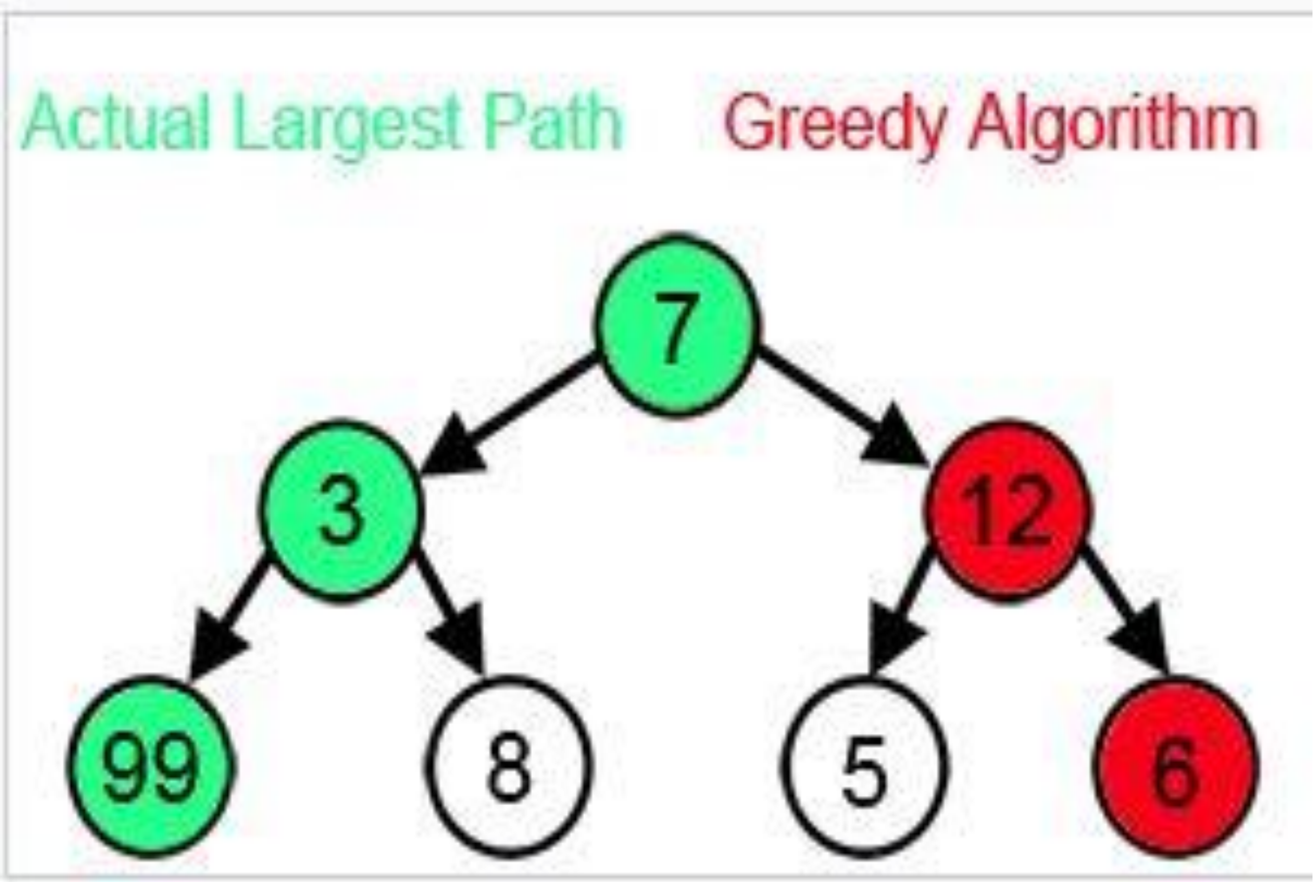
Largest Path Example Failure Case



Break

Traveling Salesman Problem

- For example, a greedy strategy for the traveling salesman problem (which is of a high computational complexity) is the following heuristic:

- "At each step of the journey, visit the nearest unvisited city." This heuristic does not intend to find a best solution, but it terminates in a reasonable number of steps; finding an optimal solution to such a complex problem typically requires unreasonably many steps.

Greedy Strategy

- We can make whatever choice seems best at the moment and then solve the subproblems that arise later.

- The choice made by a greedy algorithm may depend on choices made so far, but not on future choices or all the solutions to the subproblem.

- It iteratively makes one greedy choice after another, reducing each given problem into a smaller one. In other words, a greedy algorithm never reconsiders its choices.

Break

Difference with DP

- This is the main difference from dynamic programming, which is exhaustive and is guaranteed to find the solution. After every stage, dynamic programming makes decisions based on all the decisions made in the previous stage, and may reconsider the previous stage's algorithmic path to solution.

Break

Knap Sack Problem

- The **knapsack problem** or **rucksack problem** is a problem in combinatorial optimization: Given a set of items, each with a weight and a value, determine the number of each item to include in a collection so that the total weight is less than or equal to a given limit and the total value is as large as possible.

- It derives its name from the problem faced by someone who is constrained by a fixed-size knapsack and must fill it with the most valuable items.

- The problem often arises in resource allocation where there are financial constraints and is studied in fields such as combinatorics, computer science, complexity theory, cryptography and applied mathematics.

- The knapsack problem has been studied for more than a century, with early works dating as far back as 1897. The name "knapsack problem" dates back to the early works of mathematician Tobias Dantzig (1884–1956).

Consider the weights and values of items listed below. Note that there is only one unit of each item.

Object	O_1	O_2	O_3
Weight	5	3	2
Profit	25	24	14

Break

Module

Algorithm Concepts and Problems (Easy/Medium/Hard)

Topic

DSA by Prashant sir

Subtopic

Backtracking

Lesson

Lesson 20 of 23

Backtracking

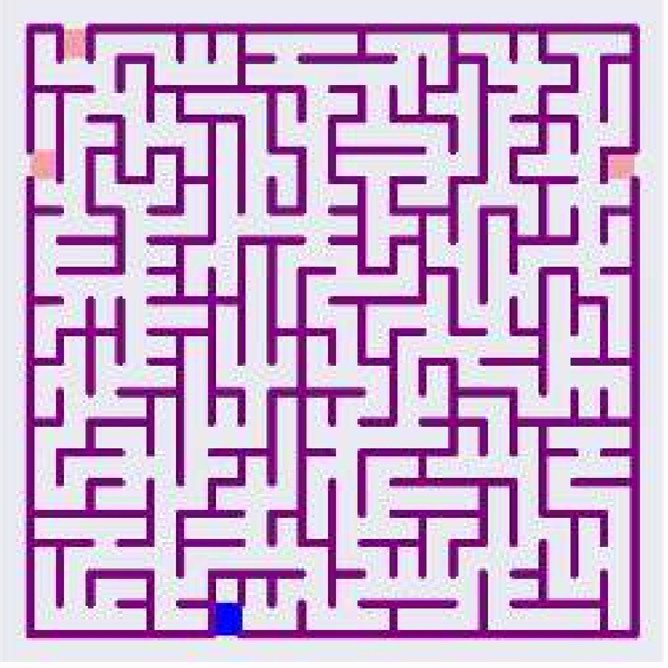
Backtracking

Backtracking

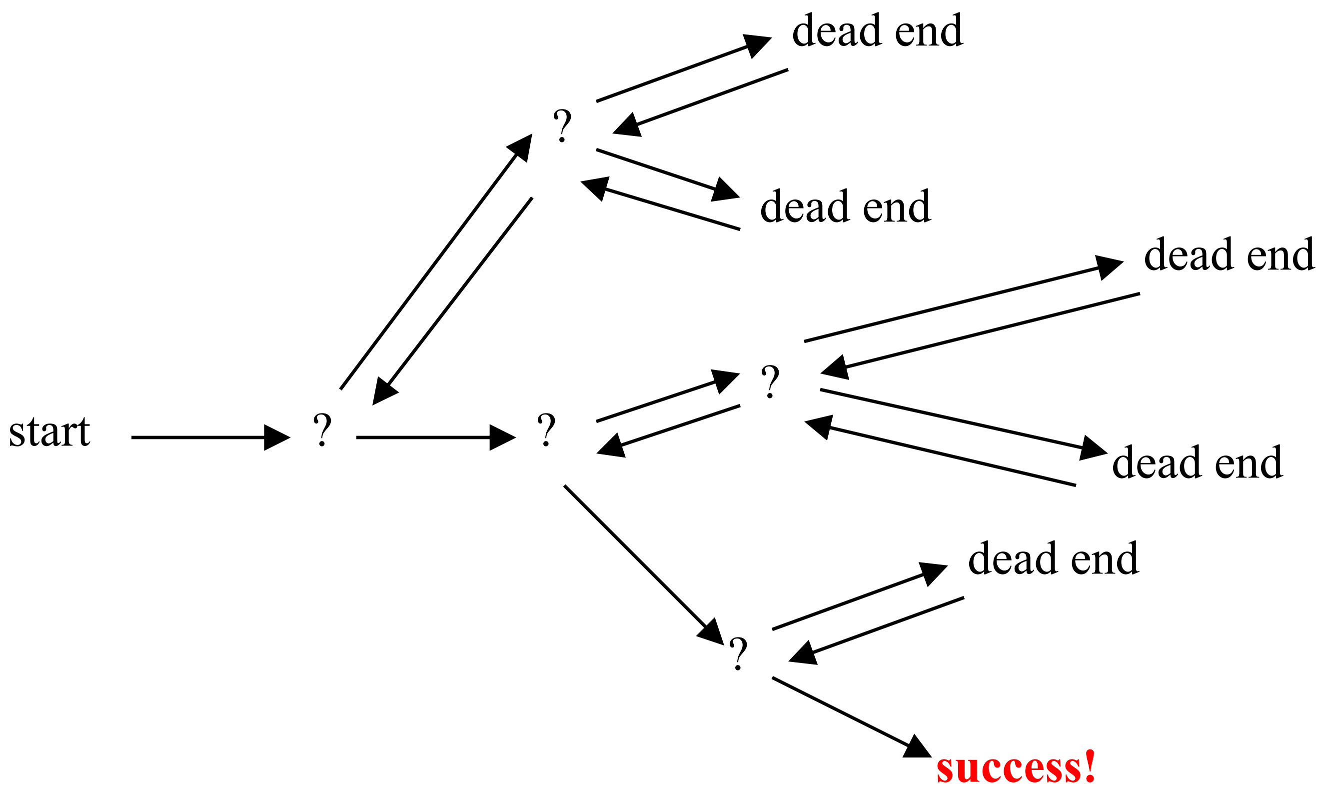
- Suppose you have to make a series of *decisions,* among various *choices,* where
  - You don't have enough information to know what to choose
  - Each decision leads to a new set of choices
  - Some sequence of choices (possibly more than one) may be a solution to your problem
- Backtracking is a methodical way of trying out various sequences of decisions, until you find one that "works"

Solving a maze

- Given a maze, find a path from start to finish
- You don't have enough information to choose correctly
- Each choice leads to another set of choices
- One or more sequences of choices may (or may not) lead to a solution
- Many types of maze problem can be solved with backtracking



Backtracking



The backtracking algorithm

- Backtracking is really quite simple--we “explore” each node, as follows:
- To “explore” node N:
    1. If N is a goal node, return “success”
    2. If N is a leaf node, return “failure”
    3. For each child C of N,
        3.1. Explore C
            3.1.1. If C was successful, return “success”
    4. Return “failure”

Module

Algorithm Concepts and Problems (Easy/Medium/Hard)

Topic

DSA by Prashant sir

Subtopic

Heaps

Lesson

Heap

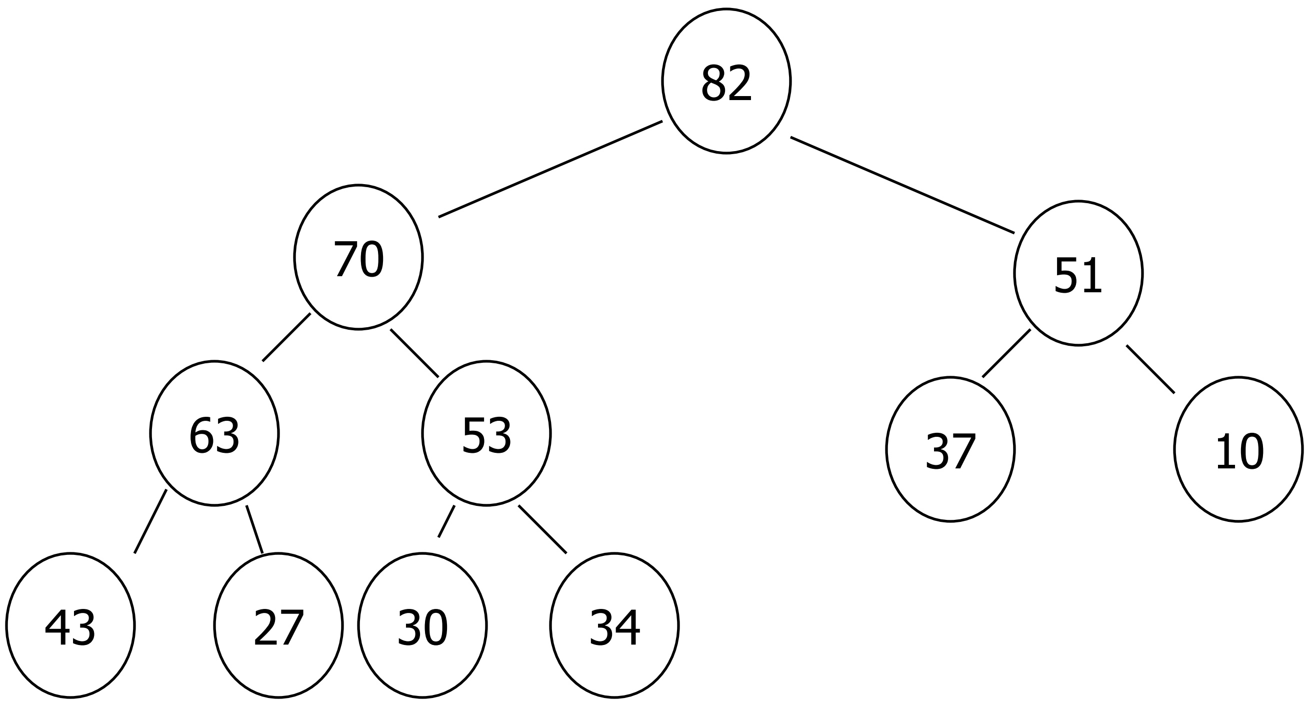
Heaps

Introduction to Heaps

Each node in a heap satisfies the ‘heap condition,’ which states that **every node’s key is larger than or equal to the keys of its children**.

Introduction

- First of all, a heap is a **<u>kind of</u>** **<u>tree</u>** that offers both insertion and deletion in $O(\log_2 n)$ time.
- Fast for insertions; not so fast for deletions.



Introduction

- ►Heaps are largely about priority queues.
- ►They are an alternative data structure to implementing priority queues (we had arrays, linked lists…)
- ►Recall the advantages and disadvantages of queues implemented as arrays
  - Insertions / deletions? O(n) … O(1)!
- ►Priority queues are critical to many real-world applications.

BREAK

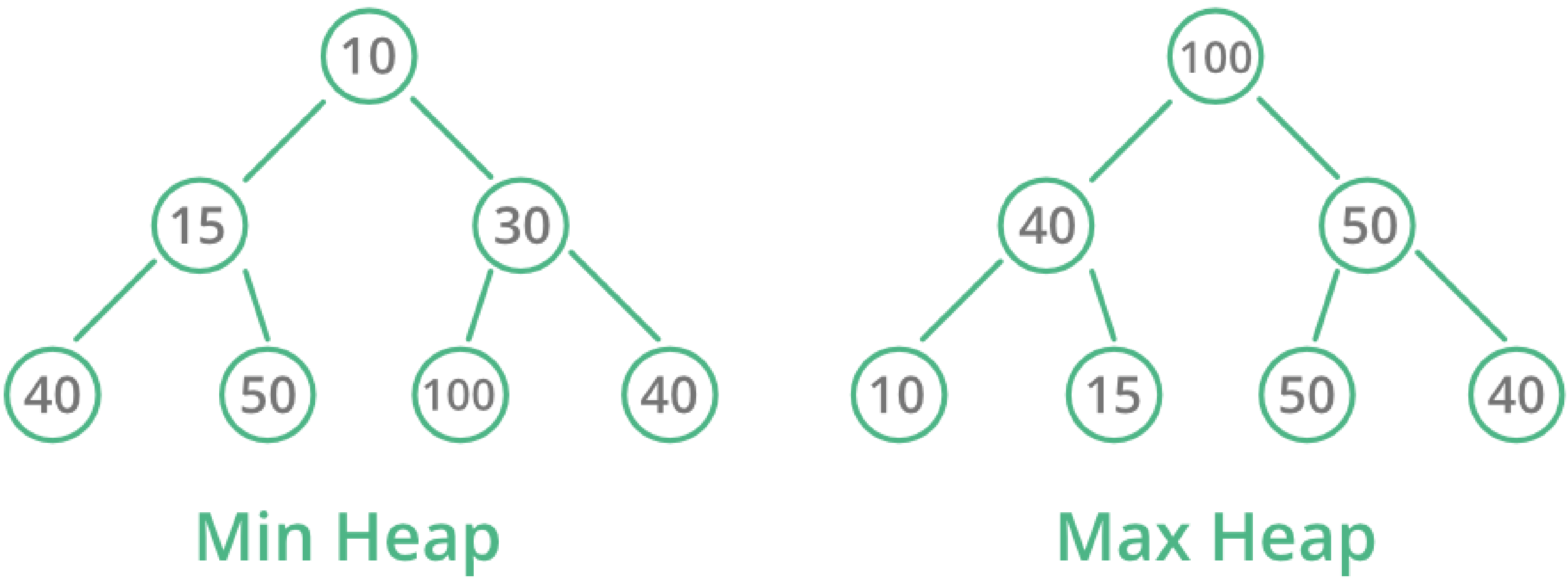
How to Implement Heaps

► Characteristics:

- U**sually** **<u>implemented</u>** as an **array**
- ➔ The heap is thus an **<u>abstraction</u>**; we can draw it to **look** like a tree, but recognize that it is the **array** that **<u>implements</u> this abstraction** and that it is stored and processed in **primary memory (RAM)**.
- No 'holes' in the array.

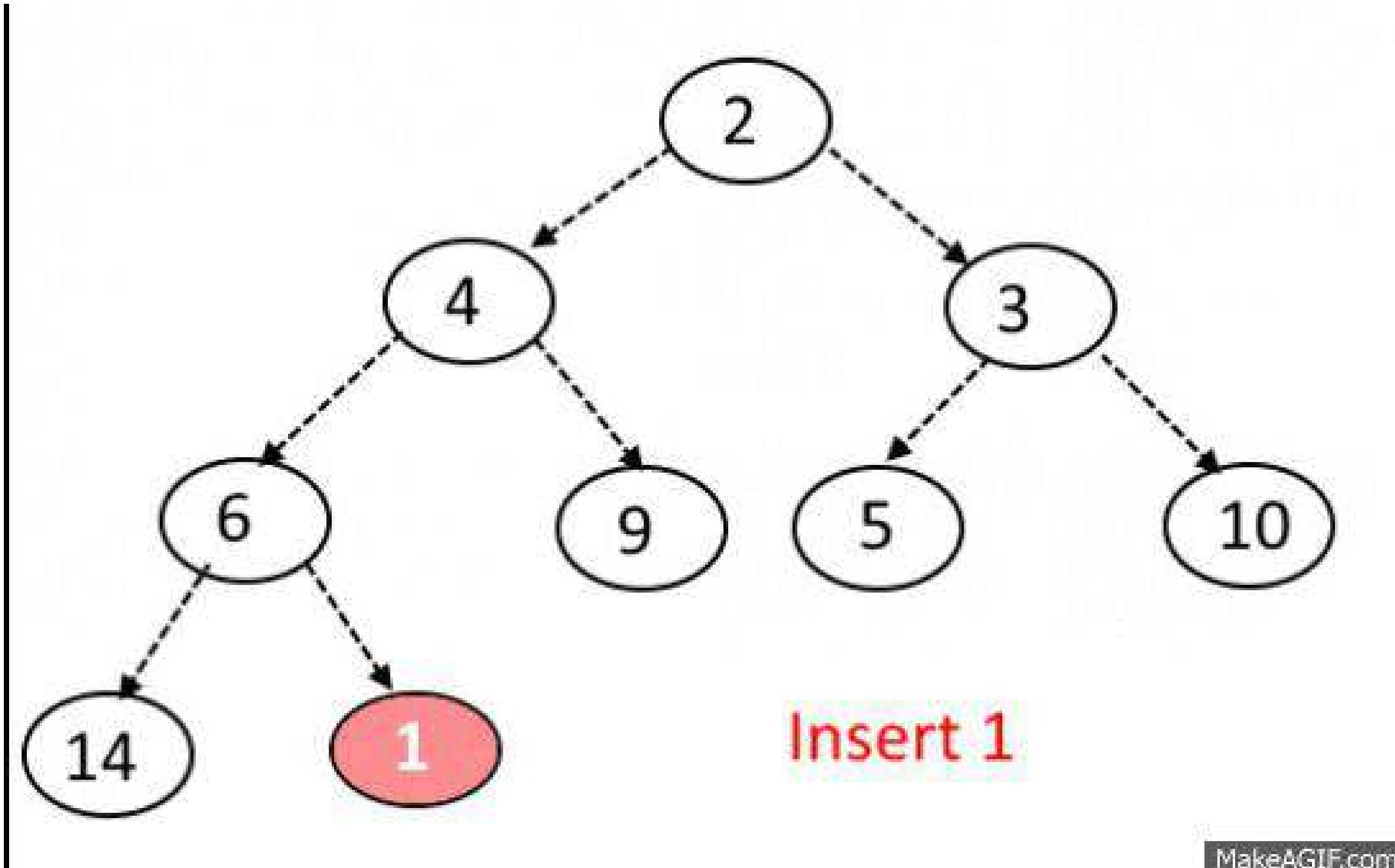
BREAK

Types of Heap

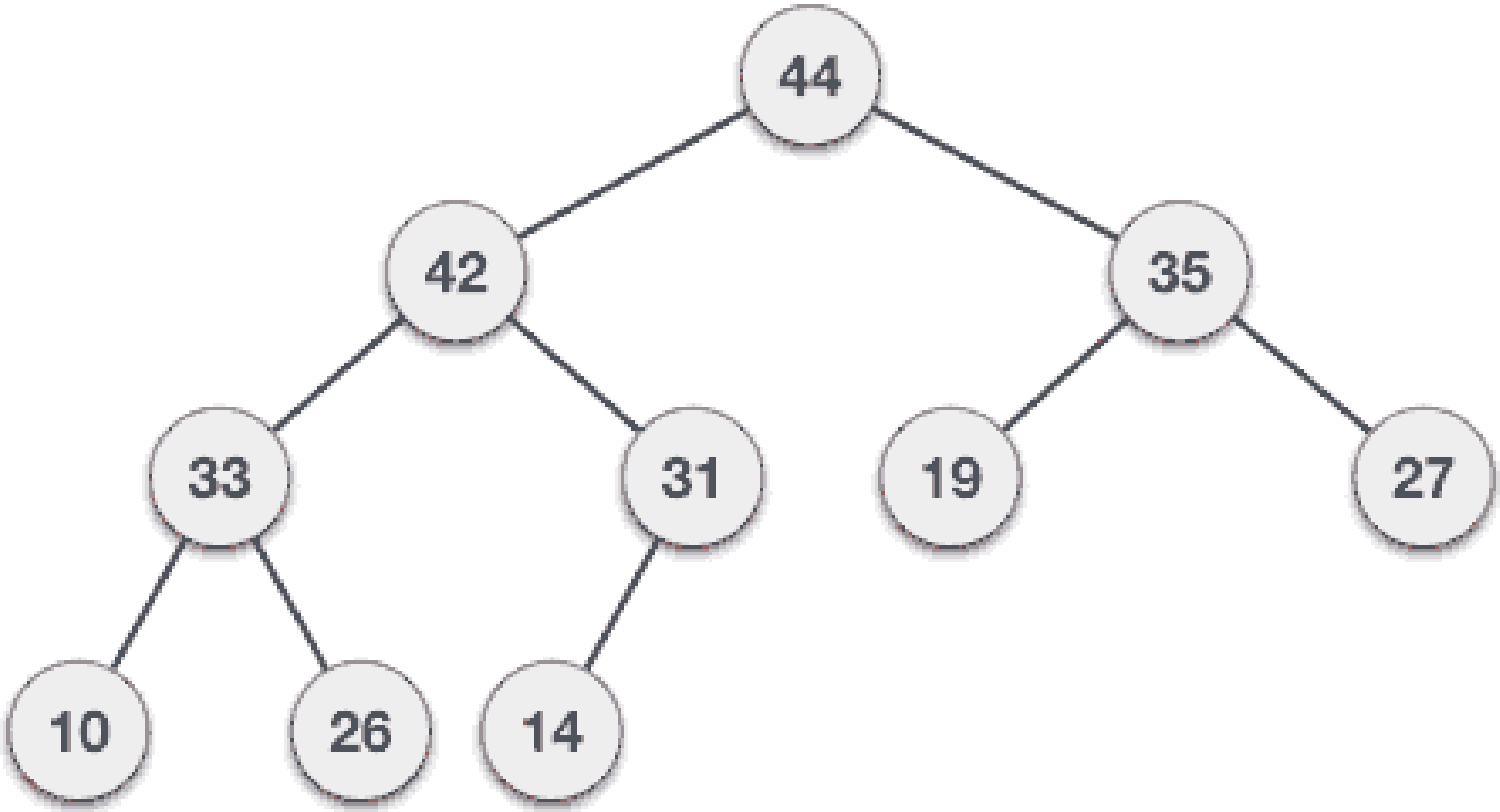


BREAK

Insertion in Min Heap



Deletion in Max Heap



BREAK

Heapsort

- Merge sort time is O(n log n)
  - ***But*** requires (temporarily) n extra storage items
- *Heapsort*
  - Works *in place:* no additional storage
  - Offers same O(n log n) performance
- Idea (not quite in-place):
  - Insert each element into a priority queue
  - Repeatedly remove from priority queue to array
    - Array slots go from 0 to n-1

Heapsort

6 5 3 1 8 7 2 4

Algorithm for In-Place Heapsort

- Build heap starting from unsorted array
- While the heap is not empty
  - Remove the first item from the heap:
    - Swap it with the last item
  - Restore the heap property

Heapsort Analysis

- Insertion cost is log i for heap of size i
  - Total insertion cost = log(n)+log(n-1)+...+log(1)
  - This is O(n log n)
- Removal cost is also log i for heap of size i
  - Total removal cost = O(n log n)

- Total cost is O(n log n)

BREAK

Module

Algorithm Concepts and Problems (Easy/Medium/Hard)

Topic

DSA by Prashant sir

Subtopic

Dynamic Programming

Lesson

DynamicProgramming

DP

(Dynamic Programming)

What is DP?

- Wikipedia definition: “method for solving complex problems by breaking them down into simpler subproblems”
- This definition will make sense once we see some examples
  - Actually, we'll only see problem solving examples today

Dynamic Programming

- Dynamic programming is a very powerful, general tool for solving optimization problems.
- Once understood it is relatively easy to apply, but many people have trouble understanding it.

BREAK

Steps for Solving DP Problems

1. Define subproblems
2. Write down the recurrence that relates subproblems
3. Recognize and solve the base cases

- Each step is very important!

Greedy Algorithms

- Greedy algorithms focus on making the best local choice at each decision point.
- For example, a natural way to compute a shortest path from x to y might be to walk out of x, repeatedly following the cheapest edge until we get to y. WRONG!
- In the absence of a correctness proof greedy algorithms are very likely to fail.

BREAK

Problem:

Let’s consider the calculation of **Fibonacci** numbers:

$F(n) = F(n-2) + F(n-1)$

with seed values $F(1) = 1$, $F(2) = 1$

or $F(0) = 0$, $F(1) = 1$

What would a series look like:

0, 1, 1, 2, 3, 4, 5, 8, 13, 21, 34, 55, 89, 144, …

Recursive Algorithm:

```
Fib(n)
{
   if (n == 0)
      return 0;

   if (n == 1)
      return 1;

  Return Fib(n-1)+Fib(n-2)
}
```

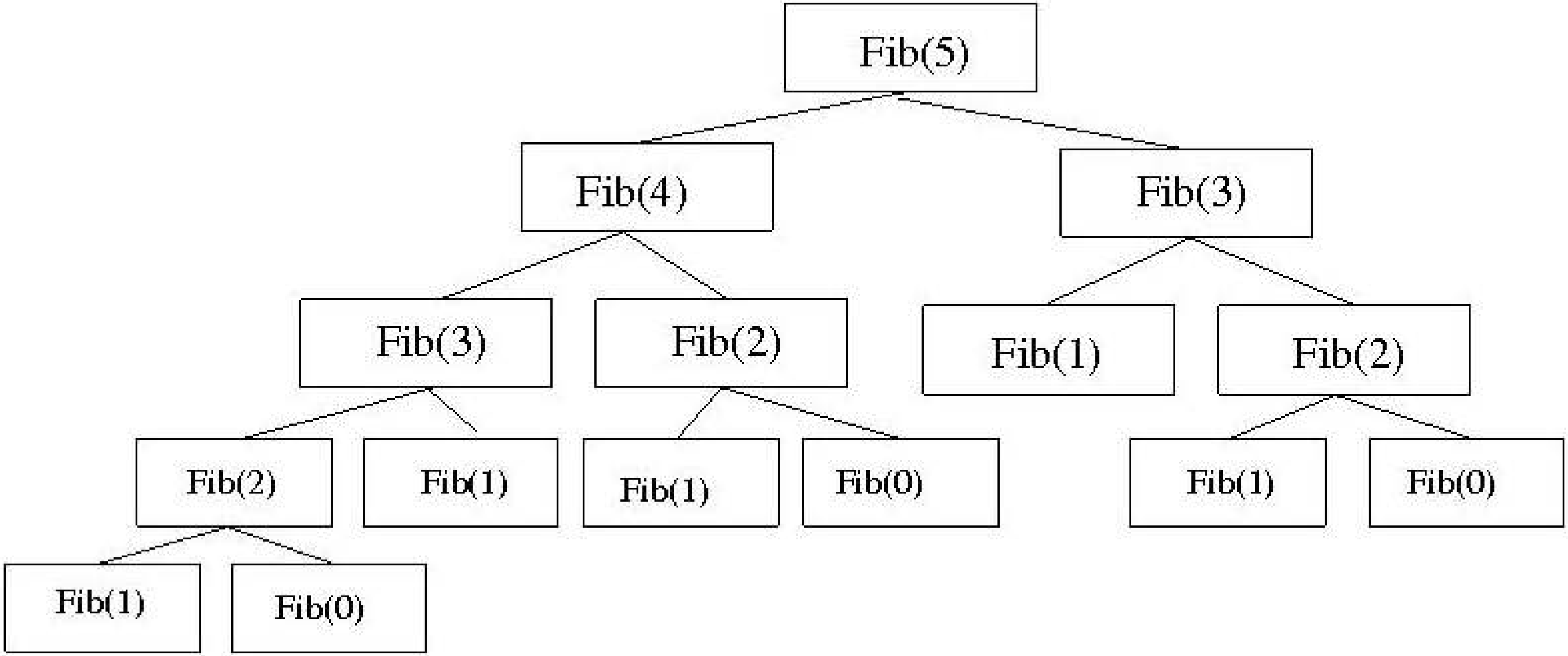
Recursive Algorithm:

```
Fib(n)
{
   if (n == 0)
      return 0;

   if (n == 1)
      return 1;

  Return Fib(n-1)+Fib(n-2)
}
```

It has a serious issue!

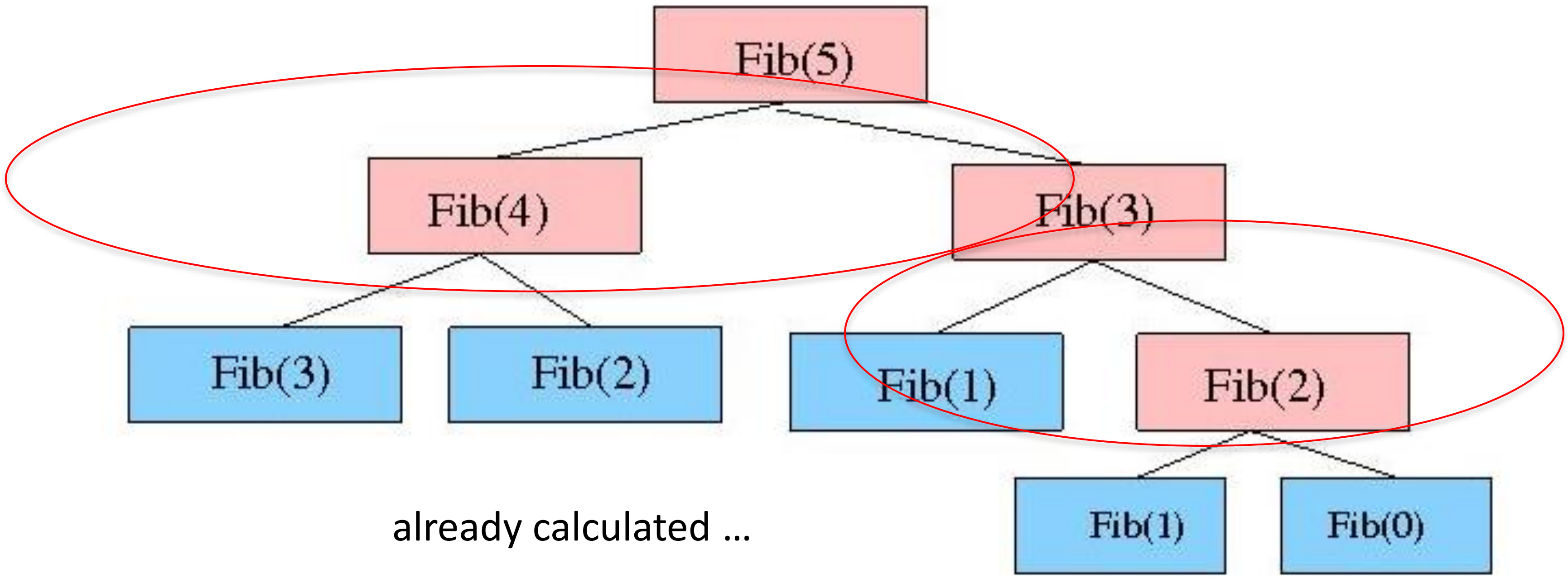
Recursion tree

What's the problem?



Memoization:

```
Fib(n)
{
   if (n == 0)
      return M[0];

   if (n == 1)
      return M[1];

   if (Fib(n-2) is not already calculated)
      call Fib(n-2);

   if(Fib(n-1) is already calculated)
      call Fib(n-1);

   //Store the ${n}^{th}$ Fibonacci no. in memory & use previous results.
   M[n] = M[n-1] + M[n-2]

   Return M[n];
}
```



already calculated …

Steps for Solving DP Problems

- Define subproblems
- Write down the recurrence that relates subproblems
- Recognize and solve the base cases

- Each step is very important.

BREAK

Approaches of Dynamic programming

- Main approach: recursive, holds answers to a sub problem in a table, can be used without recomputing.

- Can be formulated both via recursion and saving results in a table (*memoization*). Typically, we first formulate the recursive solution and then turn it into recursion plus dynamic programming via *memoization* or bottom-up.

-"*programming*" as in tabular not programming code

Comments

- Dynamic programming relies on working "from the bottom up" and saving the results of solving simpler problems
  - These solutions to simpler problems are then used to compute the solution to more complex problems
- Dynamic programming solutions can often be quite complex and tricky

Comments

- Dynamic programming is used for optimization problems, especially ones that would otherwise take exponential time
  - Only problems that satisfy the principle of optimality are suitable for dynamic programming solutions
- Since exponential time is unacceptable for all but the smallest problems, dynamic programming is sometimes essential.

BREAK

Module

Algorithm Concepts and Problems (Easy/Medium/Hard)

Topic

DSA by Prashant sir

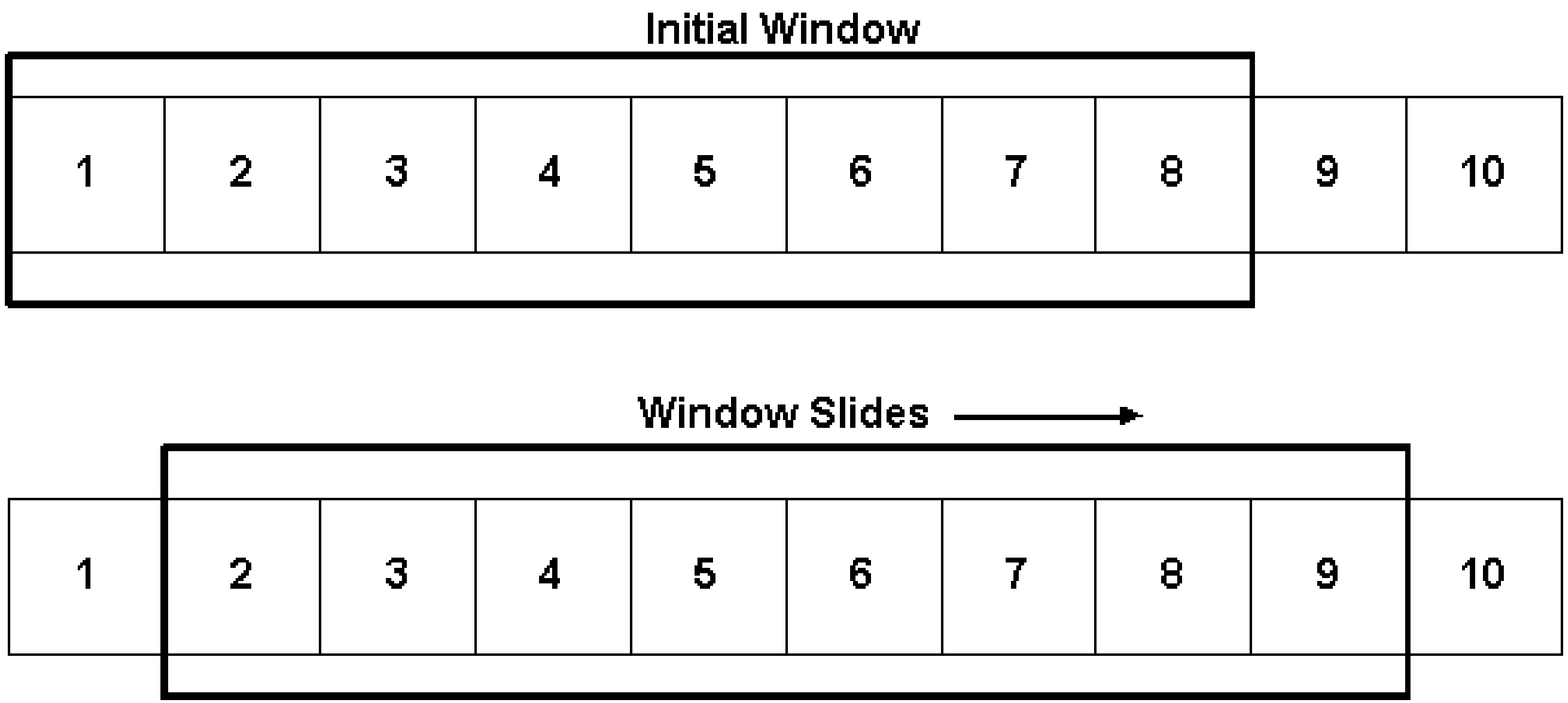
Subtopic

Sliding Window

Lesson

Sliding Window

Sliding Window


Initial Window
1
2
3
4
5
6
7
8
9
10
Window Slides
1
2
3
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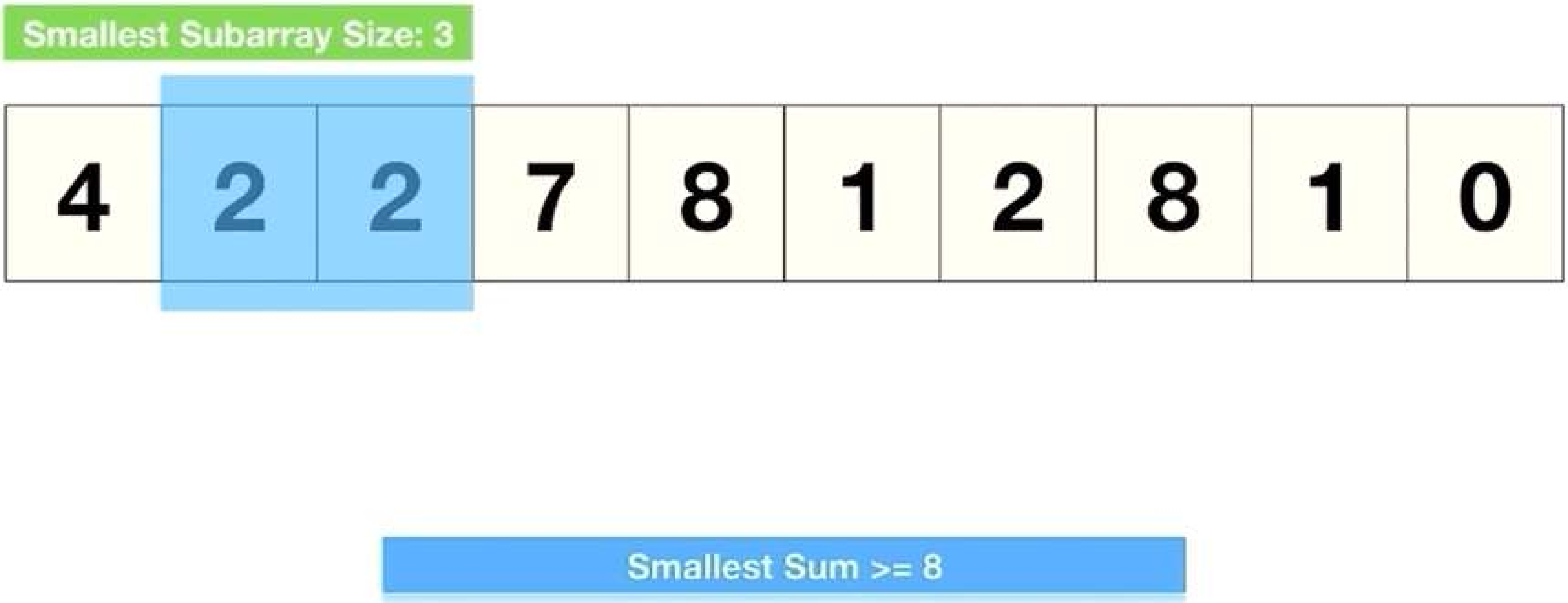
7
8
9
10


Smallest Subarray Size: 3
4
2
2
7
8
1
2
8
1
0
Smallest Sum >= 8